

Lecture 1

Lecturer: Asst. Prof. M. Mert Ankarali

1.1 Dynamical Systems

A dynamical system is a collection of “elements” for which there is a time-dependent cause and effect relationships between “variables”. The behavior of a dynamical system changes with time, usually in response to external inputs.

The elements can range from atoms and molecules to oceans and planets. Similarly, time scales can range from pico seconds to years and even decades.

1.1.1 Modeling of Dynamical Systems

Goal: Obtaining a mathematical description of the dynamic relationship between the variables of the given dynamical system/behavior. ***Always***, the mathematical description is a simplification of the real phenomena. Always remember:

“All models are wrong but some are useful” George E. Box, 1978.

Representations of (Continuous-Time) Dynamical Systems

1. Differential equations

$$a_n y^{(n)} + \dots + a_1 y' + a_0 y = b_n u^{(n)} + \dots + b_1 u' + b_0 u$$

Differential equations can model both linear and non-linear systems, time-invariant and time-varying ones.

2. Impulse response representation

$$y(t) = h(t) * u(t) = \int_{-\infty}^{\infty} h(t - \tau) u(\tau) d\tau$$

Impulse-response representation can be used to model the dynamic relation between input, $u(t)$, and output $y(t)$ signals. This representation is limited to Linear-Time-Invariant (LTI) systems.

Time varying impulse response representation can be used to model Linear-Time-Varying (LTV) systems.

3. Transfer function representation

$$Y(s) = G(s)U(s), \text{ where,}$$

$$Y(s) = \mathcal{L}\{y(t)\}, \& U(s) = \mathcal{L}\{u(t)\}$$

Transfer functions are limited to LTI systems.

4. State-space representation

$$\text{Let } x(t) \in \mathbb{R}^n, y(t) \in \mathbb{R}, u(t) \in \mathbb{R},$$

$$\dot{x}(t) = Ax(t) + Bu(t),$$

$$y(t) = Cx(t) + Du(t),$$

$$\text{where } A \in \mathbb{R}^{n \times n}, B \in \mathbb{R}^{n \times 1}, C \in \mathbb{R}^{1 \times n}, D \in \mathbb{R}$$

Similar to differential equations, we can model both linear and non-linear systems, and time-invariant and time-varying systems using state-space representations.

1.1.2 Feedback Systems

The *feedback* terms constitute the case, where dynamical systems are connected in a way that the output of each system influences its own driving input (directly or indirectly), and dynamic relations between different sub-systems are tightly coupled.

Simple causal reasoning about a feedback system is difficult (if not possible). For example, if we observe the feedback system illustrated in Fig. 1.1(B), we see that the first system influences the second, and the second system influences the first, leading to a circular argument. We can conclude that formal methods are necessary to understand the behavior of feedback systems.

Fig. 1.1 illustrates the idea of feedforward vs. feedback in block diagram topology. Open-loop and closed-loop terms are also commonly used to refer to feedforward and feedback structures, respectively. When systems connect each other in a cyclic topology, we refer to the topology as a closed-loop system. If one cuts an interconnection such that the cyclic structure is broken, the system becomes an open-loop system.

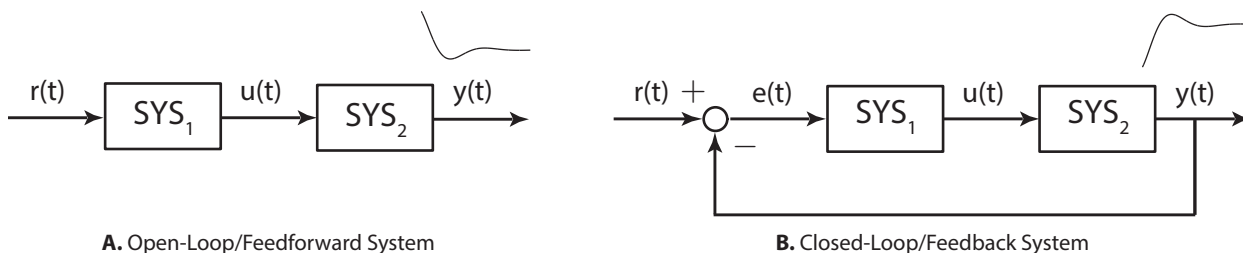


Figure 1.1: Open- and closed-loop systems

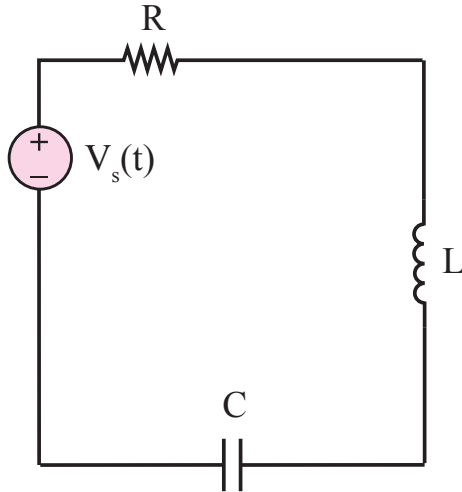
Feedback based control may not be mandatory for some control applications, yet a feedforward based control policy can be more advantageous than a feedback-based control policy in some cases. The core benefit of feedback in a control system is that feedback reduces uncertainties in systems and improves the robustness. Since uncertainties are unavoidable in real life, feedback control systems are ubiquitous in both synthetic and biological control systems.

1.2 Dynamic Modeling of Some Physical Systems

1.2.1 Electrical Circuits

Ex 1: Series RLC Circuit

Given than input is $u(t) = V_s(t)$ and output os $y(t) = V_C(t)$, find an ODE description of the given dynamical circuit.



$$\begin{aligned}
 V_L + V_R + V_C &= V_s(t) \\
 L \frac{dI}{dt} + RI + V_C &= V_s(t) \\
 L \frac{d}{dt} \left(C \frac{dV_C}{dt} \right) + R \left(C \frac{dV_C}{dt} \right) + V_C &= V_s(t) \\
 LC \ddot{V}_C + RC \dot{V}_C + V_C &= V_s(t) \\
 \ddot{y} + \frac{R}{L} \dot{y} + \frac{1}{LC} y &= \frac{1}{LC} u
 \end{aligned}$$

Find the transfer function representation of the system for the given input–output pair.

$$\begin{aligned}
 \mathcal{L} \left\{ \ddot{y} + \frac{R}{L} \dot{y} + \frac{1}{LC} y \right\} &= \mathcal{L} \left\{ \frac{1}{LC} u \right\} \\
 s^2 Y(s) + s \frac{R}{L} Y(s) + \frac{1}{LC} Y(s) &= \frac{1}{LC} U(s) \\
 G(s) = \frac{Y(s)}{U(s)} &= \frac{\frac{1}{LC}}{s^2 + \frac{R}{L}s + \frac{1}{LC}}
 \end{aligned}$$

Find a state-space representation of the system.

Let $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} y \\ \dot{y} \end{bmatrix}$, then

$$\begin{aligned}
 \dot{x}_1 &= x_2 \\
 \dot{x}_2 &= -\frac{1}{LC} x_1 - \frac{R}{L} x_2 + \frac{1}{LC} u
 \end{aligned}$$

If we put the equations in state-space form, we obtain

$$\begin{aligned}
 \dot{x} &= \begin{bmatrix} 0 & 1 \\ -\frac{1}{LC} & -\frac{R}{L} \end{bmatrix} x + \begin{bmatrix} 0 \\ \frac{1}{LC} \end{bmatrix} u \\
 y &= \begin{bmatrix} 1 & 0 \end{bmatrix} x
 \end{aligned}$$

where

$$A = \begin{bmatrix} 0 & 1 \\ -\frac{1}{LC} & -\frac{R}{L} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \frac{1}{LC} \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0$$

Now let, $z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} V_C \\ I \end{bmatrix}$, then

$$\begin{aligned} \dot{z}_1 &= \frac{1}{C} z_2 \\ \dot{z}_2 &= -\frac{1}{L} z_1 - \frac{R}{L} z_2 + \frac{1}{L} u \end{aligned}$$

If we put the equations in state-space form, we obtain

$$\begin{aligned} \dot{z} &= \begin{bmatrix} 0 & \frac{1}{C} \\ -\frac{1}{L} & -\frac{R}{L} \end{bmatrix} z + \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix} u \\ y &= \begin{bmatrix} 1 & 0 \end{bmatrix} z \end{aligned}$$

It can be seen that state-space representation of a dynamical system is not unique. Indeed there exist infinitely many state-space representations of the same system.

Lecture 2

Lecturer: Asst. Prof. M. Mert Ankarali

Conversion Between Different LTI Representations

In this lecture we will cover the conversion between different LTI representations.

2.1 ODE to TF & TF to ODE

Conversion between ODE and TF representations is trivial in both directions

$$y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_1y' + a_0y = b_nu^{(n)} + \dots + b_1u' + b_0u$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_ns^n + \dots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0}$$

2.2 State-Space to TF

Note that a SS representation of an n^{th} order LTI system has the form below.

$$\begin{aligned} \text{Let } x(t) &\in \mathbb{R}^n, \quad y(t) \in \mathbb{R}, \quad u(t) \in \mathbb{R}, \\ \dot{x}(t) &= Ax(t) + Bu(t), \\ y(t) &= Cx(t) + Du(t), \\ \text{where } A &\in \mathbb{R}^{n \times n}, \quad B \in \mathbb{R}^{n \times 1}, \quad C \in \mathbb{R}^{1 \times n}, \quad D \in \mathbb{R} \end{aligned}$$

In order to convert state-space to transfer function, we start with taking the Laplace transform of the both sides of the state-equation

$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t) \\ sX(s) &= AX(s) + BU(s) \\ sX(s) - AX(s) &= BU(s) \\ (sI - A)X(s) &= BU(s) \\ X(s) &= (sI - A)^{-1} BU(s) \end{aligned}$$

Now let's concentrate on the output equation

$$\begin{aligned} y(t) &= Cx(t) + Du(t) \\ Y(s) &= \left[C(sI - A)^{-1} B + D \right] U(s) \\ G(s) &= C(sI - A)^{-1} B + D \end{aligned}$$

2.3 ODE/TF to State-Space

Note that for a given LTI system, there exist infinitely many different SS representations. In this part, we learn two different ways converting a TF/ODE into State-Space form. For the sake of clarity, we will derive the realization for a general 3rd order LTI system.

2.3.1 Canonical Realization I

In this method of realization, we will use the fact the system is LTI. Let's consider the transfer function of the system and let's perform some LTI operations.

$$\begin{aligned}
 Y(s) &= \frac{b_3 s^3 + b_2 s^2 + b_1 s + b_0}{s^3 + a_2 s^2 + a_1 s + a_0} U(s) \\
 &= (b_3 s^3 + b_2 s^2 + b_1 s + b_0) \frac{1}{s^3 + a_2 s^2 + a_1 s + a_0} U(s) \\
 &= G_2(s) G_1(s) U(s) \text{ where} \\
 G_1(s) &= \frac{H(s)}{U(s)} = \frac{1}{s^3 + a_2 s^2 + a_1 s + a_0} \\
 G_2(s) &= \frac{Y(s)}{H(s)} = b_3 s^3 + b_2 s^2 + b_1 s + b_0
 \end{aligned}$$

As you can see we introduced an intermediate variable $h(t)$ or with a Laplace transform of $H(s)$. First transfer function has static input dynamics, operates on $x(t)$, and produces an output, i.e. $h(t)$. Second transfer function is a non-causal system and operates on $h(t)$ and produces output $x(t)$. If we write the ODEs of both systems we obtain

$$\begin{aligned}
 \ddot{h} &= -a_2 \dot{h} - a_1 h - a_0 h + u \\
 y &= b_3 \ddot{h} + b_2 \dot{h} + b_1 \dot{h} + b_0 h
 \end{aligned}$$

Now let the state-variables be $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} h \\ \dot{h} \\ \ddot{h} \end{bmatrix}$. Then, individual state equations take the form

$$\begin{aligned}
 \dot{x}_1 &= x_2 \\
 \dot{x}_2 &= x_3 \\
 \dot{x}_3 &= -a_2 x_3 - a_1 x_2 - a_0 x_1 + u
 \end{aligned}$$

and the output equation take the form

$$\begin{aligned}
 y &= b_3 (-a_2 x_3 - a_1 x_2 - a_0 x_1 + u) + b_2 x_3 + b_1 x_2 + b_0 x_1 \\
 &= (b_0 - b_3 a_0) x_1 + (b_1 - b_3 a_1) x_2 + (b_2 - b_3 a_2) x_3 + b_3 u
 \end{aligned}$$

If we re-write the equations in matrix form we obtain the state-space representation as

$$\begin{aligned}
 \dot{x} &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \\
 y &= \begin{bmatrix} (b_0 - b_3 a_0) & (b_1 - b_3 a_1) & (b_2 - b_3 a_2) \end{bmatrix} x + [b_3] u
 \end{aligned}$$

If we obtain a state-space model from this approach, the form will be in *controllable canonical form*. We will cover this later in the semester. Thus we can call this representation also as *controllable canonical realization*.

For a general n^{th} order system controllable canonical form has the following A , B , C , & D matrices

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

$$C = \begin{bmatrix} (b_0 - b_n a_0) & (b_1 - b_n a_1) & \cdots & (b_{n-1} - b_n a_{n-1}) \end{bmatrix}, \quad D = b_n$$

2.3.2 Canonical Realization II

In this method will obtain a different minimal state-space realization. The process will be different and state-space structure will have a different topology. Let's start with the transfer function and perform some grouping based on the s elements.

$$Y(s) = \frac{b_3 s^3 + b_2 s^2 + b_1 s + b_0}{s^3 + a_2 s^2 + a_1 s + a_0} U(s)$$

$$Y(s) (s^3 + a_2 s^2 + a_1 s + a_0) = (b_3 s^3 + b_2 s^2 + b_1 s + b_0) U(s)$$

$$s^3 Y(s) = b_3 s^3 U(s) + s^2 (-a_2 Y(s) + b_2 U(s)) + s (-a_1 Y(s) + b_1 U(s)) + (-a_0 Y(s) + b_0 U(s))$$

Let's multiply both sides with $\frac{1}{s^3}$ and perform further grouping

$$Y(s) = b_3 U(s) + \frac{1}{s} (-a_2 Y(s) + b_2 U(s)) + \frac{1}{s^2} (-a_1 Y(s) + b_1 U(s)) + \frac{1}{s^3} (-a_0 Y(s) + b_0 U(s))$$

$$Y(s) = b_3 U(s) + \frac{1}{s} \left[(-a_2 Y(s) + b_2 U(s)) + \frac{1}{s} \left\{ (-a_1 Y(s) + b_1 U(s)) + \frac{1}{s} (-a_0 Y(s) + b_0 U(s)) \right\} \right]$$

Let the Laplace domain representations of state variables $X(s) = \begin{bmatrix} X_1(s) \\ X_2(s) \\ X_3(s) \end{bmatrix}$ defined as

$$X_1(s) = \frac{1}{s} (-a_0 Y(s) + b_0 U(s))$$

$$X_2(s) = \frac{1}{s} \left\{ (-a_1 Y(s) + b_1 U(s)) + \frac{1}{s} (-a_0 Y(s) + b_0 U(s)) \right\}$$

$$X_3(s) = \frac{1}{s} \left[(-a_2 Y(s) + b_2 U(s)) + \frac{1}{s} \left\{ (-a_1 Y(s) + b_1 U(s)) + \frac{1}{s} (-a_0 Y(s) + b_0 U(s)) \right\} \right]$$

In this context output equation in s and time domains simply takes the form

$$Y(s) = X_3(s) + b_3 U(s) \quad \rightarrow \quad y(t) = x_3(t) + b_3 u(t)$$

Dependently the state equations (in s and time domains) take the form

$$sX_1(s) = -a_0 X_3(s) + (b_0 - a_0 b_3) U(s) \quad \rightarrow \quad \dot{x}_1 = -a_0 x_3 + (b_0 - a_0 b_3) u$$

$$sX_2(s) = X_1(s) - a_1 X_3(s) + (b_1 - a_1 b_3) U(s) \quad \rightarrow \quad \dot{x}_2 = x_1 - a_1 x_3 + (b_1 - a_1 b_3) u$$

$$sX_3(s) = X_2(s) - a_2 X_3(s) + (b_2 - a_2 b_3) U(s) \quad \rightarrow \quad \dot{x}_3 = x_2 - a_2 x_3 + (b_2 - a_2 b_3) u$$

If we re-write all equations in matrix form, we obtain the state-space representation as

$$\dot{x} = \begin{bmatrix} 0 & 0 & -a_0 \\ 1 & 0 & -a_1 \\ 0 & 1 & -a_2 \end{bmatrix} x + \begin{bmatrix} (b_0 - b_3 a_0) \\ (b_1 - b_3 a_1) \\ (b_2 - b_3 a_2) \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} x + [b_3] u$$

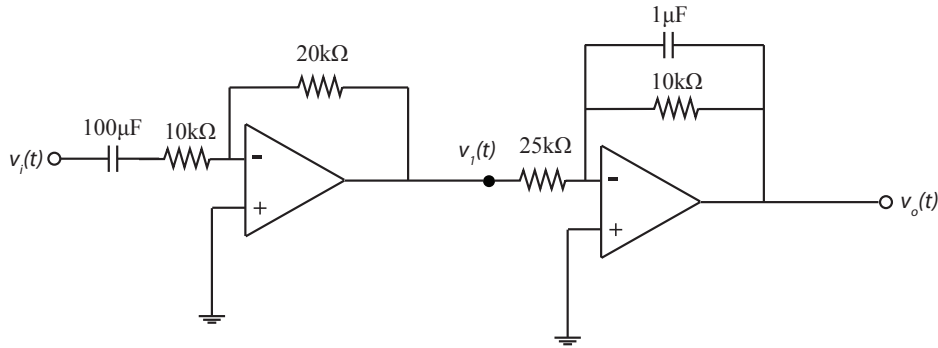
If we obtain a state-space model from this method, the form will be in *observable canonical form*. Thus we can call this representation also as *observable canonical realization*. This form and representation is the dual of the previous representation.

For a general n^{th} order system observable canonical form has the following A , B , C , & D matrices

$$A = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & -a_{n-2} \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{bmatrix}, \quad B = \begin{bmatrix} (b_0 - b_n a_0) \\ (b_1 - b_n a_1) \\ \vdots \\ (b_{n-2} - b_n a_{n-2}) \\ (b_{n-1} - b_n a_{n-1}) \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 0 & \cdots & 0 & 1 \end{bmatrix}, \quad D = b_n$$

Example:



1. Given that $u(t) = v_i(t)$ and $y(t) = v_o(t)$, compute the transfer function for the given (ideal) OPAMP circuit

Solution:

First let's compute $V_1(s)/U(s)$

$$\frac{U(s)}{\frac{10^4}{s} + 10^4} = \frac{-V_1(s)}{2 \cdot 10^4}$$

$$\frac{V_1(s)}{U(s)} = \frac{-2s}{s + 1}$$

Now let's compute $Y(s)/V_1(s)$

$$\begin{aligned}\frac{V_1(s)}{2.5 \cdot 10^4} &= -Y(s) (s \cdot 10^{-6} + 10^{-4}) \\ \frac{V_1(s)}{2.5 \cdot 10^4} &= -Y(s) (s + 100) 10^{-6} \\ \frac{Y(s)}{V_1(s)} &= \frac{-40}{s + 100}\end{aligned}$$

Hence, the transfer function of the whole system/circuit can be found as

$$\begin{aligned}G(s) &= \frac{Y(s)}{U(s)} = \frac{Y(s)}{V_1(s)} \frac{V_1(s)}{U(s)} \\ &= \frac{80s}{(s + 100)(s + 1)} \\ &= \frac{80s}{s^2 + 101s + 100}\end{aligned}$$

2. Now find a state-space representation from the given TF.

Solution:

If we follow the derivation of controllable canonical form for a second order system we obtain the following structure

$$\begin{aligned}\dot{x} &= \begin{bmatrix} 0 & 1 \\ -a_0 & -a_1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \\ y &= \begin{bmatrix} (b_0 - b_2 a_0) & (b_1 - b_2 a_1) \end{bmatrix} x + [b_2] u\end{aligned}$$

where

$$a_0 = 100, \quad a_1 = 101, \quad b_0 = 0, \quad b_1 = 80, \quad \& \quad b_2 = 0$$

Thus, the state-space representation takes the form

$$\begin{aligned}\dot{x} &= \begin{bmatrix} 0 & 1 \\ -100 & -101 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \\ y &= \begin{bmatrix} 0 & 80 \end{bmatrix} x\end{aligned}$$

3. Now re-compute the TF from the given state-space representation

Solution:

$$\begin{aligned}G(s) &= C(sI - A)^{-1}B + D \\ &= \begin{bmatrix} 0 & 80 \end{bmatrix} \begin{bmatrix} s & -1 \\ 100 & s + 101 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 80 \end{bmatrix} \frac{1}{s^2 + 101s + 100} \begin{bmatrix} s + 101 & 1 \\ -100 & s \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 80 \end{bmatrix} \begin{bmatrix} 1 \\ s \end{bmatrix} \frac{1}{s^2 + 101s + 100} \\ &= \frac{80s}{s^2 + 101s + 100}\end{aligned}$$

Take Home Problem:

1. First find state space representations of the sub-system transfer functions, i.e. $\frac{V_1(s)}{U(s)}$ and $\frac{Y(s)}{V_1(s)}$, separately.
2. Then combine the state-space representations of the sub-systems to find a state-space representation for the whole system.
3. Compute the TF from the computed state-space representation and compare it to the previous results.

EE302 - Feedback Systems

Spring 2019

Lecture 3

Lecturer: Asst. Prof. M. Mert Ankarali

3.1 Modeling of Mechanical Systems and Their Electrical Analogy

We use Kirchhoff's Current and Voltage laws to derive the dynamical (and static) relationships in Electrical Circuits. Similarly, we utilize Newton's laws of motion to derive equations of motion in (rigid body) mechanical dynamical systems.

3.1.1 Mechanical vs. Electrical Analogy Between Dependent Variables

There exist two different analogies that we can construct between electrical and mechanical systems. Mathematically, there is no difference between the two approaches. In this lecture, we will learn one of these analogies.

In electrical circuits, the core variables are Voltage, V , and current, I , whereas in translational mechanical systems, core variables are translational velocity, ν , and force, f . Similarly, in rotational mechanical systems, the core variables are angular velocity, ω , and torque τ .

Voltage, $V \iff$ Velocity, $\nu \iff$ Angular Velocity, ω

In electrical systems, voltage, also called electric potential difference, accounts for the difference in electric potential between two points. When we refer to the voltage of a node/point, we always measure it with respect to a reference point, e.g. *ground*. In mechanical systems, we measure the velocity either between two points in space, or (which is more general) with respect to an inertial reference frame, e.g. *ground* or *earth* in general. The analogy is similar with angular velocity. For this reason, we say that voltage, linear velocity, and angular velocity are the analog variables.

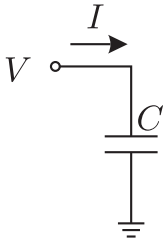
Current, $I \iff$ force, $f \iff$ Torque, τ

In electrical systems, the current is the flow (or rate of change of) of electric charge and carried by electrons in motion. Roughly speaking, based on Newton's second law, the force acting upon a (rigid) body is equal to the rate of change of momentum related to this specific force component. Momentum can be considered as an analog of the electrical charge in this case. A similar analogy can also be constructed using Torque and Angular Momentum. For this reason, we say that Current, Force, and Torque are the analog variables.

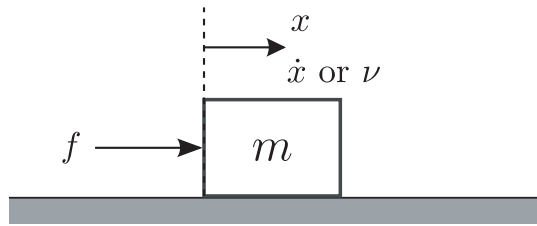
3.1.2 Capacitor C , 1-DOF Translating Body with Mass m , and 1-DOF Rotating Body with Inertia J

If we follow the analogs between the variables, we can see that ideal capacitor for which one end is connected to the ground, 1-DOF translating body with a mass of m , and 1-DOF rotating body with an inertia of J are analogs of each other. These are all ideal energy storage elements in their modeling domains and they are illustrated in the figure below.

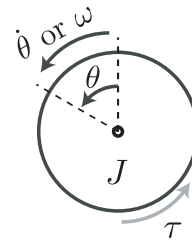
Ideal Capacitor



Ideal 1-DOF Translating Body (m)



Ideal 1-DOF Rotating Body (J)



The ODEs that govern the dynamics of these elements are provided below

$$\text{Capacitor : } C\dot{V}(t) = I(t)$$

$$\text{Mass : } m\dot{\nu} = f(t) \text{ (or } m\ddot{x} = f(t))$$

$$\text{Inertia : } J\dot{\omega} = \tau(t) \text{ (or } J\ddot{\theta} = \tau(t))$$

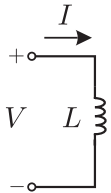
Based on these equations we can reach the following (system) parameter analogy as

$$C \equiv m \equiv J$$

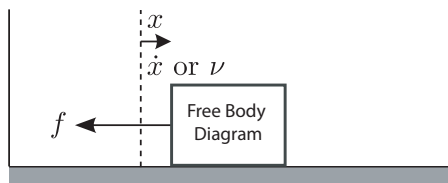
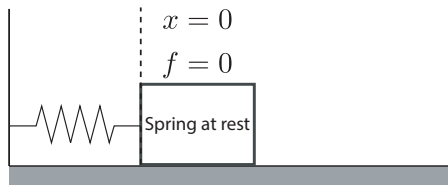
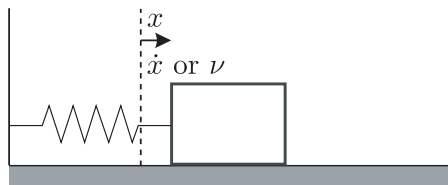
3.1.3 Inductor L , Translational Spring k , and Torsional Spring κ

If we follow the analogs between the variables we can see that Ideal inductor (L), linear translational spring (k), and linear torsional spring (κ) are analogs of each other. These are also ideal energy storage elements in their modeling domains and they are illustrated in the figure below.

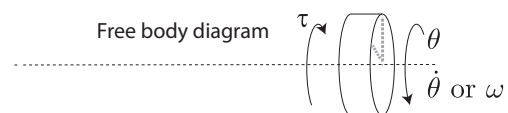
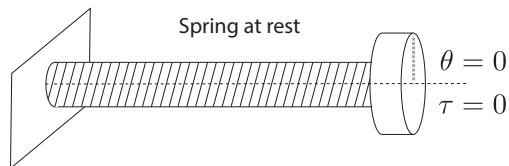
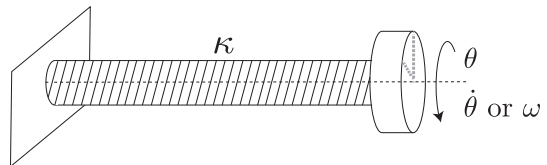
Ideal Inductor



Ideal Linear Spring



Ideal Torsional Spring



The ODEs that govern the dynamics of these elements are provided below

$$\text{Induction : } L\dot{I}(t) = V(t)$$

$$\text{TranslationalSpring : } f(t) = kx(t) \rightarrow \frac{1}{k}\dot{f}(t) = \nu(t)$$

$$\text{TorsionalSpring : } \tau(t) = \kappa\theta(t) \rightarrow \frac{1}{\kappa}\dot{\tau}(t) = \omega(t)$$

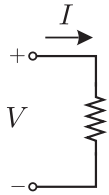
Based on these equations we can reach the following (system) parameter analogy as

$$L \equiv \frac{1}{k} \equiv \frac{1}{\kappa}$$

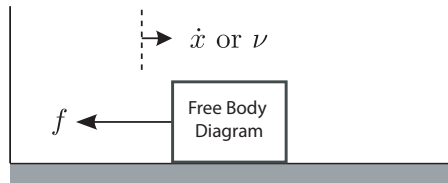
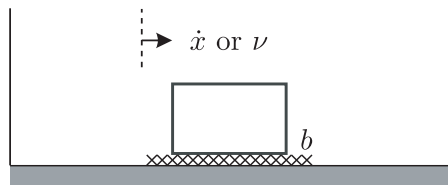
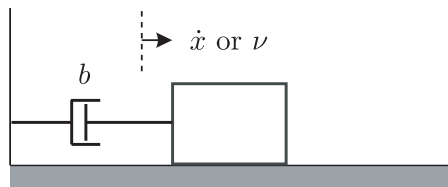
3.1.4 Resistor R , Damper (Viscous Friction) b , and Torsional Damper β

If we follow the analogs between the variables we can see that Ideal resistor (R), linear translational damper (k), and linear torsional damper (κ) are analogs of each other. These elements are ideal fully passive dissipative elements. Thus, these are memoryless (static) components as opposed to the previous elements. These elements are illustrated in the figure below.

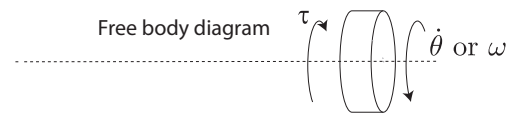
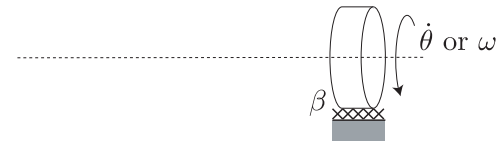
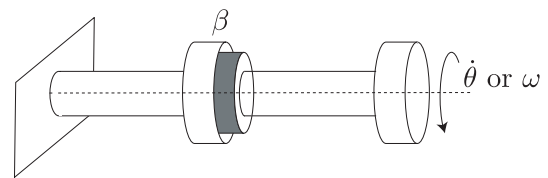
Ideal Resistor



Ideal Linear Damper (Viscous Friction)



Ideal Torsional Damper



The algebraic equations that govern the statics of these elements are provided below

$$\text{Resistor : } V(t) = RI(t)$$

$$\text{Translational Damper : } f(t) = b\dot{x}(t) \text{ or } \frac{1}{b}f(t) = \nu(t)$$

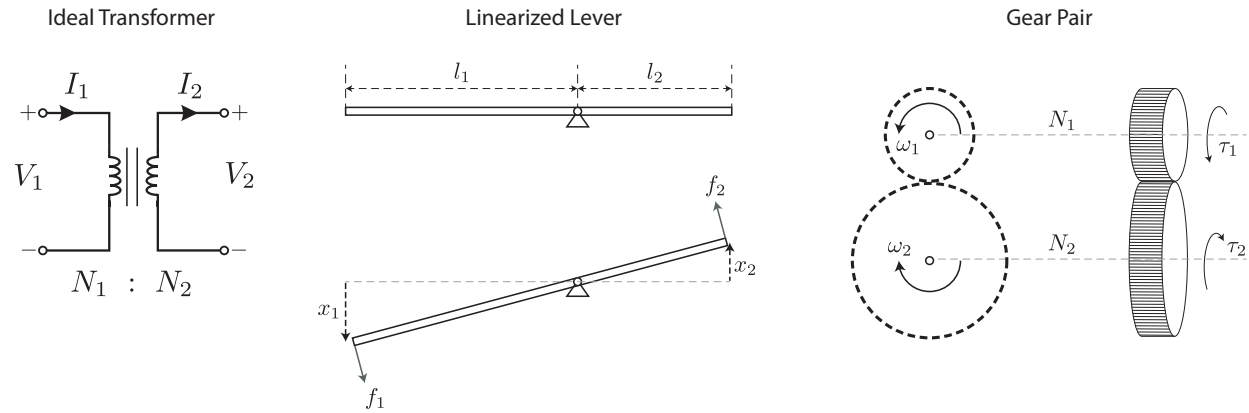
$$\text{Torsional Damper : } \tau(t) = \beta\dot{\theta}(t) \rightarrow \frac{1}{\beta}\tau(t) = \omega(t)$$

Based on these equations we can reach the following (system) parameter analogy as

$$R \equiv \frac{1}{b} \equiv \frac{1}{\beta}$$

3.1.5 Ideal Transformer, Linearized Lever, and Gear Pair

In both electrical and mechanical systems, we have transmission elements. In their ideal form, they conserve the energy after the transformation. In electrical systems, transformer is the component that achieves the transmission. In translational mechanical systems a linearized lever can achieve this under the assumption of small movements, where as for rotational systems a gear pair is one of the many solutions for mechanical transmission. These components are illustrated in the figure below.



The algebraic equations that govern the statics of these elements are provided below

$$\begin{aligned} \text{Electrical Transformer :} \quad & \frac{V_1}{N_1} = \frac{V_2}{N_2} \\ & I_1 N_1 = I_2 N_2 \end{aligned}$$

$$\begin{aligned} \text{Lever :} \quad & \frac{\nu_1}{l_1} = \frac{\nu_2}{l_2} \\ & f_1 l_1 = f_2 l_2 \end{aligned}$$

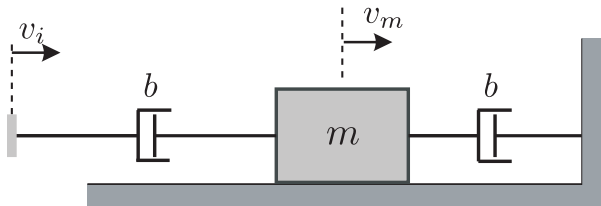
$$\begin{aligned} \text{Gear - Pair :} \quad & \frac{\omega_1}{r_2} = \frac{\omega_2}{r_1} \\ & \tau_1 r_2 = \tau_2 r_1 \end{aligned}$$

Based on these equations, it is to see that following (system) parameter analogy as

$$\frac{N_1}{N_2} \equiv \frac{l_1}{l_2} \equiv \frac{r_2}{r_1}$$

3.2 Examples

Ex 3.1. Let's consider the following translational mechanical system. The input of the system is $v_i(t)$ which is the velocity of the one side of the first damper. The output of the system is $v_m(t)$, i.e. the velocity of the mass.

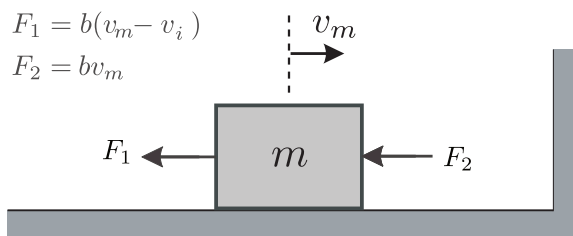


1. Given that $u(t) = v_i(t)$ and $y(t) = v_m(t)$, find the transfer function $G(s) = \frac{Y(s)}{U(s)}$

Let's first draw the FBD and then derive the equations of motion

$$F_1 = b(v_m - v_i)$$

$$F_2 = bv_m$$



$$m\dot{v}_m = -F_1 - F_2 = bv_i - 2bv_m$$

$$m\dot{y} + 2by = bu$$

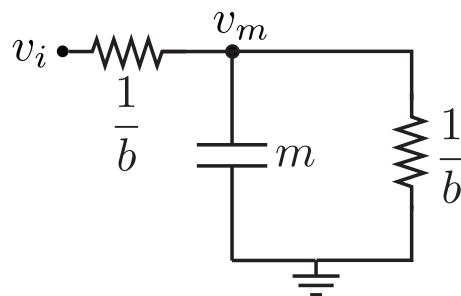
$$G(s) = \frac{Y(s)}{U(s)} = \frac{b}{ms + 2b} = \frac{b/m}{s + (2b)/m}$$

2. Now, let's construct an electrical circuit analog of the mechanical system.

Let

$$V_i \equiv v_i \quad I_i \equiv f_i$$

then we can build the circuit analog as in the illustration below.



3. Finally, compute $G(s)$ by analyzing the electrical analog circuit.

We simply apply KCL

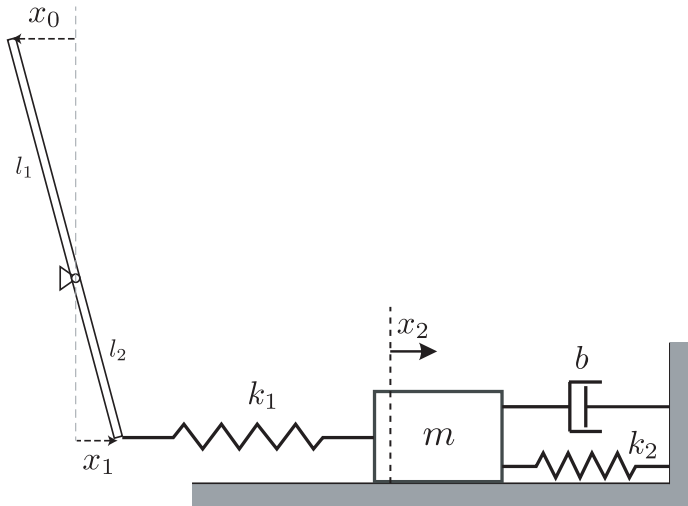
$$\frac{v_i - v_m}{1/b} = I_m + \frac{v_m}{1/b} \Rightarrow I_m = m\dot{v}_m$$

$$bv_i = m\dot{v}_m + 2bv_m \Rightarrow bu = m\dot{y} + 2by$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b/m}{s + (2b)/m}$$

Ex 3.2. Let's consider the following translational mechanical system. It is given that when the lever is in vertical position, $[x_0 \ x_1 \ x_2] = 0$ and springs are at their rest length positions.

1. Given that $u(t) = x_0(t)$ and $y(t) = x_2(t)$, find the ODE of the system dynamics.



$$m\ddot{x}_2 = -b\dot{x}_2 - k_2x_2 + k_1(x_1 - x_2)$$

$$x_1 = x_0 \frac{l_2}{l_1}$$

$$m\ddot{x}_2 + b\dot{x}_2 + (k_1 + k_2)x_2 = k_1 \frac{l_2}{l_1} x_0$$

$$\ddot{y} + \frac{b}{m}\dot{y} + \frac{k_1 + k_2}{m}y = \frac{k_1}{m} \frac{l_2}{l_1} u$$

2. Find the transfer function for same input-output pair.

$$G(s) = \frac{Y(s)}{U(s)} = \frac{\frac{k_1 l_2}{m l_1}}{s^2 + \frac{b}{m}s + \frac{k_1 + k_2}{m}}$$

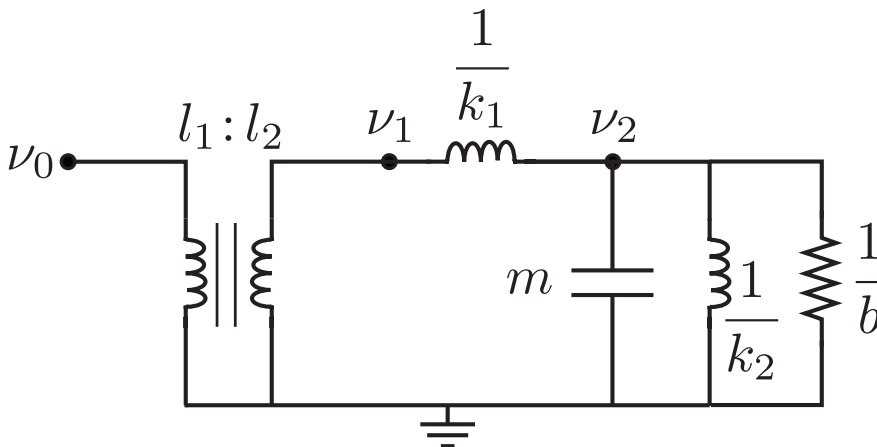
3. Now, let's construct an electrical circuit analog of the mechanical system.

Let

$$V_i \equiv \dot{x}_i$$

$$I_i \equiv f_i$$

then we can build the circuit analog as in the illustration below.



Let's also compute $\frac{V_2(s)}{V_1(s)}$ using node voltage analysis in impedance domain.

$$\begin{aligned}\frac{\mathcal{V}_2(s) - \mathcal{V}_1(s)}{\frac{s}{k_1}} + \frac{\mathcal{V}_2(s)}{\frac{1}{ms}} + \frac{\mathcal{V}_2(s)}{\frac{s}{k_2}} + \frac{\mathcal{V}_2(s)}{\frac{1}{b}} &= 0 \\ \mathcal{V}_2(s) \left[ms + b + \frac{k_1 + k_2}{s} \right] &= \mathcal{V}_1(s) \left[\frac{k_1}{s} \right] \\ \mathcal{V}_2(s) [ms^2 + bs + (k_1 + k_2)] &= \mathcal{V}_1(s) [k_1]\end{aligned}$$

Since ideal transformer has the following relation, $\mathcal{V}_1(s) = \frac{l_2}{l_1} \mathcal{V}_0(s)$, we have the following transfer function between $\mathcal{V}_0(s)$ and $\mathcal{V}_2(s)$

$$\frac{\mathcal{V}_2(s)}{\mathcal{V}_0(s)} = \frac{\frac{k_1}{m} \frac{l_2}{l_1}}{s^2 + \frac{b}{m}s + \frac{k_1 + k_2}{m}}$$

Obviously this transfer function is equal to $G(s)$ computed from directly mechanical system and considering positional variables.

4. Convert the derived ODE into a state-space form

We will solve the problem using a different approach. First let's integrate the ODE twice

$$y = \int \left[-\frac{b}{m}y + \int \left\{ -\frac{k_1 + k_2}{m}y + \frac{k_1}{m} \frac{l_2}{l_1} u \right\} dt \right] dt$$

Then let the state variable definitions be

$$\begin{aligned}x_1 &= y = \int \left[-\frac{b}{m}y + \int \left\{ -\frac{k_1 + k_2}{m}y + \frac{k_1}{m} \frac{l_2}{l_1} u \right\} dt \right] dt \\ x_2 &= \int \left\{ -\frac{k_1 + k_2}{m}y + \frac{k_1}{m} \frac{l_2}{l_1} u \right\} dt\end{aligned}$$

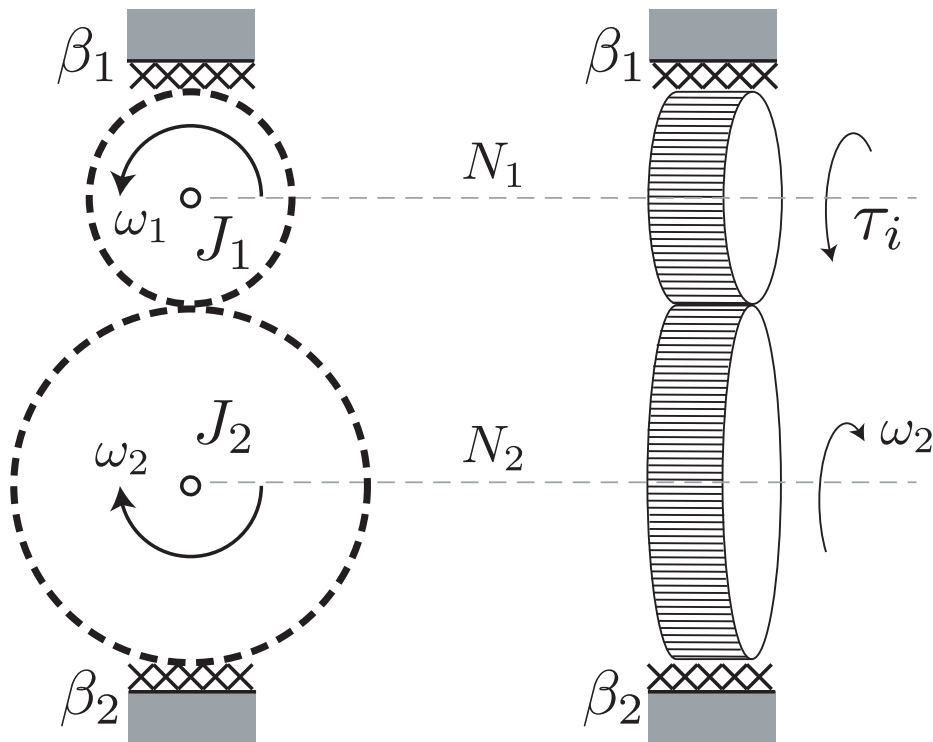
Then state-equations take the form

$$\begin{aligned}\dot{x}_1 &= \left[-\frac{b}{m}y + \int \left\{ -\frac{k_1 + k_2}{m}y + \frac{k_1}{m} \frac{l_2}{l_1} u \right\} dt \right] \\ &= -\frac{b}{m}x_1 + x_2 \\ \dot{x}_2 &= \left\{ -\frac{k_1 + k_2}{m}y + \frac{k_1}{m} \frac{l_2}{l_1} u \right\} \\ &= -\frac{k_1 + k_2}{m}x_1 + \frac{k_1}{m} \frac{l_2}{l_1} u\end{aligned}$$

If we gather the obtained state-equations in matrix form we obtain the following state-space representation

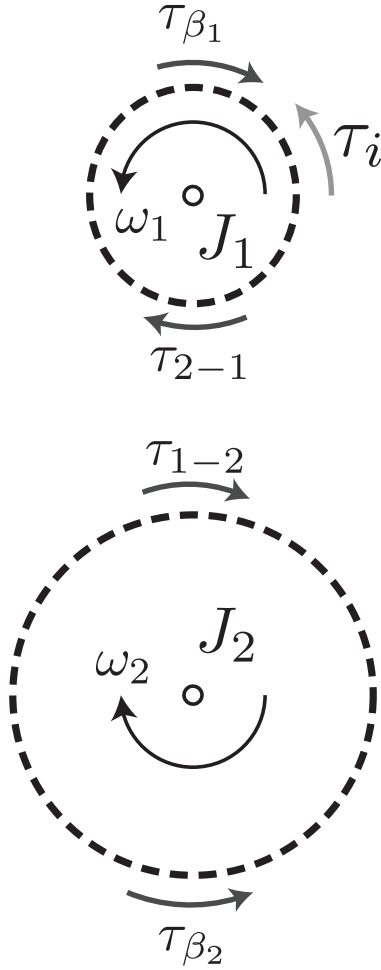
$$\begin{aligned}\dot{x} &= \begin{bmatrix} -\frac{b}{m} & 1 \\ -\frac{k_1 + k_2}{m} & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ \frac{k_1}{m} \frac{l_2}{l_1} \end{bmatrix} u \\ y &= \begin{bmatrix} 1 & 0 \end{bmatrix} x\end{aligned}$$

Ex 3.3. Let's consider the following gear system. Unlike ideal gear pair case, now each gear has its own inertia, J_1 and J_2 , as well as both gears are affected by viscous friction, β_1 and β_2 , due to mechanical contact with the environment.



1. Given that there is an external torque, τ_i , acting on the first gear is the input of the system, and the rotational speed of the second gear, ω_2 , is the output of the system, find the ODE of the gear-box dynamics.

First let's draw the free-body diagrams of both gears separately and then write the equations of motion for each body.



$$\begin{aligned}
 J_1 \dot{\omega}_1 &= \tau_i - \tau_{\beta_1} - \tau_{2-1} \\
 &= \tau_i - \beta_1 \omega_1 - \tau_{2-1} \\
 J_2 \dot{\omega}_2 &= -\tau_{\beta_2} + \tau_{1-2} \\
 &= -\beta_2 \omega_2 + \tau_{1-2}
 \end{aligned}$$

Based on the gear kinematics we know that

$$\begin{aligned}
 \omega_1 &= \frac{N_2}{N_1} \omega_2 \\
 \tau_{1-2} &= \frac{N_2}{N_1} \tau_{2-1}
 \end{aligned}$$

Thus we have the following derivations

$$\begin{aligned}
 \tau_{2-1} &= \tau_i - J_1 \frac{N_2}{N_1} \dot{\omega}_2 - \beta_1 \frac{N_2}{N_1} \omega_2 \\
 \tau_{1-2} &= \frac{N_2}{N_1} \tau_i - J_1 \left(\frac{N_2}{N_1} \right)^2 \dot{\omega}_2 - \beta_1 \left(\frac{N_2}{N_1} \right)^2 \omega_2
 \end{aligned}$$

Hence the ODE governing the dynamics can be formed as

$$\left[J_2 + J_1 \left(\frac{N_2}{N_1} \right)^2 \right] \dot{y} + \left[\beta_2 + \beta_1 \left(\frac{N_2}{N_1} \right)^2 \right] y = \frac{N_2}{N_1} u$$

It can be seen that the resultant ODE is a first order ODE. Ee can also consider the whole system as a single rotating body with an effective total inertia of $J_T = \left[J_2 + J_1 \left(\frac{N_2}{N_1} \right)^2 \right]$ and effective total viscous friction of $\beta_T = \left[\beta_2 + \beta_1 \left(\frac{N_2}{N_1} \right)^2 \right]$.

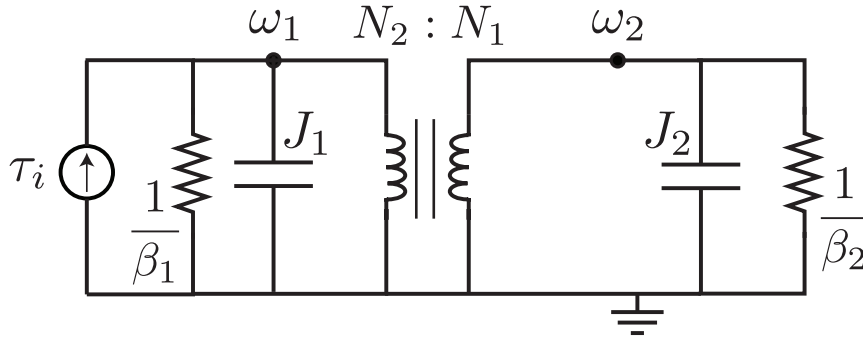
Take Home Problem: Now let's assume that output is $y(t) = \omega_1(t)$, and govern the ODE and re-compute the new effective inertia and viscous friction.

2. Compute the transfer function

$$\begin{aligned}
 G(s) &= \frac{Y(s)}{U(s)} = \frac{\frac{N_2}{N_1}}{\left[J_2 + J_1 \left(\frac{N_2}{N_1} \right)^2 \right] s + \left[\beta_2 + \beta_1 \left(\frac{N_2}{N_1} \right)^2 \right]} \\
 &= \frac{\frac{N_2}{N_1}}{J_T s + \beta_T} = \frac{\frac{N_2}{N_1} \frac{1}{J_T}}{s + \frac{\beta_T}{J_T}}
 \end{aligned}$$

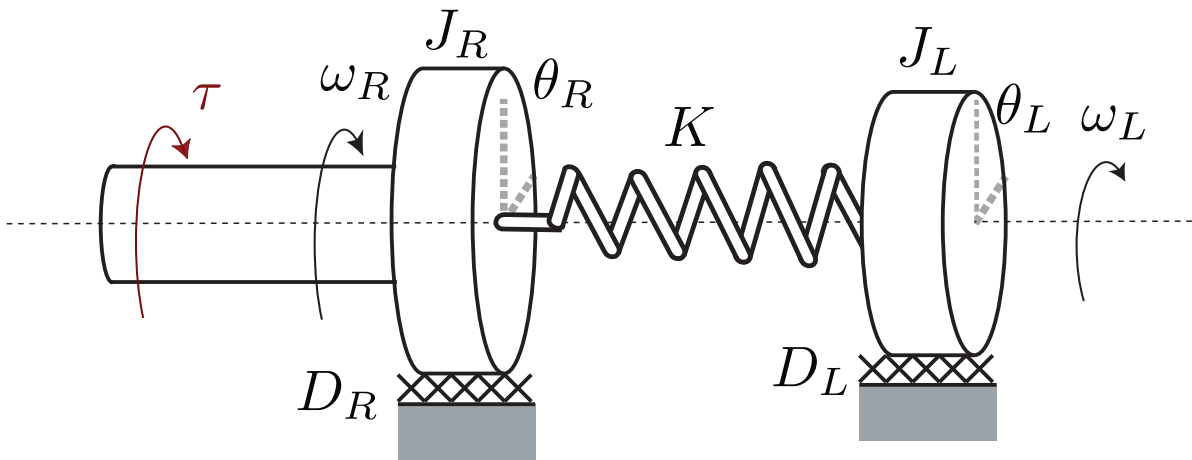
3. Now, build the electrical circuit analog of the gear-box system

The circuit diagram is given below



Take Home Problem: Solve the electrical circuit and compute the transfer function $G(s)$

Ex 3.4. The figure below illustrates the model of the target system. Input of the system is the external torque, $u(t) = \tau(t)$, acting on the R axis and the output of the system is the angular velocity of the Load, i.e. $y(t) = \omega_L(t)$



Let's use the following state-definition

$$x(t) = \begin{bmatrix} \omega_L \\ \omega_R \\ \theta_R - \theta_L \end{bmatrix}$$

and then find a state-space representation for the system under this assumption. Indeed, since output is simply equal to one of the states, output equation is trivial

$$y(t) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x(t) + [0]u(t)$$

Let's first write the differential equation on the motor axis

$$J_R \cdot \dot{\omega}_R = \tau - D_R \omega_R - \kappa(\theta_R - \theta_L)$$

If we replace the variables with state variables, we obtain.

$$\begin{aligned} J_R \cdot \dot{x}_2 &= u - D_R x_2 - K x_3 \\ \dot{x}_2 &= \frac{1}{J_R} u - \frac{D_R}{J_R} x_2 - \frac{\kappa}{J_R} x_3 \end{aligned}$$

Now let's write the differential equation on the load axis and perform the same change of variables operation

$$\begin{aligned} J_L \cdot \dot{\omega}_L &= \kappa(\theta_R - \theta_L) - D_L \omega_L \\ \dot{x}_1 &= \frac{K}{J_L} x_3 - \frac{D_L}{J_L} x_1 \end{aligned}$$

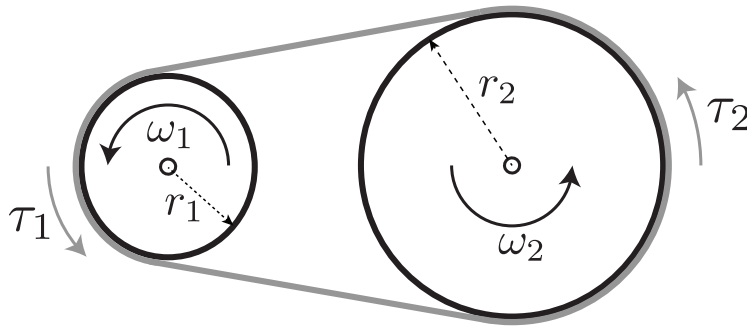
We need one more state-equation which can be derived as

$$\dot{x}_3 = \frac{d}{dt} (\theta_R - \theta_L) = \omega_R - \omega_L = x_2 - x_1$$

Based on our choice of state-definition, full state-space representation takes the form

$$\begin{aligned} \dot{x}(t) &= \begin{bmatrix} -\frac{D_L}{J_L} & 0 & \frac{\kappa}{J_L} \\ 0 & -\frac{D_R}{J_R} & -\frac{\kappa}{J_R} \\ -1 & 1 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ \frac{1}{J_R} \\ 0 \end{bmatrix} u(t) \\ y(t) &= [1 \ 0 \ 0] x(t) + [0] u(t) \end{aligned}$$

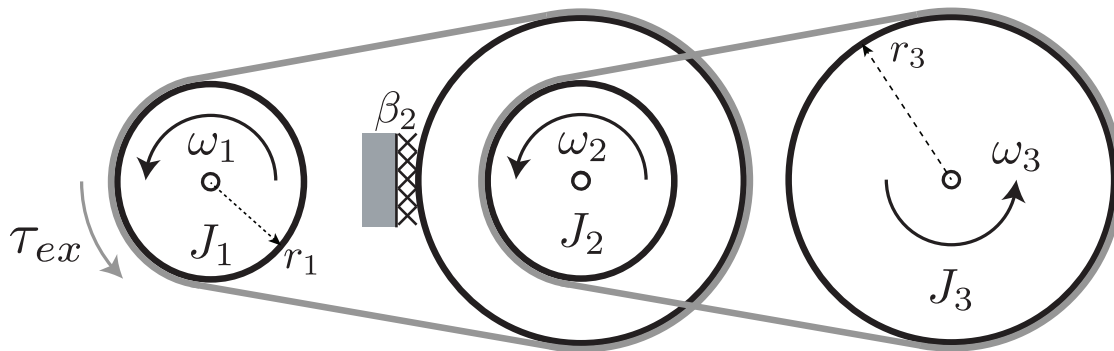
Ex 3.5. The mechanism illustration given below is an ideal belt-pulley mechanism. Fundamentally, it has the same kinematic relations with a gear pair system. The only difference is that, the direction of motion is preserved in a pulley system. r_1 and r_2 correspond to the radii of the first and second pulleys respectively.



$$\begin{aligned} \frac{\omega_1}{r_2} &= \frac{\omega_2}{r_1} \\ \frac{\tau_1}{r_1} &= \frac{\tau_2}{r_2} \end{aligned}$$

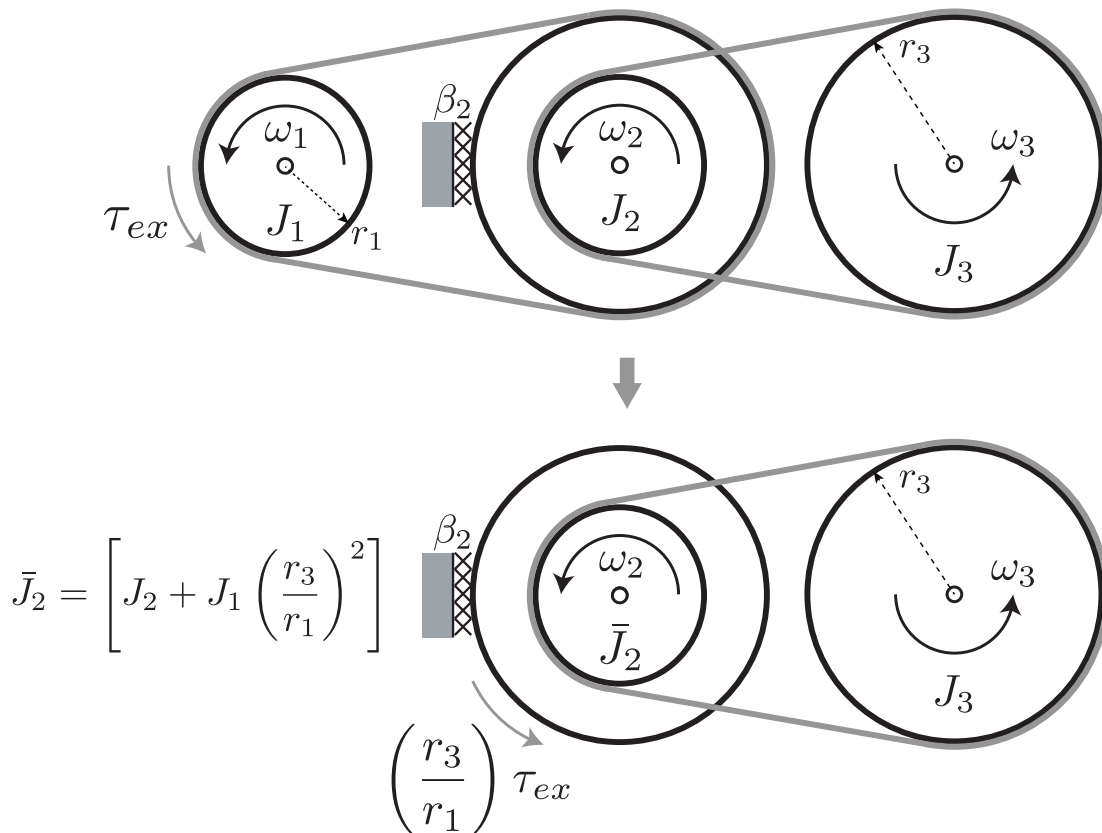
You will analyze the following belt-pulley system consisting of three pulleys and two belts. First pulley has a radius of r_1 and an inertia of J_1 . Third pulley has a radius of r_3 and an inertia of J_3 . Second pulley (in the middle) is connected to the first pulley through its outer disk, which has the same radius with the third pulley, i.e., $r_{2,o} = r_3$. Second pulley is also connected to the first pulley via its inner disk, which has the same radius with the first pulley, i.e., $r_{2,i} = r_1$. Second pulley has an inertia of J_2 (outer and inner disks move together). A linear rotational viscous damping with a damping constant β_2 also affects the motion of the second pulley.

Given that the external torque acting on the first pulley is the input, $u(t) = \tau_{ex}(t)$, and the angular velocity of the third pulley is the output, $y(t) = \omega_3(t)$, compute the transfer function of the system.

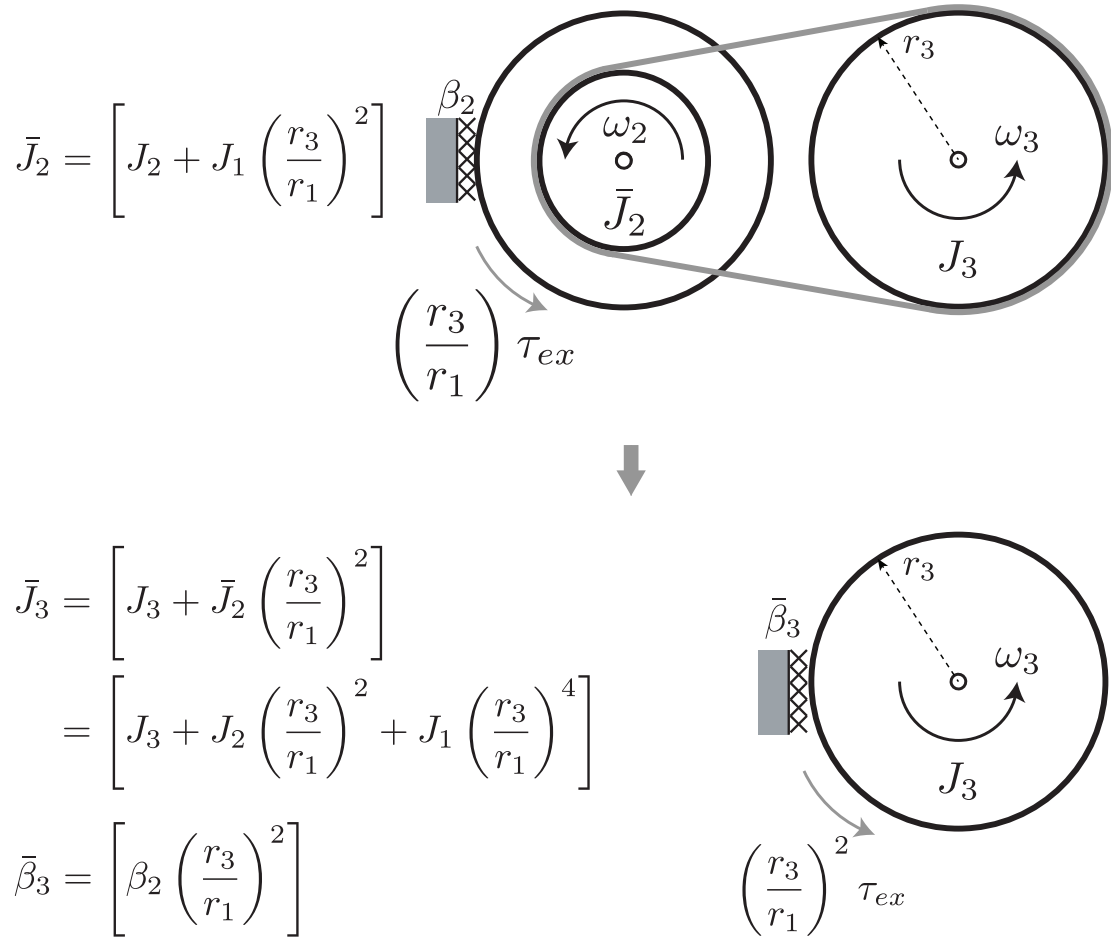


Let's solve this problem using the concept of reflected inertia, damping, and torque.

If we reflect the variables and parameters of first pulley to the second pulley we obtain



Now if we reflect the variables and parameters of the modified second pulley to the third pulley we obtain

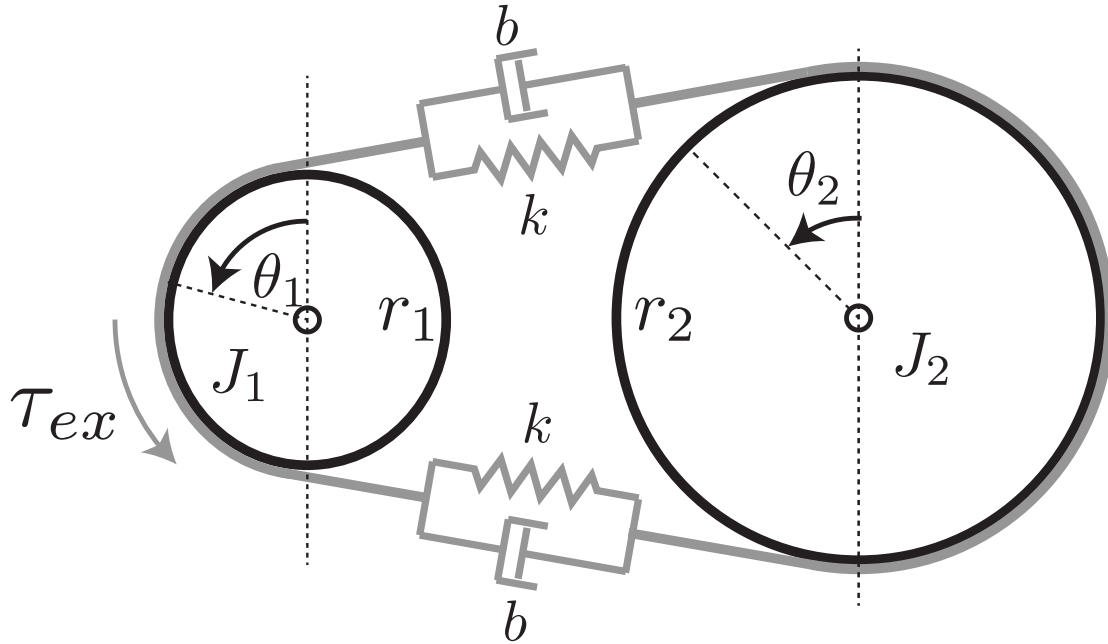


Hence, ode and transfer function of the system can be computed as

$$\bar{J}_3 \dot{\omega}_3 + \bar{\beta}_3 \omega_3 = \left(\frac{r_3}{r_1} \right)^2 \tau_{ex} \quad \rightarrow \quad \dot{y} + \frac{\bar{\beta}_3}{\bar{J}_3} y = \frac{\left(\frac{r_3}{r_1} \right)^2}{\bar{J}_3} u$$

$$\frac{Y(s)}{U(s)} = \frac{\left(\frac{r_3}{r_1} \right)^2 \frac{1}{\bar{J}_3}}{s + \frac{\bar{\beta}_3}{\bar{J}_3}}$$

Ex 3.6. In some belt-pulley applications, ignoring the elasticity of the belt can be very crude and can lead to substantial modeling errors. In order to overcome this problem, a very common method is modeling the belt with a linear (translational) spring-damper as shown in the belt-pulley mechanism below. In this mechanism, first pulley has a radius of r_1 and inertia of J_1 , where as the second pulley has a radius of r_2 and inertia of J_2 . The spring-mass dampers (above and below) that model the elasticity of the belt have spring stiffnesses of k and damping constants of b .



Given that the external torque acting on the first pulley is the input, $u(t) = \tau_{ex}(t)$, and the angular displacement of the second pulley is the output, $y(t) = \theta_2(t)$,

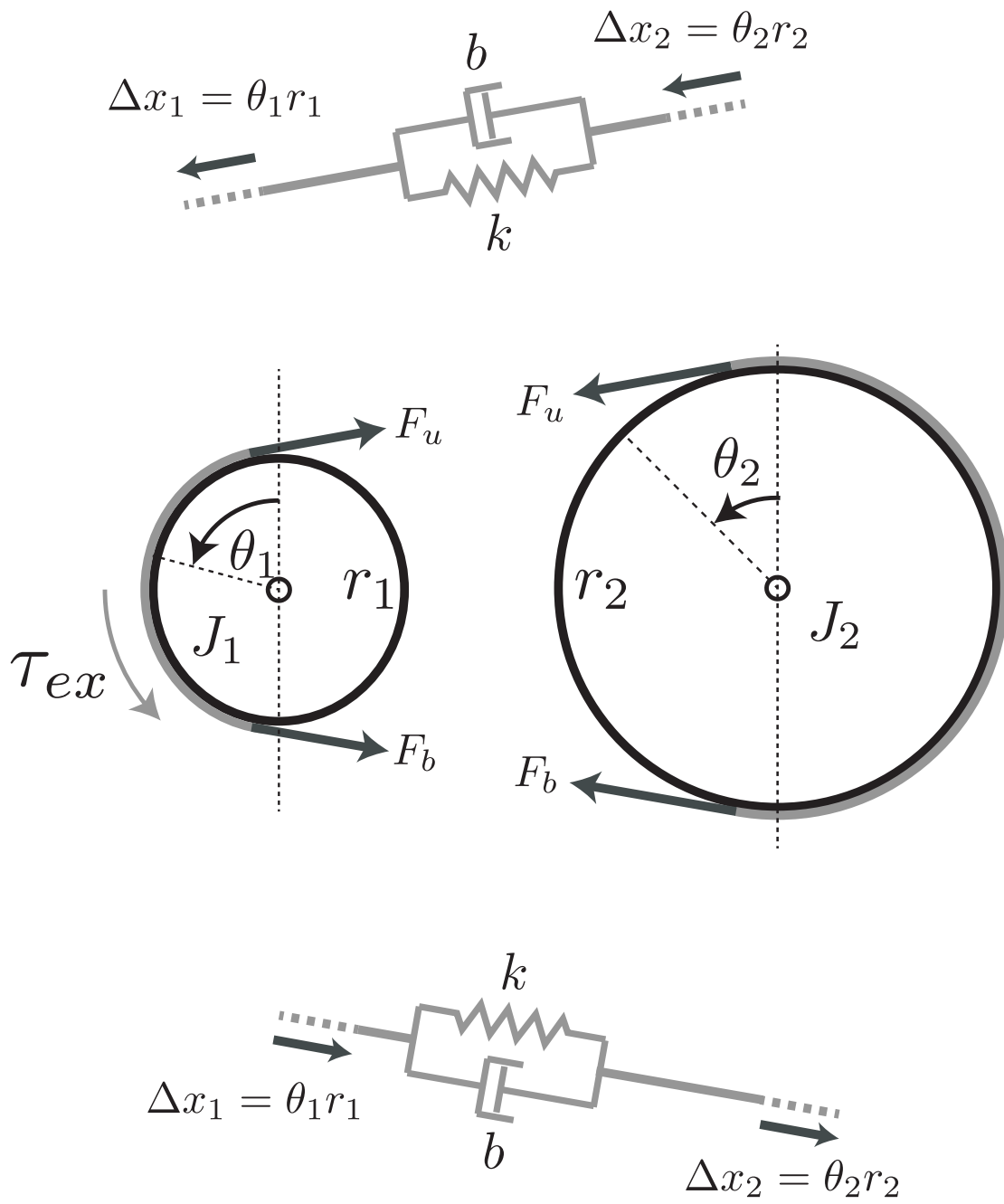
1. Find a state-space representation of the dynamics. (*Hint:* You can choose your state variables as $\mathbf{x} = [\theta_1 \ \dot{\theta}_1 \ \theta_2 \ \dot{\theta}_2]^T$).
2. Compute the transfer function, $G(s) = Y(s)/U(s)$
3. Now, let the parameters of the system be equal to following numerical values

$$r_1 = 0.05 \text{ m} , \ r_2 = 0.1 \text{ m} , \ J_1 = 0.01 \text{ kg} \cdot \text{m}^2 , \ J_2 = 0.1 \text{ kg} \cdot \text{m}^2$$

$$k = 100 \text{ N/m} , \ b = 10 \text{ N/(m} \cdot \text{s)} .$$

Simplify both the state-space and transfer function representations using these numerical values. Finally convert the state-space form to the transfer function form and verify that converted transfer function is equal to the previously computed one. (*Hint:* You can use MATLAB's `ss2tf` command for conversion).

We assume that when $[\theta_1 \ \theta_2 \ \dot{\theta}_1 \ \dot{\theta}_2] = [0 \ 0 \ 0 \ 0]$ the mechanism is at rest condition. Then let's draw the free-body diagrams



Let's first write the spring force relations

$$\begin{aligned}
 F_u &= k(\Delta x_1 - \Delta x_2) + b \frac{d}{dt}(\Delta x_1 - \Delta x_2) \\
 &= k(r_1\theta_1 - r_2\theta_2) + b(r_1\dot{\theta}_1 - r_2\dot{\theta}_2) \\
 &= kr_1\theta_1 - kr_2\theta_2 + br_1\dot{\theta}_1 - br_2\dot{\theta}_2 \\
 F_b &= k(-\Delta x_1 + \Delta x_2) + b \frac{d}{dt}(-\Delta x_1 + \Delta x_2) \\
 &= k(-r_1\theta_1 + r_2\theta_2) + b(-r_1\dot{\theta}_1 + r_2\dot{\theta}_2) \\
 &= -kr_1\theta_1 + kr_2\theta_2 - br_1\dot{\theta}_1 + br_2\dot{\theta}_2
 \end{aligned}$$

Now let's write the equations of motion of the individual bodies

$$\begin{aligned}
 J_1\ddot{\theta}_1 &= \tau_{ex} - F_u r_1 + F_b r_1 \\
 &= \tau_{ex} + (-kr_1^2\theta_1 + kr_2r_1\theta_2 - br_1^2\dot{\theta}_1 + br_2r_1\dot{\theta}_2) + (-kr_1^2\theta_1 + kr_2r_1\theta_2 - br_1^2\dot{\theta}_1 + br_2r_1\dot{\theta}_2) \\
 &= \tau_{ex} - 2kr_1^2\theta_1 + 2kr_2r_1\theta_2 - 2br_1^2\dot{\theta}_1 + 2br_2r_1\dot{\theta}_2 \\
 J_2\ddot{\theta}_2 &= F_u r_2 - F_b r_2 \\
 &= (kr_1r_2\theta_1 - kr_2^2\theta_2 + br_1r_2\dot{\theta}_1 - br_2^2\dot{\theta}_2) + (kr_1r_2\theta_1 - kr_2^2\theta_2 + br_1r_2\dot{\theta}_1 - br_2^2\dot{\theta}_2) \\
 &= 2kr_1r_2\theta_1 - 2kr_2^2\theta_2 + 2br_1r_2\dot{\theta}_1 - 2br_2^2\dot{\theta}_2
 \end{aligned}$$

1. Let $\mathbf{x} = [\theta_1 \ \dot{\theta}_1 \ \theta_2 \ \dot{\theta}_2]^T$, then we can find a state-space representation

$$\begin{aligned}
 \dot{\mathbf{x}} &= \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{2kr_1^2}{J_1} & -\frac{2br_1^2}{J_1} & \frac{2kr_1r_2}{J_1} & \frac{2br_1r_2}{J_1} \\ 0 & 0 & 0 & 1 \\ \frac{2kr_1r_2}{J_2} & \frac{2br_1r_2}{J_2} & -\frac{2kr_2^2}{J_2} & -\frac{2br_2^2}{J_2} \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ \frac{1}{J_1} \\ 0 \\ 0 \end{bmatrix} u \\
 y &= [0 \ 0 \ 1 \ 0] \mathbf{x}
 \end{aligned}$$

2. Let's take the Laplace transform of the derived differential equations

$$\begin{aligned}
 [J_1s^2 + 2br_1^2s + 2kr_1^2] \Theta_1(s) &= U(s) + [2br_2r_1s + 2kr_2r_1] Y(s) \\
 [J_2s^2 + 2br_2^2s + 2kr_2^2] Y(s) &= [2kr_1r_2 + 2br_1r_2s] \Theta_1(s)
 \end{aligned}$$

In order to simplify the expression let

$$B_1 = 2br_1^2, \ K_1 = 2kr_1^2, \ B_2 = 2br_2^2, \ K_2 = 2kr_2^2, \ B_{12} = 2br_2r_1, \ K_{12} = 2kr_1r_2$$

Then we can have

$$\begin{aligned}
 [J_1 s^2 + B_1 s + K_1] \Theta_1(s) &= U(s) + [B_{12} s + K_{12}] Y(s) \\
 [J_2 s^2 + B_2 s + K_2] Y(s) &= [B_{12} s + K_{12}] \Theta_1(s) \quad \rightarrow \quad \Theta_1(s) = \frac{[J_2 s^2 + B_2 s + K_2]}{[B_{12} s + K_{12}]} Y(s) \\
 Y(s) \left\{ [J_1 s^2 + B_1 s + K_1] \frac{[J_2 s^2 + B_2 s + K_2]}{[B_{12} s + K_{12}]} - [B_{12} s + K_{12}] \right\} &= U(s) \\
 Y(s) \left\{ [J_1 s^2 + B_1 s + K_1] \frac{[J_2 s^2 + B_2 s + K_2]}{[B_{12} s + K_{12}]} - [B_{12} s + K_{12}] \right\} &= U(s)
 \end{aligned}$$

Let $\frac{Y(s)}{U(s)} = \frac{N(s)}{D(s)}$, then

$$\begin{aligned}
 D(s) &= J_1 J_2 s^4 + (B_1 J_2 + B_2 J_1) s^3 + (-B_{12}^2 + B_1 B_2 + J_1 K_2 + J_2 K_1) s^2 \\
 &\quad + (B_1 K_2 + B_2 K_1 - 2B_{12} K_{12}) s + (-K_{12}^2 + K_1 K_2) \\
 D(s) &= J_1 J_2 s^4 + (B_1 J_2 + B_2 J_1) s^3 + (J_1 K_2 + J_2 K_1) s^2 + 0 + 0
 \end{aligned}$$

Finally the transfer function can be computed as

$$\frac{Y(s)}{U(s)} = \frac{B_{12} s + K_{12}}{J_1 J_2 s^4 + (B_1 J_2 + B_2 J_1) s^3 + (J_1 K_2 + J_2 K_1) s^2}$$

3. The state-space representation with the given coefficients take the form

$$\begin{aligned}
 \dot{\mathbf{x}} &= \begin{bmatrix} 0 & 100 & 0 & 0 \\ -50 & -5 & 100 & 10 \\ 0 & 0 & 0 & 1 \\ 10 & 1 & -20 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} u \\
 y &= \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \mathbf{x}
 \end{aligned}$$

Transfer function with the given coefficients take the form

$$G(s) = \frac{Y(s)}{U(s)} = \frac{100(s + 10)}{s^4 + 7s^3 + 70s^2}$$

A sample MATLAB code piece which converts the state-space form to the transfer function form (in terms of numerator and denominator coefficients) is provided below. It is clear that, the computed coefficients match the previous ones.

```

>> r1 = 0.05;
>> r2 = 0.1;
>> J1 = 0.01;
>> J2 = 0.1;
>> k = 100;
>> b = 10;
>>
>> A(1,:) = [0 1 0 0];
>> A(2,:) = [-2*k*r1^2/J1 -2*b*r1^2/J1 2*k*r1*r2/J1 2*b*r1*r2/J1];
>> A(3,:) = [0 0 0 1];
>> A(4,:) = [2*k*r1*r2/J2 2*b*r1*r2/J2 -2*k*r2^2/J2 -2*b*r2^2/J2];
>> A

A =

         0         1.0000         0         0
    -50.0000    -5.0000    100.0000    10.0000
         0         0         0         1.0000
    10.0000     1.0000   -20.0000    -2.0000

>> B = [0 ; 100 ; 0 ; 0];
>> C = [0 0 1 0];
>> D = 0;
>>
>> [num,denum] = ss2tf(A,B,C,D)

num =

    1.0e+03 *

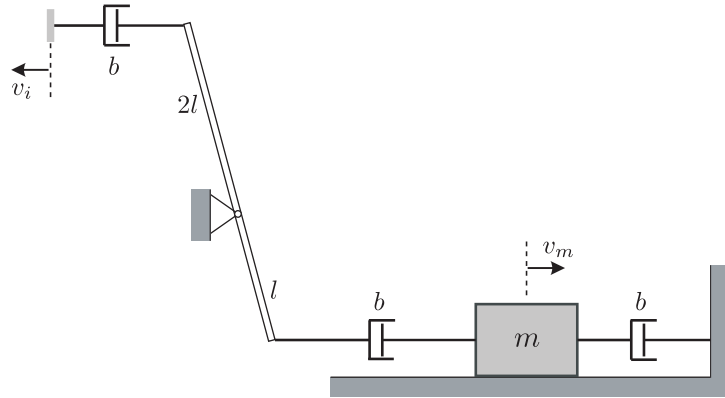
         0         0         0    0.1000    1.0000

denum =

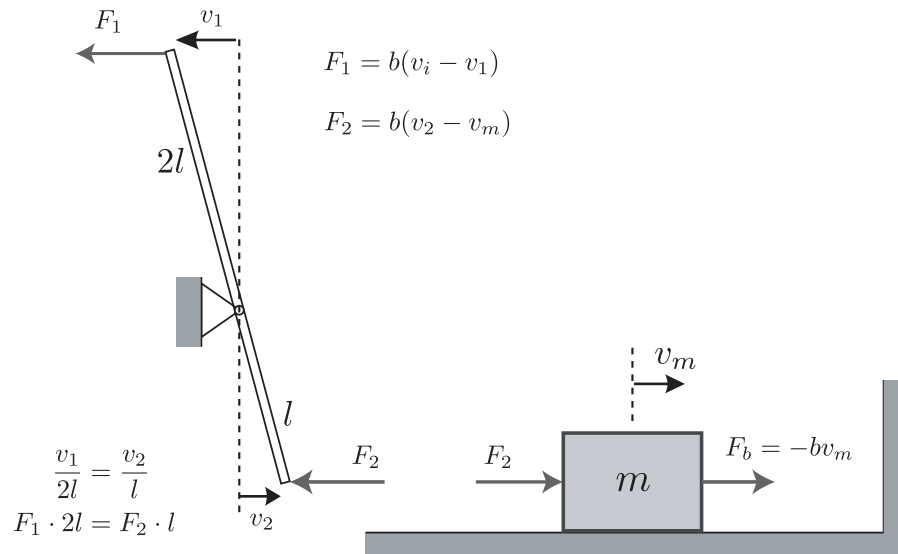
    1.0000    7.0000   70.0000    0.0000    0.0000

>> |

```

Ex 3.7. Mechanical system with ideal lever

Input-output of the system is the are $u(t) = v_i(t)$, and $y(t) = v_m(t)$ respectively. If we draw the free body diagram of the mass and draw the kinematic relations of the lever we obtain the following illustration



First let's concentrate on the lever side and try to eliminate intermediate variables

$$\begin{aligned}
 2F_1 &= F_2 \\
 2(v_i - v_1) &= (v_2 - v_m) \\
 v_2 + 2v_1 &= 2v_i + v_m \\
 v_2 &= \frac{2}{5}v_i + \frac{1}{5}v_m
 \end{aligned}$$

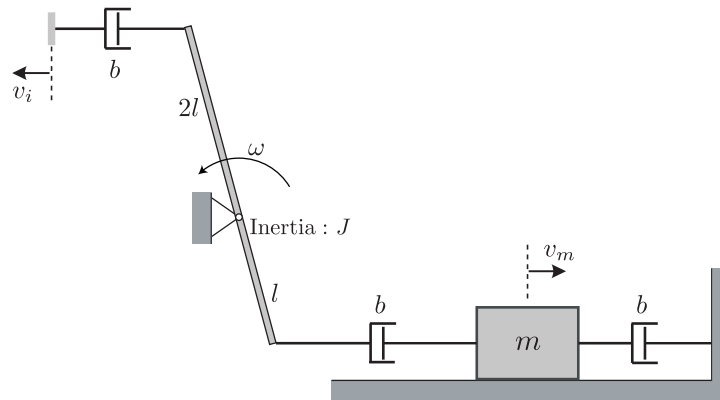
Now let's write equations of motion for the single body

$$\begin{aligned}
 m\dot{v}_M &= F_2 + F_b \\
 m\dot{v}_M &= b(v_2 - v_m) - bv_m \\
 \dot{v}_M &= \frac{2}{5}v_i + \frac{1}{5}v_m - v_m - v_m \\
 \dot{v}_M + \frac{9}{5}v_m &= \frac{2}{5}v_i \\
 \dot{y} + \frac{9}{5}y &= \frac{2}{5}u
 \end{aligned}$$

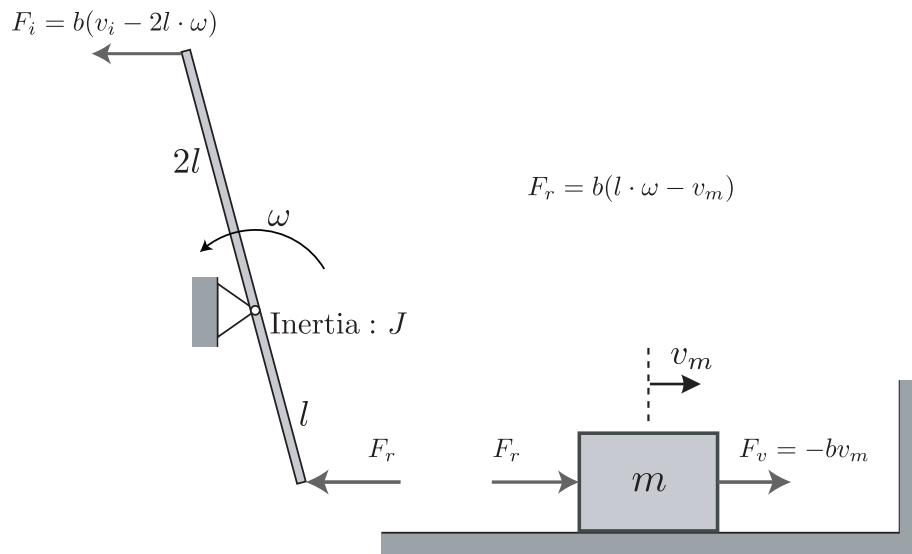
Transfer function simply takes the form

$$\frac{Y(s)}{U(s)} = \frac{2/5}{s + 9/5}$$

Ex 3.8. Mechanical system with a non-ideal lever that has an inertia of $J = ml^2$



Input-output of the system is the are $u(t) = v_i(t)$, and $y(t) = v_m(t)$ respectively. If we draw the free body diagrams of the lever and the mass, and derive the force relations we obtain



If we write equations of motion for the lever and the mass, we obtain following differential equations

$$\begin{aligned}
J\dot{\omega} &= F_i \times 2l - F_r \times l = (-5l^2b)\omega + (2lb)v_i + (lb)v_m \\
\dot{\omega} + \frac{5b}{m}\omega &= \frac{b}{ml}v_m + \frac{2b}{ml}v_i \\
\dot{\omega} + 5\omega &= \frac{1}{l}v_m + \frac{2}{l}v_i \\
, \\
m\dot{v}_m &= F_r + F_v = (-2b)v_m + (bl)\omega \\
\dot{v}_m + \frac{2b}{m}v_m &= \frac{bl}{m}\omega \\
\dot{v}_m + 2v_m &= l\omega
\end{aligned}$$

If we take the Laplace transform of the derived differential equations

$$\begin{aligned}
\Omega(s)[s+5] &= \frac{1}{l}Y(s) + \frac{2}{l}U(s) \\
Y(s)[s+2] &= l\Omega(s) \\
Y(s)[s+2][s+5] \frac{1}{l} &= \frac{1}{l}Y(s) + \frac{2}{l}2U(s) \\
Y(s)(s+2)(s+5) &= Y(s) + 2U(s) \\
\frac{Y(s)}{U(s)} &= \frac{2}{s^2 + 7s + 11}
\end{aligned}$$

Now let's find a state-space representation for the given system. The ODE representation of the transfer function has the following form

$$\ddot{y} + 7\dot{y} + 11y = 2u$$

The system is a 2^{nd} order dynamical system. Let $\mathbf{x} = [y \ \dot{y}]^T$, then

$$\begin{aligned}
\dot{x}_1 &= \dot{y} = x_2 \\
\dot{x}_2 &= \ddot{y} = -7\dot{y} - 11y + 2u = -7x_2 - 11x_1 + 2u \\
y &= x_1
\end{aligned}$$

The state-representation with the choosed state definition takes the form

$$\begin{aligned}
\dot{\mathbf{x}} &= \begin{bmatrix} 0 & 1 \\ -11 & -7 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \\
y &= \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x} + [0]u
\end{aligned}$$

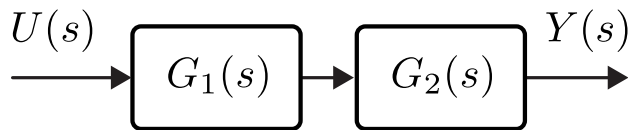
Lecture 4

Lecturer: Asst. Prof. M. Mert Ankarali

4.1 Block Diagrams & Simplifications

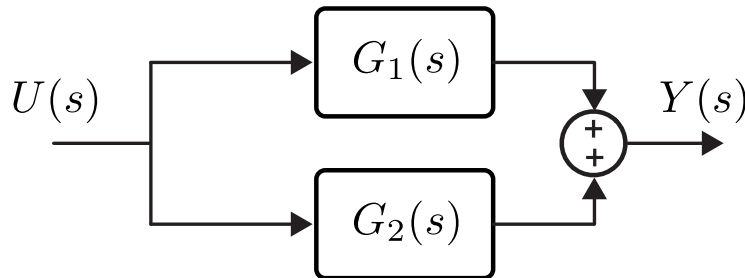
4.1.1 Fundamental Block Diagram Topologies

Cascaded (Series) Block Diagrams



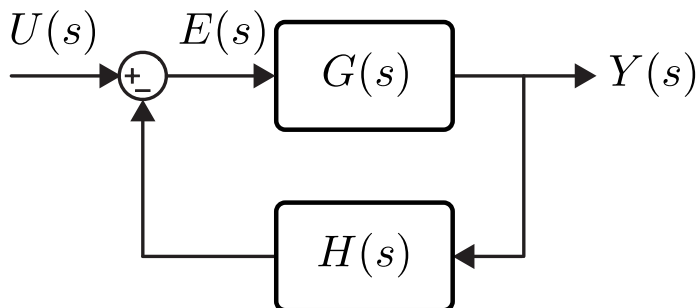
$$\frac{Y(s)}{U(s)} = \bar{G}(s) = G_1(s)G_2(s)$$

Parallel Block Diagrams



$$\frac{Y(s)}{U(s)} = \bar{G}(s) = G_1(s) + G_2(s)$$

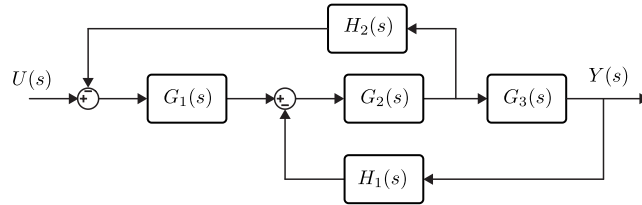
Negative Feedback Loop



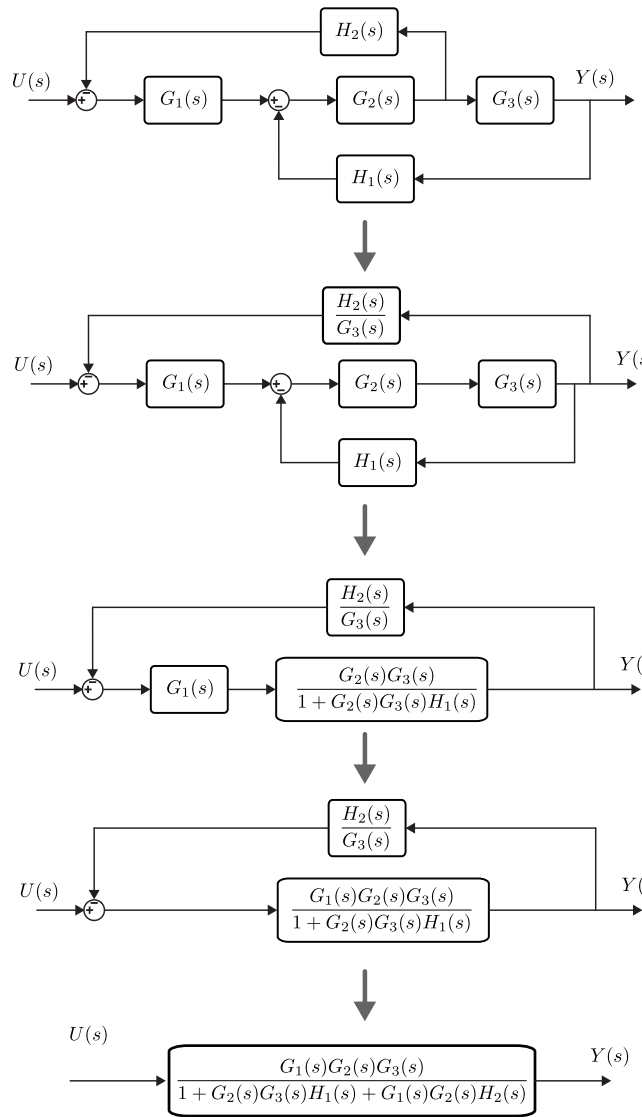
$$\begin{aligned} E(s) &= U(s) - H(s)Y(s) \\ E(s)(1 + H(s)G(s)) &= U(s) \\ \frac{Y(s)}{U(s)} = \bar{G}(s) &= \frac{G(s)}{1 + H(s)G(s)} \end{aligned}$$

4.1.2 Examples

Ex 1: Simplify the following block-diagram topology



Solution:

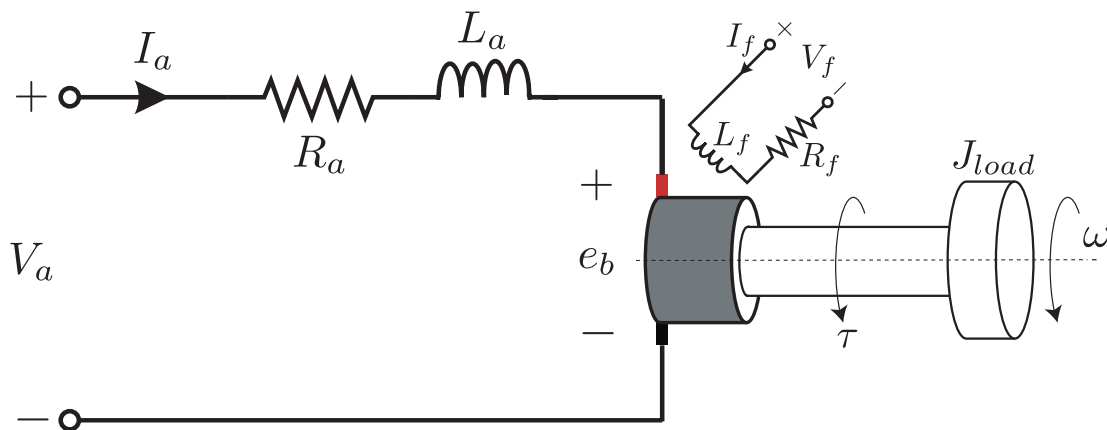


Lecture 5

Lecturer: Asst. Prof. M. Mert Ankarali

5.1 DC Motor Modeling

A general “ideal” DC motor can be modeled as in the Figure below.



The dependent and “independent” variables associated with the idealized DC motor model and important relations/equations regarding the electro-mechanical interactions are given below.

V_a	Armature voltage
i_a	Armature current
V_f	“Field voltage”
i_f	“Field current”
V_b	Back emf
ω	Rotor angular velocity
τ	Generated torque
Φ	Air-gap magnetic flux

$$\begin{aligned}\Phi(t) &= K_f I_f(t) \\ \tau(t) &= K_m \Phi(t) I_a(t) \\ e_b(t) &= K_b \omega(t)\end{aligned}$$

Note that if both $i_f(t)$ and $i_a(t)$ are non-constant the electric-motor model won’t be LTI. In order to have an LTI representation, there are two options

- Armature controlled DC motor: Φ is kept constant

$$\tau(t) = K_m \Phi I_a(t) = K_\tau^a I_a(t)$$

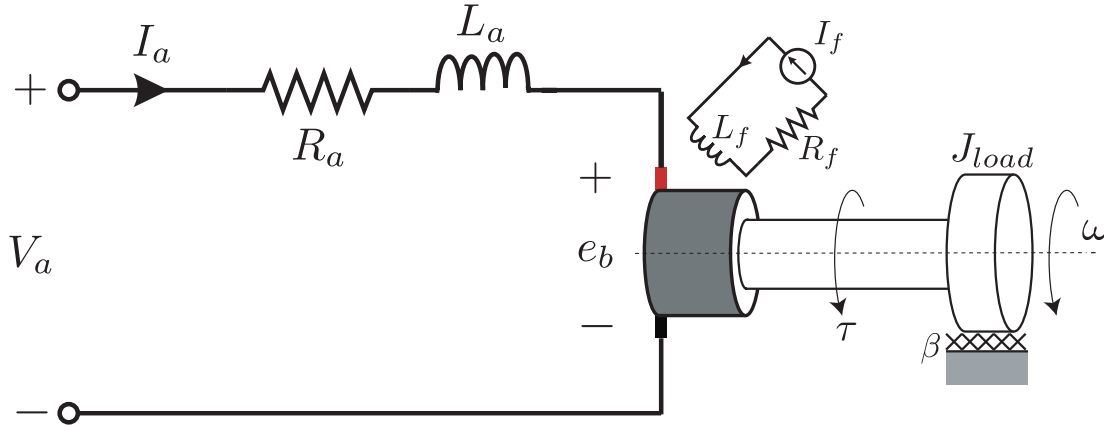
- Field controlled DC motor: i_a is kept constant

$$\tau(t) = K_m K_f I_a I_f(t) = K_\tau^f I_f(t)$$

5.1.1 Armature Controlled DC Motor

Majority of “DC” Motors are controlled (and indeed manufactured) with this approach. Either there is a permanent magnet which satisfies the constant Φ or a constant current is supplied through the coils that generates the magnetic field.

Let's model the following electro-mechanical system where the DC motor is armature controlled and given that $y(t) = \omega(t)$ and $u(t) = V_a(t)$.



We already know the transfer function of the mechanical sub system:

$$\Omega(s) = \frac{1}{Js + \beta} \mathcal{T}(s)$$

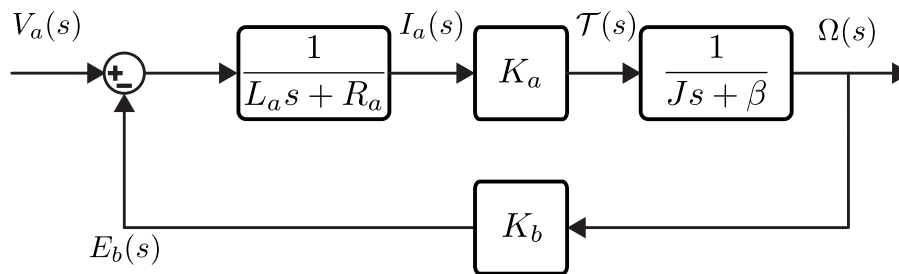
Now let's write the remaining equations in Laplace domain

$$I_a(s) = \frac{1}{L_a s + R_a} (V_a(s) - E_b(s))$$

$$\mathcal{T}(s) = K_a I_a(s)$$

$$E_b(s) = K_b \Omega(s)$$

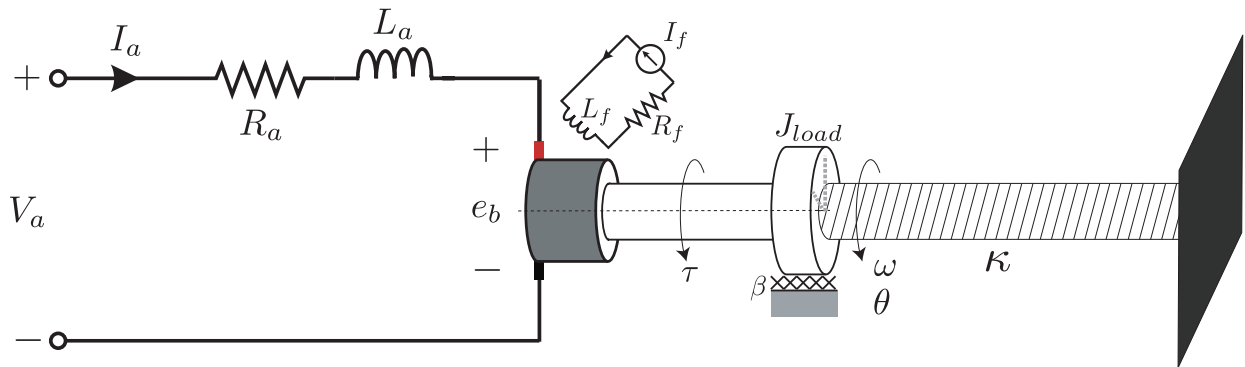
where $K_a \equiv K_\tau^a$. Now let's build a block-diagram topology



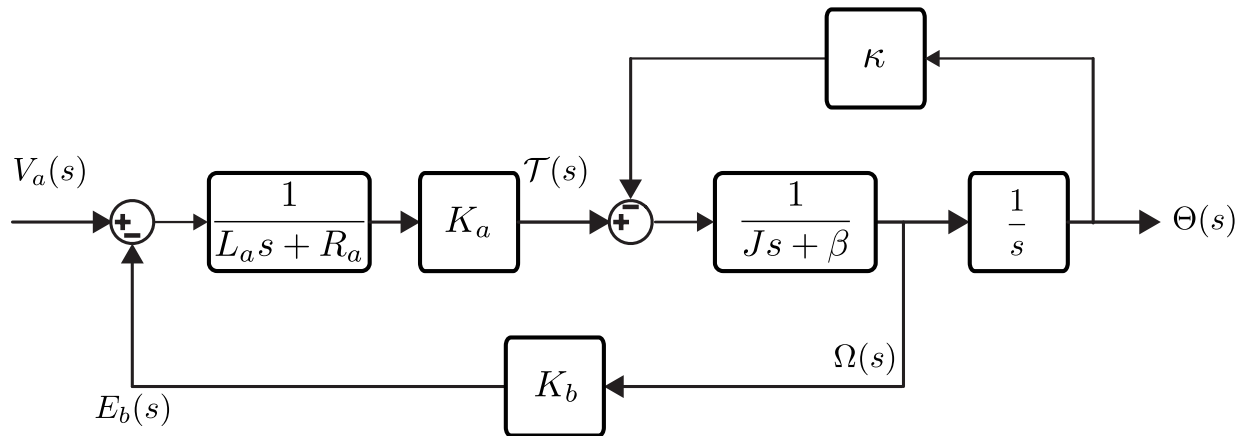
If we simplify the block diagram, we obtain the transfer function form

$$\begin{aligned} G(s) &= \frac{\Omega(s)}{V_a(s)} = \frac{\frac{K_a}{(L_a s + R_a)(J s + \beta)}}{1 + \frac{K_a K_b}{(L_a s + R_a)(J s + \beta)}} \\ &= \frac{K_a}{L_a J s^2 + (L_a \beta + R_a J) s + R_a \beta + K_a K_b} \end{aligned}$$

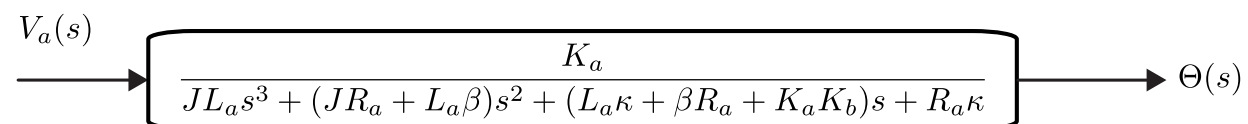
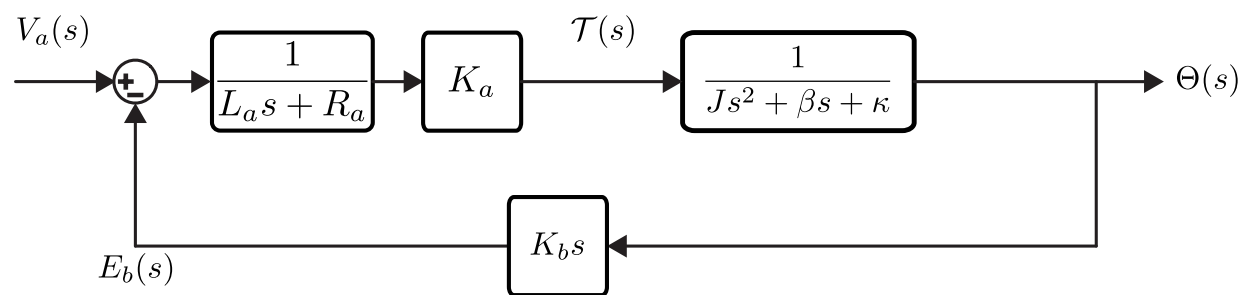
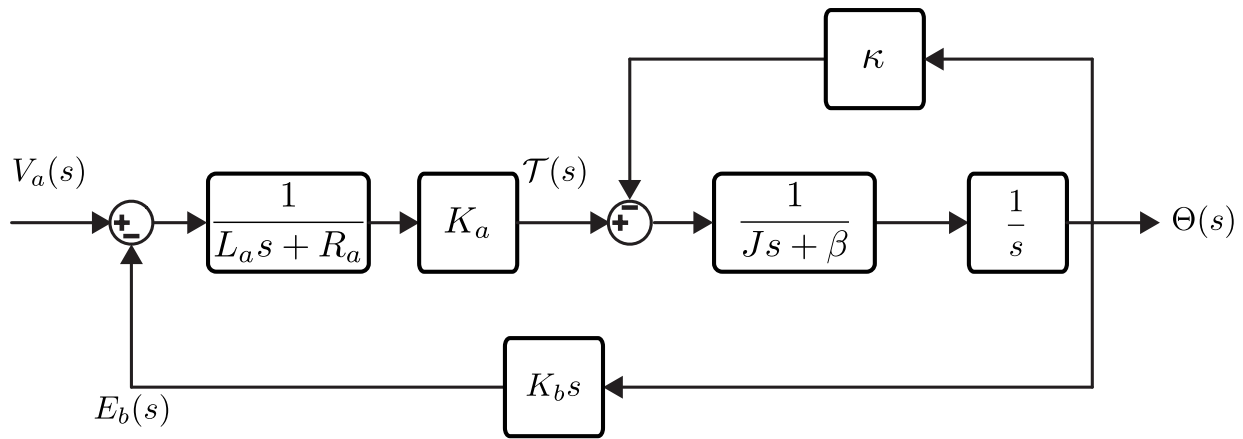
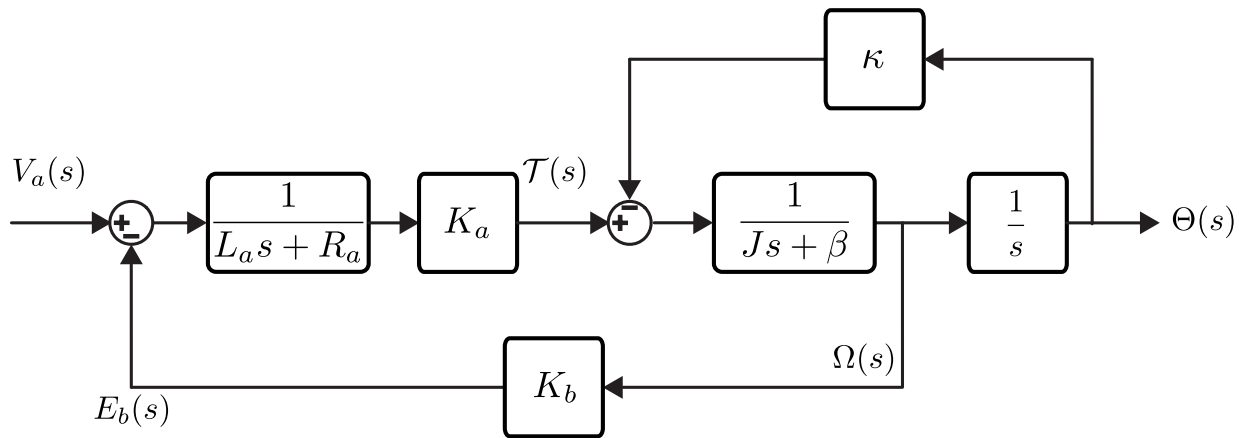
Example 1: Given that V_a is the input and θ is the output, construct a block-diagram for the following electro mechanical system and then compute the transfer function.



Solution: A block diagram topology can be constructed by modifying the previous block diagram (armature controlled DC motor without torsional spring).



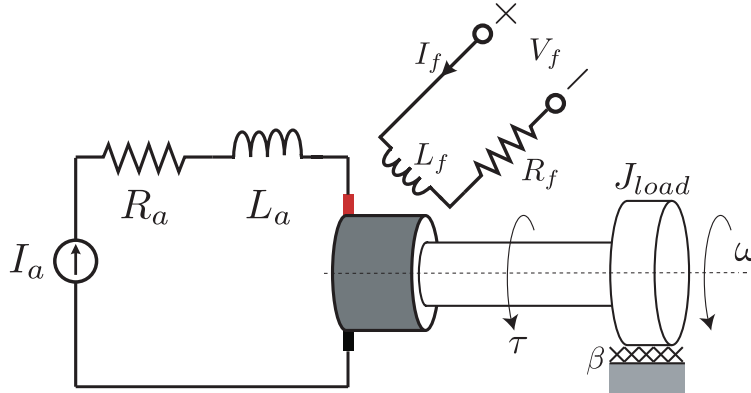
Then the transfer function can be derived using block-diagram simplification methods as given in the next page



5.1.2 Field Controlled DC Motor

In the field controlled DC motors, magnetic flux is actively controlled by adjusting electrical current/voltage. We assume that I_a is constant (LTI constraints). Since, there is no “feedback” in this field controlled DC motor model, the electrical circuit is isolated from the mechanical one.

Let's model the following electro-mechanical system where the DC motor is field controlled and given that $y(t) = \omega(t)$ and $u(t) = V_f(t)$.

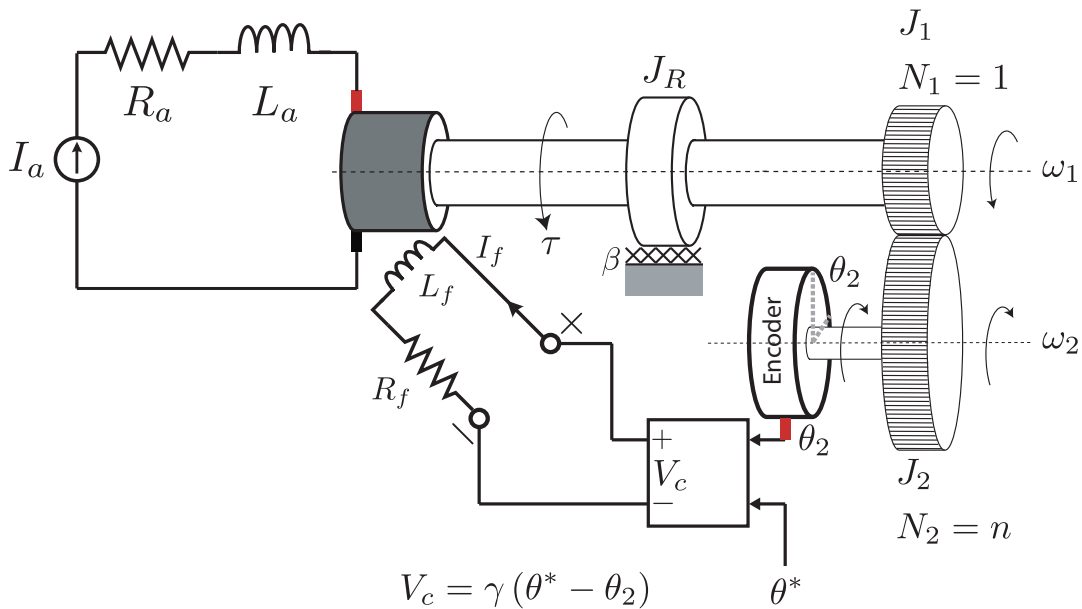


$$\begin{aligned}\frac{\Omega(s)}{\mathcal{T}(s)} &= \frac{1}{Js + \beta} \\ \frac{I_f(s)}{V_f(s)} &= \frac{1}{L_f s + R_f} \\ \mathcal{T}(s) &= K_f I_f(s)\end{aligned}$$

where $K_f \equiv K_{tau}^f$. Finally transfer function can be computed as

$$G(s) = \frac{K_f}{JL_f s^2 + (JR_f + \beta L_f)s + (\beta R_f + K_f)}$$

Example 2: Consider the following closed-loop field controlled electro-mechanical circuit. It is given that $\theta^*(t)$, i.e. reference angle signal, is the input and θ_2 , angular displacement of the second gear, is the output. In the system, there is an encoder which reads the angular displacement and sends it to a controller box. The other input of this box is the reference signal. The box produces an output voltage, $V_c = \gamma(\theta^* - \theta_2)$, and feeds it to the input terminal of the V_f . Compute the transfer function.



Solution: A block diagram topology can be constructed by modifying the previous block diagram (armature controlled DC motor without torsional spring).

Let's first find a transfer function from τ to ω_2 and θ_2 . The easiest way of computing this is using the concept of reflected inertia, damping, and torque.

$$\Omega_2(s) = \frac{\bar{\mathcal{T}}(s)}{J_T s + \beta_T} = \frac{n}{(n^2 J_R + n^2 J_1 + J_2) s + n^2 \beta} \mathcal{T}(s)$$

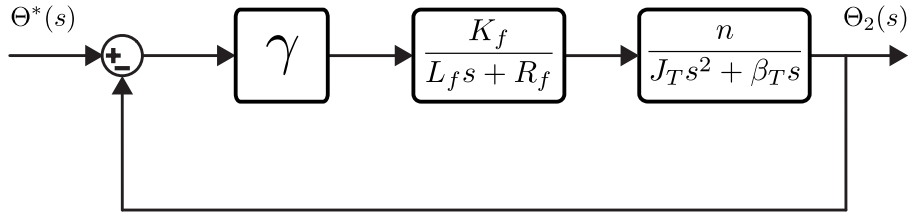
$$\Theta_2(s) = \frac{n \mathcal{T}(s)}{J_T s^2 + \beta_T s}$$

We know that Laplace domain equations for remaining parts take the form

$$\frac{\mathcal{T}(s)}{V_f(s)} = \frac{K_f}{L_f s + R_f}$$

$$V_f(s) = \gamma (\Theta^*(s) - \Theta_2(s))$$

Now let's construct a block diagram representation



Finally, transfer function can be computed as

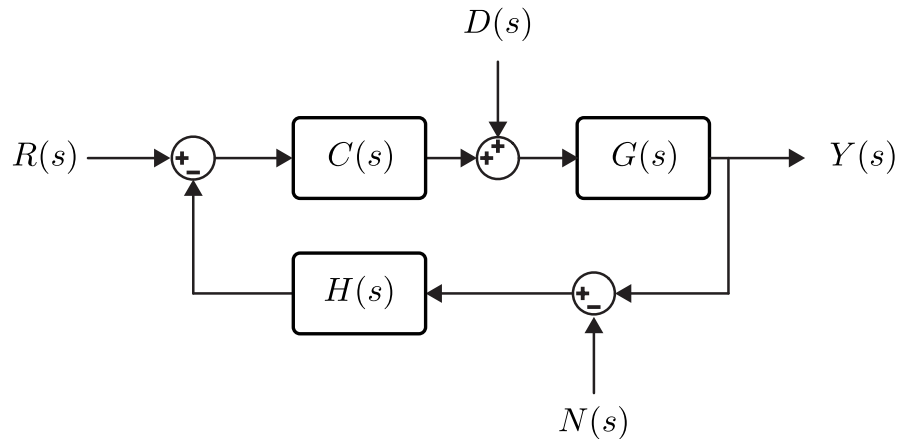
$$G(s) = \frac{\gamma K_f n}{J_T L_f s^3 + (J_T R_f + \beta_T L_f) s^2 + \beta_T R_f s + \gamma K_f n}$$

Lecture 6

Lecturer: Asst. Prof. M. Mert Ankarali

6.1 Transfer functions with multiple exogenous input

When modeling and analyzing a closed system in addition to the desired reference input signal, it is also important to model/analyze the system with unwanted disturbances and noise input signals. Let's consider the following block-diagram topology. In this closed-loop system, there exist three exogenous input signals; $r(t)$ (reference input), $d(t)$ ("disturbance" input), and $n(t)$ ("noise" input).



When modeling the response or characteristic of the system with respect to different external inputs, we assume that remaining ones are zero.

Response to $r(t)$

$$T_R(s) = \frac{Y(s)}{R(s)} = \frac{C(s)G(s)}{1 + C(s)G(s)H(s)}$$

Response to $d(t)$

$$T_D(s) = \frac{Y(s)}{D(s)} = \frac{G(s)}{1 + C(s)G(s)H(s)}$$

Response to $n(t)$

$$T_N(s) = \frac{Y(s)}{N(s)} = \frac{C(s)G(s)H(s)}{1 + C(s)G(s)H(s)}$$

If we generalize, we can write $Y(s)$ as

$$Y(s) = T_R(s)R(s) + T_D(s)D(s) + T_N(s)N(s)$$

Lets roughly analyze the desired responses under different type of inputs. Let's assume that $G(s)$ is the plant transfer function and $H(s)$ is the sensory dynamics transfer function. $C(s)$ is the transfer function of the controller.

In the ideal case, we want

- Perfect tracking of reference signal, $T_R^*(s) \approx 1$. Since it is not possible to perfectly achieve this under dynamic system constrains, we can design a “high gain” controller such that

$$T_R(s) \approx \frac{C(s)G(s)}{C(s)G(s)H(s)} \approx \frac{1}{H(s)}$$

If $H(s) \approx 1$, then we can have a high tracking performance from the system.

- Perfect rejection of disturbance signal, $T_R^*(s) \approx 0$. Similarly, we can design a “high gain” controller such that

$$T_D(s) \approx \frac{G(s)}{C(s)G(s)H(s)} \approx 0$$

It seems that the requirement on $C(s)$ is similar for good tracking and good disturbance rejection.

- Perfect rejection of noise signal, $T_N^*(s) \approx 0$. In this case, we can design a “low gain” controller (or low gain $H(s)$) such that

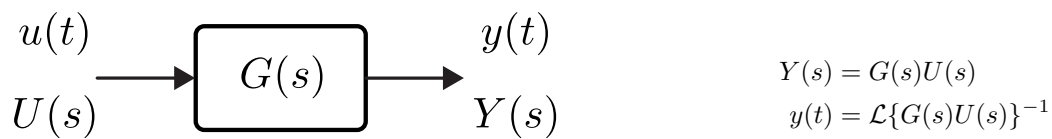
$$T_N(s) \approx \frac{C(s)G(s)H(s)}{1} \approx 0$$

It seems that requirements on $C(s)$ and $H(s)$ start conflicting when we consider both tracking performance, disturbance rejection, and noise rejection. This paradox is the most well-known limitation of feedback control systems. The basic idea is that one can not only concentrate on designing a controller $C(s)$ such that we reach excellent closed-loop tracking performance when the system suffers from uncertainties and noises. Somehow we need to design $G(s)$, $H(s)$, and even $N(s)$ and $D(s)$ together such that the whole system achieves a “good” closed-loop behavior.

Lecture 7

Lecturer: Asst. Prof. M. Mert Ankarali

7.1 Time Domain Analysis

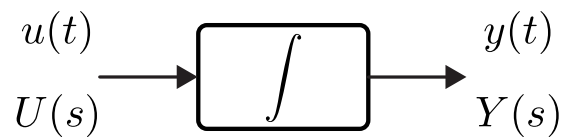
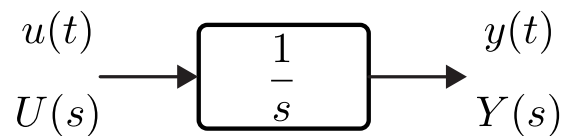


Objective:

- Calculate $y(t)$ for different $u(t)$,
- Understand the relation between the parameters and output behavior.

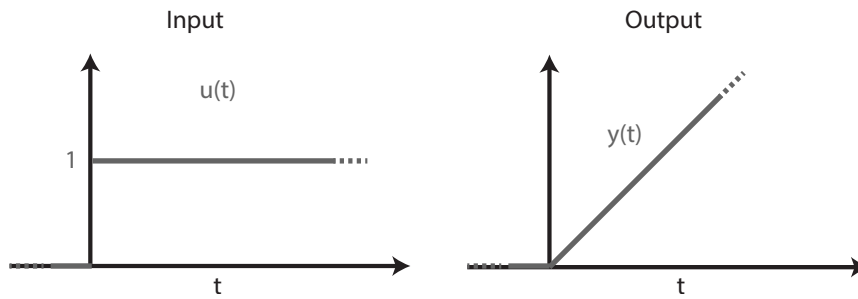
7.1.1 First Order Systems

Simplest first order system is an integrator, which is also the fundamental block for higher order systems.



Ex 1: Compute the step-response of the integrator system, $G(s) = \frac{1}{s}$.

Solution: We assume that initial conditions are zero



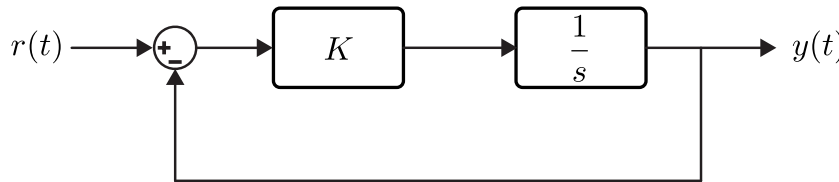
$$U(s) = \frac{1}{s}$$

$$Y(s) = G(s)U(s) = \frac{1}{s^2}$$

$$y(t) = \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\}$$

$$= t, \text{ for } t \geq 0$$

Ex 2: Compute the step-response of the following first order system



$$G(s) = \frac{K}{s + K}$$

$$Y(s) = \frac{K}{s(s + K)} = \frac{K}{s(s + K)}$$

$$= \frac{A}{s} + \frac{B}{s + K}$$

We can compute A and B as

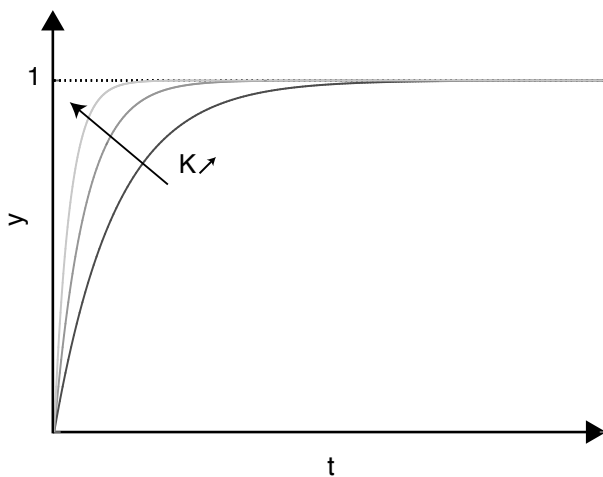
$$A = \lim_{s \rightarrow 0} [sY(s)] = \frac{K}{K} = 1$$

$$B = \lim_{s \rightarrow -K} [(s + K)Y(s)] = \frac{K}{-K} = -1$$

Then, we can compute $y(t)$ as

$$y(t) = \mathcal{L}^{-1}\left\{\frac{1}{s} - \frac{1}{s + K}\right\}$$

$$= [1 - e^{-Kt}] , \text{ for } t \geq 0$$



Proportional Controller

$$\lim_{t \rightarrow \infty} y(t) = 1$$

$$\lim_{t \rightarrow \infty} e(t) = \lim_{t \rightarrow \infty} (y(t) - u(t)) = 0$$

Zero steady-state error $\forall K > 0$
 "Convergence speed" $\nearrow$ as $K \nearrow$

Ex 2: Find the unit-ramp response for the same system

$$Y(s) = \frac{K}{s^2(s+K)}$$

$$= \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+K}$$

A , B , and C can be computed as

$$C = \lim_{s \rightarrow -K} [(s+K)Y(s)] = \frac{1}{K}$$

$$B = \lim_{s \rightarrow 0} [s^2 Y(s)] = 1$$

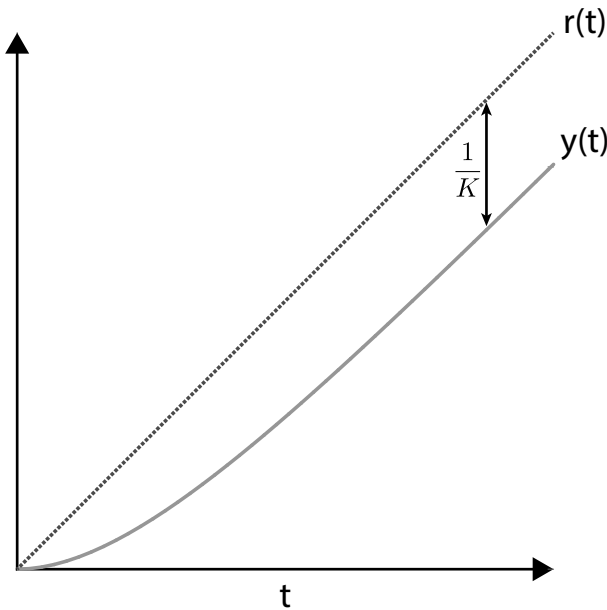
$$A = \lim_{s \rightarrow 0} \frac{d}{ds} [s^2 Y(s)] = \lim_{s \rightarrow 0} \frac{d}{ds} \left[\frac{K}{(s+K)} \right] = \lim_{s \rightarrow 0} \left[\frac{-K}{(s+K)^2} \right] = \frac{-1}{K}$$

Then, we can compute $y(t)$ as

$$y(t) = \frac{-1}{K} + t + \frac{1}{K} e^{-Kt}, \text{ for } t \geq 0$$

$$y(t) = t + \frac{-1}{K} \left[1 - \frac{1}{K} e^{-Kt} \right], \text{ for } t \geq 0$$

Note that $r(t) = t$, for $t \geq 0$.



Proportional Controller

$$e(t) = r(t) - y(t) = \frac{1}{K} \left[1 - \frac{1}{K} e^{-Kt} \right]$$

$$\lim_{t \rightarrow \infty} e(t) = \frac{1}{K}$$

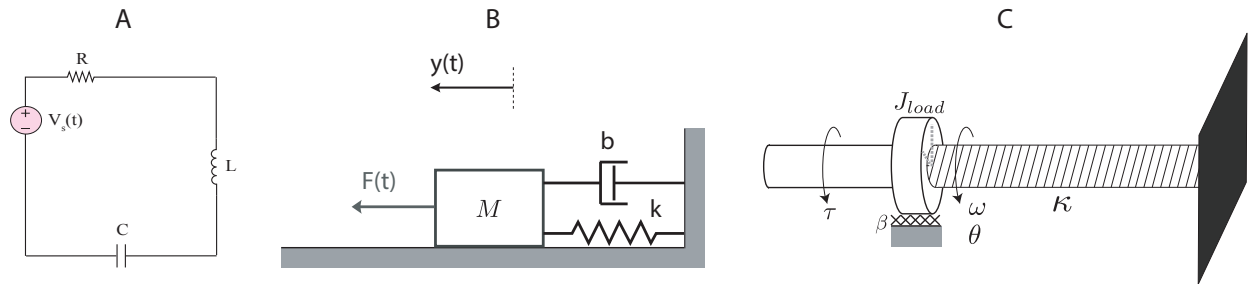
Non-zero steady-state error
Steady-state error $\searrow$ as $K \nearrow$

Lecture 8

Lecturer: Asst. Prof. M. Mert Ankarali

8.1 Second Order Systems

Let's derive the transfer functions for the following electrical and mechanical systems



$$G_A(s) = \frac{V_C(s)}{V_s(s)} = \frac{\frac{1}{LC}}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

$$G_B(s) = \frac{Y(s)}{F(s)} = \frac{\frac{1}{M}}{s^2 + \frac{b}{M}s + \frac{k}{M}}$$

$$G_C(s) = \frac{\Theta(s)}{\mathcal{T}(s)} = \frac{\frac{1}{J}}{s^2 + \frac{\beta}{J}s + \frac{\kappa}{J}}$$

Most of the (passive) second order systems, can be put into the following the standard form

$$G(s) = K_{DC} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

where

$0 < \omega_n$	Undamped natural frequency
$0 < \zeta$	Damping ratio
K_{DC}	DC Gain

Accordingly for the systems that we analyzed previously we have the following relations

$$A: \quad \omega_n = \sqrt{\frac{1}{LC}}, \quad \zeta = \frac{R}{2} \sqrt{\frac{C}{L}}, \quad K_{DC} = 1$$

$$B: \quad \omega_n = \sqrt{\frac{k}{M}}, \quad \zeta = \frac{b}{2} \sqrt{\frac{1}{Mk}}, \quad K_{DC} = \frac{1}{k}$$

$$C: \quad \omega_n = \sqrt{\frac{\kappa}{J}}, \quad \zeta = \frac{b}{2} \sqrt{\frac{1}{J\kappa}}, \quad K_{DC} = \frac{1}{\kappa}$$

Definition: Given a transfer function, $G(s) = \frac{N(s)}{D(s)}$, where $N(s)$ and $D(s)$ are polynomials in s . Roots of $N(s)$ are called “zeros” of $G(s)$, and roots of $D(s)$ are called the “poles” of $G(s)$.

Behavior of the output $y(t)$ are majorly determined by the pole locations.

8.1.1 Step Response Types for the Second Order System in Standard Form

Given that $G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$, the poles can be computed as

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

Case 1: When $\zeta = 0$, the system becomes undamped and $G(s)$ takes the form

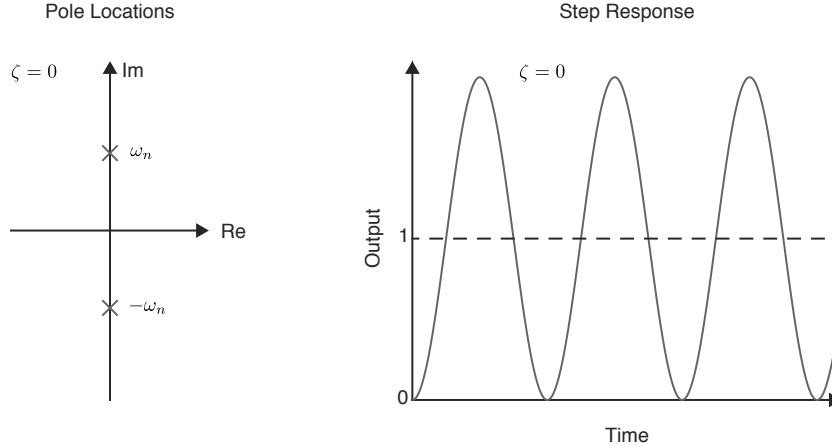
$$G(s) = \frac{\omega_n^2}{s^2 + \omega_n^2}$$

We can compute the step-response as

$$Y(s) = G(s)U(s) = \frac{1}{s} \frac{\omega_n^2}{s^2 + \omega_n^2} = \frac{1}{s} - \frac{s}{s^2 + \omega_n^2}$$

$$y(t) = 1 - \cos(\omega_n t) \quad \text{for } t > 0$$

Pole locations and step response when $\zeta = 0$ (undamped), is illustrated in the Figure below



Case 2: When $\zeta = 1$, the system becomes “critically” damped and $G(s)$ takes the form

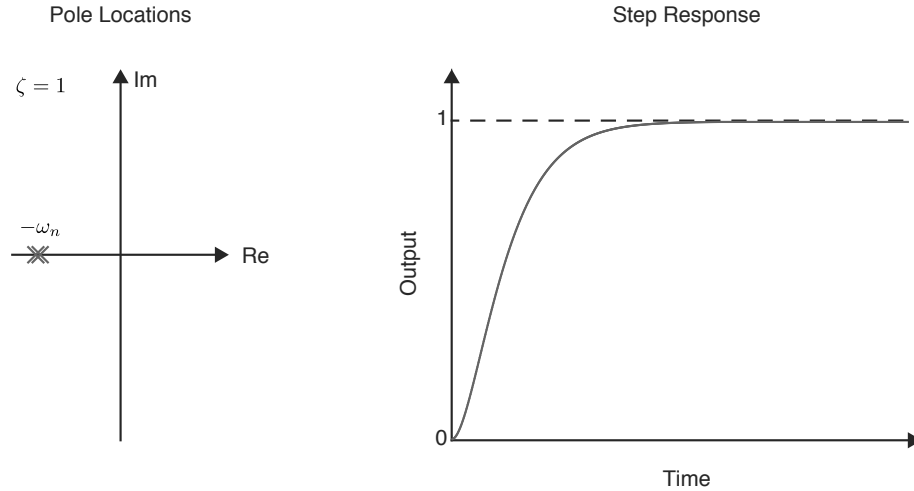
$$G(s) = \frac{\omega_n^2}{s^2 + 2\omega_n s + \omega_n^2} = \frac{\omega_n^2}{(s + \omega_n)^2}$$

We can compute the step-response as

$$Y(s) = G(s)U(s) = \frac{1}{s} \frac{\omega_n^2}{(s + \omega_n)^2} = \frac{1}{s} - \frac{1}{s + \omega_n} + \frac{\omega_n}{(s + \omega_n)^2}$$

$$y(t) = 1 - e^{-\omega_n t} - \omega_n t e^{-\omega_n t}$$

Pole locations and step response when $\zeta = 1$ (critically damped), is illustrated in the Figure below



Case 3: When $\zeta > 1$, the system becomes over damped and there exist two real roots

$$p_1 = -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1} > -\omega_n$$

$$p_2 = -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1} < -\omega_n$$

$G(s)$ can be written in terms of s_1 and s_2

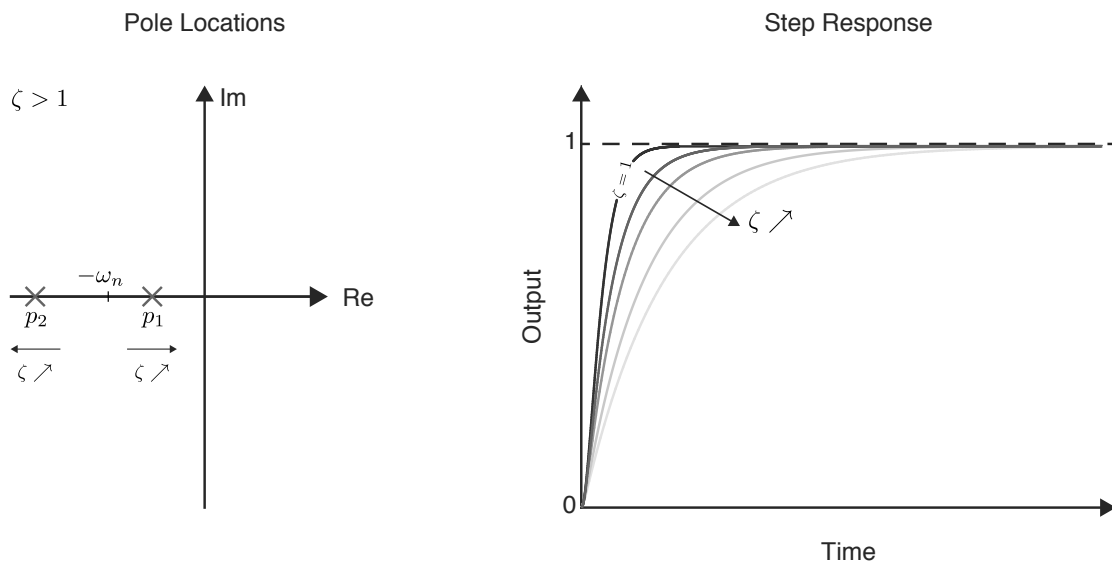
$$G(s) = \frac{p_1 p_2}{(s + p_1)(s + p_2)}$$

where it is easy to see that $p_1 p_2 = \omega_n^2$. Finally, we can compute the step-response as

$$Y(s) = G(s)U(s) = \frac{p_1 p_2}{s(s + p_1)(s + p_2)}$$

$$y(t) = 1 + \frac{p_2}{p_1 - p_2} e^{p_1 t} - \frac{p_1}{p_1 - p_2} e^{p_2 t}$$

Pole locations and step response when $\zeta > 1$ (over damped), is illustrated in the Figure below



Case 3: When $0 < \zeta < 1$, the system becomes under damped and there exist two complex conjugate roots.

$$p_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

$$|p_{1,2}| = \omega_n$$

Let $\sigma = \zeta\omega_n$ and $\omega_d = \omega_n\sqrt{1-\zeta^2}$ (which is called damped natural frequency), then we know that general solution of the ODE solution takes the form

$$y(t) = y_p(t) + C_1e^{-\sigma t} \cos(\omega_d t) + C_2e^{-\sigma t} \sin(\omega_d t)$$

Steady-state conditions leads that $y_p(t) = 1$. Then we can compute remaining coefficients from zero initial conditions constraints

$$\begin{aligned} y(0) = 0 & \rightarrow C_1 = -1 \\ \dot{y}(0) = 0 & \rightarrow \frac{d}{dt} [e^{-\sigma t} [-\cos(\omega_d t) + C_2 \sin(\omega_d t)]]|_{t=0} = 0 \\ & [-\sigma e^{-\sigma t} [-\cos(\omega_d t) + C_2 \sin(\omega_d t)] + e^{-\sigma t} [\omega_d \sin(\omega_d t) + C_2 \omega_d \cos(\omega_d t)]]|_{t=0} = 0 \\ & [\sigma + C_2 \omega_d] = 0 \\ C_2 & = -\frac{\sigma}{\omega_d} = -\frac{\zeta}{\sqrt{1-\zeta^2}} \end{aligned}$$

Finally output, $y(t)$, takes the form

$$y(t) = 1 - e^{-\sigma t} \left[\cos(\omega_d t) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin(\omega_d t) \right]$$

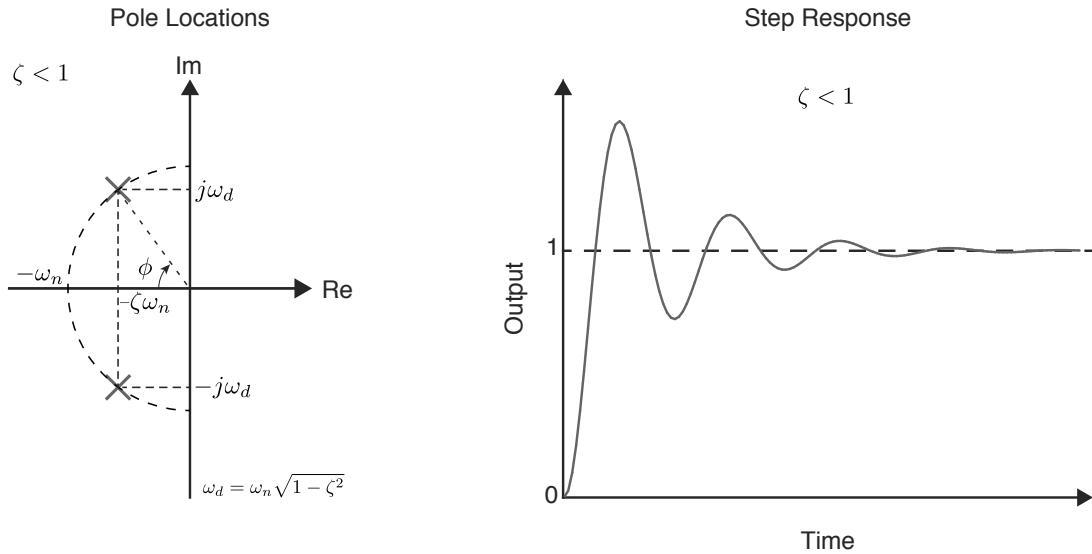
If we combine cos and sin terms into a single sin with phase shift we obtain

$$\begin{aligned} y(t) &= 1 - \frac{e^{-\sigma t}}{\sqrt{1-\zeta^2}} \left[\sqrt{1-\zeta^2} \cos(\omega_d t) + \zeta \sin(\omega_d t) \right] \\ &= 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \phi) \quad t \geq 0 \end{aligned}$$

where

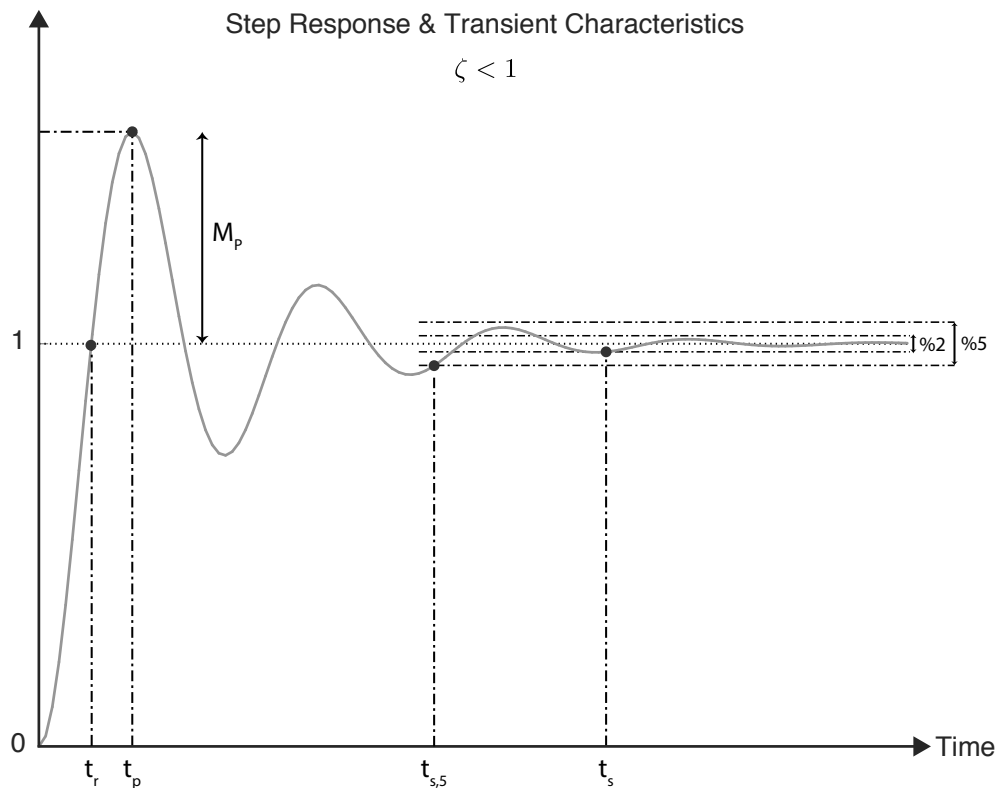
$$\begin{aligned} \sin(\omega_d t + \phi) &= \sin(\phi) \cos(\omega_d t) + \cos(\phi) \sin(\omega_d t) \\ \cos(\phi) &= \zeta \\ \sin(\phi) &= \sqrt{1-\zeta^2} \\ \tan(\phi) &= \frac{\sqrt{1-\zeta^2}}{\zeta} \end{aligned}$$

Pole locations and step response when $\zeta > 1$ (under damped), is illustrated in the Figure below



8.1.2 Transient Response Characteristics for Underdamped Second Order Systems in Standard Form

Important transient characteristics and performance metrics for 2^{nd} order underdamped systems are illustrated in the Figure below.



Rise Time (t_r): The first time instant the response intersects the $y = 1$ line.

$$\begin{aligned}
 y(t_r) &= 1 \\
 1 &= 1 - \frac{e^{-\zeta\omega_n t_r}}{\sqrt{1-\zeta^2}} \sin(\omega_d t_r + \phi) \\
 \pi &= \omega_d t_r + \phi \\
 t_r &= \frac{\pi - \phi}{\omega_d}
 \end{aligned}$$

Peak Time (t_p): The first time instant the response makes a peak

$$\begin{aligned}
 \left[\frac{dy}{dt} \right]_{t_p} &= 0 \\
 \frac{d}{dt} \left[-\frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \left[\sqrt{1-\zeta^2} \cos(\omega_d t) + \zeta \sin(\omega_d t) \right] \right]_{t_p} &= 0 \\
 \left[\zeta\omega_n e^{-\zeta\omega_n t} \left[\sqrt{1-\zeta^2} \cos(\omega_d t) + \zeta \sin(\omega_d t) \right] - e^{-\zeta\omega_n t} \left[-\omega_d \sqrt{1-\zeta^2} \sin(\omega_d t) + \zeta\omega_d \cos(\omega_d t) \right] \right]_{t_p} &= 0 \\
 \left[\left[\zeta\sqrt{1-\zeta^2} \cos(\omega_d t) + \zeta^2 \sin(\omega_d t) \right] - \left[-(1-\zeta^2) \sin(\omega_d t) + \zeta\sqrt{1-\zeta^2} \cos(\omega_d t) \right] \right]_{t_p} &= 0 \\
 \left[[\zeta^2 \sin(\omega_d t) + (1-\zeta^2) \sin(\omega_d t)] \right]_{t_p} &= 0 \\
 \sin(\omega_d t_p) &= 0 \\
 t_p &= \frac{\pi}{\omega_d}
 \end{aligned}$$

Maximum Overshoot (M_p): The maximum amount by which the response exceeds the value 1.

$$\begin{aligned}
 M_p &= y(t_p) - 1 \\
 &= \left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \left[\sqrt{1-\zeta^2} \cos(\omega_d t) + \zeta \sin(\omega_d t) \right] \right]_{t_p} - 1 \\
 &= \left[\frac{e^{-\zeta\omega_n \frac{\pi}{\omega_d}}}{\sqrt{1-\zeta^2}} \sqrt{1-\zeta^2} (-1) \right] \\
 M_p &= e^{-\pi \frac{\zeta}{\sqrt{1-\zeta^2}}} = e^{\frac{-\pi}{\tan \phi}}
 \end{aligned}$$

Maximum Percentage Overshoot (MP_p) is simply calculated as $MP_p = M_p 100$.

Settling Time (t_s): The earliest time instant such that $|y(t) - 1| \leq 0.02s$ or $(|y(t) - 1| \leq 0.05s)$ for all $t \geq t_s$. Actual, settling time is very difficult to compute analytically (not so hard with numerical simulations). Thus we use following approximations.

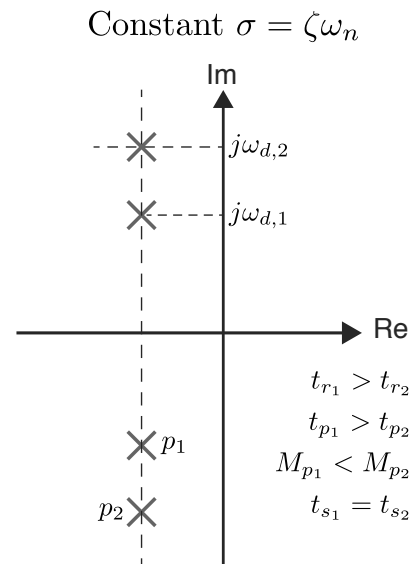
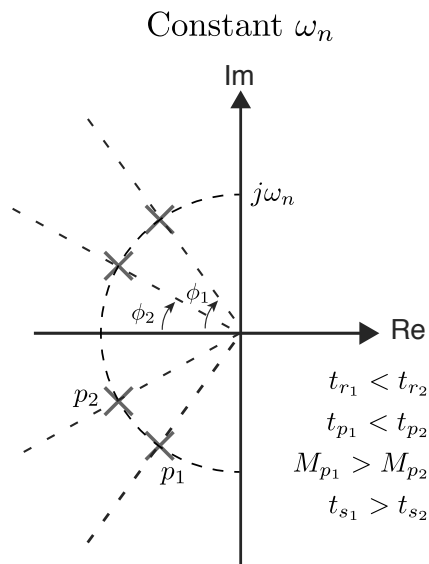
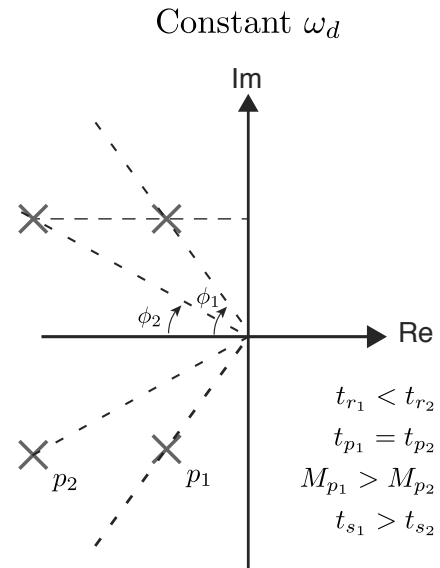
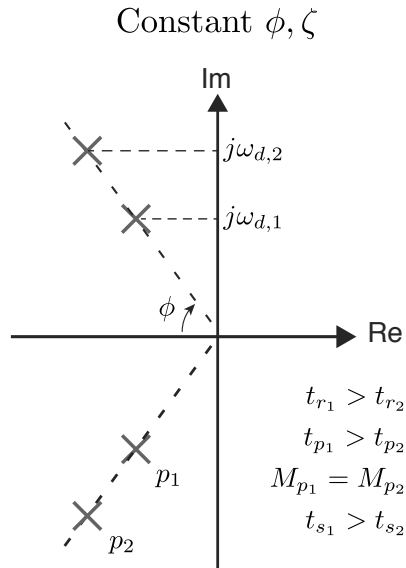
$$\begin{aligned}
 t_{s,5} &= \frac{3}{\zeta\omega_n} \quad \%5 \\
 t_{s,2} &= \frac{4}{\zeta\omega_n} \quad \%2
 \end{aligned}$$

Example 1:

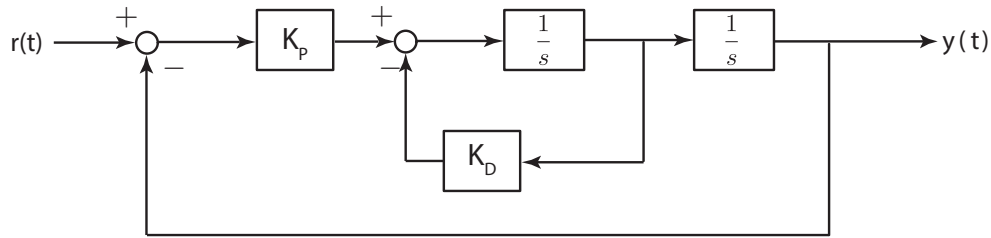
In this problem, we perform four different pole re-location cases. During re-locations we keep some parameters constant. Specifically

1. p_1 moved to a new location p_2 by keeping ζ (and ϕ) constant.
2. p_1 moved to a new location p_2 by keeping ω_d constant.
3. p_1 moved to a new location p_2 by keeping ω_n constant.
4. p_1 moved to a new location p_2 by keeping $\sigma = \zeta\omega_n$ constant.

For each four cases, explain what happens rise time, peak time, maximum overshoot, and settling time.



Example 2: Consider the following closed-loop system



Design K_P and K_D gains such that, maximum percent overshoot is less than %4.32, and settling time (%2) is less than 1 s.

Solution: Lets compute the closed-loop transfer function. In order to do that, first derive the transfer function from $E(s)$ to $Y(s)$ which is called feed-forward transfer function.

$$\frac{Y(s)}{E(s)} = K_P \left[\frac{1/s}{1 + K_D/s} \right] \frac{1}{s} = \frac{K_P}{s(s + K_D)}$$

we can derive the $G(s)$ as

$$G(s) = \frac{\frac{K_P}{s(s+K_D)}}{1 + \frac{K_P}{s(s+K_D)}} = \frac{K_P}{s^2 + K_D s + K_P}$$

We can see that with K_P and K_D gains we have total control on the characteristic equation. Now let's analyze the performance requirements.

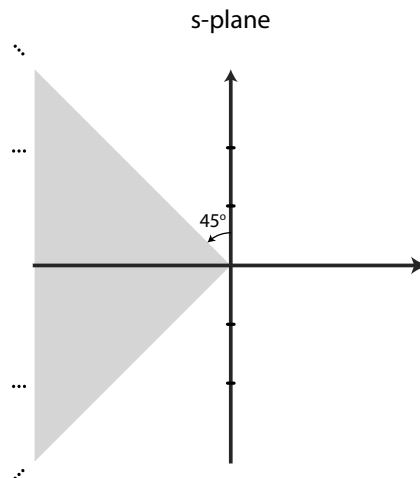
First requirement state that Maximum percent overshoot is less than %4.32, which means that $M_P < 0.0432$. Let's find a condition on ζ or ϕ ,

$$M_P = e^{-\pi \frac{\zeta}{\sqrt{1-\zeta^2}}} = e^{-\pi / \tan \phi} < 0.0432$$

$$-\pi / \tan \phi < -\pi$$

$$\tan \phi < 1$$

The region on s -plane that satisfy the M_P requirement is illustrated below.

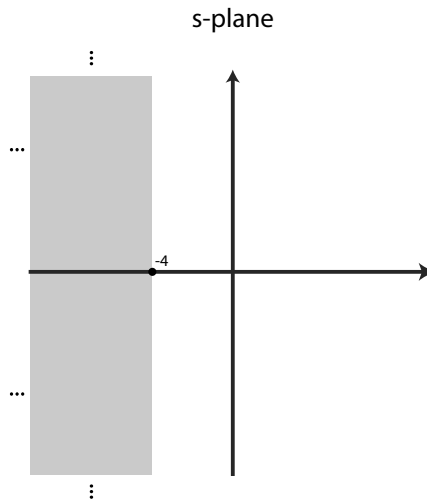


Second requirement state that settling time ($\%2$) is less than 1 s, which implies

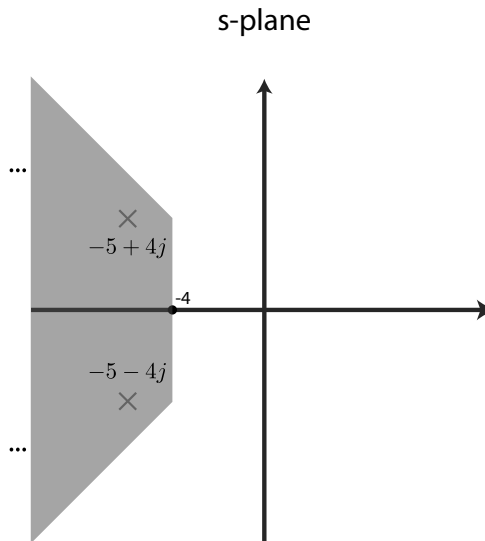
$$t_{s,2} = \frac{4}{\zeta\omega_n} = \frac{4}{\sigma} < 1$$

$$\sigma > 4$$

The region on s -plane that satisfy the settling time requirement is illustrated below.



If we combine the requirements, we obtain the following region of possible pole locations.



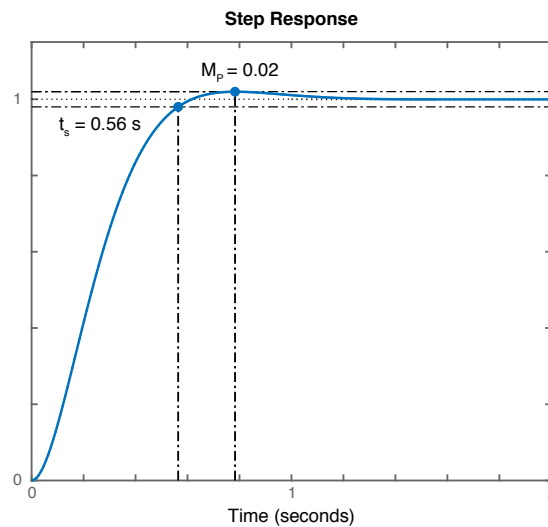
Based on these requirements let's choose $p_{1,2} = -5 \pm 4j$ as desired pole locations. We can then compute the desired characteristic equation and then find the associated controller gains as

$$d^*(s) = (s + 5 + 4j)(s + 5 - 4j) = s^2 + 10s + 41$$

$$K_P = 41$$

$$K_D = 10$$

If we plot the step-response, we can illustrate the performance and check if we can meet the requirements.



Transient Specifications for Over-damped Case

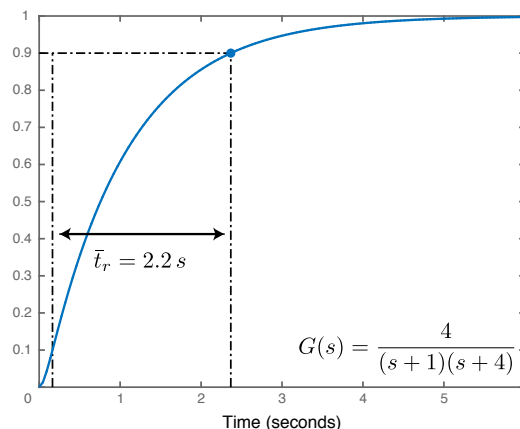
- Obviously, there is no over-shoot in over-damp case, thus $M_P = 0$.
- For settling time we can use the same approximate formula by considering the dominant/slowest pole, i.e.

$$t_{s,5} = \frac{3}{\sigma_{min}} \quad \%5$$

$$t_{s,2} = \frac{4}{\sigma_{min}} \quad \%2$$

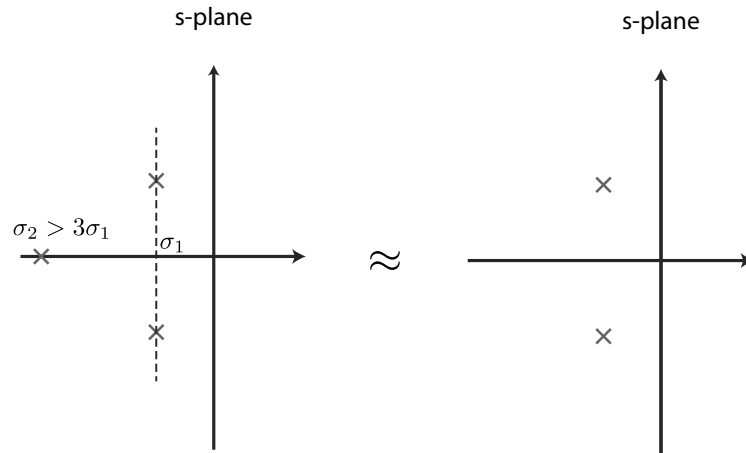
- Since $y(t)$ only crosses the $y = 1$ as $t \rightarrow \infty$, t_r definition is not applicable for over-damped case. Instead, a different rise time definition can be used (which is applicable for both over-damped, under-damped, critically-damped systems, as well as first order systems). $\bar{t}_r$ is the time for $y(t)$ to go from 0.1 to 0.9. It is pretty hard to compute this time analytically, thus in general numerical and/or graphical methods are used.

Rise-time concept is illustrated in the figure below, for an example system.



8.2 Higher Order Systems

In general, the poles closer to the $j\omega$ axis determine the behavior of the system. If the “distance” between the poles that are close to the $j\omega$ axis and other poles is high, then they dominate the behavior and we call them dominant poles. For example, figure given below illustrates a third order system, where we have two complex conjugate roots and one real root. In this case, the magnitude of the real pole is more than three times of the magnitude of the real part of the complex conjugate poles, thus the systems acts like a second order system.



Lecture 9

Lecturer: Asst. Prof. M. Mert Ankarali

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9.1 Steady-State Response Analysis

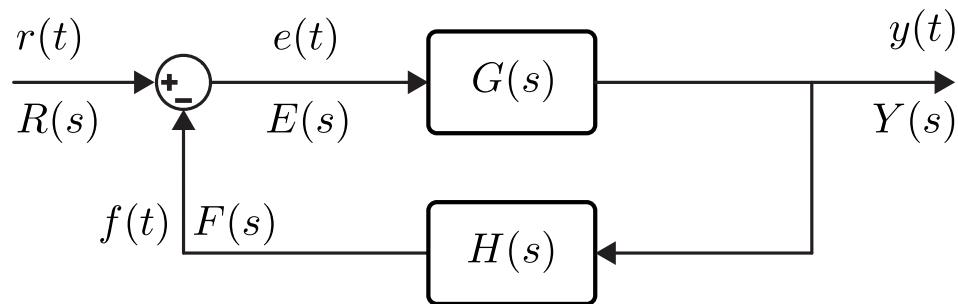
Fundamental concept that we need to perform steady-state response analysis of a control system is the final value theorem. Given a continuous time signal $x(t)$ and its Laplace transform $X(s)$, if $x(t)$ is convergent signal, final value theorem states that

$$\lim_{t \rightarrow \infty} x(t) = \lim_{s \rightarrow 0} [sX(s)]$$

$$x_{ss} = \lim_{s \rightarrow 0} [sX(s)]$$

9.1.1 Tracking Performance

The most important steady-state performance condition for a control system is the tracking performance under steady-state conditions. Let's consider the following fundamental feedback topology.



In order to achieve a good tracking performance, obviously the error signal $e(t)$ need to be small. Accordingly, steady-state tracking performance is determined by the steady-state error of the closed-loop system, that we can compute using final value theorem as

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} [se(s)]$$

Let's compute $E(s)/R(s)$, i.e. transfer function from the reference input to the error signal,

$$E(s) = R(s) - E(s)G(s)H(s),$$

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G(s)H(s)}$$

Note that $G(s)H(s)$ is the transfer function from the error signal $E(s)$ to the signal which is fed to the negative terminal of the main difference operator, i.e. $F(s)$. This transfer function is called the feed-forward or open-loop pulse transfer function of the closed-loop control system. For this system,

$$\frac{F(s)}{E(s)} = G_{OL} = G(s)H(s)$$

Then $E(s)$ can be written as

$$E(s) = R(s) \frac{1}{1 + G_{OL}(s)}$$

It is obvious that first requirement on steady-state error performance is that closed-loop system have to be stable. Now let's analyze specific but fundamental input scenarios.

Unit-Step Input

We know that $r(t) = h(t)$ and $R(s) = \frac{1}{s}$ then we have

$$\begin{aligned} e_{ss} &= \lim_{s \rightarrow 0} \left[s R(s) \frac{1}{1 + G_{OL}(s)} \right] \\ &= \lim_{s \rightarrow 0} \left[s \frac{1}{s} \frac{1}{1 + G_{OL}(s)} \right] \\ e_{ss} &= \frac{1}{1 + \lim_{s \rightarrow 0} G_{OL}(s)} \end{aligned}$$

If the DC gain of the system (also called static error constant) is constant, i.e. $\lim_{s \rightarrow 0} G_{OL}(s) = K_{DC}$ then the steady state error can be computed as

$$e_{ss} = \frac{1}{1 + K_{DC}}$$

It is obvious that

$$\begin{aligned} e_{ss} &\neq 0 \quad \text{if} \quad |K_{DC}| < \infty \\ e_{ss} &\rightarrow 0 \quad \text{if} \quad K_{DC} \rightarrow \infty \end{aligned}$$

At this point, it could be helpful to introduce the concept of system *type*, to generalize the steady-state error analysis.

Definition: Let's write the open-loop transfer function of a closed-loop system in the following standard form

$$G_{OL}(s) = \frac{K}{s^N} \frac{b_0 s^m + \dots + b_{m-1} s + 1}{a_0 s^n + \dots + a_{n-1} s + 1}$$

The closed-loop system is called as **Type N** system, where N is the # of integrators in the open-loop transfer function (OLTF).

Based on these results, we can have the following conclusions regarding steady-state error for unit-step input

- If $G_{OL}(0) = K_P$, $|K_P| < \infty$, then

$$e_{ss} = 1/(1 + K_P)$$

These are **Type 0** (or **Type N** $N \leq 0$) systems. We observe a bounded steady-state error and it is possible to reduce the error by increasing the static gain constant K_P .

- If $G_{OL}(0) = \infty$, then $e_{ss} = 0$. In other words, for **Type** $N > 0$ systems, the steady-state error is perfectly zero .

Now let's summarize the steady-state error conditions

- Type $N \leq 0$: $e_{ss} = \frac{1}{1+K_P}$
- Type $N > 0$: $e_{ss} = 0$

Unit-Ramp Input

We know that $r(t) = th(t)$ and $R(s) = \frac{1}{s^2}$ then we have

$$\begin{aligned}
 e_{ss} &= \lim_{s \rightarrow 0} \left[sR(s) \frac{1}{1 + G_{OL}(s)} \right] \\
 &= \lim_{s \rightarrow 0} \left[s \frac{1}{s^2} \frac{1}{1 + G_{OL}(s)} \right] \\
 &= \frac{1}{\lim_{s \rightarrow 0} s^{-1} G_{OL}(s)} \\
 e_{ss} &= \frac{1}{\lim_{s \rightarrow 0} \frac{K}{s^{N-1}} \frac{b_0 s^m + \dots + b_{m-1} s + 1}{a_0 s^n + \dots + a_{n-1} s + 1}}
 \end{aligned}$$

Based on this result we can have the following steady-state error conditions for the unit-ramp input based on the type condition of the system

- Type $N < 1$: $e_{ss} \rightarrow \infty$
- Type $N = 1$: $e_{ss} = \frac{1}{K_v}$
- Type $N > 1$: $e_{ss} = 0$

where K_v is called the velocity error constant.

Unit-Quadratic (Acceleration) Input

We know that $r(t) = \frac{1}{2}t^2h(t)$ and $R(s) = \frac{1}{s^3}$ then we have

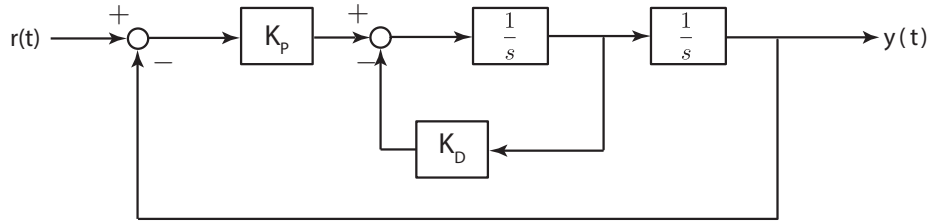
$$\begin{aligned}
 e_{ss} &= \lim_{s \rightarrow 0} \left[sR(s) \frac{1}{1 + G_{OL}(s)} \right] \\
 &= \lim_{s \rightarrow 0} \left[s \frac{1}{s^3} \frac{1}{1 + G_{OL}(s)} \right] \\
 &= \frac{1}{\lim_{s \rightarrow 0} s^{-2} G_{OL}(s)} \\
 e_{ss} &= \frac{1}{\lim_{s \rightarrow 0} \frac{K}{s^{N-2}} \frac{b_0 s^m + \dots + b_{m-1} s + 1}{a_0 s^n + \dots + a_{n-1} s + 1}}
 \end{aligned}$$

Based on this result we can have the following steady-state error conditions for the unit-ramp input based on the type condition of the system

- Type $N < 2$: $e_{ss} \rightarrow \infty$
- Type $N = 2$: $e_{ss} = \frac{1}{K_a}$
- Type $N > 2$: $e_{ss} = 0$

where K_a is called the acceleration (parabolic) error constant.

Example 1: Compute the $G_{OL}(s)$ for the following closed-loop system and define its **Type**. After that, compute the steady-state errors to unit-step, unit-ramp, a and unit-quadratic inputs.



Solution:

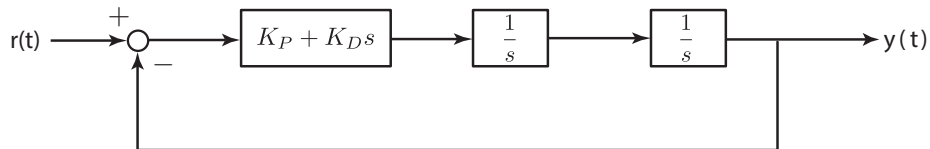
$$G_{OL}(s) = \frac{K_P}{s(s + K_D)}$$

$$\text{Type 1} \quad , \quad K_v = \frac{K_P}{K_D}$$

Then the steady-state errors are computed as

- Unit-step: $e_{ss} = 0$
- Unit-ramp: $e_{ss} = \frac{K_D}{K_P}$
- Unit-acceleration: $e_{ss} = \infty$

Example 2: Compute the $G_{OL}(s)$ for the following closed-loop system and define its **Type**. After that, compute the steady-state errors to unit-step, unit-ramp, a and unit-quadratic inputs.



Solution:

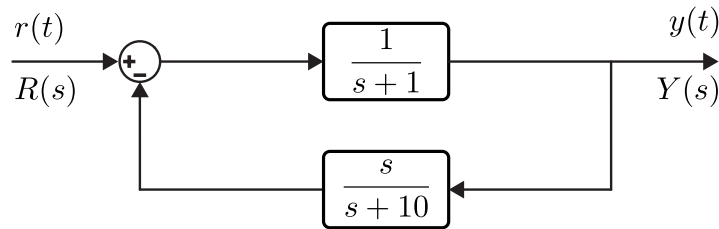
$$G_{OL}(s) = \frac{K_P + K_D s}{s^2}$$

$$\text{Type 2} \quad , \quad K_a = K_P$$

Then the steady-state errors are computed as

- Unit-step: $e_{ss} = 0$
- Unit-ramp: $e_{ss} = 0$
- Unit-acceleration: $e_{ss} = \frac{1}{K_a}$

Example 3: Compute the $G_{OL}(s)$ for the following closed-loop system and define its **Type**. After that, compute the steady-state errors to unit-step, unit-ramp, a and unit-quadratic inputs.



Solution:

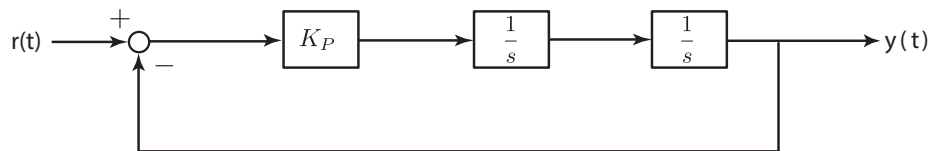
$$G_{OL}(s) = \frac{s}{(s+1)(s+10)}$$

Type **-1** , $K_P = 0$

Then the steady-state errors are computed as

- Unit-step: $e_{ss} = 1$
- Unit-ramp: $e_{ss} = \infty$
- Unit-acceleration: $e_{ss} = \infty$

Example 4: Compute the steady-state error to unit-step input for the following system.



Bad Solution:

$$G_{OL}(s) = \frac{K_p}{s^2}$$

Type **2**

$$e_{ss} = 0 \quad \text{?????}$$

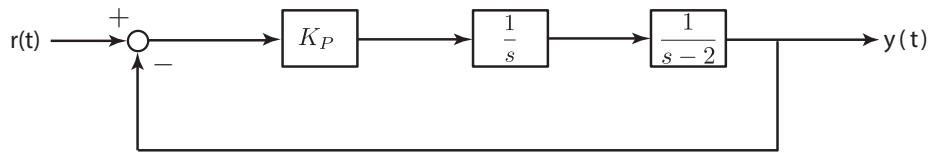
Good Solution: Let's compute $Y(s)$ and then $y(t)$,

$$Y(s) = \frac{\frac{K_p}{s^2}}{1 + \frac{K_p}{s^2}} R(s) = \frac{K_p}{s(s^2 + K_p)}$$

$$y(t) = 1 - \cos(Kt) \quad t > 0$$

Error function takes the form $e(t) = \cos(Kt)$ which does not have a limit, i.e., there is no e_{ss} . If closed-loop transfer function has poles on imaginary axis then, we can not apply final value theorem.

Example 5: Compute the steady-state error to unit-step input for the following system when $K_P = 1$.



Good Solution :) Let's check if $y(t)$ is a convergent signal

$$Y(s) = \frac{\frac{1}{s(s-2)}}{1 + \frac{1}{s(s-2)}} R(s) = \frac{1}{s(s^2 - 2s + 1)}$$

$$y(t) = te^t - e^t + 1 \quad t > 0$$

Error function takes the form $e(t) = e^t - te^t$, thus

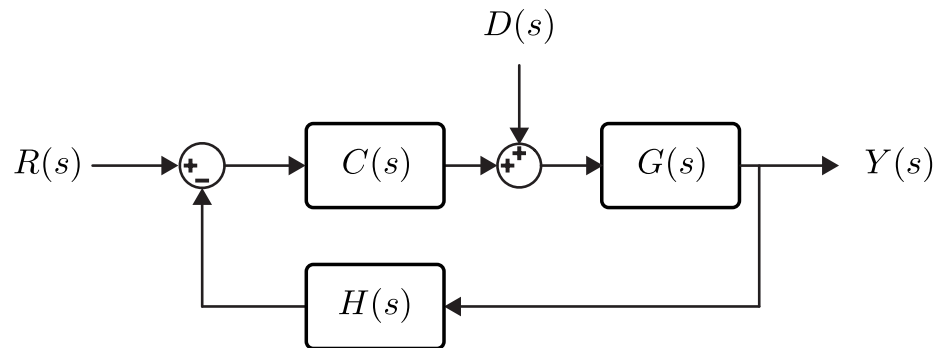
$$e_{ss} = \left| \lim_{t \rightarrow \infty} e(t) \right| = \infty$$

In conclusion, If closed-loop transfer function has poles on imaginary axis or open right half-plane then, we can not apply final value theorem.

9.1.2 Stead-State Response to Disturbances

When analyzing the steady-state response of a system in addition to the desired response to the reference input, it is also important to analyze the response to unwanted disturbances and noises.

Let's analyze the steady-state performance of the following topology which is perturbed by a disturbance input, $d(t)$.

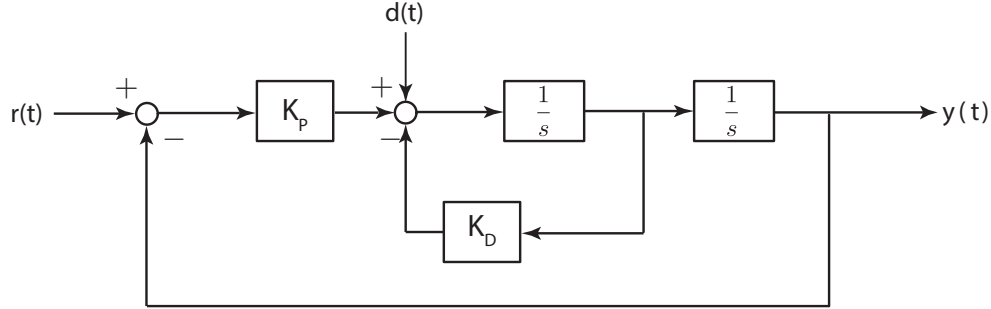


In order to analyze the response to the disturbance $d(t)$, we assume $r(t) = 0$ (which is just fine due to the linearity). Let's first find the pulse transfer function from $D(s)$ to $Y(s)$.

$$\begin{aligned} T_D(s) &= \frac{Y(s)}{D(s)} = \frac{G(s)}{1 + C(s)G(s)H(s)} \\ &= \frac{G(s)}{1 + G_{OL}(s)} \end{aligned}$$

Note that $Y(s)$ depends on both $G_{OL}(s)$ (OLTF) and $G(s)$ (Plant TF). If one wants to generalize the steady-state disturbance rejection performance, he/she needs to analyze the conditions for both $G_{OL}(s)$ and $G(s)$. Moreover, for a different topology and type of disturbance, we can have very different conditions. For this reason, in order to analyze steady-state disturbance/noise rejection performance, it is better to use fundamentals and apply final value theorem.

Example 6: The following closed-loop system is affected by a disturbance input $d(t)$. Compute the steady-state performance/response to a unit step disturbance input.



Solution: Lets compute $Y(s)/D(s)$

$$Y(s) = (D(s) - Y(s)K_P) \frac{1}{s(s + K_D)}$$

$$Y(s) \left[1 + \frac{K_P}{s(s + K_D)} \right] = D(s) \frac{1}{s(s + K_D)}$$

$$\frac{Y(s)}{D(s)} = \frac{\frac{1}{s(s+K_D)}}{\frac{s^2+K_Ds+K_P}{s(s+K_D)}} = \frac{1}{s^2 + K_Ds + K_P}$$

Now let's compute y_{ss} ,

$$y_{ss} = \lim_{s \rightarrow 0} [sY(s)] = \lim_{s \rightarrow 0} \left[sD(s) \frac{1}{s^2 + K_Ds + K_P} \right]$$

$$= \lim_{s \rightarrow 0} \left[s \frac{1}{s} \frac{1}{s^2 + K_Ds + K_P} \right]$$

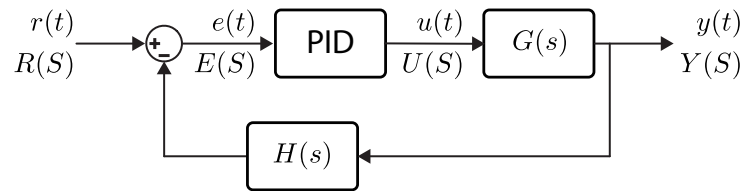
$$= \frac{1}{K_P}$$

We can see that even if same system has 0 steady-state error when the reference signal is step-like input, the error under unit-step disturbance is not zero, i.e., $y_{ss} = 1/K_P$. One can improve the disturbance rejection performance by increasing the K_P gain.

Lecture 10

Lecturer: Asst. Prof. M. Mert Ankarali

10.1 PID Control Policy



A PID controller has the following forms in time and Laplace domain

$$u(t) = K_P e(t) + K_D \frac{d}{dt} e(t) + K_I \int_0^t e(t) dt$$

$$U(s) = \left[K_P + K_D s + K_I \frac{1}{s} \right] E(s)$$

In this lecture we will analyze the effects of PID coefficients on the transient and steady-state performance on a 2nd order plant.

10.1.1 Proportional (P) Controller

Let's assume that $H(s) = 1$ and $G(s)$ is a second order transfer function in general form

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n + \omega_n^2}$$

Since $C(s) = K_P$, let's first analyze the steady-state error performance

$$G_{OL}(s) = \frac{K_P \omega_n^2}{s^2 + 2\zeta\omega_n + \omega_n^2}$$

$$\text{Type 0} \quad K_p = K_P$$

Thus steady-state error for unit-step and unit-ramp inputs can be found as

- Unit step: $e_{ss} = \frac{1}{1+K_P}$, i.e. $K_P \nearrow \Rightarrow e_{ss} \searrow$
Unit ramp: $e_{ss} = \infty$

To sum up, higher K_P provides better steady-state performance. Now let's analyze transient performance.

$$T(s) = \frac{Y(s)}{R(s)} = \frac{K_P \omega_n^2}{s^2 + 2\zeta\omega_n + (1 + K_P)\omega_n^2}$$

$$\bar{\omega}_n = \sqrt{1 + K_P}\omega_n$$

$$\bar{\zeta} = \zeta \frac{1}{\sqrt{1 + K_P}}$$

We can see that

$$K_p \nearrow \Rightarrow \omega_n \nearrow$$

$$K_p \nearrow \Rightarrow \zeta \searrow$$

If the plant is an over-damped system (i.e. $\zeta > 1$), then increasing ω_n and decreasing ζ should have a positive net effect on the closed-loop performance.

On the other hand if the plant is an under-damped system, we can observe the following relations

$$\bar{\omega}_d = \bar{\omega}_n \sqrt{1 - \bar{\zeta}^2} = \omega_n \sqrt{1 + K_P} \sqrt{1 - \frac{\zeta^2}{1 + K_P}} = \omega_n \sqrt{1 + K_P - \zeta^2}$$

$$\bar{\zeta} \bar{\omega}_n = \frac{\zeta}{\sqrt{1 + K_P}} \omega_n \sqrt{1 + K_P} = \zeta \omega_n$$

We can see that

$$K_p \nearrow \Rightarrow \omega_d \nearrow$$

$$K_p \nearrow \Rightarrow \zeta \omega_n =$$

In other words real part of the complex conjugate poles are unchanged, yet imaginary part deviates from the real axis. From previous lectures we know that in this scenario

$$K_p \nearrow \Rightarrow M_p \nearrow$$

In conclusion, If the plant is an under-damped system (i.e. $\zeta < 1$), then increasing the P gains has a negative effect on the closed-loop performance (transient).

Example 1: Let $G(s) = \frac{1}{(s+0.5)(s+5)}$ (and over-damp plant), then compute the unit-step steady state error for $K_p = 2$ and $K_P = 5$.

$$e_2 = \frac{1}{1 + 2/2.5} \approx 0.55, (\%55)$$

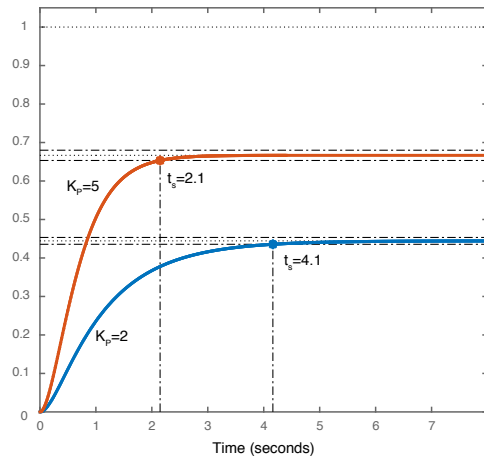
$$e_5 = \frac{1}{1 + 5/2.5} \approx 0.33, (\%33)$$

Obviously, steady-state performance is better with $K_p = 5$ compared to $K_p = 2$. Now compute the closed-loop poles for $K_p = 1$ and $K_P = 5$, and estimate associated settling times (%2).

$$T_2(s) = \frac{2}{s^2 + 5.5s + 4.5} \rightarrow p_1^{(2)} = -1, p_2^{(2)} = -4.5 \quad t_s^{(2)} \approx 4s$$

$$T_5(s) = \frac{5}{s^2 + 5.5s + 7.5} \rightarrow p_1^{(2)} = -2.5, p_2^{(2)} = -3 \quad t_s^{(2)} \approx 1.6s$$

We can see that $K_P = 5$ provides a better transient performance compared to $K_P = 2$. Now let's draw step-responses and verify these observations



We verify that both steady-state and transient performance increases with larger K_P . However, we can also see that settling time estimation for $K_P = 5$ has a larger error, which is expected since the poles are close to each other thus violating the dominant pole assumption.

Example 2: Let $G(s) = \frac{1}{s^2 + 4s + 5}$ (and under-damped plant), then compute the unit-step steady state error for $K_P = 3$ and $K_P = 8$.

$$e_3 = \frac{1}{1 + 3/5} \approx 0.625, (\%62.5)$$

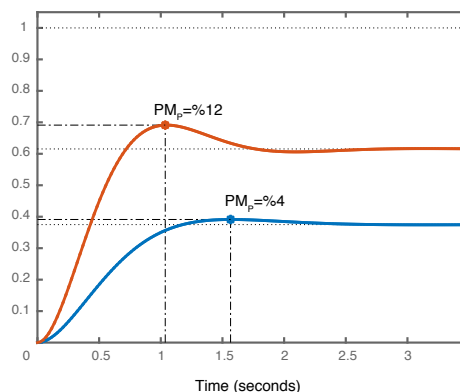
$$e_8 = \frac{1}{1 + 8/5} \approx 0.385, (\%38.5)$$

Obviously, steady-state performance is better with $K_P = 8$ compared to $K_P = 3$. Now compute the closed-loop poles for $K_P = 3$ and $K_P = 8$, and estimate associated maximum overshoots

$$T_3(s) = \frac{3}{s^2 + 4s + 8} \rightarrow p_{1,2}^{(3)} = -2 \pm 2j \quad M_P = M_P = e^{-\pi/\tan \phi_3} = e^{-\pi} \approx 0.04 (\%4)$$

$$T_8(s) = \frac{5}{s^2 + 4s + 13} \rightarrow p_{1,1}^{(8)} = -2 \pm 3j \quad M_P = e^{-\pi/\tan \phi_8} = e^{-\pi/2/3} \approx 0.12 (\%12)$$

We can see that $K_P = 3$ provides a better transient performance compared to $K_P = 8$, since both frequency oscillates and overshoot is increased with a higher P gain. Now let's draw step-responses and verify these observations.



We verify that steady-state performance is better with a larger P gain. However, transient performance is worse with a larger P-gain. This implies that if the plant is an under-damped plant, then P controller has some serious limitations.

10.1.2 Proportional Derivative (PD) Controller

In the classical form of PD controller, $C(s)$, takes the form

$$C(s) = K_P + K_D s$$

Let's first analyze the effect of K_D term on steady-state performance on the same case (2^{nd} order plant in standard form)

$$G_{OL}(s) = \frac{K_P \omega_n^2 + K_D \omega_n^2 s}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$

$$\text{Type 0 } K_p = K_P$$

Thus steady-state error for unit-step and unit-ramp inputs can be found as

- Unit step: $e_{ss} = \frac{1}{1+K_P}$, i.e. $K_P \nearrow \Rightarrow e_{ss} \searrow$
Unit ramp: $e_{ss} = \infty$

Obviously, K_D has no effect on steady-state performance. Now let's analyze the closed-loop transfer function

$$\begin{aligned} T(s) &= \frac{Y(s)}{R(s)} = \frac{K_P \omega_n^2 + K_D \omega_n^2 s}{s^2 + 2\zeta \omega_n s + \omega_n^2 + K_D \omega_n^2 s + K_P \omega_n^2} \\ &= \frac{K_P \omega_n^2 + K_D \omega_n^2 s}{s^2 + (2\zeta \omega_n + K_D \omega_n^2)s + (1 + K_P)\omega_n^2} \\ &, \\ \bar{\omega}_n &= \sqrt{1 + K_P} \omega_n \\ \bar{\zeta} &= \frac{\zeta + K_D \omega_n / 2}{\sqrt{1 + K_P}} \end{aligned}$$

We can see that

$$\begin{aligned} K_P \nearrow &\Rightarrow \omega_n \nearrow \ \& \ \zeta \searrow \\ K_D \nearrow &\Rightarrow \omega_n = \ \& \ \zeta \nearrow \end{aligned}$$

In other words, for a second order system, we have full control on closed-loop pole locations with PD control policy. Since, a high K_P is required/preferred for steady-state performance (which causes the system to have overshoot and oscillatory behavior), K_D term can be used to suppress oscillations and overshoot. Note that in the closed-loop transfer function, numerator part has a zero due to $K_D \omega_n^2 s$ term, which implies that the closed-loop transfer function is not in standard 2^{nd} order form. One should note the fact that, the existence of closed-loop zero can affect the accuracy of our closed-loop transient performance metric calculations (most probably deviations will be minor).

Example 3: Let $G(s) = \frac{1}{s^2 + 4s + 5}$ (and under-damp plant). Design a PD controller such that steady-state error to unit-step input is around %20 and the maximum percentage overshoot is less than %4.

Solution: We first design K_D based on steady-state requirement then choose K_D based on the over-shoot requirement.

$$e_{ss} = \frac{1}{1 + K_P/5} = 0.2$$

$$K_P = 20$$

Now let's compute the closed-loop transfer function with $K_P = 20$.

$$T(s) = \frac{20 + K_D s}{s^2 + (4 + K_D)s + 25}$$

Let $K_D = 4$, then the closed-loop transfer function and associated poles are computed as

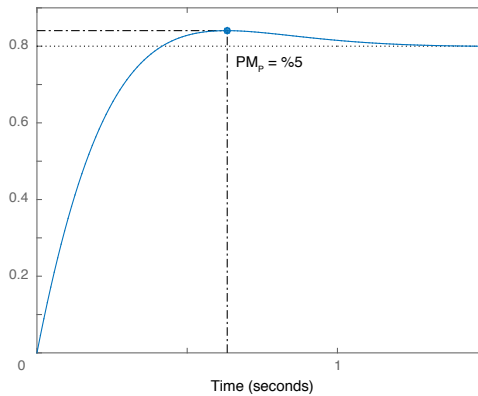
$$T(s) = \frac{20 + 4s}{s^2 + 8s + 25}$$

$$p_{1,2} = -4 \pm 3j$$

We can estimate the overshoot based on the pole locations

$$PM_P = \%100e^{-\pi/\tan\phi} = e^{-\pi 4/3} = \%1.5 < \%4$$

Let's plot the step-response of the resultant system and check if we can meet the specifications.

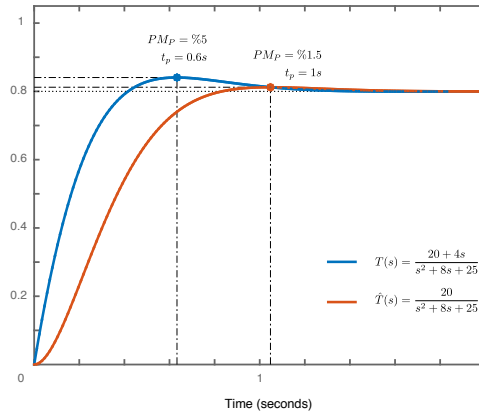


We can observe that the computed over-shoot is %5, and indeed higher than the requirement. Moreover, the gap between estimated and numerically computed over-shoot is around %3.5. Obviously, the $K_D s = 4s$ term in the numerator effects the output behavior and its effect should be maximum when $0 < t < t_s$. In order to see how $K_D s = 4s$, let's compare step responses of following transfer functions

$$T(s) = \frac{20 + 4s}{s^2 + 8s + 25}$$

$$\hat{T}(s) = \frac{20}{s^2 + 8s + 25}$$

$T(s)$ and $\hat{T}(s)$ share the same poles and DC gain, but $\hat{T}(s)$ is in standard form, thus has no zeros.



We can see that there exist non-negligible differences between two transfer functions, which clearly shows that the numerator dynamics can substantially affect the response. In this context, we should think of the transient performance metrics and associated approximate estimation formulas as heuristics which guides the design of controllers.

10.1.3 Integral (I) Controller

Practical use of Integral controller (alone) is extremely rare, however we will analyze this case to better understand the effect of Integral action on more useful PI and PID topologies.

In the pure Integral controller, $C(s)$, takes the form

$$C(s) = K_I \frac{1}{s}$$

Let's first analyze the steady-state performance where the plant is a 2nd system in standard

$$G_{OL}(s) = \frac{K_I \omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

Type 1 $K_v = K_I$

Thus steady-state error for unit-step and unit-ramp inputs can be find as

- Unit step: $e_{ss} = 0$
- Unit ramp: $e_{ss} = \frac{1}{K_I}$, i.e. $K_I \nearrow \Rightarrow e_{ss} \searrow$

Basic idea is very clear, Integral action increases the type of the system by introducing an extra pole at the origin (also increases the total system order). Thus for a Type 0 plant, it completely eliminates the steady-state error for step-like inputs, and provides a constant steady-state error for ramp-like inputs.

Let's compute the closed-loop transfer function

$$T(s) = \frac{K_I \omega_n^2}{s^3 + 2\zeta\omega_n s^2 + \omega_n^2 s + K_I \omega_n^2}$$

The closed loop system is now a third order system and thus harder to analyze (we need new analysis tools). Moreover, we have only one parameter K_I and the closed-loop system has three poles. One may easily guess that it is very hard to obtain a good performance with a pure Integral controller. Sometimes it may be even quite hard to obtain a stable behavior.

Example 4: Let's analyze the influence of an I controller on a first order plant in order to better understand the positive and negative effects of I action. Let $G(s) = \frac{1}{s+2}$.

The steady-state value of $y(t)$ (for unit-step input) for the uncontrolled plant is $y_{ss} = 1/2$, thus we can say that steady-state error under unit-step input is $e_{ss} = 0.5$, where as for unit ramp it is easy to show that $e_{ss} = \infty$. On the other hand, we can estimate the settling time (%2) for the uncontrolled plant as $t_s \approx 2$.

Now let's first analyze the steady-state performance under $C(s) = \frac{K_I}{s}$,

$$G_{OL}(s) = \frac{K_I}{s(s+2)}$$

Type 1 $K_v = K_I/2$

- Unit step: $e_{ss} = 0$
- Unit ramp: $e_{ss} = \frac{2}{K_I}$, i.e. $K_I \nearrow \Rightarrow e_{ss} \searrow$

Obviously, steady-state performance improvement is significant (structurally). Now, let's compute closed-loop transfer function

$$T(s) = \frac{K_I}{s^2 + 2s + K_I}$$

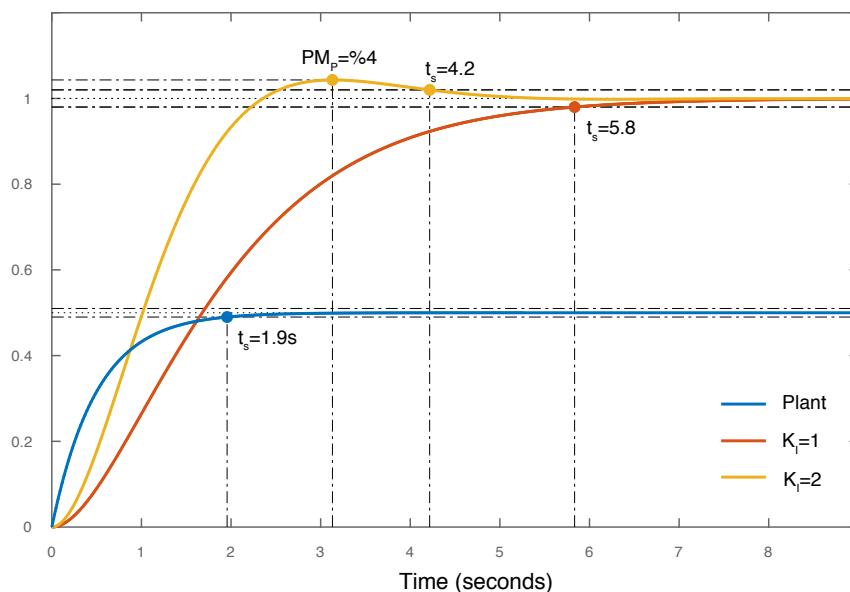
Now let's analyze the transient performance (settling time and maximum overshoot) for $K_I = 1$ and $K_I = 2$ and compare them w.r.t. uncontrolled plant

$$K_I = 1 \rightarrow \zeta = 1, p_{1,2} = -1 \rightarrow M_P = 0, t_s \approx 4s$$

$$K_I = 2 \rightarrow \zeta = 1/\sqrt{2}, p_{1,2} = -1 \pm j \rightarrow M_P = 0.04, t_s \approx 4s$$

We can clearly see that integral action has a negative effect on transient performance. $K_I = 1$ case has a worse settling time value than the original plant; Moreover we start to observe overshoot at the output when we increase integral gain to $K_I = 2$ case.

In order to illustrate these analytic observations, we plotted the step responses of original plant, closed-loop system with $K_I = 1$ and closed-loop system with $K_I = 2$.



It is clear that PI controller eliminates the steady-state error, but we can also observe that transient performance is substantially degraded. In both cases settling time is worse than the original plant, and for case $K_I = 2$, overshoot is clear from the figure. One interesting result is that settling time performance for $K_I = 1$ (critically damped) is worse than $K_I = 2$ (under-damped) even though the approximate formula provides the same estimate. The reason is that for the critically damped case, the approximation underestimates the settling time.

In this context, we can conclude that a little bit overshoot could be good for the closed-loop system from the perspective of settling time.

10.1.4 Proportional Integral (PI) Controller

PI controller is commonly used in practical applications. In general, if a single P controller is satisfactory for transient requirements, but one seeks perfect steady-state performance for unit-step like inputs, *PI* is the first choice to test (compared to PID).

In the PI controller, $C(s)$, takes the form

$$C(s) = K_P + \frac{K_I}{s}$$

Let's first analyze the steady-state performance again with the plant is a 2^{nd} system in standard

$$G_{OL}(s) = \frac{K_P \omega_n^2 s + K_I \omega_n^2}{s(s^2 + 2\zeta \omega_n s + \omega_n^2)}$$

Type 1 $K_v = K_I$

Thus steady-state error for unit-step and unit-ramp inputs can be found as

- Unit step: $e_{ss} = 0$
- Unit ramp: $e_{ss} = \frac{1}{K_I}$, i.e. $K_I \nearrow \Rightarrow e_{ss} \searrow$

In other words *PI* and *I* controllers have same steady-state performance characteristics. Now, let's compute the closed-loop transfer function

$$T(s) = \frac{K_P \omega_n^2 s + K_I \omega_n^2}{s^3 + 2\zeta \omega_n s^2 + (1 + K_P) \omega_n^2 s + K_I \omega_n^2}$$

In this case, the closed-loop transfer function has three poles, and we have two parameters for tuning. Even though it provides a much better framework than an I controller, still we need different tools to tune the K_I and K_P gains.

Example 5: Let's compare PI and I controllers using the same plant in previous example, $G(s) = \frac{1}{s+2}$. We know that PI controller has the transfer function form $C(s) = \frac{K_I}{s}$ and steady-state error characteristics can be derived as

$$G_{OL}(s) = \frac{sK_P + K_I}{s(s+2)}$$

Type 1 $K_v = K_I/2$

Unit step : $e_{ss} = 0$

Unit ramp : $e_{ss} = \frac{2}{K_I}$

which are exactly same with the I controller. Now, let's compute closed-loop transfer function

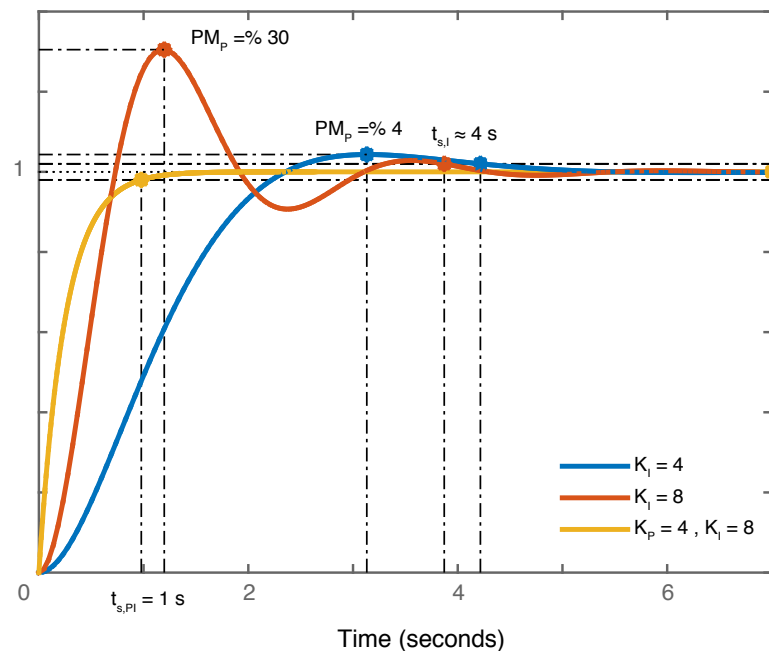
$$T(s) = \frac{K_D s + K_I}{s^2 + (2 + K_D)s + K_I}$$

Now let's choose $K_I = 8$ and $K_P = 4$, and estimate maximum over-shoot and settling time for the closed-loop plant.

$$p_1 = -2, p_2 = -4 \rightarrow M_P = 0, t_s \approx 2s$$

These PI gains can match the settling time value of the original plant without any over-shoot (since closed-loop TF is over-damped).

Let's illustrate this analytic observations, by plotting the step responses of the closed-loop system with only integral controller with $K_I = 2$, the closed-loop system with only integral controller with $K_I = 8$, and the closed-loop system with PI controller with $K_P = 4, K_I = 8$.



In this illustration, we can see that settling time for both I controllers is around 4s, however settling time for the PI controller is 1s which is even better than our estimation. The gap between settling time estimates come from the effect of zero introduced by the PI controller. If one carefully analyzes the closed-loop transfer function, he/she can see that a pole-zero cancellation occurs (which may not be a good feature for practical reasons). Technically, the closed-loop transfer function is reduced to a first order system in this case (????).

We also see that when PI and I controllers has the same K_I gain, the over-shoot in I controller is %30 which is very high.

10.1.5 Proportional Integral Derivative (PID) Controller

PID controller technically combines the advantages of the PD and PI controllers, with the trade of increased parameter and implementation complexity. We know that PID controller has the following transfer function

form

$$C(s) = K_P + K_D s + \frac{K_I}{s}$$

Let's first analyze the steady-state performance again where the plant is a 2nd system in standard form

$$G_{OL}(s) = \frac{(K_D s^2 + K_P s + K_I) \omega_n^2}{s(s^2 + 2\zeta \omega_n s + \omega_n^2)}$$

Type 1 $K_v = K_I$

Thus steady-state error for unit-step and unit-ramp inputs can be find as

- Unit step: $e_{ss} = 0$
- Unit ramp: $e_{ss} = \frac{1}{K_I}$, i.e. $K_I \nearrow \Rightarrow e_{ss} \searrow$

In other words *PID*, *PI*, and *I* controllers share the same steady-state performance characteristics. Now, let's compute the closed-loop transfer function

$$T(s) = \frac{(K_D s^2 + K_P s + K_I) \omega_n^2}{s^3 + (2\zeta \omega_n + K_D \omega_n^2) s^2 + (1 + K_P) \omega_n^2 s + K_I}$$

We can see that the closed-loop transfer function has three poles, and we have three parameters to tune. If we have no limits on gains, we can place the closed-loop poles to any desired location. However, numerator has now two zeros, thus it is now harder to predict the effect of closed-loop zeros on the output behavior.

Example 4: Let $G(s) = \frac{1}{s^2 + 4s + 5}$, Design a PID controller such that we observe zero unit-step steady state error, maximum percent overshoot is less than %5, and settling time is around 1s.

Solution: We already know that PID controller for a Type 0 plant completely eliminates the unit-step steady state error. Now let's compute closed-loop transfer function

$$T(s) = \frac{K_D s^2 + K_P s + K_I}{s^3 + (4 + K_D) s^2 + (5 + K_P) s + K_I}$$

One way of choosing appropriate pole locations for a third-order closed-loop system is placing one of the poles (a real one) far away from the other poles such that closed-loop system shows a second order like behavior. We require that maximum overshoot of %5 and settling time is around 1s. Let the dominant poles of the closed-loop system be

$$p_{1,2} = -4 \pm 3j$$

Settling time and maximum overshoot associated with this closed-loop poles can be estimated as

$$t_s \approx 1s$$

$$M_P \approx e^{-\pi/4} = 0.015$$

These estimates satisfy the requirements. Let $p_3 = 3(-4) = -12$, then desired characteristic equation takes the form

$$d^*(s) = (s + 12)(s^2 + 8s + 25) = s^3 + 20s^2 + 121s + 300$$

Then, we can compute the PID gains as

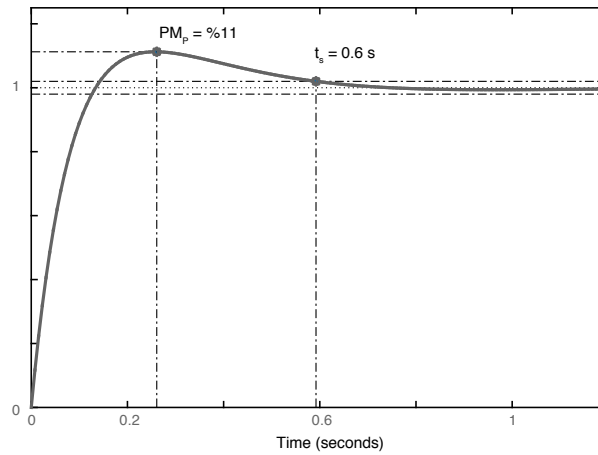
$$K_P = 116$$

$$K_I = 300$$

$$K_D = 16$$

The first thing, we can observe is that the quantitative values of the gains seem to be much larger than the gains that we played before. In general, if we want to improve the performance of both steady-state and transient characteristics by implementing PID topology instead of P, PI, or PD the gain values would go up which may cause practical problems and potentially can be costly in terms of energetic performance.

Now let's plot the step response of the closed-loop system and try to verify these analytic observations



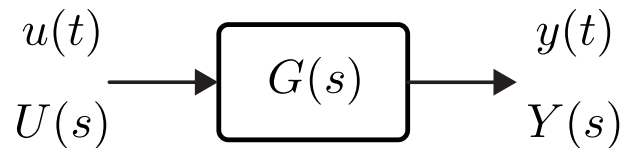
We can see that the settling time in numerical simulation is much better than our estimate $0.6s < 1s$, however numerical percentage over-shoot is higher than our estimate, $\%11 > \%1.5$, and does not meet our specifications. The core reason behind this is that it is basically harder to tune a PID controller compared to P, PD, and PI controllers due to increased parametric complexity and (may be more importantly) 2nd numerator dynamics introduced with the PID control policy.

Lecture 11

Lecturer: Asst. Prof. M. Mert Ankarali

11.1 Stability

A SISO system is called BIBO (bounded-input-bounded-output) stable, if the output will be bounded for **every** input to the system that is bounded.



If we use the impulse response representation for an LTI SISO system, i.e.. $y(t) = \int_0^t g(t-\tau)u(\tau)d\tau$ (we assume that system is causal), the system is BIBO stable if and only if its impulse response is absolutely integrable.

$$\text{BIBO Stable} \Leftrightarrow \int_0^{\infty} |g(t)|dt < B < \infty$$

However, we generally use transfer function representation in this course. Let

$$\frac{Y(s)}{U(s)} = G(s) = \frac{N(s)}{D(s)}$$

In this context, a rational transfer function representation is BIBO stable if and only if, the poles of $G(s)$ (or roots of $D(s)$) are strictly located in the open left half s -plane.

Ex: Show that $G(s) = \frac{1}{s}$ is not BIBO stable

Solution: Let's first check the impulse response condition

$$g(t) = 1, t \geq 0$$

$$\lim_{t \rightarrow \infty} \int_0^t |g(\tau)|d\tau = \lim_{t \rightarrow \infty} \int_0^t 1d\tau = \infty$$

Thus the system is BIBO unstable. The system has a single pole at the origin, so we already know that it is BIBO unstable. Now, let's find a specific bounded input such that output is unbounded. Let $u(t)$ be the unit-step input then

$$Y(s) = \frac{1}{s^2} \Rightarrow y(t) = t, t \geq 0$$

$$\lim_{t \rightarrow \infty} |y(t)| = \infty$$

Obviously, we obtained an un-bounded output.

Ex: Let $G(s) = \frac{1}{s^2+1}$. Find a bounded input, such that output is unbounded. **Solution:** Let $u(t) = \cos t, t \geq 0$, then

$$Y(s) = G(s)U(s) = \frac{s}{(s^2 + 1)^2}$$

$$y(t) = t \sin t, t \geq 0$$

Now we obtain an un-bounded output due to the resonance effect.

11.1.1 Stability of First & Second Order Systems

In order to gain some intuition about how to check stability of general rational LTI systems, we will analyze first and second order systems.

The transfer function of a first order system has the form

$$D(s) = a_0 s + a_1$$

$$a_0 > 0, \text{ w.l.g}$$

The single pole of the system and associated stability condition can be derived as

$$p = -\frac{a_1}{a_0}$$

$$\text{Stable} \Leftrightarrow a_1 > 0$$

Now lets analyze second order systems. Transfer function of a second order system has the following form

$$D(s) = a_0 s^2 + a_1 s + a_2$$

$$a_0 > 0, \text{ w.l.g}$$

If we carefully analyze $\text{Sign}[D(0)]$, we can see that

- $\text{Sign}[D(0)] < 0$, one pole is located in the open-left half plane, where as the other one is located in the open- right half plane.
- $\text{Sign}[D(0)] = 0$, there exist at least one pole at the origin.

In this context, we can derive the first condition (necessary but not sufficient)

$$\text{Stable} \Rightarrow a_2 > 0$$

Under this condition, we can re-write the characteristic equation in a more standard form

$$D(s) = a_0 \left(s^2 + \frac{a_1}{a_0} s + \frac{a_2}{a_0} \right)$$

$$D(s) = a_0 (s^2 + 2\zeta\omega_n s + \omega_n^2)$$

where

$$\omega_n = \sqrt{a_2/a_0} > 0$$

$$\zeta = \frac{a_1}{2a_0\omega_n}$$

From the analysis of second order system in standard form, we know that when

- $\zeta < 0$, system poles are located in the open right-half plane
- $\zeta = 0$, system poles are located on the imaginary axis
- $\zeta > 0$, system poles are located in the open left-half plane

It is also easy to see that $\zeta > 0 \Leftrightarrow a_1 > 0$. As a result, we can derive a necessary and sufficient condition on BIBO stability.

A second order system with $D(s) = a_0 s^2 + a_1 s + a_2$ (with $a_0 > 0$), is BIBO stable if and only if $a_i > 0$, $\forall i \in \{1, 2\}$.

11.1.2 Routh's Stability Criterion

Routh's Stability Criterion (a.k.a Routh–Hurwitz stability criterion) is a mathematical test that is a necessary and sufficient condition for the stability of an LTI system. The Routh test is a very computationally efficient algorithm for the test of absolute LTI system stability.

In this test, one first constructs the routh table for a given

$$D(s) = a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} + \cdots + a_{n-2} s^2 + a_{n-1} s^1 + a_n$$

which has $n + 1$ rows.

s^n	a_0	a_2	a_4	a_6	$\cdots$
s^{n-1}	a_1	a_3	a_5	a_7	$\cdots$
s^{n-2}	b_1	b_2	b_3	b_4	$\cdots$
s^{n-3}	c_1	c_2	c_3	c_4	$\cdots$
s^{n-4}	d_1	d_2	d_3	d_4	$\cdots$
$\vdots$	$\vdots$	$\vdots$			
s^2	e_1	e_2	0	$\cdots$	
s^1	f_1	0	0	$\cdots$	
s^0	g_0	0	0	$\cdots$	

where the coefficients b_i are computed with

$$\begin{aligned}
 b_1 &= -\det \left(\begin{bmatrix} a_0 & a_2 \\ a_1 & a_3 \end{bmatrix} \right) / a_1 = \frac{a_1 a_2 - a_0 a_3}{a_1} \\
 b_2 &= -\det \left(\begin{bmatrix} a_0 & a_4 \\ a_1 & a_5 \end{bmatrix} \right) / a_1 = \frac{a_1 a_4 - a_0 a_5}{a_1} \\
 b_3 &= -\det \left(\begin{bmatrix} a_0 & a_6 \\ a_1 & a_7 \end{bmatrix} \right) / a_1 = \frac{a_1 a_6 - a_0 a_7}{a_1} \\
 &\vdots
 \end{aligned}$$

coefficients c_i are computed with

$$\begin{aligned} c_1 &= -\det \left(\begin{bmatrix} a_1 & a_3 \\ b_1 & b_2 \end{bmatrix} \right) / b_1 = \frac{b_1 a_3 - a_1 b_2}{b_1} \\ c_2 &= -\det \left(\begin{bmatrix} a_1 & a_5 \\ b_1 & b_3 \end{bmatrix} \right) / b_1 = \frac{b_1 a_5 - a_1 b_3}{b_1} \\ c_3 &= -\det \left(\begin{bmatrix} a_1 & a_7 \\ b_1 & b_4 \end{bmatrix} \right) / b_1 = \frac{b_1 a_7 - a_1 b_4}{b_1} \\ &\vdots \end{aligned}$$

Coefficients d_i are computed with

$$\begin{aligned} d_1 &= -\det \left(\begin{bmatrix} b_1 & b_2 \\ c_1 & c_2 \end{bmatrix} \right) / c_1 = \frac{c_1 b_2 - b_1 c_2}{c_1} \\ d_2 &= -\det \left(\begin{bmatrix} b_1 & b_3 \\ c_1 & c_3 \end{bmatrix} \right) / c_1 = \frac{c_1 b_3 - b_1 c_3}{c_1} \\ d_3 &= -\det \left(\begin{bmatrix} b_1 & b_4 \\ c_1 & c_4 \end{bmatrix} \right) / c_1 = \frac{c_1 b_4 - b_1 c_4}{c_1} \\ &\vdots \end{aligned}$$

Other coefficients are computed with the same structure with the computation flow of d_i 's.

Result 1: The system is NOT BIBO stable if $\exists i, s.t., a_i \leq 0$. In other words, since we assumed $a_0 > 0$, all other coefficients have to be strictly positive. This is a necessary, but not sufficient condition.

Result 2: The system is BIBO stable, if and only if all the coefficients in the first row strictly positive. Routh test provides a necessary and sufficient condition.

Result 3: The # of roots of $D(s)$ with positive real parts is equal to the # of sign changes in the first column (shown in table below) of the Routh array.

a_0
a_1
b_1
c_1
d_1
$\vdots$
e_1
f_1
g_0

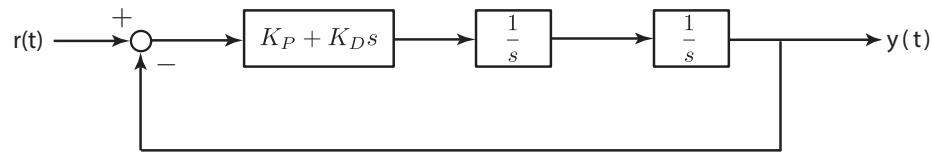
Ex: Let $D(s) = s^4 + 2s^3 + 3s^2 + 4s + 5$, is this system is stable. If not what is the number of poles with positive real parts.

Solution: Let's build the Routh table

s^4	1	3	5	0
s^3	2	4	0	0
s^2	$\frac{2 \cdot 3 - 1 \cdot 4}{2} = 1$	$\frac{2 \cdot 5 - 1 \cdot 0}{2} = 5$	0	0
s^1	$\frac{1 \cdot 4 - 2 \cdot 5}{1} = -6$	0	0	0
s^0	5	0	0	0

sign changes is equal to 2, thus the system is unstable and 2 out of 4 poles are located in the open right-half plane. If we compute the poles numerically using a programming environment, we can find that $p_{1,2} = -1.29 \pm 0.86j$ and $p_{3,4} = 0.29 \pm 1.4j$, which indeed verifies the Routh table result.

Ex: Consider the following closed-loop system, find the set of K_P and K_D gains such that closed-loop system is stable

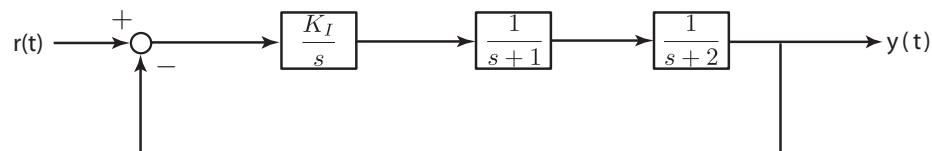


Solution: Denominator of the closed-loop transfer function of the system has the following form $D(s) = s^2 + K_D s + K_P$. Now, let's construct the Routh table for the $D(s)$

s^2	1	K_P
s^1	K_D	0
s^0	K_P	0

In order the closed-loop system to be stable, # sign changes in the first column must be equal to zero, thus the closed-loop system is BIBO stable if and only if $K_P > 0$ and $K_D > 0$.

Ex: Consider the following closed-loop system, find the set of K_I gains such that closed-loop system is stable



Solution: Denominator of the closed-loop transfer function of the system has the following form

$$D(s) = s^3 + 3s^2 + 2s + K_I$$

Now, let's construct the Routh table for the $D(s)$

s^3	1	2
s^2	3	K_I
s^1	$\frac{6 - K_I}{3}$	0
s^0	K_I	0

In order the closed-loop system to be stable, # sign changes in the first column must be equal to zero, thus the closed-loop system is BIBO stable if and only if $K_I \in (0, 6)$.

11.1.2.1 Routh Hurwitz Stability Test: Special Cases

Case I: Let's analyze the stability of " $D(s) = s^3 - 3s + 2$ ". Since $a_1 = 0$ and $a_2 = -3$, we know that the system is not BIBO stable. Let's construct the Routh table to verify this result.

s^3	1	-3
s^2	0	2
s^1	?	
s^0	?	

We can see that one of the coefficients in the Routh array is zero, and we can not complete the Routh table. If a coefficient in the Routh array is zero, we know from the Routh Hurwitz test that the system is BIBO unstable. However, what can we do, if we want to compute the # poles with positive real parts.

Solution Type 1: Let's find a new $\bar{D}(s) = (s + \alpha)D(s)$, $\alpha > 0$. We know that $D(s)$ and $\bar{D}(s)$ has the same # poles with positive real parts. Then we can construct a Routh table for $\bar{D}(s)$ to seek an answer.

Let $\bar{D}(s) = (s + 2)(s^3 - 3s + 2) = s^4 + 2s^3 - 3s^2 - 4s + 4$, the Routh table takes the form

s^4	1	-3	4
s^3	2	-4	0
s^2	-1	4	0
s^1	4	0	
s^0	4	0	

sign changes in the Routh array of $\bar{D}(s)$ is equal to 2, so we can conclude that $D(s)$ has 2 poles with positive real parts. If we compute the poles of $D(s)$, we find that $p_1 = -2$, $p_{2,3} = 1$, which verifies our finding.

Solution Type 2: Replace 0 element with an infinitesimal but non-zero element $\epsilon > 0$.

s^3	1	-3
s^2	ϵ	2
s^1	$-\frac{3\epsilon+2}{\epsilon}$	0
s^0	2	

sign changes in the perturbed Routh array of $D(s)$ is equal to 2, so we can conclude that $D(s)$ has 2 poles with positive real parts.

Solution Type 3: Replace s with $1/q$

$$D(s)|_{s=1/q} = q^{-3} - 3q^{-1} + 2 = q^{-3} (2q^2 - 3q^2 + 1)$$

Now let's define $\bar{D}(q) = q^3 D(s)|_{s=1/q} = 2q^3 - 3q^2 + 1$. Note that we simply flip the coefficients of $D(s)$ to find the coefficients of $\bar{D}(q)$. It is easy to see that if p_i is a pole of $D(s)$, then $1/p_i$ is a pole of $\bar{D}(q)$. Let $p_i = \sigma + j\omega$, then

$$\begin{aligned} \frac{1}{p_i} &= \frac{1}{\sigma + j\omega} = \frac{\sigma - j\omega}{\sigma^2 + \omega^2} \\ &= \frac{\sigma}{\sigma^2 + \omega^2} - j \frac{\omega}{\sigma^2 + \omega^2} \\ \text{Sign}(\text{Re}\{p_i\}) &= \text{Sign}\left(\text{Re}\left\{\frac{1}{p_i}\right\}\right) \end{aligned}$$

We can see that $D(s)$ and $\bar{D}(q)$ have same number of stable and unstable poles. Thus, we can perform a Routh Hurwitz test on $\bar{D}(q)$

s^3	2	0
s^2	-3	1
s^1	2/3	0
s^0	1	

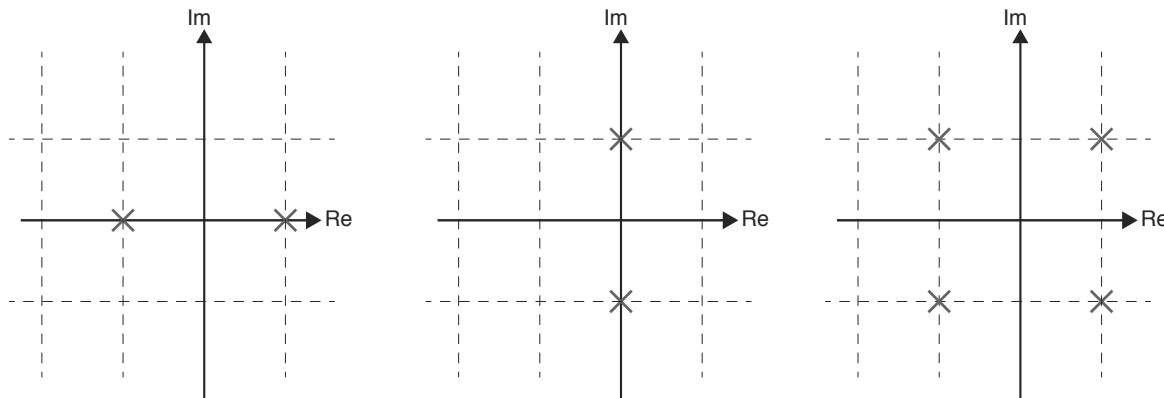
sign changes in the Routh array of $\bar{D}(q)$ is equal to 2, so we can conclude that $D(s)$ has 2 poles with positive real parts.

Case II:

Ex: Analyze the stability of $D(s) = s^4 + 2s^3 + 2s^2 + 2s + 1$ using Routh table

s^4	1	2	1
s^3	2	2	0
s^2	1	1	0
s^1	0	0	
s^0	?	?	

We can see that one row in Routh table is completely zero. Thus we know that system is indeed BIBO unstable. However, what we can do in order to find number of unstable poles. This happens when there exists roots of equal magnitude located radially opposite in s -plane, i.e. symmetric w.r.t. origin. These cases are illustrated in the figure below.



This case happens always after an even row, in this example right after s^2 row. In such cases, we compute the Auxiliary polynomial, $A(s) = s^2 + 1$ in this example. After that, we compute its derivative, $A'(s) = 2s$ in this example, and use the coefficient of $A'(s)$ in replacement of the zero row and compute the Routh array accordingly. This process is illustrated below

s^4	1	2	1	
s^3	2	2	0	
s^2	1	1	0	$\rightarrow A(s) = s^2 + 1$
s^1	2	0		$\leftarrow A'(s) = 2s$
s^0	1	0		

sign changes in the Routh array of $D(s)$ with Auxiliary polynomial is equal to 0, so we can conclude that $D(s)$ has 0 poles with positive real parts. This means that all unstable poles are located on the imaginary axis. If we compute the poles numerically we find that $p_{1,2} = -1$ and $p_{3,4} = \pm j$ which verifies our finding. Indeed, we can see that problematic roots are the roots of $A(s)$.

Ex: Compute # unstable poles of $D(s) = s^3 + 2s^2 - s - 2$ using Routh table

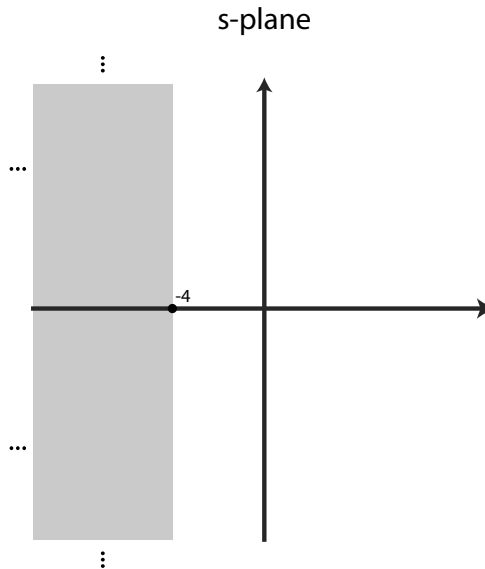
s^3	1	-1	
s^2	2	-2	$\rightarrow A(s) = 2s^2 - 2$
s^1	4	0	$\leftarrow A'(s) = 4s$
s^0	-2	0	

sign changes in the Routh array of $D(s)$ with Auxiliary polynomial is equal to 1, so we can conclude that $D(s)$ has 1 pole with positive real part. If we compute the poles numerically we find that $p_1 = -1$, $p_2 = -2$ and $p_3 = 1$ which verifies our finding. Indeed, we can see that problematic roots is one of the roots of $A(s)$.

This case occurs when there are roots of equal magnitude lying radially opposite in the s-plane.

11.1.2.2 Routh Hurwitz Stability Test: Relative Stability

Let's assume that we not only interested in the absolute stability of a system, but also we would like to test whether all poles are located in a region where the real part of the whole poles has an upper bound of -4 . Similarly we can think that, we define a performance region based on a settling time requirement.



We can still use Routh Hurwitz test to compute the # poles for which real parts is lower than -4 . However, we need to perform a change of variables.

Replace s with $z - 4$

$$D(s)|_{s=z-4} = D(z)$$

If we apply a Routh Hurwitz test on $D(z)$, we compute the # poles of $D(z)$ with positive real parts. Based on change of variables, we defined

$$\text{If } \text{Re}\{z_i\} > 0 \Leftrightarrow \text{Re}\{s_i\} > -4$$

where z_i and s_i are poles in associated planes.

Ex: Let $D(s) = s^3 + 8s^2 + 19s + 12$, first test if the system is BIBO stable using Routh Hurwitz test. If the system is BIBO stable, then check if the real parts of all the poles are located in the region $\sigma \in (-\infty, -2)$.

Solution: First test the absolute stability using Routh test on $D(s)$

s^3	1	19
s^2	8	12
s^1	17.5	0
s^0	12	0

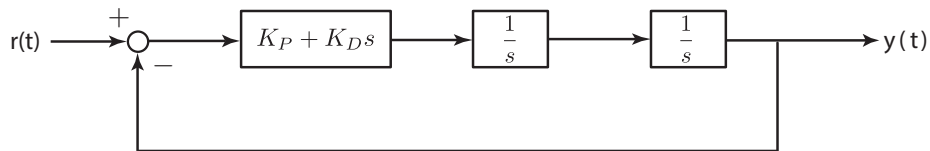
Since all of the coefficients of the Routh table are positive, the system is BIBO stable. Now check the relative stability by applying Routh test on “ $D(z) = D(s)|_{s=z-2} = z^3 + 2z^2 - z - 2$ ”,

s^3	1	-1	
s^2	2	-2	$\rightarrow A(z) = 2z^2 - 2$
s^1	4	0	$\leftarrow A'(z) = 4z$
s^0	-2	0	

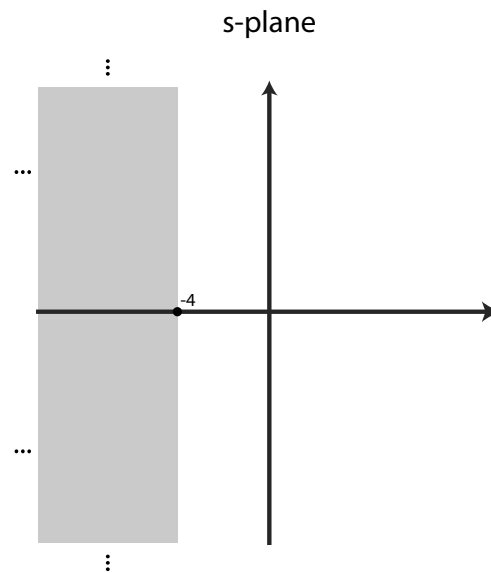
Since in the Routh table (with Auxiliary polynomial), there exist one sign change, $D(z)$ has one pole in the open right half z-plane. This means that $D(s)$ has a single pole where its real part is located in the $\sigma \in (-2, 0)$ region.

If we compute the roots of $D(s)$ numerically, we find that $p_1 = -4$, $p_2 = -3$, and $p_3 = -1$. This verifies that the system is BIBO stable but, only two of the poles are located in the $\sigma \in (-\infty, -2)$ region.

Ex: Consider the closed-loop system that we analyzed previously in terms of absolute stability.



Now let's fix $K_P = 20$, and find the set of K_D gains such that closed-loop poles are located in the desired (gray) region illustrated below.



Solution:

We can use Routh Hurwitz test on

$$\begin{aligned}
 D(z) &= D(s)|_{s=z-4} = (z-4)^2 + K_D(z-4) + 20 \\
 &= z^2 + (K_D - 8)z + (36 - 4K_D)
 \end{aligned}$$

in order to solve this problem. In order to achieve the desired pole locations, $D(z)$ can not have any poles

s^2	1	$36 - 4K_D$
s^1	$K_D - 8$	0
s^0	$36 - 4K_D$	0

with positive real parts, thus the coefficients of the Routh array has to be positive.

$$K_D > 8$$

$$K_D < 9$$

In conclusion, desired pole locations require that $K_D \in (8, 9)$.

EE302 - Feedback Systems

Spring 2019

Lecture 12

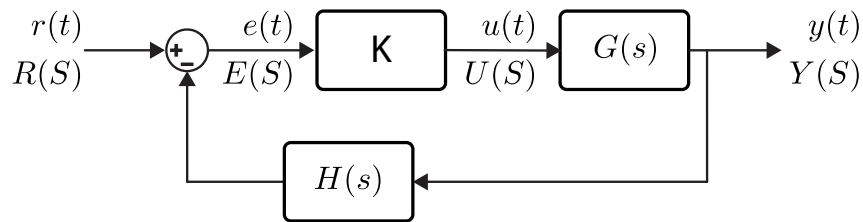
Lecturer: Asst. Prof. M. Mert Ankarali

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12.1 Root Locus

In control theory, root locus analysis is a graphical analysis method for investigating the change of closed-loop poles/roots of a system with respect to the changes of a system parameter, commonly a gain parameter $K > 0$.

In order to better understand the root locus and derive fundamental rules, we start with the following basic feedback topology where the controller is a P-controller with a gain K .



The closed loop transfer function of this basic control system is

$$\frac{Y(s)}{R(s)} = \frac{KG(s)}{1 + KG(s)H(s)} = \frac{KG(s)}{1 + KG_{OL}(s)}$$

where the poles of the closed loop system are the roots of the characteristic equation

$$1 + KG_{OL}(s) = 0$$

$$1 + K \frac{n(s)}{d(s)} = 0$$

The goal is deriving the qualitative and quantitative behavior of closed-loop pole “paths” for **positive** gain K that solves the equation $1 + KG_{OL}(s) = 0$ (or $1 + K \frac{n(s)}{d(s)} = 0$).

12.1.1 Angle and Magnitude Conditions

Let's analyze the characteristic equation

$$KG_{OL}(s) = -1 \quad , \text{ or } \quad K \frac{n(s)}{d(s)} = -1 \quad , \text{ or } \quad K \frac{(s - z_1) \cdots (s - z_M)}{(s - p_1) \cdots (s - p_N)} = -1$$

Let's derive the **magnitude condition** given that $K > 0$,

$$K|G_{OL}(s)| = 1 \quad , \text{ or } \quad K \left| \frac{n(s)}{d(s)} \right| = 1 \quad , \text{ or } \quad K \frac{|s - z_1| \cdots |s - z_M|}{|s - p_1| \cdots |s - p_N|} = 1$$

Now let's derive the **angle condition** given that $K > 0$,

$$\begin{aligned} \angle[G_{OL}(s)] &= \pi(2k + 1) \quad , \text{ or } \quad \angle[n(s)] - \angle[d(s)] = \pi(2k + 1) \\ & , \text{ or } \quad \angle[s - z_1] \cdots \angle[s - z_M] - (\angle[s - p_1] \cdots \angle[s - p_N]) = \pi(2k + 1), \quad k \in \mathbb{Z} \end{aligned}$$

For a given K , s values that satisfy both magnitude and angle conditions are located on the root loci. These constitutes the most fundamental knowledge regarding the root locus analysis.

How we can check whether a candidate s^* is in the root -locus or not. If we analyze the angle condition, we can see that it is independent from the parameter K . However, If we focus on the magnitude condition, we can see that

$$K = \frac{1}{|G_{OL}(s^*)|} = \left| \frac{d(s^*)}{n(s^*)} \right| = \frac{|s^* - p_1| \cdots |s^* - p_N|}{|s^* - z_1| \cdots |s^* - z_M|}$$

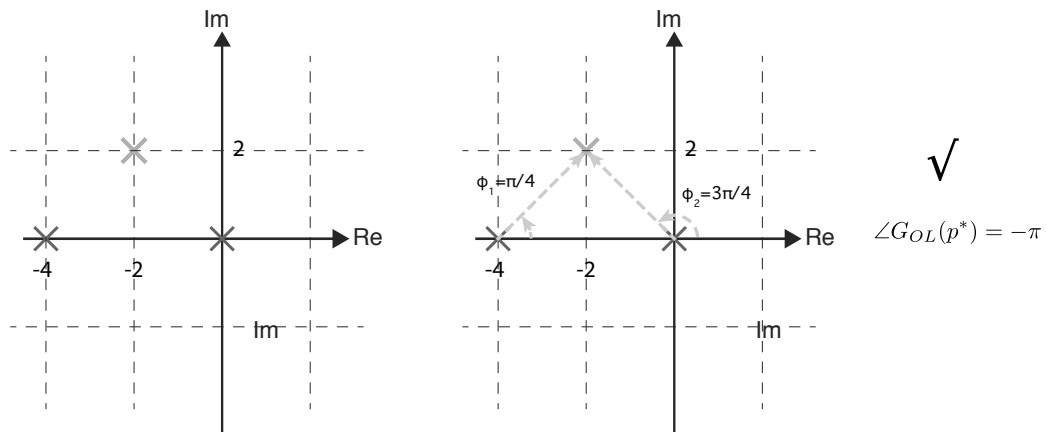
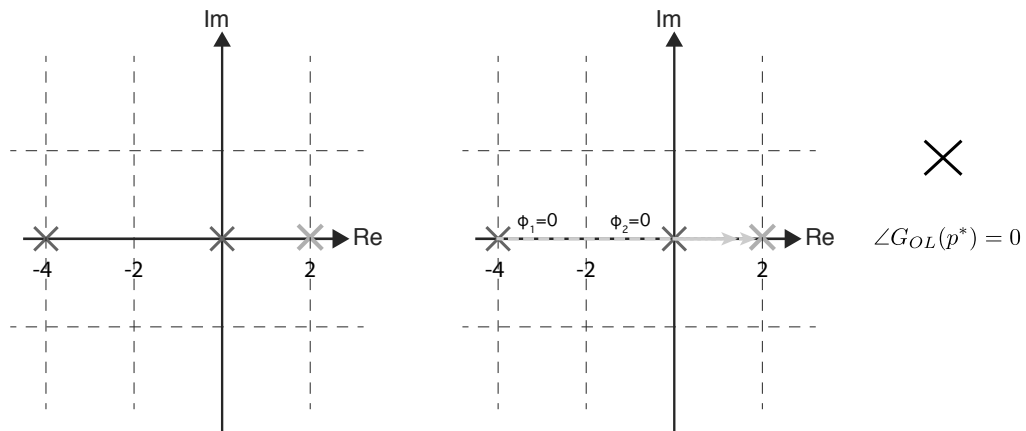
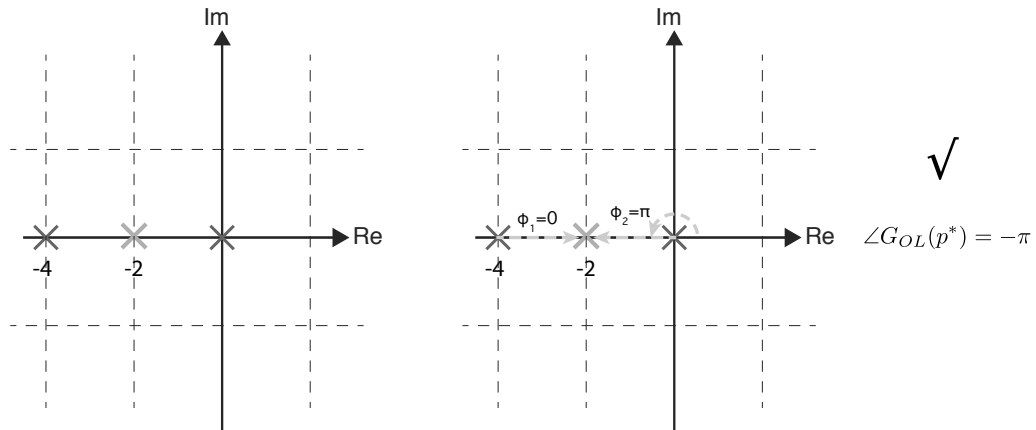
which implies that for every s^* candidate (that is not a pole or zero), we can indeed compute a gain K value.

In conclusion, only angle condition is used for testing whether a point is in the root-locus or not. On the other hand, will use the magnitude condition to compute the value of gain K , if we find that a candidate s^* is in the root locus based on the angle condition.

Ex: It is given that $G_{OL}(s) = \frac{1}{s(s+4)}$. Determine if the following pole candidates are on the root-locus or not

$$p_1^* = -2, \quad p_2^* = 2, \quad p_3^* = -2 + 2j$$

Solution: We only test the angle condition. Solutions are illustrated on the s -planes provided below



12.1.2 Rules and procedure for constructing root loci

1. Compute the zeros poles of the open-loop transfer function and write the characteristic equation of closed-loop system.

$$1 + KG_{OL}(s) = 0$$

$$1 + K \frac{n(s)}{d(s)} = 0$$

$$1 + K \frac{(zs - z_1) \cdots (zs - z_M)}{(s - p_1) \cdots (s - p_N)} = 0$$

2. Root loci has N separate branches. Since,

$$[(s - p_1) \cdots (s - p_N)] + K [(zs - z_1) \cdots (zs - z_M)] = 0$$

$$d(s) + Kn(s) = 0$$

has N number of roots for all K .

3. Root loci starts from poles of $G_{OL}(s)$ and
 - (a) M branches terminates at the zeros of $G_{OL}(s)$
 - (b) $N - M$ branches terminates at ∞ (implicit zeros of $G_{OL}(s)$)

It is relatively easy to understand this

$$d(s) + Kn(s) = 0$$

$$K \rightarrow 0 \Rightarrow [d(s) + Kn(s) = 0 \rightarrow [d(s) = 0]$$

$$K \rightarrow \infty \Rightarrow [d(s) + Kn(s) = 0 \rightarrow [n(s) = 0]$$

4. Root loci on the real axis determined by open-loop zeros and poles. $s = \sigma \in \mathbb{R}$ then, based on the angle condition we have

$$\text{Sign}[G_{OL}(\sigma)] = -1$$

Let's first analyze the effect of complex conjugate pole/zero (and double pole/zero on real axis) pairs on the equation above. Let $\sigma^* \in \mathbb{R}$ is the candidate location and complex conjugate poles has the following form $p_{1,2} = \sigma \pm j\omega$

$$\text{Sign}[(\sigma^* + \sigma - j\omega)(\sigma^* + \sigma + j\omega)] = \text{Sign}[(\sigma^* + \sigma)^2 + \omega^2] = 1$$

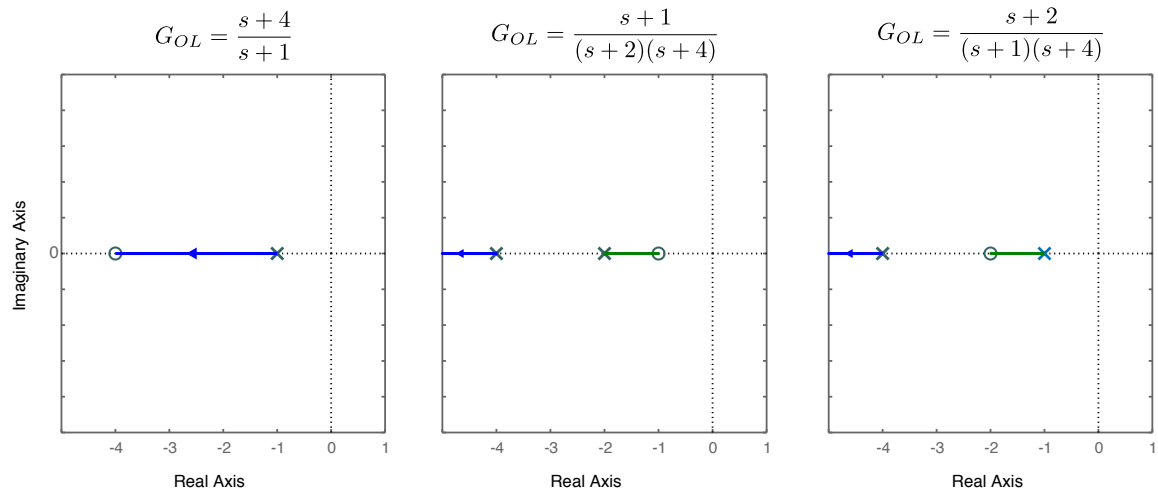
We can see that complex conjugate zero/pole pairs have not effect on angle condition for the roots on the real axis. Then for the remaining ones we can derive the following condition

$$\text{Sign}[G_{OL}(\sigma)] = \prod_{i=1}^{\bar{M}} \text{Sign}[\sigma - z_i] \prod_{j=1}^{\bar{N}} \text{Sign}[\sigma - p_j] = -1$$

which means that for ODD number of poles + zeros $\text{Sign}[\sigma - p_i]$ and $\text{Sign}[\sigma - z_i]$ must be negative for satisfying this condition for that particular σ to be on the root-locus. We can summarize the rule as

If the test point σ on real axis has ODD numbers of poles and zeros in its right, then this point is located on the root-locus.

Ex: The figure below illustrates the root locus plots of three different transfer functions.



5. Asymptotes

- $N - M$ branches goes to infinity. Thus, there exist $N - M$ many asymptotes
- For large s we can have the following approximation

$$K \frac{(s - z_1) \cdots (s - z_M)}{(s - p_1) \cdots (s - p_N)} \approx \frac{K}{s^{N-M}}$$

$$\angle \left[\frac{K}{s^{N-M}} \right] = -(N - M) \angle [s] = \pi(2k + 1), \quad k \in \mathbb{Z}$$

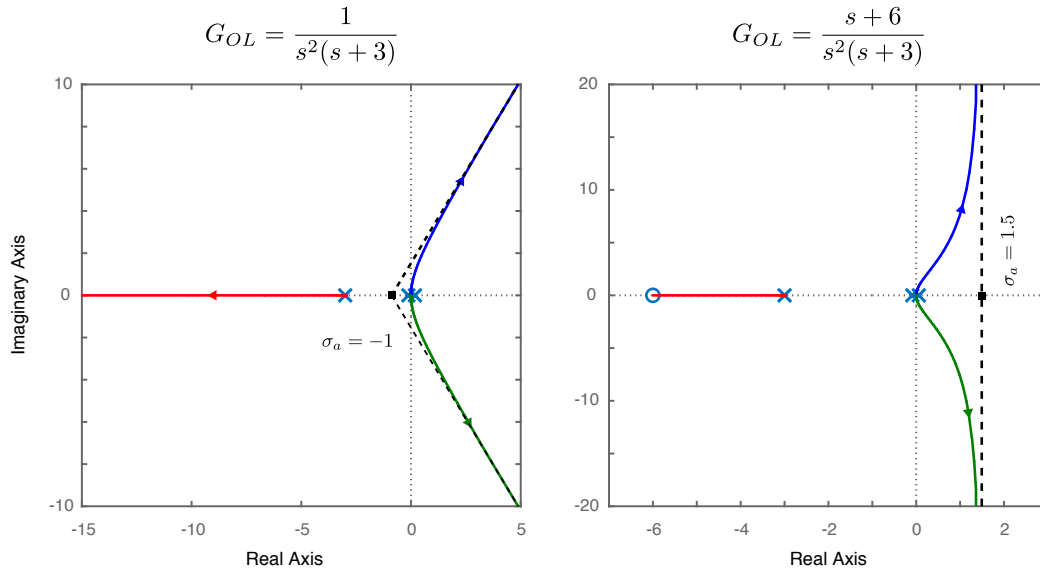
$$\phi_a = \frac{\pm \pi(2k + 1)}{N - M}, \quad k \in \{1, \dots, N - M\}$$

- Real axis intercept σ_c can be computed as

$$\sigma_c = \frac{\sum p_i - \sum z_i}{N - M}$$

This can be derived via a different approximation (see textbook)

Ex: The figure below illustrates the root locus plots of two different transfer functions.



6. Breakaway and break-in points on real axis. When s is real $s = \sigma$, $\sigma \in \mathbb{R}$, we have

$$1 + KG_{OL}(\sigma) = 0$$

Note that break-in and breakaway points corresponds to double roots. Thus, if σ_b is a break-away or break-in point we have

$$\begin{aligned} 1 + KG_{OL}(\sigma_b) &= 0 \\ K \left[\frac{d}{d\sigma} G_{OL}(\sigma) \right]_{\sigma_b} &= 0 \end{aligned}$$

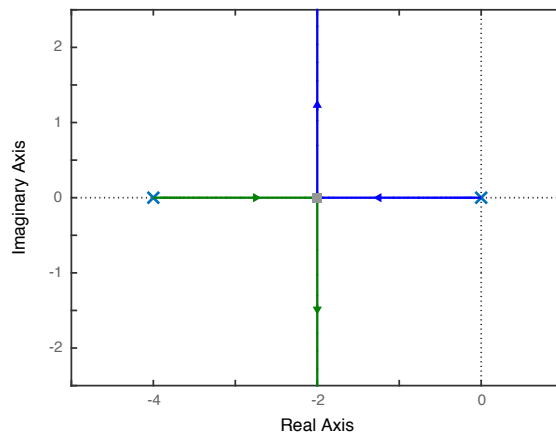
Thus, we conclude that break-in or break-away points satisfy the following conditions

$$\left[\frac{dG_{OL}(\sigma)}{d\sigma} \right]_{\sigma=\sigma_b} = 0, \quad K(\sigma_b) = \frac{-1}{G_{OL}(\sigma_b)}, \quad K(\sigma_b) > 0$$

We can derive two corollary conditions for computing σ_b as

$$\left[\frac{d}{d\sigma} \frac{1}{G_{OL}(\sigma)} \right]_{\sigma=\sigma_b} = 0 \quad \text{or,} \quad \left[\left(\frac{d}{d\sigma} N(\sigma) \right) D(\sigma) - \left(\frac{d}{d\sigma} D(\sigma) \right) N(\sigma) \right]_{\sigma_b} = 0$$

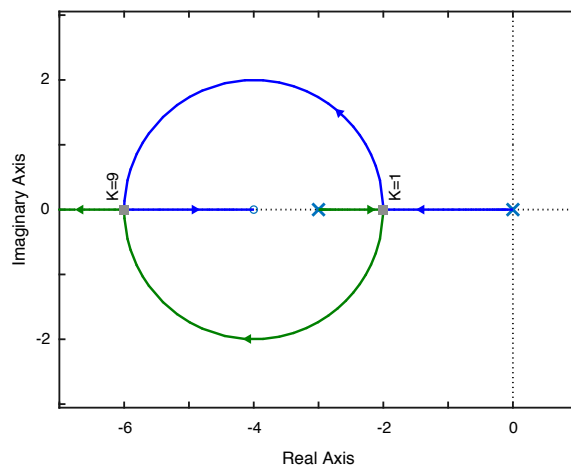
Ex: Draw the root locus diagram for $G_{OL}(s) = \frac{1}{s(s+4)}$, compute the real axis intercept σ_c and break away point (with the associated gain value).



$$\sigma_c = -2$$

$$\sigma_{ba} = -2, \quad K(\sigma_{ba}) = 4$$

Ex: Draw the root locus diagram for $G_{OL}(s) = \frac{s+4}{s(s+3)}$, compute the break-away and break-in points (with the associated gain values).



$$\sigma_b^2 + 8\sigma_b + 12 = 0$$

$$\sigma_{b-a} = -2, \quad K(\sigma_{b-a}) = 1$$

$$\sigma_{b-in} = -6, \quad K(\sigma_{b-in}) = 9$$

7. Finding the imaginary axis crossings.

These crossings are particularly important, since at these crossings (generally) stability changes. At these points the poles become purely imaginary $p_{1,2} = \pm j\omega$. If we insert this into characteristic equation we get

$$1 + KG_{OL}(j\omega) = 0$$

$$D(j\omega) + KN(j\omega) = 0$$

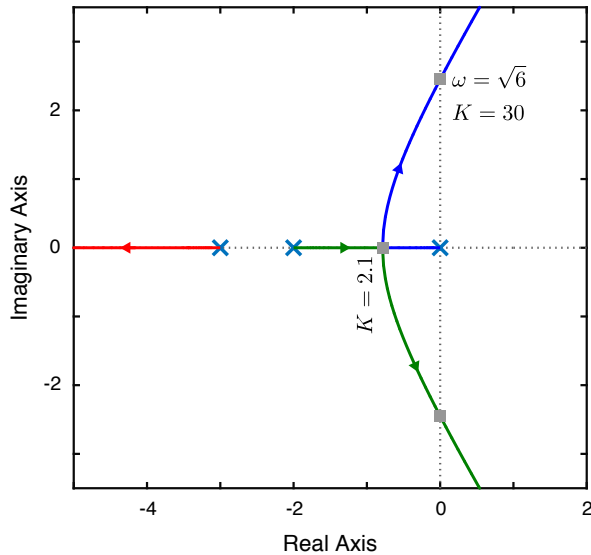
$$\text{Re}\{D(j\omega) + KN(j\omega)\} = 0$$

$$\text{Im}\{D(j\omega) + KN(j\omega)\} = 0$$

Note that depending on the order of the system, solving the above equation can be most computationally very heavy.

Second way of finding the imaginary axis crossing is to apply the Routh-Hurwitz criteria. Note that at these crossings the system becomes unstable. Using this fact, we can first construct a Routh table for the closed-loop characteristic equation and then derive the K values where a change of stability occurs. After that, we can use the computed critical K values to derive the pole locations on the imaginary axis..

Ex: Draw the root locus diagram for $G_{OL}(s) = \frac{1}{s(s+2)(s+3)}$, compute the break-away point and imaginary axis crossings (with the associated gain values).



Break-away point

$$3\sigma_b^2 + 10\sigma_b + 6 = 0$$

$$\sigma_{b,1} = -0.8 \quad , \quad K(\sigma_{b,1}) = 2.1 > 0 \rightarrow \text{OK}$$

$$\sigma_{b,2} = -2.5 \quad , \quad K(\sigma_{b,2}) = -0.6 < 0 \rightarrow \text{NO}$$

Imaginary axis crossing

$$D(j\omega) + K(j\omega) = 0$$

$$(j\omega)^3 + 5(j\omega)^2 + 6(j\omega) + K = 0$$

$$(K - 5\omega^2) + (6\omega - \omega^3)j = 0$$

$$\Rightarrow \omega = \sqrt{6} \quad , \quad K = 30$$

Now let's find the imaginary axis crossings using the Routh table. The characteristic equation for this system is $s^3 + 5s^2 + 6s + K$, then the Routh table takes the form

s^3	1	6	
s^2	5	K	
s^1	$\frac{30-K}{5}$	0	
s^0	K	0	

We know that in order for the system to be stable $K \in (0, 30)$, since we only consider positive K values, when $K = 30$ system stability changes (from stable to unstable). Let $K = 30$ and re-form the Routh table.

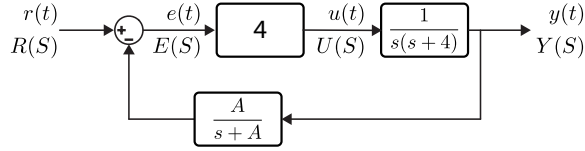
s^3	1	6	
s^2	5	30	$\rightarrow A(s) = 5s^2 + 30$
s^1	10	0	$\leftarrow A'(s) = 10s$
s^0	30	0	

Based on the Routh table, we conclude that when $K = 30$, system becomes unstable, the unstable poles are located on the imaginary axis, and their locations can be found using the Auxiliary polynomial as

$$A(s) = 0 \rightarrow p_{1,2} = \pm\sqrt{6}j$$

12.1.3 Root-locus with respect to different parameters

Let's consider the following feedback control system. We wonder the location of closed-loop poles with respect to the parameter A which does not directly fit to the classical form of root-locus.



Let's first compute the closed-loop TF and analyze the characteristic equation.

$$\begin{aligned}\frac{Y(s)}{R(s)} &= \frac{KG(s)}{1 + KG(s)H(s)} \\ &= \frac{s + A}{s(s + 4)(s + A) + 4A} \\ &= \frac{s + A}{s^3 + (A + 4)s^2 + 4As + 4A}\end{aligned}$$

Now let's organize the characteristic equation

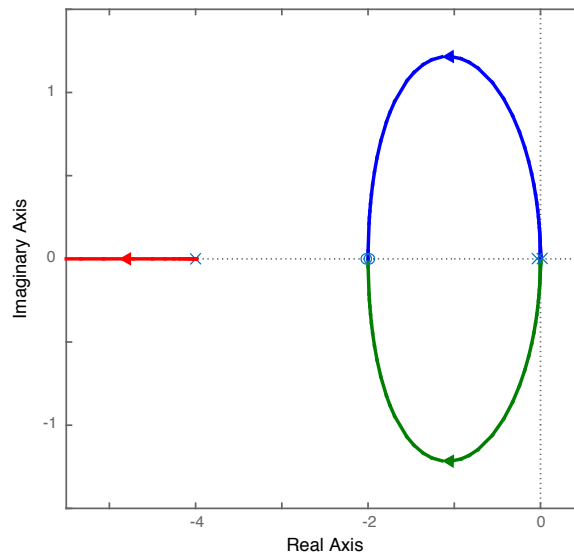
$$(s^3 + 4s^2) + A(s^2 + 4s + 4) = 0$$

If we divide the characteristic equation by $(s^3 + 4s^2)$ we obtain

$$\begin{aligned}1 + A \frac{s^2 + 4s + 4}{s^3 + 4s^2} &= 0 \\ 1 + A\bar{G}_{OL}(s) &= 0\end{aligned}$$

Now if we consider $\bar{G}_{OL}(s)$ as the open-loop transfer function and draw the root-locus, then we would derive the dependence of the roots to the parameter A .

Root-locus of the system w.r.t parameter $A > 0$ is given below.



EE302 - Feedback Systems**Spring 2019****Lecture 13***Lecturer: Asst. Prof. M. Mert Ankarali*

13.1 PID Control Analysis & Design with Root-Locus

Let's assume that we would like you to design a controller for the following second order plant transfer function

$$G(s) = \frac{1}{(s+1)(s+3)}$$

The requirements of the closed-loop system are

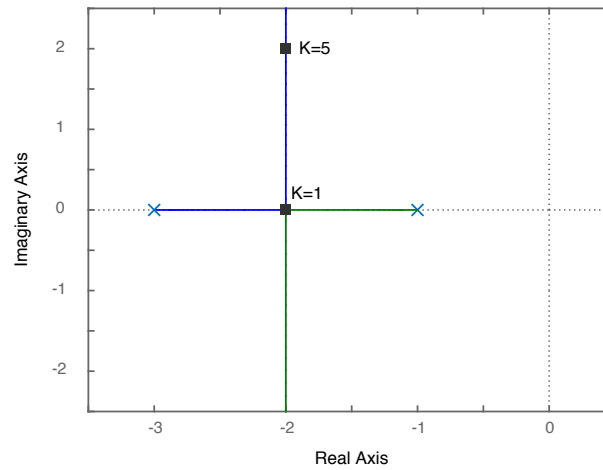
- Minimum possible settling time (%2)
- Minimum steady-state error
- Minimum possible over-shoot

13.1.1 Proportional (P) Controller

Let's first design a P controller, $C(s) = K$. Let's start with steady-state error performance. Plant is a type-one system and controller is a static gain thus unit step and unit ramp steady-state error can be computed as

$$\begin{aligned}\text{Step : } e_{ss} &= \frac{1}{1 + K_P/3} \\ \text{Step : } e_{ss} &= \infty\end{aligned}$$

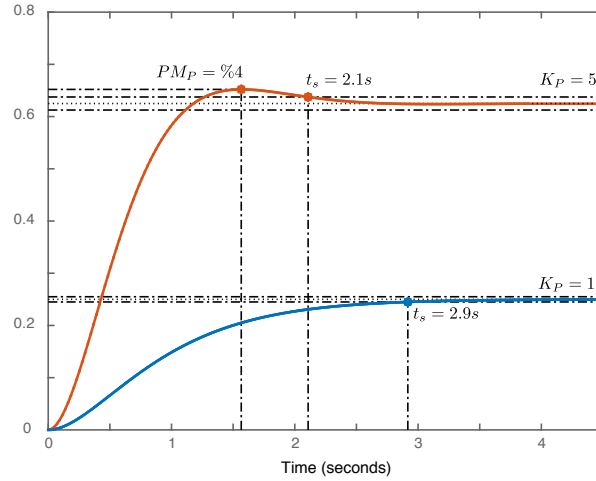
We can see that unit-ramp error is ∞ regardless of K_P , where as we can reduce the steady-state error by increasing proportional gain K_P . Now let's draw root locus and comment on settling time and over-shoot performance.



Based on root-locus plot we can see that best convergence rate achieved with $\sigma = -2$, for which the approximate settling time is $2s$. When $K = 1$, system becomes critically damped. When we further increase the gain, real part of the poles do not change, however oscillations starts and grows with K_P . Obviously, we should choose a gain $K_P > 1$, however after that point there is a trade-off between over-shoot and settling time performance. Let's choose two candidate locations, $p_{1,2}^{(1)} = -2$ and $p_{1,2}^{(1)} = -2 \pm 2j$. Table below details the gain values at these pole locations, unit-step steady-state errors, and estimated settling time and maximum over-shoot values.

$p_{1,2}$	K_P	e_{ss}	t_s	M_P
-2	1	0.75	2	0
$-2 \pm 2j$	5	0.375	2	0.04

We verify that both steady-state and transient performance increases with larger K_P . However, we can also see that settling time estimation for $K_P = 5$ has a larger error, which is expected since the poles are close to each other thus violating the dominant pole assumption. If we consider the values in the table, it seems to be reasonable to choose $K_P = 5$ since it provides a much better steady-state performance, of course if existing of a small over-shoot is not a major problem for the design. Now let's plot step-responses of the closed loop system for both cases.



We verify most of our theoretical findings with the only exception about the settling time for $K_P = 1$, which is larger than the estimation. In conclusion, $K_P = 5$ is a good choice for overall requirements.

13.1.2 Proportional Derivative (PD) Controller

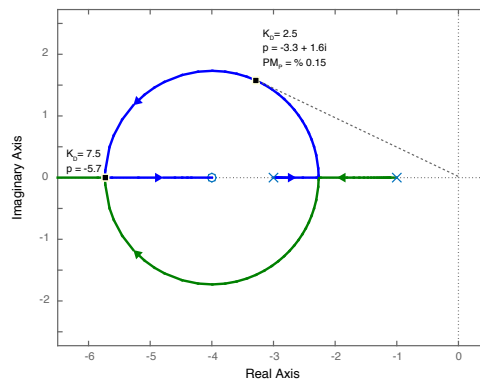
Now let's design a PD controller. First make some change of variables and re-write the PD controller in a different form.

$$C(s) = K_P + K_D s = K_D \left(s + \frac{K_P}{K_D} \right) = K_D (s + \alpha)$$

We can see that PD controller introduces an extra zero to the open loop transfer function. Let's compute the steady-state error performance.

- Unit step: $e_{ss} = \frac{1}{1+K_P/3} = \frac{1}{1+\alpha K_D/3}$,
- Unit ramp: $e_{ss} = \infty$

Technically in classical PD form K_D has no effect on steady-state performance. However if we adopt the second form with α and fix α first, then as we increase K_D steady-state error will decrease. Now let's fix $\alpha = 4 = \frac{K_P}{K_D}$, and draw the root-locus.

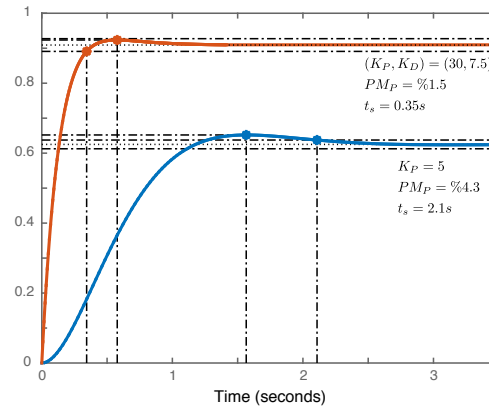


We think that the improvement of added zero is very clear. If we carefully look at the root-locus, we can see that at the pole location that gives the worst maximum over-shoot performance, maximum percent over-shoot is only %0.15, moreover over-shoot completely disappears when $K_D \geq 7.5$. The best settling/convergence time performance occurs at the break-in point which has an approximate gain of $K_D = 7.5$,

Table below details the P and D gain values at this pole locations, closed-pole locations, unit-step steady-state error, and estimated settling time value.

K_P	K_D	p	e_{ss}	$t_s[s]$
30	7.5	-5.7	0.09	0.7

Now let's compare this PD controller (with $(K_P, K_D) = (30, 7.5)$) and the previously designed P controller (with $(K_P, K_D) = (5, 0)$) by simulating the step responses.



We can see that PD control policy outperforms the P controller in every category.

On the other hand, interestingly we expected no over-shoot with that PD controller, yet we still observe some over-shoot at the output. In addition to this, actual settling time is approximately half of our previous estimation. The discrepancies between the transient characteristics between simulation and estimated values occurs due to the effect of extra zero in the closed-loop transfer function.

13.1.3 Integral (I) Controller

Now let's analyze how a pure integral controller affects the proposed system using root-locus analysis.

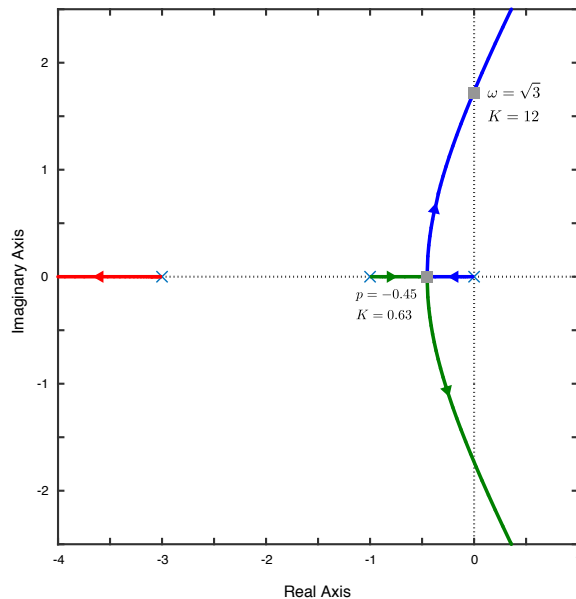
$$C(s) = \frac{K_I}{s}, \quad G(s) = \frac{1}{(s+1)(s+3)}$$

$$G_{OL}(s) = \frac{1}{s(s+1)(s+3)}, \quad K = K_I$$

We already know that steady-state performance is affected positively with an Integral controller

- Unit step: $e_{ss} = 0$
- Unit ramp: $e_{ss} = \frac{3}{K_I}$

Now let's analyze the affects on transient performance and stability using root-locus diagram



Break-away point

$$3\sigma_b^2 + 8\sigma_b + 3 = 0$$

$$\sigma_{b,1} = -0.45 \rightarrow \text{OK} \quad , \quad K = 0.63$$

$$\sigma_{b,2} = -2.2 \rightarrow \text{NO}$$

Imaginary axis crossing

$$D(j\omega) + K_I N(j\omega) = 0$$

$$(j\omega)^3 + 4(j\omega)^2 + 3(j\omega) + K_I = 0$$

$$(K - 4\omega^2) + (3\omega - \omega^3)j = 0$$

$$\Rightarrow \omega = \sqrt{3} \quad , \quad K = 12$$

We can make the following observations

- Closed-loop system becomes unstable for $K_I > 6$. Note that with a P controller closed loop system was always stable for $K_P > 0$
- Best settling time that can be achieved with an I controller is approximately, $t_s = 9s$, which is quite bad compared to the simple P-controller.

In conclusion, integral action improves steady-state performance, but it degrades the stability and transient performance.

13.1.4 Proportional Integral (PI) Controller

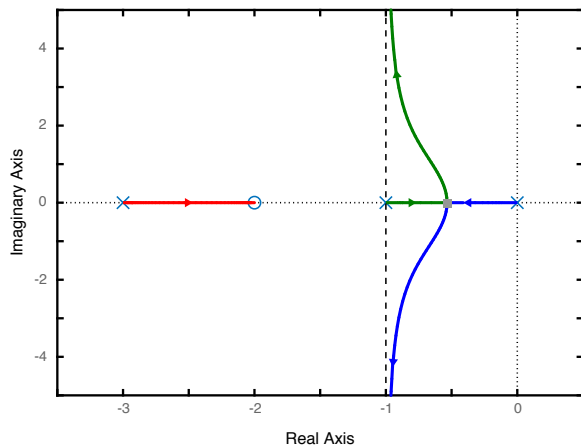
Now let's design a PI controller. First make some change of variables and re-write the PI controller in a different form.

$$C(s) = K_P + K_I \frac{1}{s} = K_P \left(\frac{s + K_I/K_P}{s} \right) = K_P \left(\frac{s + \alpha}{s} \right)$$

We can see that PI controller introduces an extra zero to the open loop transfer function and a pole at the origin. We know that the steady-state error performance characteristics of a PI controller is same with an I controller.

Now let's fix $\alpha = 2 = \frac{K_I}{K_P}$, and draw the root-locus w.r.t K_P . Note that

$$G_{OL}(s) = \frac{s + 2}{s(s + 1)(s + 3)}$$



Break-away point

$$2\sigma_b^3 + 10\sigma_b^2 + 16\sigma_b + 6 = 0$$

$$\sigma_{b,1} = -0.53 \rightarrow \text{OK} \quad , \quad K = 0.42$$

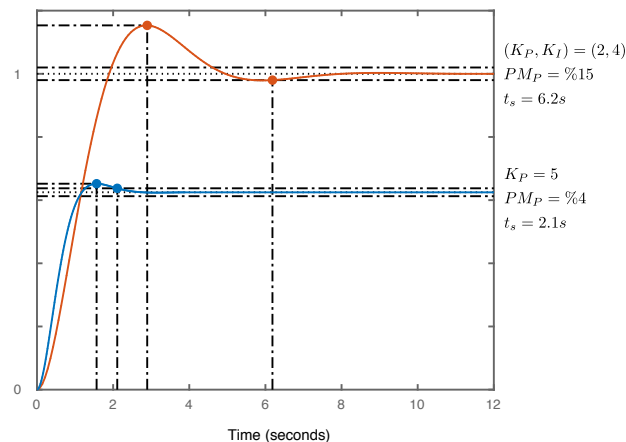
$$\sigma_{b,2} = -2.2 \pm 0.8j \rightarrow \text{NO}$$

We can make the following observations

- Closed-loop system is stable for $K_P > 0$ for this choice of α .
- Best settling time that can be achieved with this PI controller is approximately, $t_s = 4s$, but this is achieved when $K \rightarrow \infty$.
- Best settling time value with this PI controller is the approximately double of the best settling time value of the P controller.
- System is approximately acts like a second-order over-damped system when $K_P \in (0, 0.42)$, where as acts like a a second-order under-damped system when $K_P \in (0.42, \infty)$.
- Oscillations and over-shoot increases as we increase K_P . This there is a trade-off between settling time and over-shoot performance.

In conclusion, PI controller has superior steady-state performance compared to P and PD controllers, however there exist substantial transient performance drop even compared to simple P controller.

Now let's choose $K_P = 2$, then we can find that $K_I = 4$. Now let's compare this PI controller $(K_P, K_I) = (2, 4)$ with our previously chosen P controller $(K_P, K_I) = (5, 0)$ by simulating the step-responses.



It is obvious that while PI controller has a superior steady-state performance, P controller provides much better transient characteristics.

Practice Question: Choose α from this set $\{4, -1.2, 0.8\}$, and draw the root-locus diagrams for each α . Comment on the results in terms of overall transient performance based on the root locus diagrams.

13.1.5 Proportional Integral Derivative (PID) Controller

If one is not satisfied with the transient characteristics of a P-controller but also would like to eliminate steady-state error for the unit step input, he/she can design a PID controller.

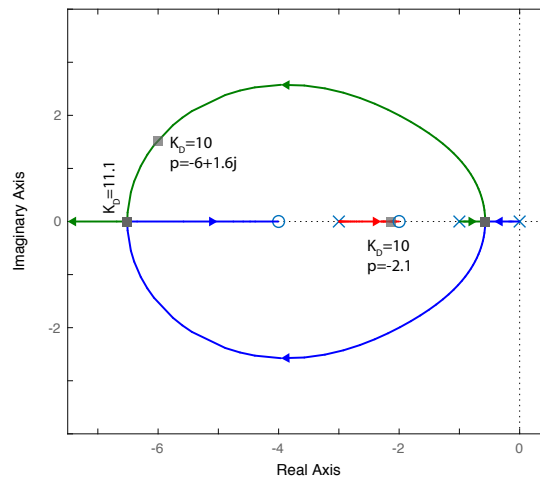
First make some change of variables and re-write the PID controller in a different form.

$$\begin{aligned} C(s) &= K_D s + K_P + K_I \frac{1}{s} = K_D \left(s + (K_P/K_D) + (K_I/K_D) \frac{1}{s} \right) \\ &= K_D \left(s^2 + (K_P/K_D)s + (K_I/K_D) \right) \frac{1}{s} \end{aligned}$$

We can see that a PID controller has a second order numerator dynamics, and it is possible to have two real zeros, or two complex conjugate zeros. In general, PID gains are chosen such that numerator has two zeros

$$C(s) = K_D \frac{(s + \alpha)(s + \beta)}{s}$$

We know that PID controller has same steady-state performance characteristics with PI and I controllers. Now let's choose $\alpha = 2$ and $\beta = 4$ and draw the root locus w.r.t. K_D .



Unlike the root-locus plots of previous cases, it is harder to have an idea about what is good for transient performance. It seems that as $K_D \nearrow$, one pole on the real axis move towards imaginary axis and stops at $\sigma = -2$ when $K \rightarrow \infty$, where as other two poles deviates from the imaginary axis when $K_D \nearrow$. Let's choose $K_D = 10$ which marked on the root-locus plot. In this case poles of the closed-loop system takes the form $p_1 = -2.1$ and $p_{2,3} = -6 \pm 1.6j$. We may conclude that the system is approximately first order since the single pole is much closer to the real-axis. In this case, we expect settling time as $t \approx 1.9s$. Note that this settling time value is slightly better then the best settling time value that is satisfied with a P-controller. Moreover we expect that over-shoot would be negligible (less than %0.001).

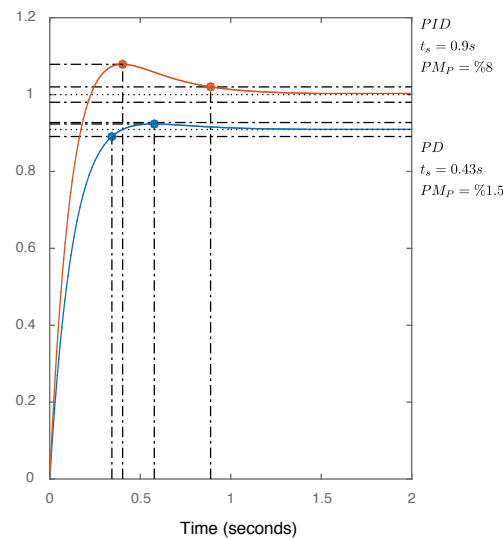
When $K_D = 10$, we have

$$s^2 + (K_P/K_D)s + (K_I/K_D) = (s + 2)(s + 4) = s^2 + 6s + 8$$

$$K_P = 60$$

$$K_I = 80$$

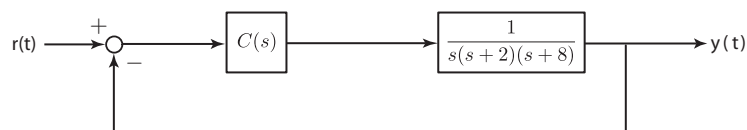
Now let's simulate this PID controller and previously designed PD controller and compare the transient and steady state performances.



We can clearly see the qualitative differences between the designed PD and PID controllers. For this specific controllers, we can see that PD has much better transient performance and PID is superior in steady-state performance. Note that PID completely eliminates the steady-state error and it is certainly possible to improve the steady-state performance of PD controller by increasing its gain.

On the other hand in PID controller, if we locate both zeros to the left of the pole at -3 location (**left as a practice**). we would obtain better root-locus picture (in terms of pole locations with better transient performance). However, the main trade-off will be the substantially increased gain values, which is already much larger than the designed P and PD controllers.

Example: Design a controller $C(s)$ for the following feedback-system such that dominant closed-loop poles has a damping ratio of $\zeta = 1/\sqrt{2}$ and damped natural frequency of $\omega_d = 2\text{rad/s}$.



Solution: Since there is no steady-state requirement and the goal is to locate the dominant poles to a specific location, first natural choice of a controller is a PD controller. We know that PD controller can be

written in the following forms

$$C(s) = K_P + K_D s = K_D(s + \alpha)$$

and the open-loop transfer function takes the form

$$G_{OL}(s) = K_D \frac{s + \alpha}{s(s + 2)(s + 8)}$$

Desired pole location can be explicitly computed as

$$\tan \phi = \frac{\sqrt{1 - \zeta^2}}{\zeta} = 1$$

$$p_{1,2} = -2 \pm 2j$$

Now let's try to compute α using the angle condition since we want the specified pole to be on the root-locus

$$\angle[G(s)]_{s=p} = (2k + 1)\pi \quad k \in \mathbb{Z}$$

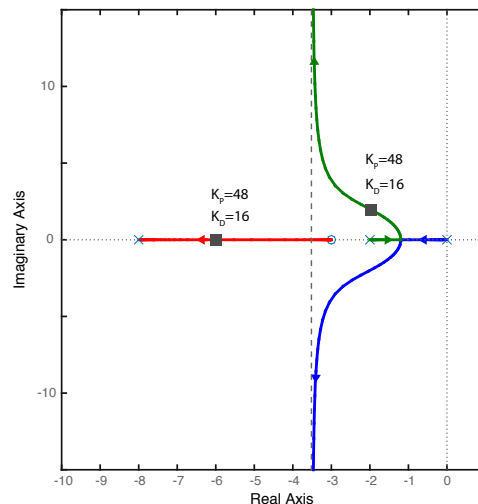
$$\begin{aligned} \angle[G(p)] &= \angle[p + \alpha] - (\angle[p] + \angle[p + 2] + \angle[p + 8]) \\ &= \angle[(-2 + \alpha) \pm 2j] - \left(\frac{3\pi}{4} + \frac{\pi}{2} + \arctan(1/3) \right) \\ &= \angle[(-2 + \alpha) \pm 2j] + \frac{3\pi}{4} - \arctan(1/3) \end{aligned}$$

$$\angle[(-2 + \alpha) \pm 2j] = \frac{\pi}{4} - \arctan(1/3)$$

$$\Rightarrow \frac{2}{\alpha - 2} = \arctan\left(\frac{\pi}{4} + \arctan(1/3)\right) = 2$$

$$\Rightarrow \alpha = 3$$

Note that direct complex algebra could provide a simpler computational process. Now given that $\alpha = 3$, let's draw the root locus. We can see that the desired dominant pole locations are satisfied when $K_P = 48$ and $K_D = 16$.



Example: Design a controller for the following feedback-system such that dominant closed-loop poles has a damping ratio of $\zeta = 1/\sqrt{2}$ and damped natural frequency of $\omega_d = 2\text{rad/s}$, system has zero unit step steady-state error.



Solution: Since the requirement imposes zero steady-state error we have to implement an I action. Let's test a PI controller first.

$$C(s) = K_P + K_I \frac{1}{s} = K_I \frac{s + \alpha}{s}$$

and the open-loop transfer function takes the form

$$G_{OL}(s) = K_I \frac{s + \alpha}{s(s+1)(s+3)}$$

Now let's try to compute α using the angle condition since we want the specified pole to be on the root-locus

$$\begin{aligned} \angle[G(s)]_{s=p} &= (2k+1)\pi \quad k \in \mathbb{Z} \\ , \\ \angle[G(p)] &= \angle[p + \alpha] - (\angle[p] + \angle[p+1] + \angle[p+3]) \\ &= \angle[p + \alpha] - \left(\frac{3\pi}{4} + \pi - \arctan(2) + \arctan(2) \right) \\ &= \angle[p + \alpha] + \frac{\pi}{4} \\ , \\ \angle[(-2 + \alpha) \pm 2j] &= \frac{3\pi}{4} \Rightarrow \alpha = 0 \quad (???????) \end{aligned}$$

This implies that a PI controller can not satisfy the requirements. Now let's try to design a PID controller.

$$\begin{aligned} C(s) &= K_D s + K_P + K_I \frac{1}{s} = K_D \frac{s^2 + (K_P/K_D)s + (K_I/K_D)}{s} \\ &= K_D \frac{(s + \alpha)(s + \beta)}{s} \end{aligned}$$

Now we have two parameters to satisfy angle condition. Let's simplify the process and let $\alpha = \beta$, thus the PID controller has the following form

$$C(s) = K_D \frac{(s + \alpha)^2}{s}$$

Now let's try to compute α using the angle condition

$$\angle[G_{OL}(s)]_{s=p} = (2k+1)\pi \quad k \in \mathbb{Z}$$

$$\begin{aligned} \angle[G_{OL}(p)] &= 2\angle[p+\alpha] - (\angle[p] + \angle[p+1] + \angle[p+3]) \\ &= 2\angle[p+\alpha] - \left(\frac{3\pi}{4} + \pi - \arctan(2) + \arctan(2)\right) \\ &= 2\angle[p+\alpha] + \frac{\pi}{4} \end{aligned}$$

$$\angle[(-2+\alpha) \pm 2j] = \frac{3\pi}{8} \Rightarrow \alpha = 2.8284$$

Now let's compute K_D using magnitude condition

$$K_D = \frac{1}{|G_{OL}(-2+2j)|} = 3.018$$

then we can compute K_P and K_I gains

$$K_D(s^2 + (K_P/K_D)s + (K_I/K_D)) = K_D(s+\alpha)^2 = K_D(s^2 + 5.657s + 8)$$

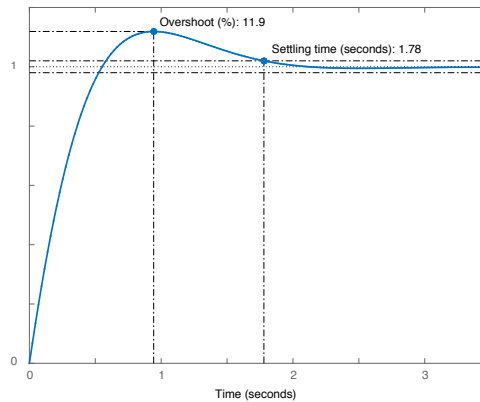
$$K_I = 24.12$$

$$K_P = 17.07$$

Let's compute closed-loop transfer function and associated closed loop-poles

$$\begin{aligned} T(s) &= \frac{3.018s^2 + 17.07s + 24.12}{s^3 + 7.018s^2 + 20.07s + 24.12} \\ p_{1,2} &= -2 \pm 2j \\ p_3 &= -3.0144 \end{aligned}$$

We can see that the poles that are closer to the origin are placed at desired locations, but we can also see that third pole is not far away enough to conclude the dominance. Now let's plot the step-response of the closed-loop system,



Based on the dominant pole locations, we would expect a settling time value of $t_s \approx 2s$, while the actual settling time is approximately 1.8s. On the other hand the overshoot expectation based on the pole location is $PM_P = 4\%$, whereas in the simulation we observe 12% maximum overshoot. Note that, we observed similar "errors" due to closed-loop zero dynamics even in the cases where system has only two complex-conjugate poles. For this reason, PID controller fairly satisfies the design requirements.

Lecture 14

Lecturer: Asst. Prof. M. Mert Ankarali

14.1 Frequency Response Techniques in Control Systems

Let's assume $u(t)$, $y(t)$, and $G(t)$ represents the input, output, and transfer function representation of an input-output continuous time system.

In order to characterize frequency response of a dynamical system, the test signal is

$$u(t) = e^{j\omega t}$$

which is an artificial complex periodic signal with a frequency of ω . The Laplace transform of $u(t)$ takes the form

$$U(s) = \mathcal{L}\{e^{j\omega t}\} = \frac{1}{s - j\omega}$$

Response of the system in s-domain is given by

$$Y(s) = G(s)U(s) = G(s)\frac{1}{s - j\omega}$$

Assuming that $G(s)$ is a rational transfer function we can perform a partial fraction expansion

$$Y(s) = \frac{a}{s - j\omega} + [\text{terms due to the poles of } G(s)]$$

$$a = \lim_{s \rightarrow j\omega} [(s - j\omega)Y(s)] = G(j\omega)$$

$$Y(s) = \frac{G(j\omega)}{s - j\omega} + [\text{terms due to the poles of } G(s)]$$

Taking the inverse Laplace transform yields

$$y(t) = G(j\omega)e^{j\omega t} + \mathcal{L}^{-1}[\text{terms due to the poles of } G(s)]$$

If we assume that the system is “stable” or system is a part of closed loop system and closed loop behavior is stable then at steady state we have

$$\begin{aligned} y_{ss}(t) &= G(j\omega)e^{j\omega t} \\ &= |G(j\omega)|e^{j\omega t + \angle[G(j\omega)]} \\ &= Me^{j\omega t + \theta} \end{aligned}$$

In other words complex periodic signal is scaled and phase shifted based on the following operators

$$\begin{aligned} M &= |G(j\omega)| \\ \theta &= \angle G(j\omega) \end{aligned}$$

It is very easy to show that for a general real time domain signal $u(t) = \sin(\omega t + \phi)$, the output $y(t)$ at steady state is computed via

$$y_{ss}(t) = M \sin(\omega t + \phi + \theta)$$

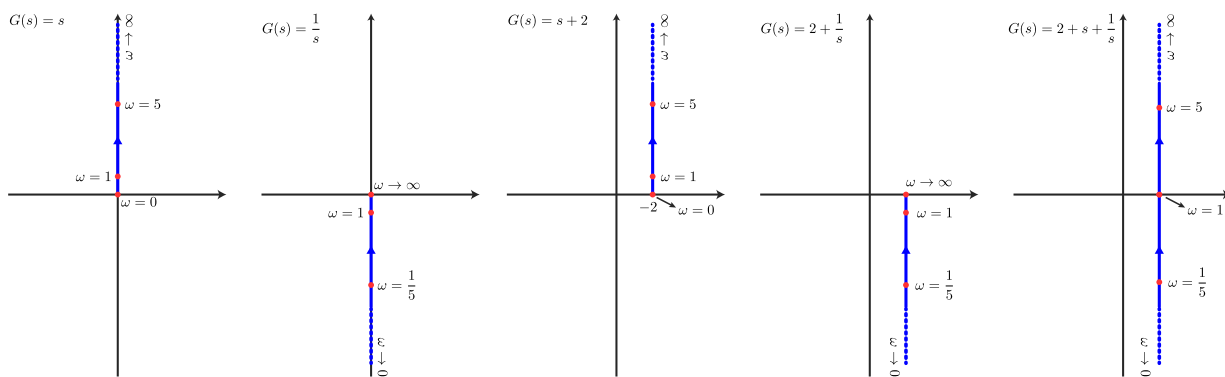
14.1.1 Plotting Frequency Response: Polar Plot

We can consider the frequency response function $G(j\omega)$ as a mapping from positive $j\omega$ axis to a curve in the complex plane. In polar plot, we draw the frequency response function starting from $\omega = 0$ (or $\omega \rightarrow 0^+$) to $\omega \rightarrow \infty$.

Let's draw the polar plots of

$$G_1(s) = s, \quad G_2(s) = \frac{1}{s}, \quad G_3(s) = s + 2$$

$$G_4(s) = 2 + \frac{1}{s}, \quad G_5(s) = 2 + s + \frac{1}{s}$$



Now let's draw the polar plots of

$$G_1(s) = \frac{1}{s+1}, \quad G_2(s) = \frac{s}{s+1}$$

Let's analyze $G_1(j\omega)$ for $\omega \in [0, \infty)$

$$G_1(j\omega) = \frac{1}{j\omega + 1} = \frac{1 - j\omega}{\omega^2 + 1} = \frac{1}{\omega^2 + 1} - \frac{\omega}{\omega^2 + 1}j$$

$$|G_1(j\omega)| = \frac{1}{\sqrt{1 + \omega^2}}$$

$$\angle[G_1(j\omega)] = \arctan(-\omega)$$

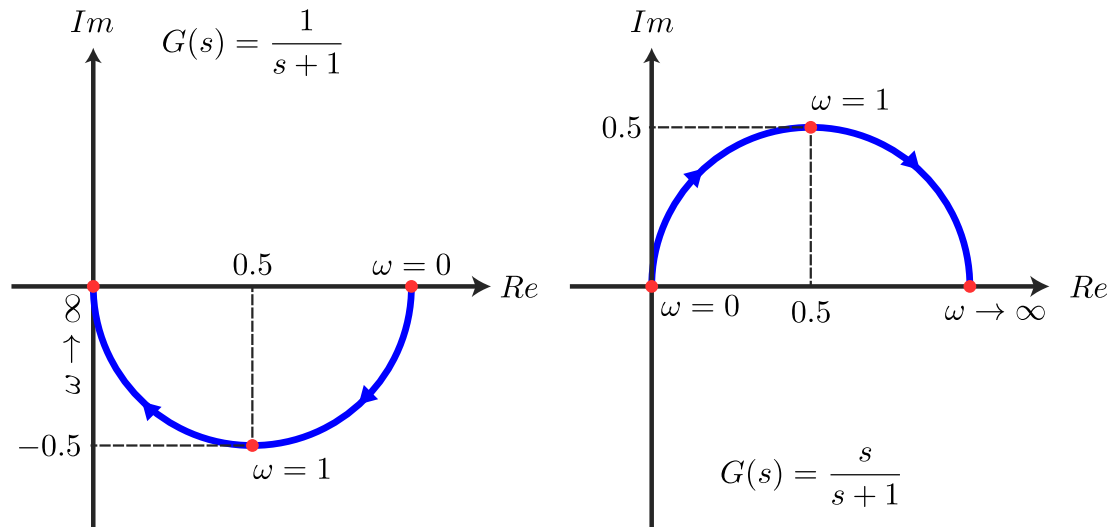
Now let's analyze $G_2(j\omega)$ for $\omega \in [0, \infty)$

$$G_2(j\omega) = \frac{j\omega}{j\omega + 1} = \frac{j\omega + \omega^2}{\omega^2 + 1} = \frac{\omega^2}{\omega^2 + 1} + \frac{\omega}{\omega^2 + 1}j$$

$$|G_2(j\omega)| = \sqrt{\frac{\omega^2}{1 + \omega^2}}$$

$$\angle[G_2(j\omega)] = \arctan(1/\omega)$$

Polar plots of $G_1(s)$ and $G_2(s)$ are illustrated below.

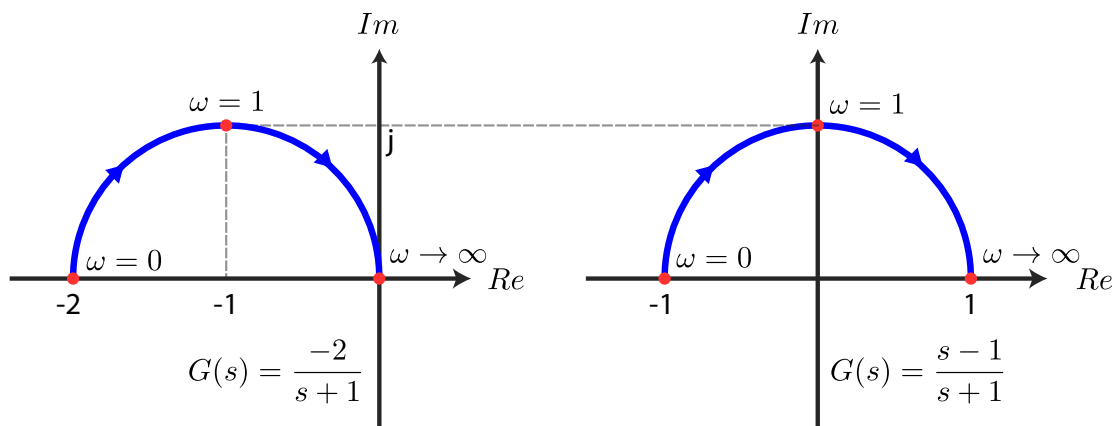


Now let's draw the polar plot of $G(s) = \frac{s-1}{s+1}$ (note that there is a zero in open-right half plane). Note that

$$G(s) = 1 - \frac{2}{s+1}$$

$$G(j\omega) = 1 - 2 \left(\frac{1}{\omega^2 + 1} - \frac{\omega}{\omega^2 + 1} j \right)$$

Process of polar plot drawing of $G(s) = \frac{s-1}{s+1}$ is illustrated below



Now let's draw the polar plot of $G(s) = \frac{1}{(s+1)^2}$

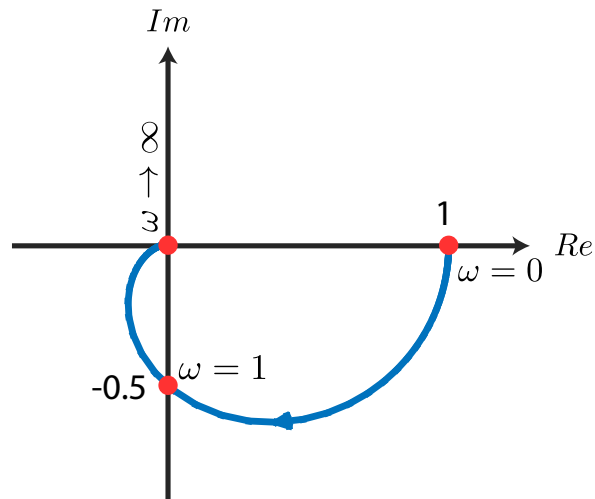
$$G(j\omega) = \frac{1}{(j\omega + 1)^2} = \frac{(-j\omega + 1)^2}{(\omega^2 + 1)^2}$$

$$= [(1 - \omega^2) + j(-2\omega)] \frac{1}{(\omega^2 + 1)^2}$$

Some important points and associated features on the polar plot can be computed as

$$\begin{aligned}\omega \rightarrow 0 &\Rightarrow G(j\omega) = 1 \\ \omega \rightarrow 1 &\Rightarrow G(j\omega) = -0.5j \\ \omega \rightarrow \infty &\Rightarrow |G(j\omega)| \rightarrow 0 \quad \& \quad \angle[G(j\omega)] \rightarrow -\pi\end{aligned}$$

Resultant polar plot is illustrated below



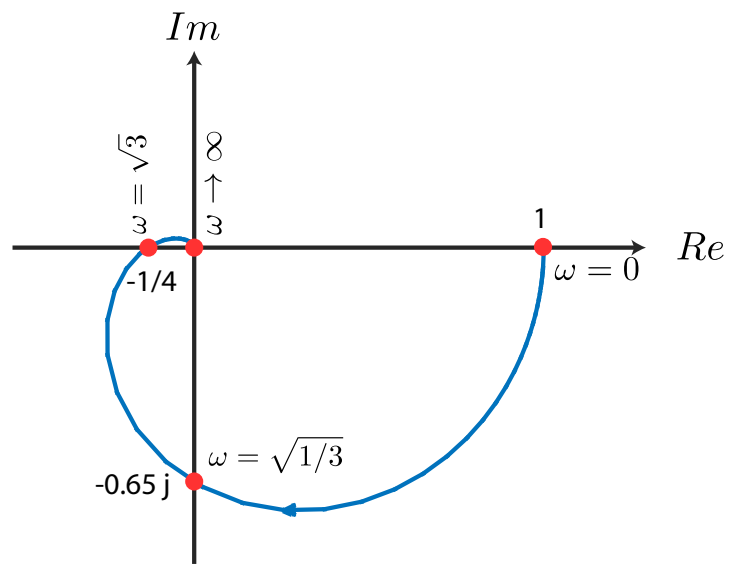
Now let's draw the polar plot of $G(s) = \frac{1}{(s+1)^3}$

$$\begin{aligned}G(j\omega) &= \frac{1}{(j\omega + 1)^3} = \frac{(-j\omega + 1)^3}{(\omega^2 + 1)^3} \\ &= [(1 - 3\omega^2) + j(\omega^3 - 3\omega)] \frac{1}{(\omega^2 + 1)^3}\end{aligned}$$

Some important points and associated features on the polar plot can be computed as

$$\begin{aligned}\omega \rightarrow 0 &\Rightarrow G(j\omega) = 1 \\ \omega \rightarrow \sqrt{1/3} &\Rightarrow G(j\omega) = -0.65j \\ \omega \rightarrow \sqrt{3} &\Rightarrow G(j\omega) = -1/4 \\ \omega \rightarrow \infty &\Rightarrow |G(j\omega)| \rightarrow 0 \quad \& \quad \angle[G(j\omega)] \rightarrow \pi/2\end{aligned}$$

Resultant polar plot is illustrated below



Lecture 15

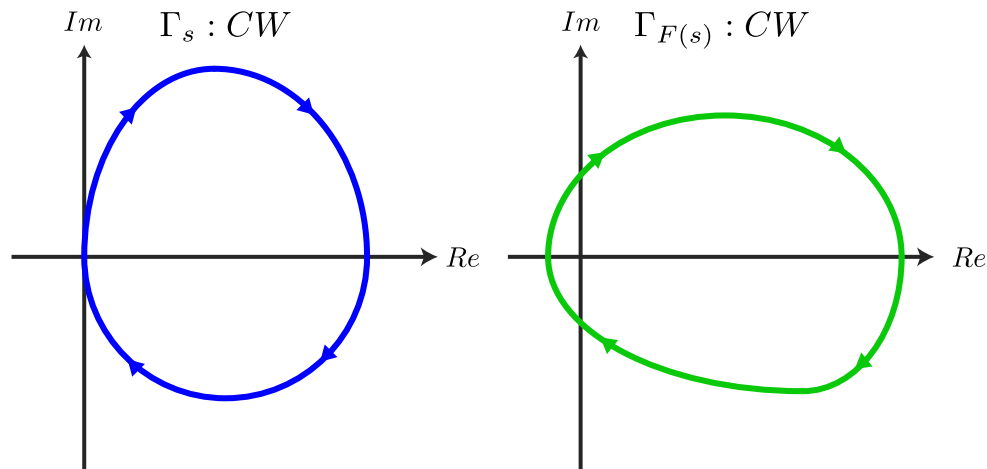
Lecturer: Asst. Prof. M. Mert Ankarali

15.1 Nyquist Stability Criterion

Nyquist stability criterion is another method to investigate the stability (absolute and relative) of a dynamical system (feed-forward and feed-back). It is based on the frequency response characteristics of a system.

Definition: A contour Γ_s is a closed path with a direction in a complex plane.

Remark: A continuous function $F(s)$ maps a contour Γ_s in s -plane to another contour $\Gamma_{F(s)}$ in $F(s)$ plane. The figure below illustrates a clock-wise contour Γ_s and its map $\Gamma_{F(s)}$ which is also clock-wise in this example.



Theorem: Cauchy's Argument Principle

Given $F(s) = \frac{N(s)}{D(s)}$ and a CW contour Γ_s in s -plane which does not cross any zeros (roots of $N(s)$) or poles (roots of $D(s)$) of $F(s)$, then the mapped contour $\Gamma_{F(s)}$ will encircle the origin

$$N = Z - P \quad \text{times}$$

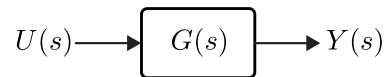
N : # CW encirclements of origin

Z : # zeros $F(s)$

P : # poles $F(s)$

15.1.1 Nyquist Stability Criterion for Feedforward Systems

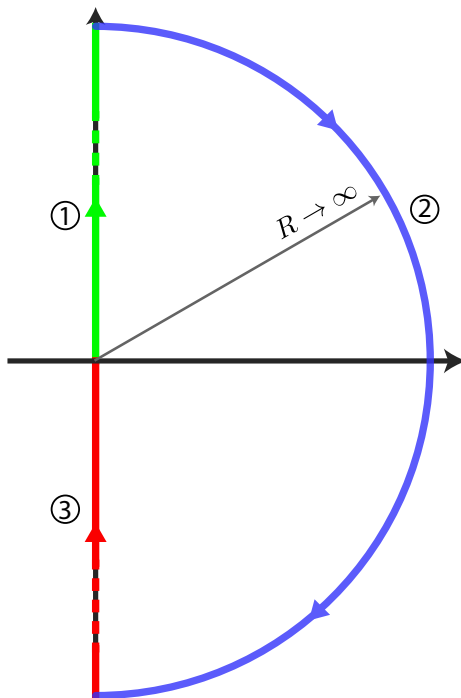
Let's assume that we want to analyze the stability of the following input-output system



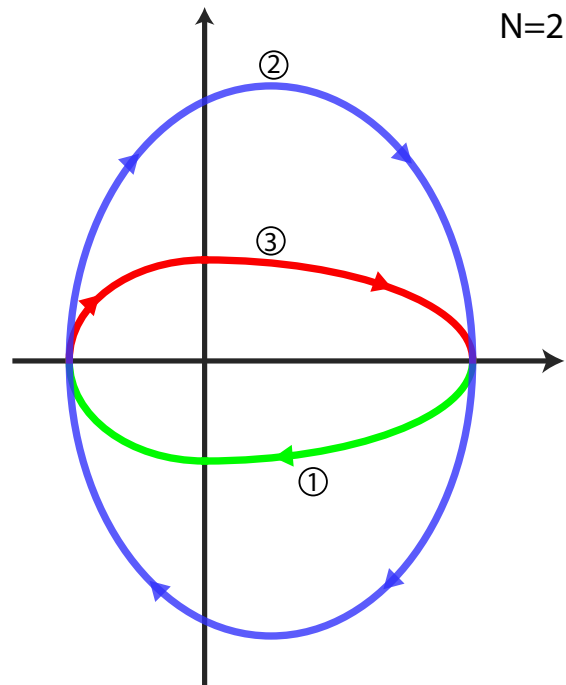
and we know that $G(s) = \frac{N(s)}{D(s)}$ has no zeros or poles on the imaginary axis. Since we are interested in finding the unstable poles, Nyquist contour/path, Γ_s , is defined in a way that it covers the whole open-right half plane. As illustrated in the Figure below, Nyquist contour is technically a half-circle for which the radius, $R \rightarrow \infty$. After that, one can draw the Nyquist plot, which is the mapped contour $\Gamma_{G(s)}$, and count the # of CW encirclements of the origin (N). Based on Cauchy's Argument Principle we know that

- $P = Z - N$, where
- N : # CW encirclements of origin by $\Gamma_{G(s)}$
- Z : # zeros of $G(s)$ with positive real parts
- P : # poles of $G(s)$ with positive real parts

Nyquist Contour



Nyquist Plot



In this example illustration $N = 2$, which implies that total number of unstable poles of $G(s)$ is given by $P = Z + 2$, if Z , i.e. number of zeros with positive real parts, is known then we can compute number of unstable poles P . Alternatively, since in a feedforward system only denominator part determines the stability, instead of $G(s)$, one can draw the Nyquist plot of $\tilde{G}(s) = \frac{1}{D(s)}$.

Ex: Let's analyze the feedforward stability of $G(s) = \frac{s-1}{s+1}$ using Nyquist plot.

Solution: Based on the Nyquist contour we have three major paths

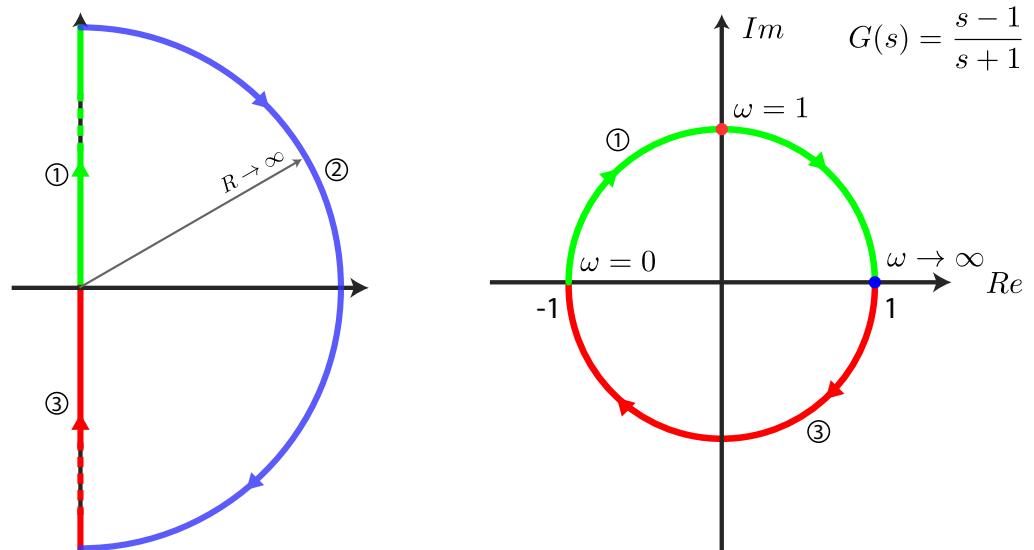
1. This is the polar plot that we covered in the previous lecture, where we plot $G(j\omega)$, where $\omega : 0 \rightarrow \infty$
2. This is the infinite radius circular path. In this case if we write s in polar form, we get $s = Re^{j\theta}$ where $R \rightarrow \infty$, and $\theta : \pi/2 \rightarrow -\pi/2$. Then we can derive that

$$G(Re^{j\theta}) = \frac{Re^{j\theta} - 1}{Re^{j\theta} + 1} \approx \frac{Re^{j\theta}}{Re^{j\theta}} \\ \Rightarrow |G(Re^{j\theta})| \approx 1, \quad \angle[G(Re^{j\theta})] \approx 0$$

In other words, this whole path in the Nyquist plot is concentrated around $1 + 0j$ point in the complex plane.

3. This path is the mapping of the negative imaginary axis, i.e $G(-j\omega)$, where $\omega : \infty \rightarrow 0$. Obviously since $G(-j\omega)$ is complex conjugate of $G(j\omega)$ this path is symmetric to the polar plot with respect to the real axis. Note that direction of this path is reverse of the direction of the polar plot.

If we follow the procedure, we obtain the following Nyquist plot.



We can see from the derived Nyquist plot that $N = 1$, and we know that the system has 1 zero with positive real part $Z = 1$. The total number of unstable poles is equal to $P = Z - N = 1 - 1 = 0$, thus the system is obviously stable.

Ex: Let's analyze the feedforward stability of $G(s) = \frac{1}{s+1}$ using Nyquist plot.

Solution: Now let's analyze the Nyquist paths

1. This is the polar plot that we covered in the previous lecture, where we plot $G(j\omega)$, where $\omega : 0 \rightarrow \infty$. In this case, the behavior when $\omega \rightarrow \infty$ is important. Now let's assume that $\omega \rightarrow R$ and $R \gg 1$

$$G_1(jR) \approx \frac{1}{R^2} - \frac{1}{R}j$$

$$\angle[G_1(j\omega)] \approx -\pi/2$$

2. Now we should be careful with mapping the infinite radius circular path. Again let $s = Re^{j\theta}$ and $\theta : \pi/2 \rightarrow -\pi/2$. Then we can derive that

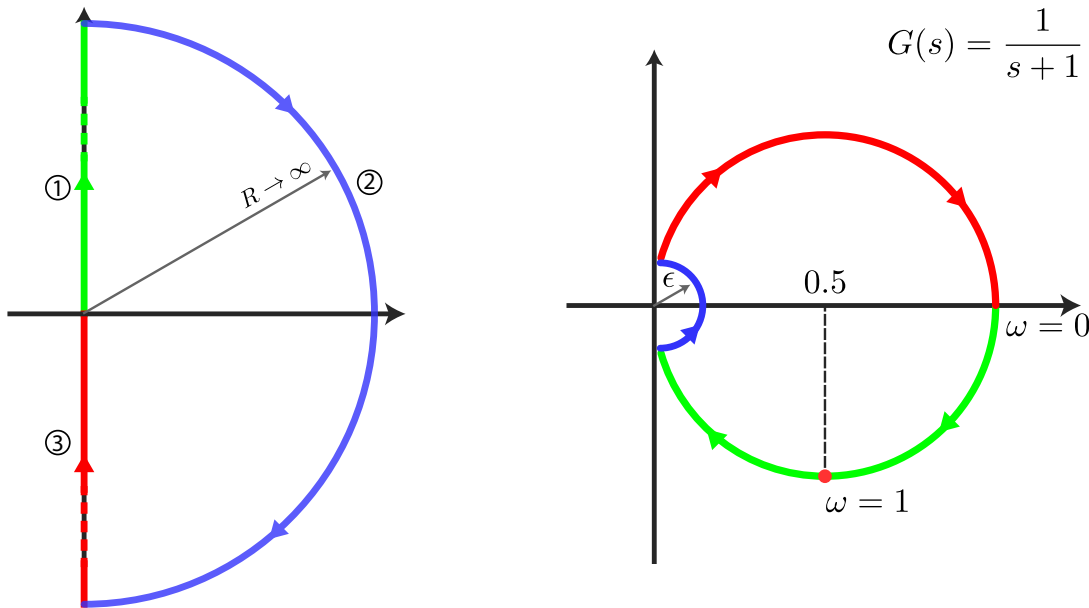
$$G(Re^{j\theta}) \approx \frac{1}{Re^{j\theta}} = \frac{e^{j(-\theta)}}{R}$$

$$\Rightarrow |G(Re^{j\theta})| \approx \epsilon \ll 1, \quad \angle[G(Re^{j\theta})] \approx -\theta$$

Note that when $\theta : \pi/2 \rightarrow -\pi/$, the infinite-small contour around origin rotates in CCW direction.

3. Last path (mapping of negative imaginary axis) is again is the conjugate of the polar plot with reverse direction.

If we follow the procedure, we obtain the following Nyquist plot.



We can see from the derived Nyquist plot that $N = 0$, and we know that the system has no zero with positive real part $Z = 0$. The total number of unstable poles is equal to $P = Z - N = 0 - 0 = 0$, thus the system is obviously stable.

Ex: Analyze the feedforward stability of $G(s) = \frac{1}{(s+1)^2}$ using Nyquist plot.

Solution: First let's analyze the Nyquist paths

1. This is the polar plot that we covered in the previous lecture, where we plot $G(j\omega)$, where $\omega : 0 \rightarrow \infty$. In this case, the behavior when $\omega \rightarrow \infty$ is important. Now let's assume that $\omega \rightarrow R$ and $R \gg 1$

$$G_1(jR) \approx -\frac{1}{R^2} - \frac{1}{R^3}j \quad , \quad \angle[G_1(j\omega)] \approx -\pi$$

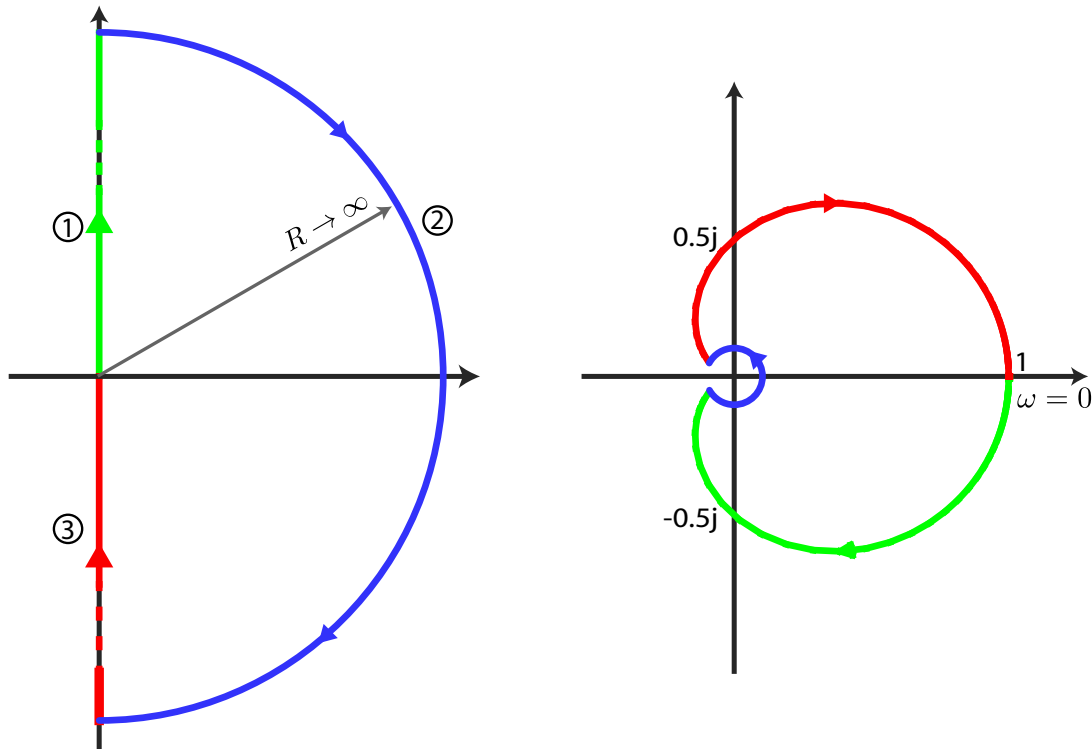
2. Again, we should be careful with mapping of the infinite radius circular path. $s = Re^{j\theta}$ and $\theta : \pi/2 \rightarrow -\pi/2$. Then we can derive that

$$\begin{aligned} G(Re^{j\theta}) &\approx \frac{1}{R^2 e^{j2\theta}} = \frac{e^{j(-2\theta)}}{R^2} \\ \Rightarrow |G(Re^{j\theta})| &\approx \epsilon^2 \ll 1 \quad , \quad \angle[G(Re^{j\theta})] \approx -2\theta \end{aligned}$$

Note that when $\theta : \pi/2 \rightarrow -\pi/2$, the infinite-small contour around origin rotates in CCW direction.

3. Last path (mapping of negative imaginary axis) is again the conjugate of polar plot with reverse direction.

If we follow the procedure, we obtain the following Nyquist plot.



We can see from the derived Nyquist plot that $N = 0$, and we know that the system has no zero with positive real part $Z = 0$. The total number of unstable poles is equal to $P = Z - N = 0 - 0 = 0$, thus the system is obviously stable.

Ex: Analyze the feedforward stability of $G(s) = \frac{1}{(s+1)^3}$ using Nyquist plot.

Solution: First let's analyze the Nyquist paths

1. This is the polar plot that we have covered in the previous lecture, where we plot $G(j\omega)$, where $\omega : 0 \rightarrow \infty$. In this case, the behavior when $\omega \rightarrow \infty$ is important. Now let's assume that $\omega \rightarrow R$ and $R \gg 1$

$$G(jR) \approx -\frac{3}{R^4} + \frac{1}{R^3}j \quad \angle[G(j\omega)] \approx \pi/2$$

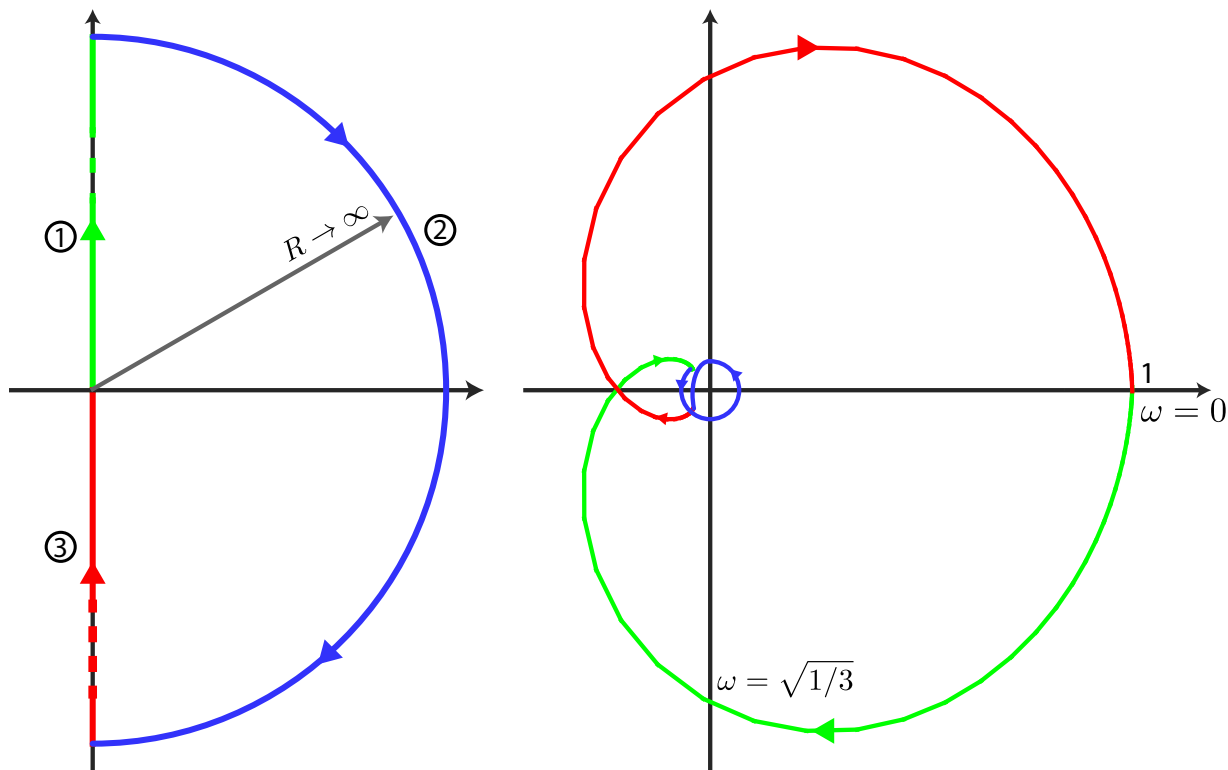
2. Again, we should be careful with mapping of the infinite radius circular path. $s = Re^{j\theta}$ and $\theta : \pi/2 \rightarrow -\pi/2$. Then we can derive that

$$\begin{aligned} G(Re^{j\theta}) &\approx \frac{1}{R^3 e^{j3\theta}} = \frac{e^{j(-3\theta)}}{R^3} \\ \Rightarrow |G(Re^{j\theta})| &\approx \epsilon^3 \ll 1, \quad \angle[G(Re^{j\theta})] \approx -3\theta \end{aligned}$$

Note that when $\theta : \pi/2 \rightarrow -\pi/2$, the infinite-small contour around origin rotates in CCW direction.

3. Last path (mapping of negative imaginary axis) is again the conjugate of polar plot with reverse direction.

If we follow the procedure, we obtain the following Nyquist plot.



If carefully analyze the encirclements around the origin, we can see that **net** encirclement around the origin is $N = 1 - 1$ (or $N = 2 - 2$). Based on this, we verify that the total number of unstable poles is equal to $P = Z - N = 0 - 0 = 0$, thus the system is stable.

Ex: Analyze the feedforward stability of $G(s) = \frac{1}{(s-1)(s+2)}$ using Nyquist plot.

Solution: In the polar plot we draw $G(j\omega)$ where $\omega : 0 \rightarrow \infty$

$$G(j\omega) = \frac{1}{(j\omega - 1)(j\omega + 2)} = \frac{-(j\omega + 1)(-j\omega + 2)}{(\omega^2 + 1)(\omega^2 + 4)} = [-(2 + \omega^2) - \omega j] \frac{1}{(\omega^2 + 1)(\omega^2 + 4)}$$

$$\forall \omega \in (0, \infty), \text{Re}\{G(j\omega)\} < 0 \text{ \& } \text{Im}\{G(j\omega)\} < 0$$

Let's derive where the polar plot approaches when $\omega \rightarrow 0$ and $\omega \rightarrow \infty$

$$\lim_{\omega \rightarrow 0} G(j\omega) = -0.5 + 0j$$

$$\lim_{\omega \rightarrow \infty} |G(j\omega)| = 0 \quad , \quad \lim_{\omega \rightarrow \infty} \angle[G(j\omega)] = -\pi$$

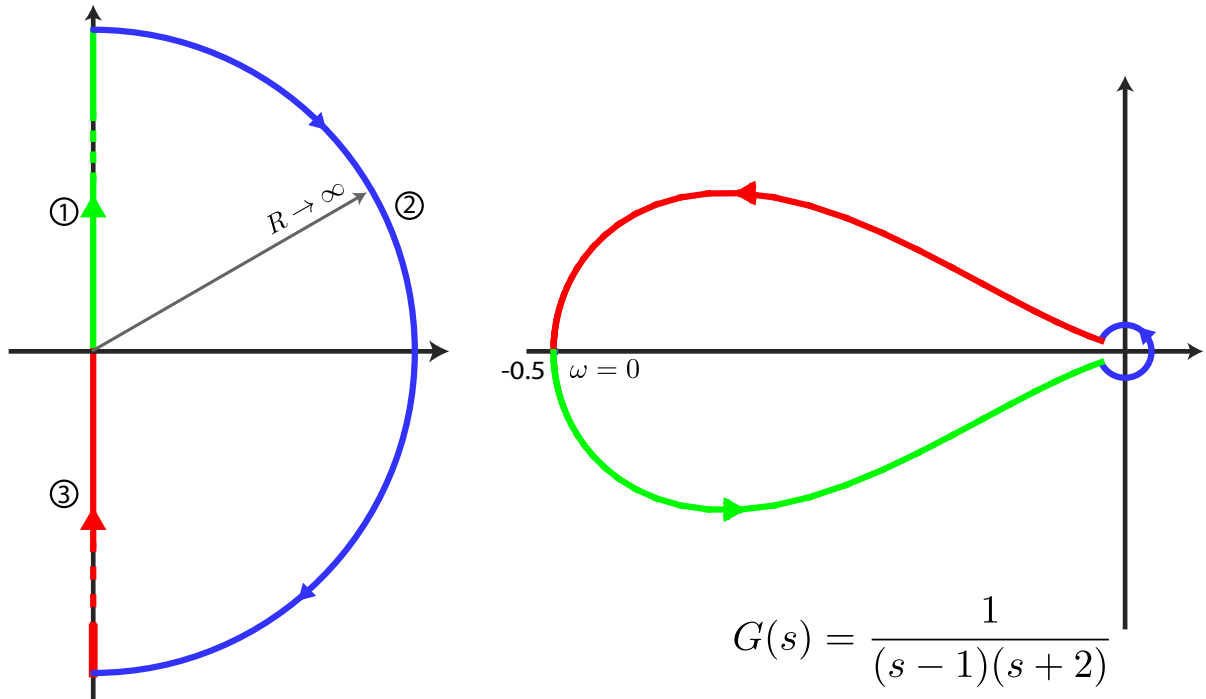
Now let's derive the Nyquist plot conditions for the infinite circle on the Nyquist contour. In this path $s = Re^{j\theta}$ where $R \rightarrow \infty$ (or $R \gg 1$) and $\theta : \pi/2 \rightarrow -\pi/2$ (in CW direction). Thus,

$$G(Re^{j\theta}) \approx \frac{1}{R^2 e^{j2\theta}} = \frac{1}{R^2} e^{j(-2\theta)}$$

$$|G(Re^{j\theta})| \approx \epsilon \ll 1$$

$$\angle[G(Re^{j\theta})] = -2\theta \quad \theta : \pi/2 \rightarrow -\pi/2$$

Note that associated Nyquist plot around origin turns approximately 2π radians in CCW direction. Last path is the mapping of negative imaginary axis which is simply conjugate of the polar plot with reverse direction.



If carefully analyze the encirclements around the origin, we can see that **net** CW encirclement around the origin is $N = -1$. Based on this, we verify that the total number of unstable poles is equal to $P = Z - N = 0 - (-1) = 1$, thus the system is indeed unstable.

Lecture 16

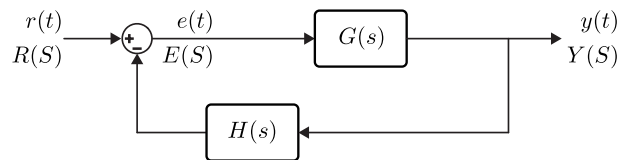
Lecturer: Asst. Prof. M. Mert Ankarali

16.1 Nyquist Stability Criterion for Feedback Systems

Even though we illustrated how to apply Nyquist stability criterion for feedforward systems in the previous lecture (for teaching the details of Nyquist plot and some basics), in control theory and applications Nyquist plot and Nyquist stability test are majorly used for analyzing feedback topologies.

16.1.1 Nyquist Stability for Feedback Systems

The figure below illustrates the fundamental feedback system topology for a SISO system



We know that the closed-loop transfer function, $T(s)$, for this system has the following form

$$T(s) = \frac{G(s)}{1 + G(s)H(s)} = \frac{G(s)}{1 + G_{OL}(s)}$$

where $G_{OL}(s)$ is the open-loop transfer function for the given topology. We know that poles of $T(s)$ are the roots that satisfy $1 + G_{OL}(s) = 0$. Now let's define an analytic function $F(s) = 1 + G_{OL}(s)$, analyze its relation with the open-loop and closed-loop transfer functions.

$$F(s) = 1 + G_{OL}(s) = 1 + \frac{N(s)}{D(s)} = \frac{D(s) + N(s)}{D(s)}$$

where

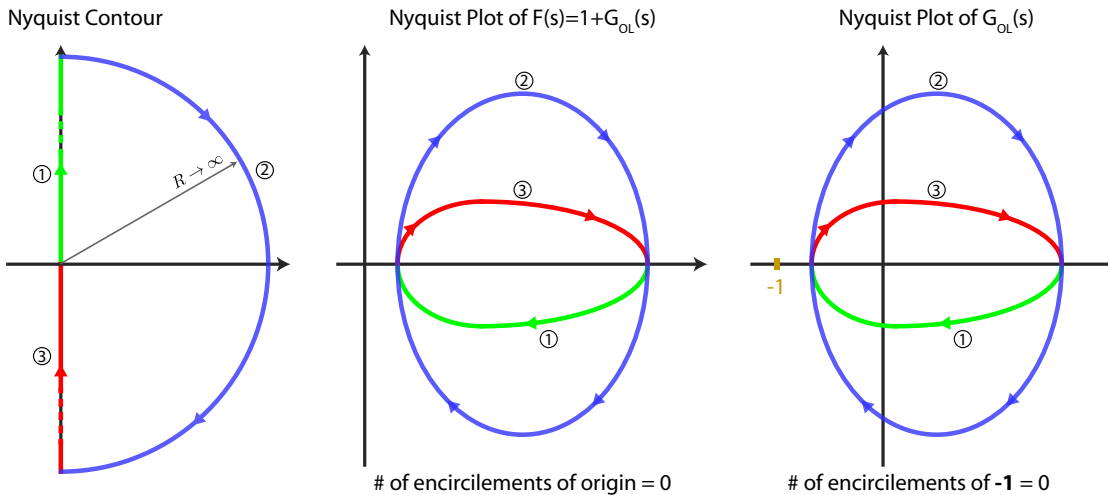
- Roots of $N(s)$ are open-loop zeros
- Roots of $D(s)$ are open-loop poles

Moreover

- **Poles** of $F(s)$ are the roots of $D(s)$, hence they constitutes the **open-loop poles**.
- **Zeros** of $F(s)$ are the poles of $T(s)$, hence they constitutes the **closed-loop poles**.

Note that open-loop zeros and poles are “known”, and goal is to investigate/analyze the number of unstable poles of $T(s)$ which is equal to the number of zeros of $F(s)$ with positive real parts

Now let's assume that we derive Nyquist plots of $F(s) = 1 + G_{OL}(s)$ and $G_{OL}(s)$ and obtain $\Gamma_{F(s)}$ and $\Gamma_{G_{OL}(s)}$. The figure below provides an illustrative example of a Nyquist contour, Nyquist plot of $F(s)$, and Nyquist plot of $G_{OL}(s)$.



First simple observation that we have to pay attention is that in order to obtain $\Gamma_{G_{OL}(s)}$ we shift $\Gamma_{F(s)}$ on real axis to the left (by 1), similarly in order to obtain $\Gamma_{F(s)}$ we shift $\Gamma_{G_{OL}(s)}$ on real axis to the right ((by 1)). From this fact we can derive the following equality which is critical for stability analysis

$$N = \# \text{ of CW encirclements of origin by } \Gamma_{F(s)} = \# \text{ of CW encirclements of } (-1 + 0j) \text{ by } \Gamma_{G_{OL}(s)}.$$

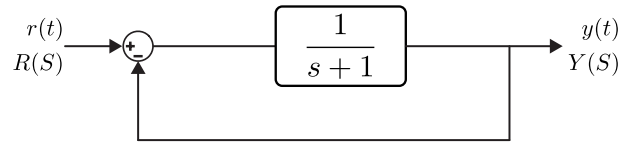
For example In this illustration above $N = 0$. If we apply Cauchy's Principle argument for $F(s)$, we can derive that

- $N = Z_F - P_F$
- P_F : # poles of $F(s)$ with positive real parts, which is indeed equal to the # poles of $G_{OL}(s)$ with positive real parts which is a “known” quantity.
- Z_F : # zeros of $F(s)$ with positive real parts, which is indeed equal to the # unstable poles of $T(s)$ which is the desired output.

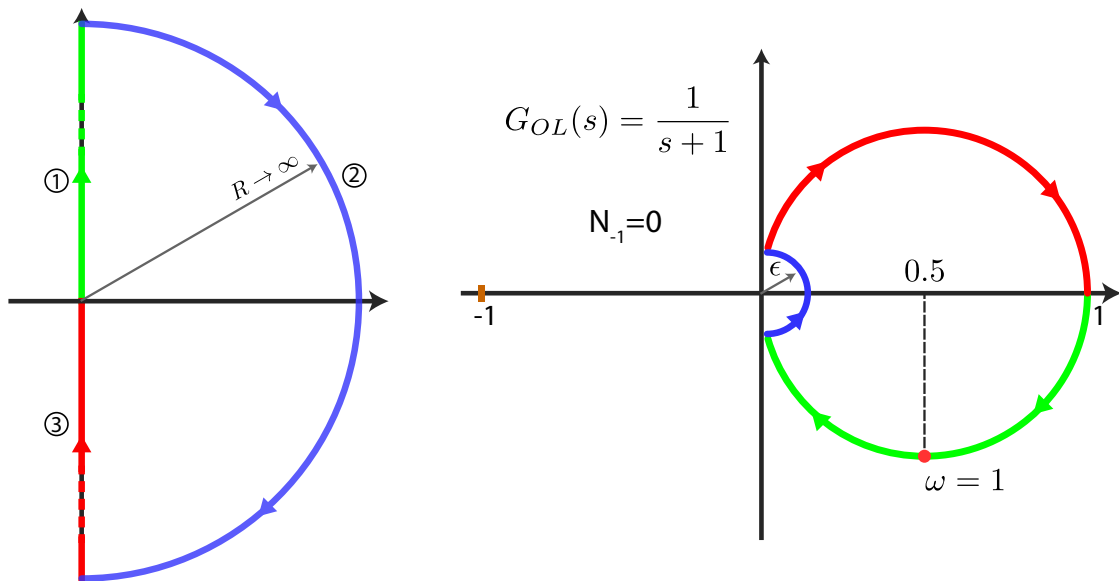
In conclusion, in order to analyze the closed-loop stability of a unity-feedback feed-back systems we apply the following procedure

- Draw the Nyquist plot of $G_{OL}(s)$
- Compute $N = \# \text{ CW encirclements of } (-1 + 0j) \text{ by } \Gamma_{G_{OL}(s)}$
- Compute $P_F = P_{OL} = \# \text{ open-loop poles with positive real parts}$
- Finally compute $\mathbf{P}_{CL} = Z_F = N + P_F = \mathbf{N} + \mathbf{P}_{OL} = \# \text{ unstable closed-loop poles}$

Ex: Analyze the stability of the following feedback system using Nyquist plot.



Solution: For this given system $G_{OL}(s) = \frac{1}{s+1}$. In the previous lecture we already derived the Nyquist plot for $\frac{1}{s+1}$ which is illustrated below



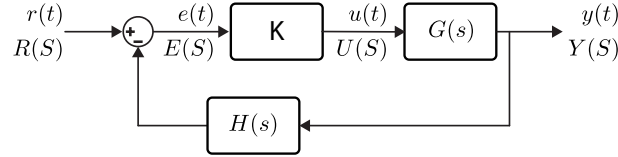
We can conclude from the derived Nyquist plot and open-loop transfer function

- $N = 0$
- $P_{OL} = 0$ (since there is no open-loop unstable pole)
- Finally compute $P_{CL} = 0 = \#$ unstable closed-loop poles.

Thus the closed-loop system is indeed stable.

16.1.2 Nyquist Stability for Feedback Systems with Varying DC Gains

Now let's consider the following feedback system topology for a SISO system where DC gain of the open-loop transfer function is adjusted with a gain parameter K (e.g. P controller).



We would like to test the stability of the closed-loop system for different values of K , moreover we would like to derive the range of K values that makes the closed-loop system stable. Indeed, we don't need to re-draw the Nyquist plot for each K that we want to test.

The analytic function, $F(s)$, that we adopt for analyzing stability can be written in the form

$$F(s) = 1 + G_{OL}(s) = 1 + K \frac{N(s)}{D(s)}$$

where

- Roots of $N(s)$ are open-loop zeros
- Roots of $D(s)$ are open-loop poles
- Variable gain parameter K

In order to analyze the stability of Nyquist plot, in the previous section we showed that we can draw the Nyquist plot $G_{OL}(s)$ and analyze the # encirclements of $(-1 + 0j)$.

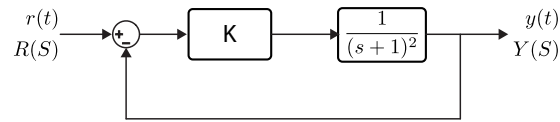
Now, in this case we will derive the Nyquist plot of $\frac{N(s)}{D(s)}$, i.e. $\Gamma_{\frac{N(s)}{D(s)}}$. Note that $\Gamma_{G_{OL}(s)} = K \Gamma_{\frac{N(s)}{D(s)}}$, i.e. we simply scale the Nyquist plot of $\frac{N(s)}{D(s)}$ with K (which can be either positive or negative) to find the Nyquist plot of $G_{OL}(s)$. It is very straightforward to conclude that

$$N = \# \text{ of CW encirclements of } (-1 + 0j) \text{ by } \Gamma_{G_{OL}(s)} = \# \text{ of CW encirclements of } \left(-\frac{1}{K} + 0j\right) \text{ by } \Gamma_{\frac{N(s)}{D(s)}}.$$

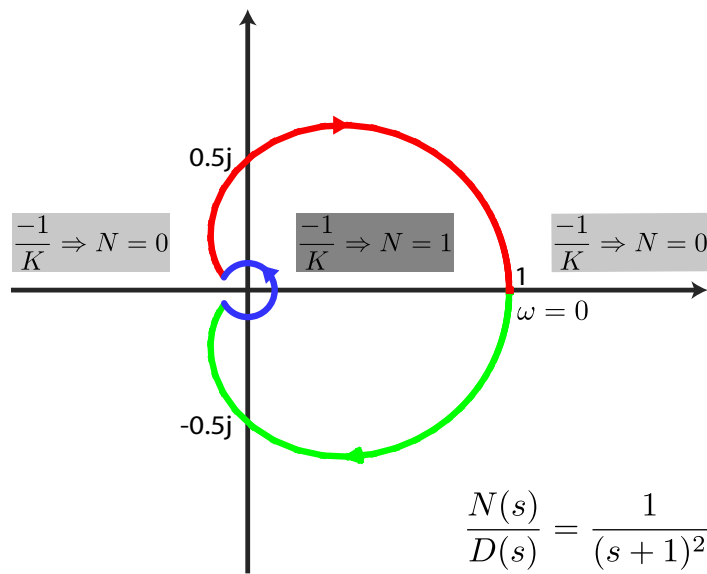
To sum-up, in order to analyze the closed-loop stability of a feedback systems for which we have a variable gain parameter K in the open-loop transfer function, we apply the following procedure

- Draw the Nyquist plot of $\frac{N(s)}{D(s)}$ where $G_{OL}(s) = K \frac{N(s)}{D(s)}$
- Compute $N = \#$ CW encirclements of $\left(-\frac{1}{K} + 0j\right)$ by $\Gamma_{\frac{N(s)}{D(s)}}$
- Compute $P_{OL} = \#$ open-loop poles with positive real parts, i.e. # roots of $D(s)$.
- Finally compute $P_{CL} = N + P_{OL} = \#$ unstable closed-loop poles

Ex: Find the range of K values that makes the following closed-loop system stable.



Solution: For this given system $\frac{N(s)}{D(s)} = \frac{1}{(s+1)^2}$. In the previous lecture we already derived the Nyquist plot for $\frac{1}{(s+1)^2}$ which is illustrated below



We can see from the derived Nyquist plot that the closed path divides the real axis in three different parts. If we analyze these regions separately, we can then find a complete range of K values that makes the closed loop-system stable

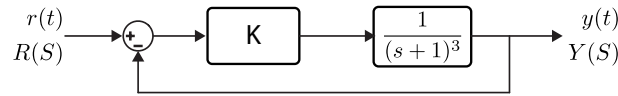
$$\begin{aligned} \frac{-1}{K} \in (-\infty, 0) &\rightarrow K \in [0, \infty) \Rightarrow N = 0 \\ \frac{-1}{K} \in (0, 1) &\rightarrow K \in (-\infty, -1) \Rightarrow N = 1 \\ \frac{-1}{K} \in (1, \infty) &\rightarrow K \in (-1, 0] \Rightarrow N = 0 \end{aligned}$$

Note that since number of open-loop unstable poles is equal to 0, we have $P_{OL} = 0$. Thus, system is BIBO stable if and only if $N = 0$. Whereas, for the region when $N = 1$, there always exist one unstable pole.

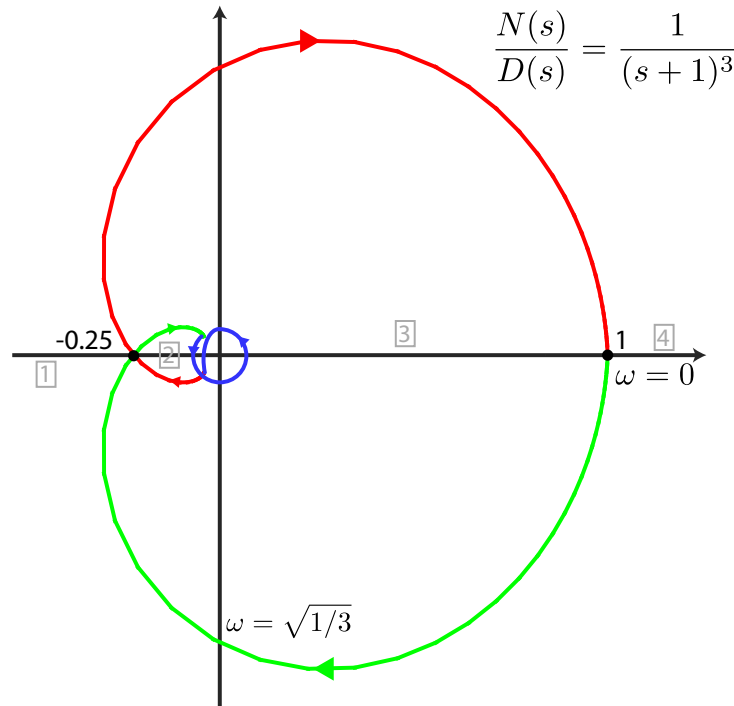
Note that for $K = -1$, we don't have a conclusion, indeed system is BIBO unstable since in this case there exist a pole on the imaginary axis. As a result

$$K \in (-1, \infty) \Leftrightarrow \text{BIBO - Stable}$$

Ex: Find the range of K values that makes the following closed-loop system stable. Also, for the K values that makes the closed-loop system unstable find the number of unstable poles.



Solution: For this given system $\frac{N(s)}{D(s)} = \frac{1}{(s+1)^3}$. In the previous lecture we already derived the Nyquist plot for $\frac{1}{(s+1)^2}$ which is illustrated below



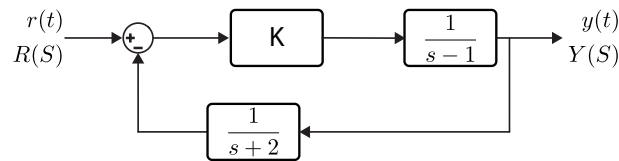
As we can see from the Nyquist plot of $\frac{1}{(s+1)^3}$, it divides the region into four different parts. If we analyze these regions separately, we can then find a complete range of K values that makes the closed loop-system stable, as well as find the number of unstable poles for the unstable cases.

1. $\frac{-1}{K} \in (-\infty, \frac{-1}{4}) \rightarrow K \in [0, 4) \Rightarrow (N = 0, P_{CL} = 0)$, closed-loop system is stable
2. $\frac{-1}{K} \in (\frac{-1}{4}, 0) \rightarrow K \in (4, \infty) \Rightarrow (N = 2, P_{CL} = 2)$. Unstable CL system with 2 unstable poles
3. $\frac{-1}{K} \in (0, 1) \rightarrow K \in (-\infty, -1) \Rightarrow (N = 1, P_{CL} = 1)$. Unstable CL system with 1 unstable pole
4. $\frac{-1}{K} \in (1, \infty) \rightarrow K \in (-1, 0) \Rightarrow (N = 0, P_{CL} = 0)$, closed-loop system is stable

As a result

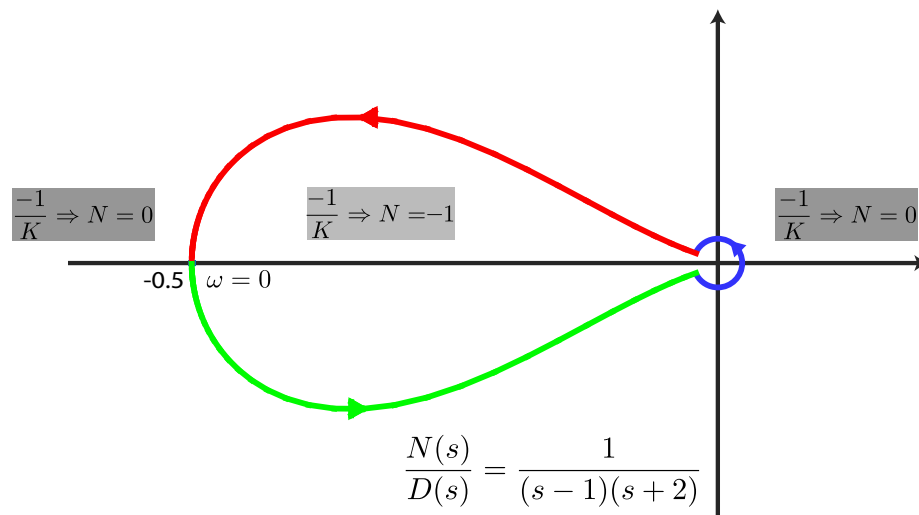
$$K \in (-1, 4) \Leftrightarrow \text{BIBO - Stable}$$

Ex: Find the range of K values that makes the following closed-loop system stable. Also, for the K values that makes the closed-loop system unstable find the number of unstable poles.



Solution: For this given system $\frac{N(s)}{D(s)} = \frac{1}{(s-1)(s+2)}$. Note that for this system $P_{OL} = 1$.

In the previous lecture we already derived the Nyquist plot for $\frac{1}{(s-1)(s+2)}$ which is illustrated below



We can see that the Nyquist plot divides the real axis in three different parts. If we analyze these regions separately, we can then find a complete range of K values that makes the closed loop-system stable

1. $\frac{-1}{K} \in (-\infty, \frac{-1}{2}) \rightarrow K \in (0, 2) \Rightarrow N = 0 \rightarrow P_{CL} = 1$. Unstable CL-system with 1 unstable pole.
2. $\frac{-1}{K} \in (\frac{-1}{2}, 0) \rightarrow K \in (2, \infty) \Rightarrow N = -1 \rightarrow P = 1 - 1 = 0$. Stable CL-system
3. $\frac{-1}{K} \in (0, \infty) \rightarrow K \text{ in } (-\infty, 0) \Rightarrow N = 0 \rightarrow P = 1$. Unstable CL-system with 1 unstable pole.

As a result

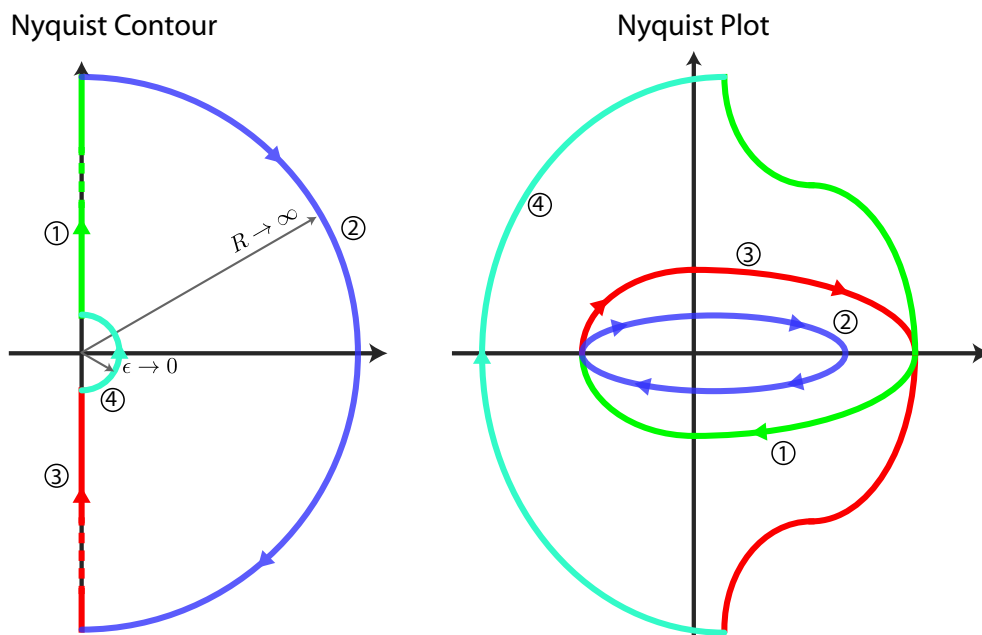
$$K \in (2, \infty) \Leftrightarrow \text{BIBO - Stable}$$

16.1.3 Nyquist Plot and Stability Test with Open-Loop Poles/Zeros on the Imaginary Axis

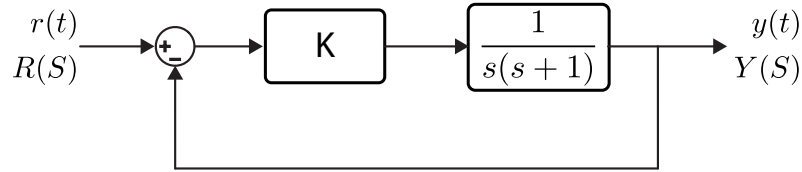
If you remember, when we introduced Nyquist contour and Nyquist stability test we assumed that there is no pole/zero on the imaginary axis. However, it is especially very common to have poles at the origin since it corresponds to simple integrator. In this course, we will explicitly cover the case when there exist a pole or zero at the origin.

Let's assume that $G_{OL}(s)$ has a pole (or zero, or multiple poles) at the origin. We simply modify the Nyquist contour by adding an infinitesimal notch at the origin to the original Nyquist contour.

The figure below illustrates this modified Nyquist contour and an illustrative Nyquist plot.



Ex: Find the range of K values that makes the following closed-loop system stable.



Solution: For this given system, $\frac{N(s)}{D(s)} = \frac{1}{s(s+1)}$. Note, there exist a pole at the origin, thus we need to utilize the modified Nyquist Contour. In the modified Nyquist contour there exist 4 major paths, and we need to draw the Nyquist plot based on these 4 paths.

1. This is the polar plot, where we need to plot $G(j\omega)$, where $\omega : 0 \rightarrow \infty$.

$$G(j\omega) = \frac{1}{j\omega(j\omega + 1)} = \frac{-j - \omega}{\omega(\omega^2 + 1)} = \frac{-1}{\omega^2 + 1} + \frac{-1/\omega}{\omega^2 + 1}j$$

Some observations about polar plot

$$\begin{aligned} \operatorname{Re}\{G(j\omega)\} &< 0 \quad \& \quad \operatorname{Im}\{G(j\omega)\} < 0 \quad , \quad \forall \omega > 0 \\ \lim_{\omega \rightarrow 0} \operatorname{Re}\{G(j\omega)\} &= -1 \quad \& \quad \lim_{\omega \rightarrow 0} \operatorname{Im}\{G(j\omega)\} = -\infty \\ \lim_{\omega \rightarrow \infty} |G(j\omega)| &= 0 \quad \& \quad \lim_{\omega \rightarrow \infty} \angle[G(j\omega)] = -\pi \end{aligned}$$

2. Note that since we are now only interested in the circulation around $-1/K$. The origin in the Nyquist plot corresponds to $|K| \rightarrow \infty$, so that it is really not critical to correctly draw this detail for feedback systems. However, let's draw this detail for this example for practice. $s = Re^{j\theta}$ and $\theta : \pi/2 \rightarrow -\pi/2$. Then we can derive that

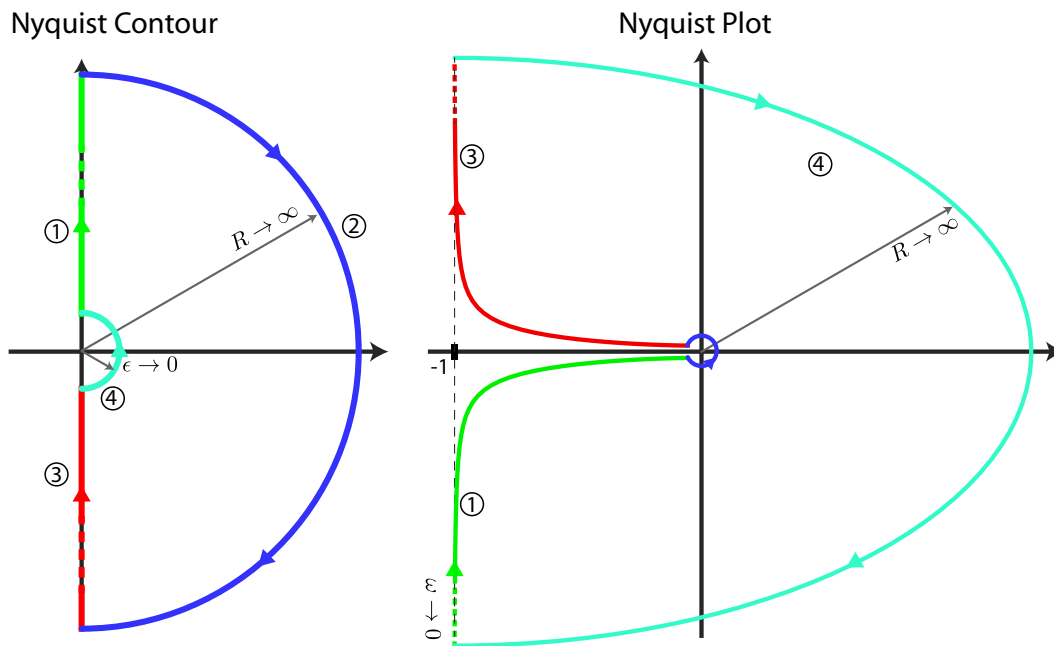
$$\begin{aligned} G(Re^{j\theta}) &\approx \frac{1}{R^2 e^{j2\theta}} = \frac{e^{j(-2\theta)}}{R^2} \\ \Rightarrow |G(Re^{j\theta})| &\approx \epsilon^2 \ll 1 \quad , \quad \angle[G(Re^{j\theta})] \approx -2\theta \end{aligned}$$

Note that when $\theta : \pi/2 \rightarrow -\pi/2$, the infinite-small contour around origin rotates in CCW direction.

3. This part is simply the conjugate of polar plot with reverse direction.
4. Now this part is new, since we are dealing with this path since there is a pole at the origin. $s = \epsilon e^{j\phi}$, where $\epsilon \rightarrow 0$ and $\phi : -\pi/2 \rightarrow \pi/2$ (CCW direction). Then we can derive that

$$\begin{aligned} G(\epsilon e^{j\phi}) &\approx \frac{1}{\epsilon \epsilon^{j\phi}} = Re^{j(-\phi)} \\ \Rightarrow |G(\epsilon e^{j\phi})| &\approx R \ll 1 \quad , \quad \angle[G(\epsilon e^{j\phi})] \approx -\phi \end{aligned}$$

As a result, we obtain the following Nyquist plot for $\frac{N(s)}{D(s)} = \frac{1}{s(s+1)}$

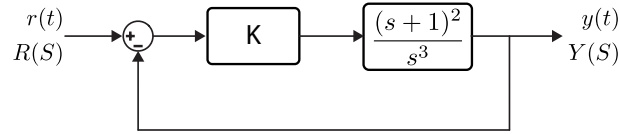


Note that the open-loop transfer function has no unstable poles, i.e. $P_{OL} = 0$. We can see from the Nyquist plot divides the real axis in two different parts. If we analyze these regions separately, we can then find a range of K values that makes the closed loop-system stable

1. $\frac{-1}{K} \in (-\infty, 0) \rightarrow K \in (0, \infty) \Rightarrow N = 0 \rightarrow P_{CL} = 0$. System is stable
2. $\frac{-1}{K} \in (0, \infty) \rightarrow K \in (-\infty, 0) \Rightarrow N = 1 \rightarrow P_{CL} = 1$. Unstable system with 1 unstable pole

As a result, $K \in (0, \infty) \Leftrightarrow$ BIBO-Stable

Ex: Find the range of K values that makes the following closed-loop system stable.



Solution: For this given system, $\frac{N(s)}{D(s)} = \frac{(s+1)^2}{s^3}$. Note, there exist three repeated pole at the origin, thus we need to utilize the modified Nyquist Contour.

In the modified Nyquist contour there exist 4 major paths, and this we need to draw Nyquist plot based on these 4 paths.

1. This is the polar plot, where we need to plot $G(j\omega)$, where $\omega : 0 \rightarrow \infty$.

$$G(j\omega) = \frac{(j\omega + 1)^2}{(j\omega)^3} = \frac{(1 - \omega^2) + 2\omega j}{-j\omega^3} = \frac{(1 - \omega^2)j - 2\omega}{\omega^3} = \frac{-2}{\omega^2} + \frac{1 - \omega^2}{\omega^3}j$$

Some observations about polar plot

$$\begin{aligned} \operatorname{Re}\{G(j\omega)\} &< 0 \\ \operatorname{Im}\{G(j\omega)\} &> 0, \quad \forall \omega \in (0, 1) \\ \operatorname{Im}\{G(j\omega)\} &< 0, \quad \forall \omega \in (1, \infty) \\ [G(j\omega)]_{\omega=1} &= -2 + 0j \\ \lim_{\omega \rightarrow 0} |G(j\omega)| &= \infty \quad \& \quad \lim_{\omega \rightarrow 0} \angle[G(j\omega)] = \pi/2 \\ \lim_{\omega \rightarrow \infty} |G(j\omega)| &= 0 \quad \& \quad \lim_{\omega \rightarrow \infty} \angle[G(j\omega)] = -\pi/2 \end{aligned}$$

2. Note that since we are interested in the circulation around $-1/K$ and origin in the Nyquist plot corresponds to $|K| \rightarrow \infty$, it is really not critical to correctly draw this detail. Thus we will omit it for this problem.
3. This part is simply the conjugate of polar plot with reverse direction.
4. This part is very critical for stability analysis $s = \epsilon e^{j\phi}$, where $\epsilon \rightarrow 0$ and $\phi : -\pi/2 \rightarrow \pi/2$ (CCW direction). Then we can derive that

$$\begin{aligned} G(\epsilon e^{j\phi}) &\approx \frac{1}{\epsilon^3 e^{j3\phi}} = R^3 e^{j(-3\phi)} \\ \Rightarrow |G(\epsilon e^{j\phi})| &\approx R^3 \rightarrow \infty, \quad \angle[G(\epsilon e^{j\phi})] \approx -3\phi \end{aligned}$$

Note that this path is a 1.5 circle that turns in CW direction

As a result, we obtain the following Nyquist plot for $\frac{N(s)}{D(s)} = \frac{(s+1)^2}{s^3}$

Lecture 17

Lecturer: Asst. Prof. M. Mert Ankarali

17.1 Stability Margins: Gain & Phase Margin

We already know that a binary stability metric is not enough to characterize the system performance and that we need metrics to evaluate how stable the system is and its robustness to perturbations. Using root-locus techniques we talked about some “good” pole regions which provides some specifications about stability and closed-loop performance.

Another common and powerful method is to use stability margins, specifically gain and phase margins, based on the frequency domain analysis of a feedback-system.

Phase and gain margins are derived from the Nyquist stability criterion and it is relatively easy to compute them only from the Polar Plot or Bode diagrams for a class of systems.

In this part of the course, we assume that

- Open-loop transfer function of the feedback system is a *minimum-phase* system, i.e.
 - No poles/zeros in the Open Right Half Plane
 - $\lim_{\omega \rightarrow \infty} [\frac{G_{OL}(s)}{s}]_{s=j\omega} = 0$
- The feed-back system is Type 0 – 2
- Phase plot of Bode diagrams of $G_{OL}(j\omega)$ crosses the -180° line at most once. (Equivalently: Polar plot of $G_{OL}(j\omega)$ crosses the negative real-axis at most once.)

Gain Margin

For a stable-system the *gain margin*, g_m , of a system is defined as the smallest amount that the open loop gain can be increased before the closed loop system goes unstable.

In terms of Nyquist & polar plot, we simply choose point, σ_{pc} where the polar plot crosses the negative-real axis and gain margin is simply equal to $g_m = \frac{1}{\sigma_p}$.

Alternatively, the gain margin can be computed based on the frequency where the phase of the loop transfer function $G_{OL}(j\omega)$ is -180° . Let ω_p represent this frequency, called the phase crossover frequency. Then the gain margin for the system is given by

$$\angle[G_{OL}(j\omega_p)] = \pm -180^\circ \Rightarrow g_m = \frac{1}{|G_{OL}(j\omega_p)|} \quad \text{or} \quad G_m = -20 \log_{10} |G_{OL}(j\omega_p)|$$

where G_m is the gain margin in dB scale. If the phase response never crosses the -180° , i.e. $\text{Re}\{G(j\omega)\} \geq 0 \forall \omega \in [0, \infty]$, gain margin is simply ∞ . Higher the gain margin is more robust and stable closed-loop system is.

Phase Margin

The *phase margin* is the amount of “phase lag” required to reach the (Nyquist) stability limit.

In terms of Nyquist & polar plot, we simply choose point, where the polar plot crosses the unit-circle, and phase margin is simply the “angular distance” between this point and the critical point $-1 + 0j$ in CW direction.

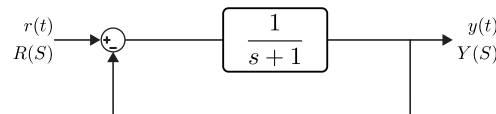
Alternatively, let ω_{gc} be the gain crossover frequency, the frequency where the loop transfer function satisfies $|G_{OL}(j\omega_g)| = 1$ (i.e. unit magnitude). The phase margin is given by

$$|G_{OL}(j\omega_g)| = 1 \Rightarrow \phi_m = \pi + \angle G_{OL}(j\omega_{gc})$$

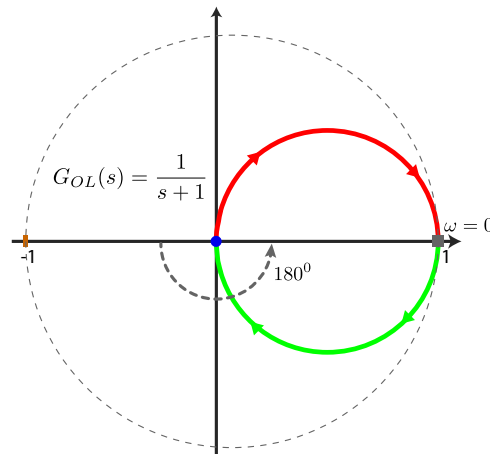
Higher the phase margin is more robust and stable closed-loop system is. Moreover, negative phase simply shows that the closed-loop system is indeed unstable.

Note that if the $G(j\omega)$ is strictly inside the unit-circle, then we can not compute the phase-crossover frequency which simply implies that $\phi_m = \infty$.

Ex: Compute the gain margin and phase margin for the following closed-loop system

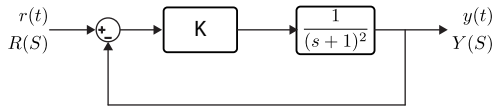


We already derived the Nyquist plot for this system



We can see that the Real part of the polar plot is always positive, thus $g_m = \infty$. Where as the polar plot crosses the unit circle only when $\omega = 0$, thus $\phi_m = 180^\circ$.

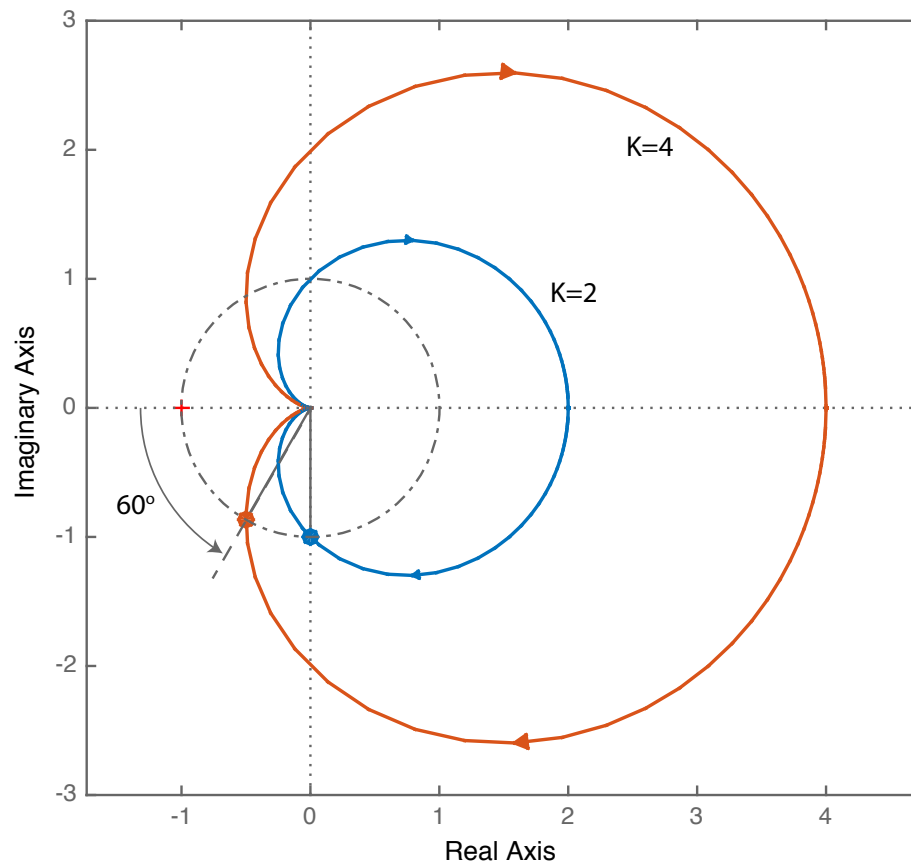
Ex: Compute the gain margin and phase margin for the following closed-loop system for $K = 2$ and $K = 4$.



Nyquist plots for both gain cases is illustrated in the Figure below. We can see from the illustration that

$$K = 2 \Rightarrow \phi_m = 90^\circ \text{ \& } g_m = \infty$$

$$K = 4 \Rightarrow \phi_m = 60^\circ \text{ \& } g_m = \infty$$



Now let's try to compute the phase margins analytically. Let's start with $K = 2$

$$|G(j\omega_g)| = 1 \rightarrow \frac{2}{\omega_g^2 + 1} = 1 \rightarrow \omega_g = 1$$

$$\angle[G(j)] = -2\angle[j + 1] = -90^\circ$$

$$\phi_m = 90^\circ$$

Now let's compute the phase margin for $K = 4$

$$|G(j\omega_g)| = 1 \rightarrow \frac{4}{\omega_g^2 + 1} = 1 \rightarrow \omega_g = \sqrt{3}$$

$$\angle[G(j\sqrt{3})] = -2\angle[\sqrt{3}j + 1] = -120^\circ$$

$$\phi_m = 60^\circ$$

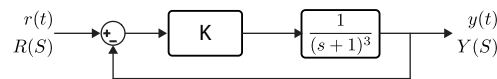
Now let's compute the closed-loop transfer function and compare the damping coefficients for both gain cases

$$T_2 = \frac{2}{s^2 + 2s + 3} \rightarrow \zeta_2 = \frac{1}{\sqrt{3}}$$

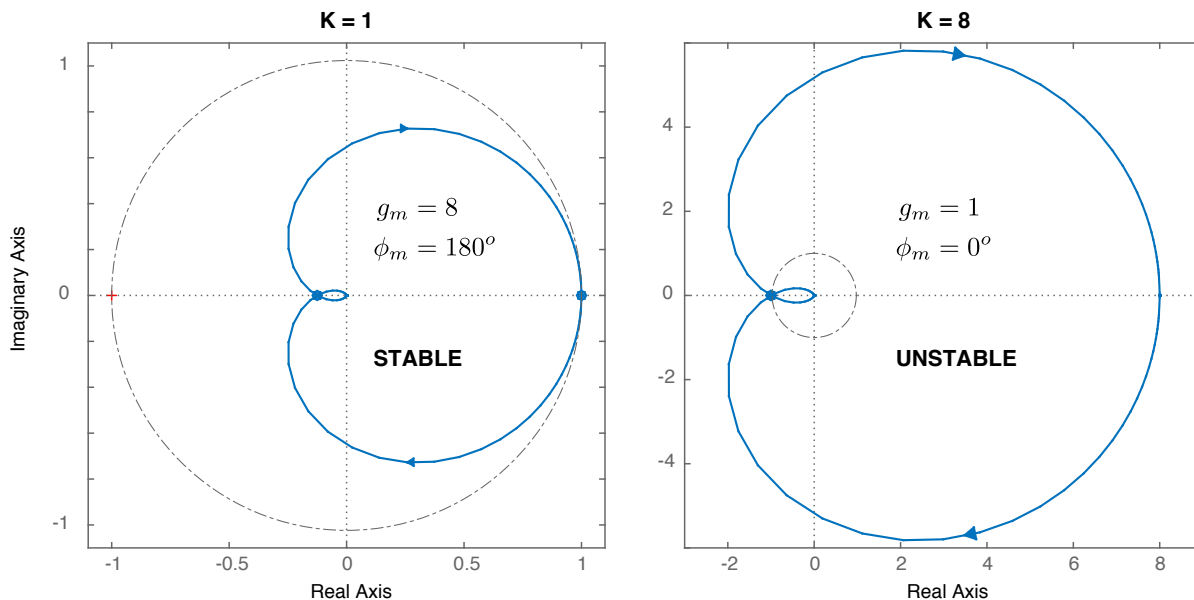
$$T_4 = \frac{4}{s^2 + 2s + 5} \rightarrow \zeta_2 = \frac{1}{\sqrt{5}}$$

We can see that as we decrease the phase margin from 90° to 60° , we also decrease the damping ratio which results in increased maximum-overshoot. In general good phase margin provides good transient performance in time domain.

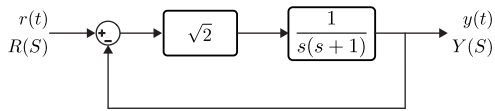
Ex: Compute the gain margin and phase margin for the following closed-loop system for $K = 1$ and $K = 8$ and comment on the stability of the system for both cases.



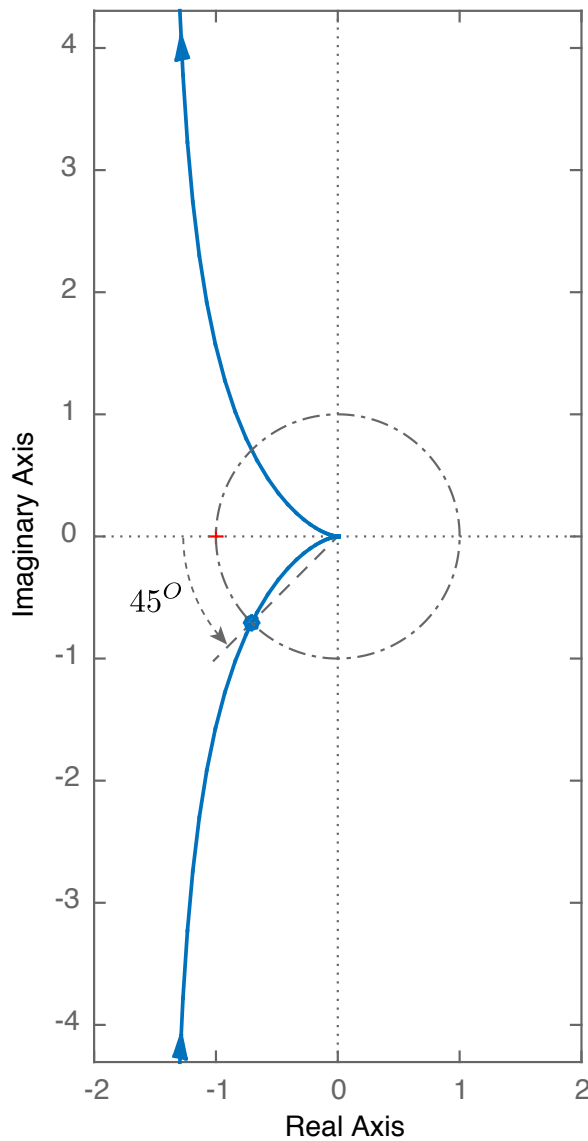
We already derived the Nyquist plot for the case $K = 1$, now let's illustrate both Nyquist plots side-by-side.



Ex: Compute the gain margin and phase margin for the following closed-loop system



We already derived the Nyquist plot for the case $K = 1$, now we have a different gain. Figure below illustrates the zoomed Nyquist plot (which is the important part for gain and phase margin computations).



$$\begin{aligned}
 |G(j\omega_g)| &= 1 \rightarrow |G(j\omega_g)|^2 = 1 \\
 \frac{2}{\omega^2(\omega^2 + 1)} &= 1 \\
 \rightarrow \omega_g &= 1 \\
 \angle[G(j)] &= -(90^\circ + 45^\circ) \\
 \phi_m &= 45^\circ \\
 g_m &= \infty
 \end{aligned}$$

Lecture 18

Lecturer: Asst. Prof. M. Mert Ankarali

18.1 The Bode Plot

Previously, we showed how to illustrate the frequency response function of an LTI system, $G(j\omega)$, using the polar plot and the Nyquist plot. In the Bode Plot gain, $|G(j\omega)|$ and phase response, $\angle[G(j\omega)]$, of the system are illustrated separately as a function of frequency, ω . In both diagrams logarithmic scale is used for frequency axis. On the other hand, in magnitude axis we use a special logarithmic scale in dB units, where as for phase axis we use linear scale. Specifically Magnitude in dB scale, M_{dB} and phase response, ϕ of a transfer function, $G(j\omega)$ are computed as

$$M_{dB}(\omega) = 20\log_{10}|G(j\omega)|$$

$$\phi(\omega) = \angle[G(j\omega)]$$

Now let's write $G(s)$ in pole-zero-gain form and analyze the magnitude (dB) and phase functions

$$G(s) = K \frac{(s - z_1) \cdots (s - z_N)}{(s - p_1) \cdots (s - p_M)}$$

$$,$$

$$M_{dB}\{G(s)\} = 20\log_{10}|G(j\omega)|$$

$$= 20\log_{10}K + [20\log_{10}|(j\omega - z_1)| + \cdots + 20\log_{10}(j\omega - z_N)]$$

$$- [20\log_{10}|(j\omega - p_1)| + \cdots + 20\log_{10}(j\omega - p_M)]$$

$$= K_{dB} + [M_{dB}\{s - z_1\} \cdots M_{dB}\{s - z_N\}] - [M_{dB}\{s - p_1\} \cdots M_{dB}\{s - p_M\}]$$

$$,$$

$$\phi\{G(s)\} = \angle[G(j\omega)]$$

$$= (\angle[j\omega - z_1]) + \cdots + \angle[j\omega - z_N] - (\angle[j\omega - p_1] + \cdots + \angle[j\omega - p_M])$$

$$= (\phi\{s - z_1\} + \cdots + \phi\{s - z_N\}) - (\phi\{s - p_1\} + \cdots + \phi\{s - p_M\})$$

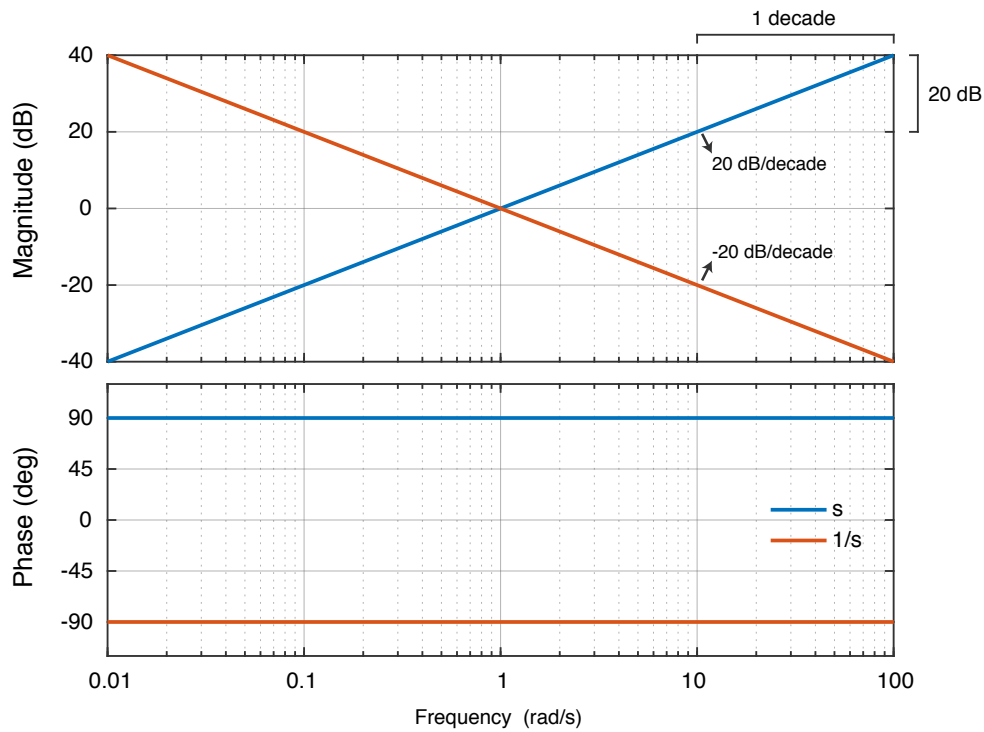
In conclusion, in order to obtain a bode diagram, we can first find phase and magnitude (dB) arguments associated with each pole/zero/gain separately for a given frequency. After that, final magnitude (dB) and phase arguments of $G(s)$ are found by simply adding (and subtracting) the individual components.

Bode Plots of s and $\frac{1}{s}$: Let's write the magnitude and phase functions

$$s \Rightarrow M_{dB}(\omega) = 20\log_{10}(\omega) \quad \& \quad \phi(\omega) = 90^\circ$$

$$\frac{1}{s} \Rightarrow M_{dB}(\omega) = -20\log_{10}(\omega) \quad \& \quad \phi(\omega) = -90^\circ$$

If we illustrate the responses in bode plot, we obtain the following Figure



Bode Plots of First-Order Forms

First let's analyze the phase and magnitude (dB) response of $G(s) = s + 1$

$$M_{dB}(\omega) = 20\log_{10}|G(j\omega)| = 20\log_{10}(\omega^2 + 1)^{1/2} = 10\log_{10}(\omega^2 + 1)$$

$$\phi(\omega) = \arctan \omega$$

Now we will approximate the gain and phase curves using piece-wise continuous straight lines. First approximate the magnitude response

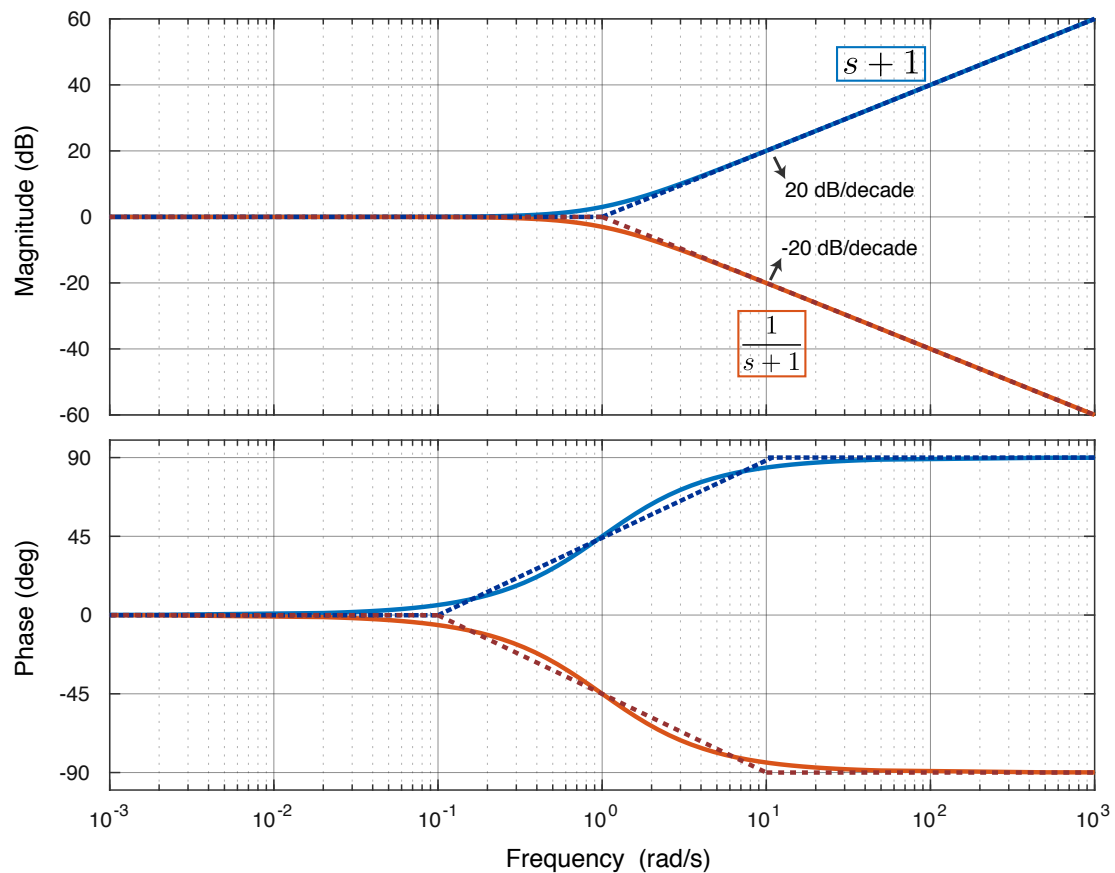
$$\text{Low - Frequency} \Rightarrow M_{dB}(\omega) \approx 0 \text{ dB}$$

$$\text{High - Frequency} \Rightarrow M_{dB}(\omega) \approx 20\log_{10}(\omega)$$

Note that high-frequency and low-frequency approximations intersect at $\omega = 1 \text{ rad/s}$ and $M_{dB} = 0 \text{ dB}$ point. Now let's approximate the phase response

$$\begin{aligned}\text{Low - Frequency} &\Rightarrow \phi \approx 0^\circ \\ \text{High - Frequency} &\Rightarrow \phi \approx 90^\circ \\ \text{Medium - Frequency} &\Rightarrow \phi \approx 45^\circ + 45^\circ \log_{10}(\omega)\end{aligned}$$

Note that, low-frequency and mid-frequency approximations intersect when $\omega = 0.1 \text{ rad/s}$, whereas high-frequency and mid-frequency approximations intersect when $\omega = 10 \text{ rad/s}$. The corner frequency of this “system” is $\omega_c = 1 \text{ rad/s}$. Note that it is very easy to obtain the bode approximations of $G(s) = \frac{1}{s+1}$, if we know the bode approximations of $G(s) = s+1$. We simply multiply both magnitude (dB) and phase responses of $s+1$ with (-1) . The figure below illustrates the original bode plots (solid curves) of $G_1(s) = (s+1)$ and $G_2(s) = \frac{1}{s+1}$ as well as their approximations (dashed lines).



Now let's analyze the phase and magnitude (dB) response of

$$G(s) = Ts + 1 = \frac{s+a}{a} \text{ where } a = \frac{1}{T}$$

Magnitude and phase functions can be obtained as

$$\begin{aligned}M_{dB} &= 20 \log_{10}|G(j\omega)| = 20 \log_{10}(T^2\omega^2 + 1)^{1/2} = 10 \log_{10}(T^2\omega^2 + 1) \\ \phi &= \arctan T\omega\end{aligned}$$

If we follow a similar approach, we can approximate the gain curves as

$$\text{Low - Frequency : } \omega \leq \frac{1}{T} = a \Rightarrow M_{dB} \approx 0 \text{ dB}$$

$$\text{High - Frequency : } \omega \geq \frac{1}{T} = a \Rightarrow M_{dB} \approx 20 \log_{10}(T\omega)$$

Note that high-frequency and low-frequency approximations intersect at $\omega = a \text{ rad/s} = \frac{1}{T} \text{ rad/s}$ and $M_{dB} = 0 \text{ dB}$ point. Now let's approximate the phase response

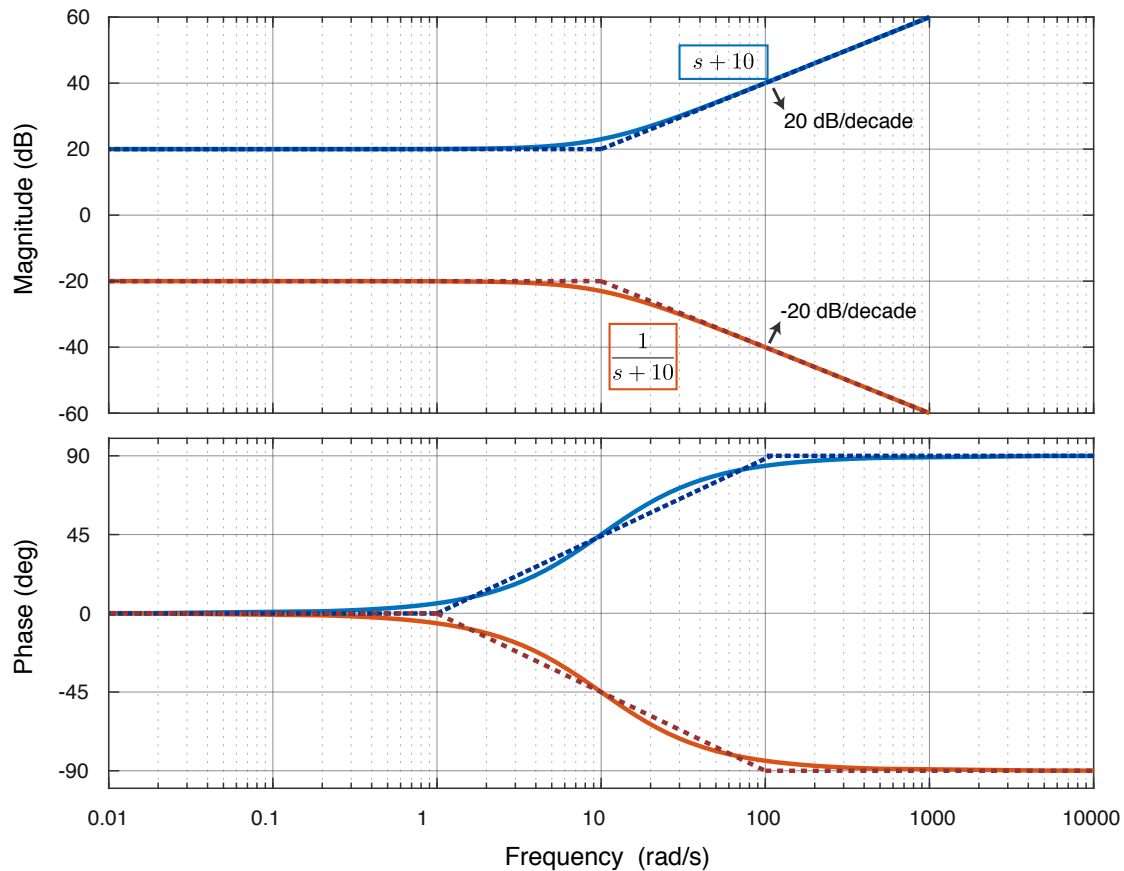
$$\text{Low - Frequency : } \omega \leq \frac{0.1}{T} = 0.1a \Rightarrow \phi \approx 0^\circ$$

$$\text{High - Frequency : } \omega \geq \frac{10}{T} = 10a \Rightarrow \phi \approx 90^\circ$$

$$\text{Medium - Frequency} \Rightarrow \phi \approx 45^\circ + 45^\circ \log_{10}(\omega)$$

Note that low-frequency and mid-frequency approximations intersect when $\omega = 0.1a \text{ rad/s}$, whereas high-frequency and mid-frequency approximations intersect when $\omega = 10a \text{ rad/s}$. The corner frequency of this system is $\omega_c = a \text{ rad/s} = 1/T \text{ rad/s}$. Note that in order to obtain the bode plot of $Ts + 1$, we simply shift the bode plot of $s + 1$ (in ω axis).

Ex: The figure below illustrates the original bode plots (solid curves) of $G_1(s) = (s + 10)$ and $G_2(s) = \frac{1}{s+10}$ as well as their approximations (dashed lines).



Bode Plots of Second-Order Forms

A unity gain standard second order system can be written in the form

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{1}{\frac{1}{\omega_n^2}s^2 + \frac{2\zeta}{\omega_n}s + 1}$$

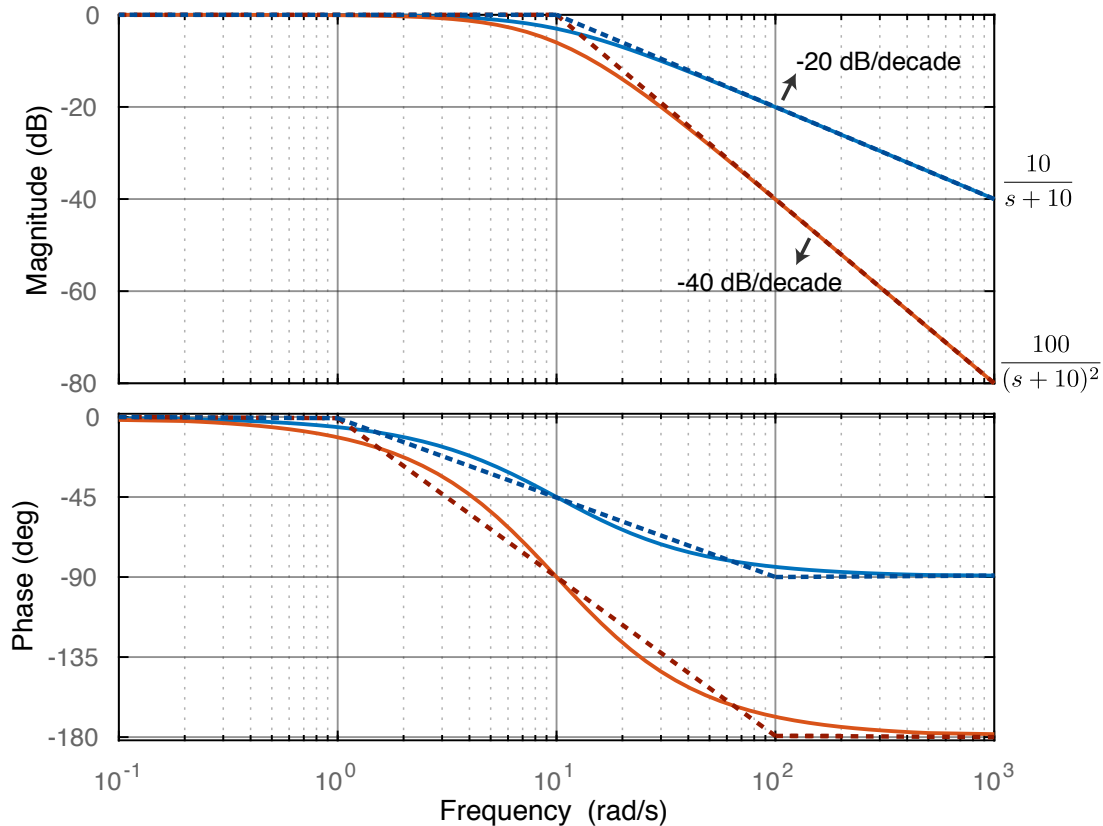
Case 1 : $\zeta = 1$ First let's analyze the bode plots for the critically damped case,

$$G(s) = \frac{1}{\frac{1}{\omega_n^2}s^2 + \frac{2\zeta}{\omega_n}s + 1} = \frac{1}{(\frac{s}{\omega_n} + 1)^2}$$

We can easily observe that Magnitude and phase functions can be obtained as

$$\begin{aligned} M_{dB}\{G(s)\} &= 20 \log_{10}|G(j\omega)| = 20 \log_{10} \left| \frac{1}{\frac{s}{\omega_n} + 1} \right|^2 \\ &= 2M_{dB} \left\{ \frac{1}{\frac{s}{\omega_n} + 1} \right\} \\ \phi\{G(s)\} &= 2\phi \left\{ \frac{1}{\frac{s}{\omega_n} + 1} \right\} \end{aligned}$$

Ex: The figure below illustrates the original bode plots (solid curves) of $G_1(s) = \frac{10}{s+10}$ and $G_2(s) = \frac{100}{(s+10)^2}$ as well as their approximations (dashed lines).



Case 2: $\zeta > 1$ Over-damped case is simply the combination of two identical first order systems.

Ex: Let's analyze the the bode plots for the following system

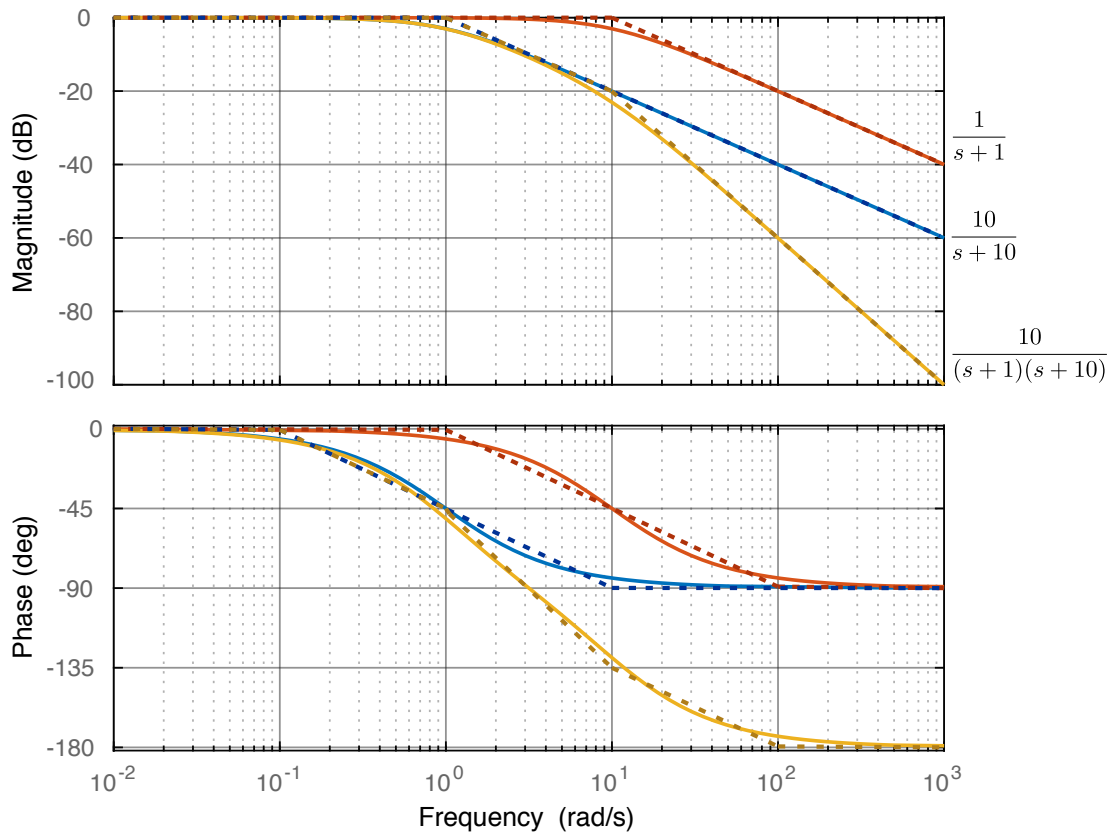
$$G(s) = \frac{10}{(s+1)(s+10)} = \frac{1}{s+1} \frac{10}{s+10}$$

We can easily observe that

$$M_{dB}\{G(s)\} = M_{dB}\left\{\frac{1}{s+1}\right\} + M_{dB}\left\{\frac{10}{s+10}\right\}$$

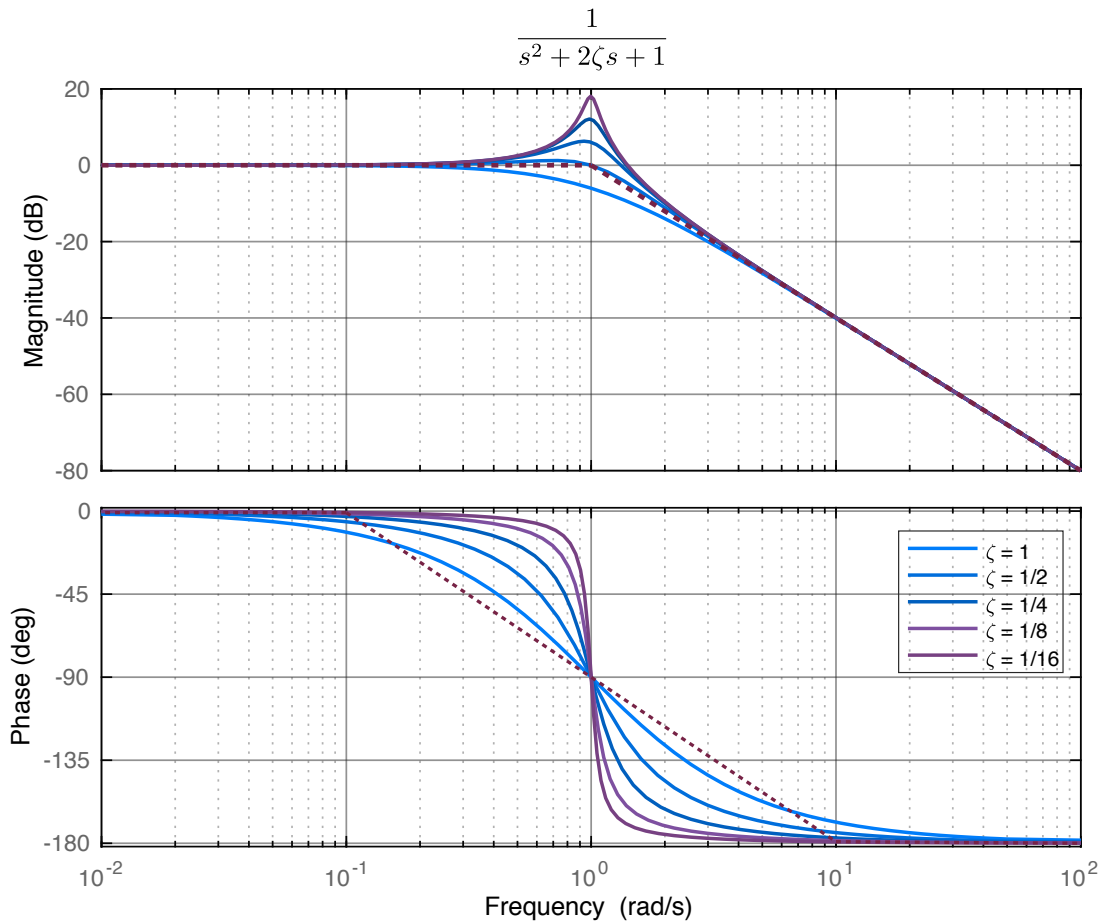
$$\phi\{G(s)\} = \phi\left\{\frac{1}{s+1}\right\} + \phi\left\{\frac{10}{s+10}\right\}$$

The figure below illustrates the original bode plots (solid curves) of $G_1(s) = \frac{1}{s+1}$, $G_2(s) = \frac{10}{s+10}$ and $G_4(s) = \frac{10}{(s+1)(s+10)}$ as well as their approximations (dashed lines).



Case 3 : $\zeta < 1$ For under-damped systems, corner frequencies of the piece-wise linear approximations are unchanged. Thus, we use same approximation with the critically damped case. However, as damping ratio increases we may observe larger differences between the actual bode plot and the approximations.

Ex: The figure below illustrates the original bode plots (solid curves) of $G_1(s) = \frac{10}{s+10}$ and $G_2(s) = \frac{100}{(s+10)^2}$ as well as their approximations (dashed lines).



We can see that best phase matching between the actual bode plot and approximation is achieved when $\zeta = 1$, however surprisingly best magnitude matching between the actual bode plot and approximation is achieved when $\zeta = 2$. Indeed, best match between the actual and approximate bode plots in magnitude is achieved when $\zeta = 1/\sqrt{2}$.

18.2 Gain & Phase Margin from Bode Plots

We already know that for a feedback system, phase and gain margins can be computed based on the Frequency Response characteristics of the open-loop transfer function, $G_{OL}(j\omega)$ (under some assumption regarding the system properties).

Specifically, we can compute the phase crossover frequency, ω_{pc} and the gain margin, g_m (linear scale) and G_m (dB scale), as

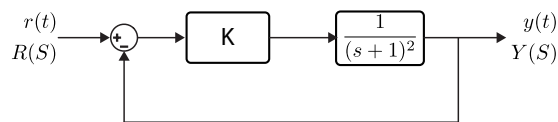
$$\angle[G_{OL}(j\omega_p)] = \pm -180^\circ \Rightarrow g_m = \frac{1}{|G_{OL}(j\omega_{pc})|} \quad \text{or} \quad G_m = -20 \log_{10} |G_{OL}(j\omega_{pc})|$$

where as gain crossover frequency, ω_{gc} , and the phase margin can be computed as

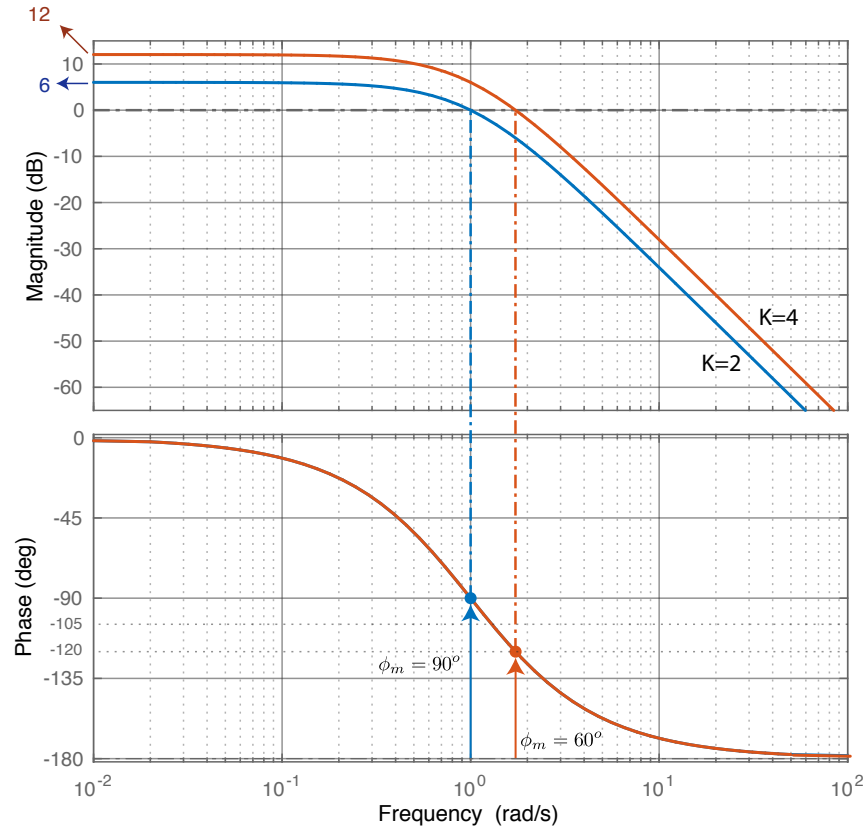
$$|G_{OL}(j\omega_{gc})| = 1 \quad \text{or} \quad M_{dB}\{G_{OL}(j\omega_{gc})\} = 0 \text{ dB} \Rightarrow \phi_m = \pi + \angle G_{OL}(j\omega_{gc})$$

Indeed it is generally easier to derive the phase and gain margin of a system from the bode plots compared to the Nyquist & polar plots.

Ex: Compute the phase margin for the following closed-loop system for $K = 2$ and $K = 4$, both from the approximate and actual bode plots.



Actual bode plots for both gains are illustrated below.

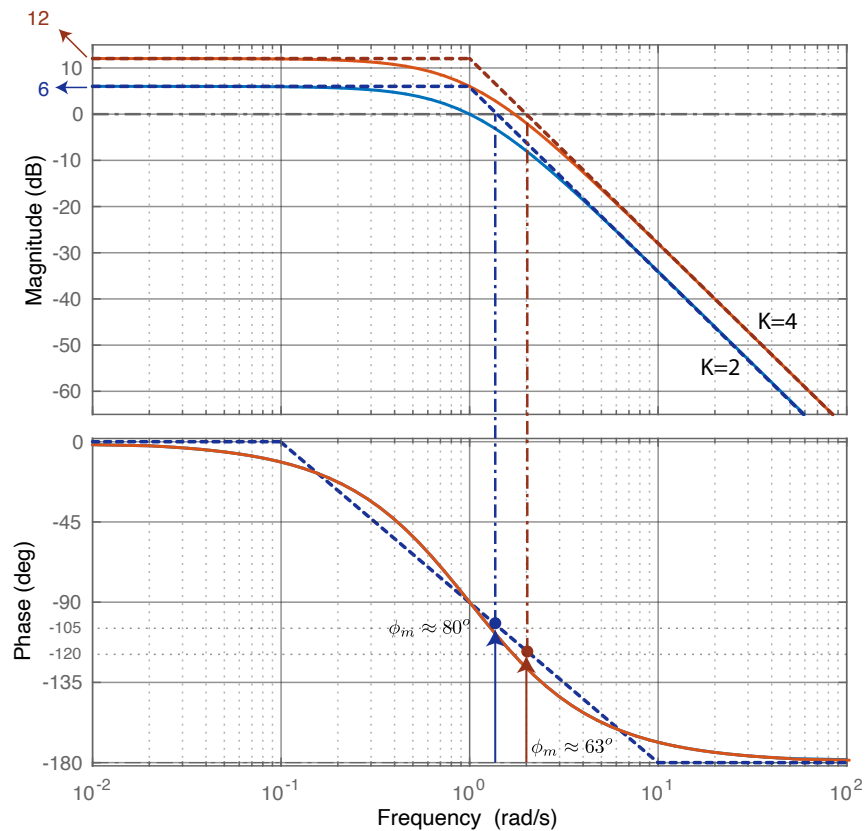


If we label the gain crossover frequencies and find the corresponding phase values, we can easily compute the phase margins as

$$\begin{aligned} K = 2 &\Rightarrow \phi_m = 90^\circ \quad (\omega_{gc} = 1 \text{ rad/s}) \\ K = 4 &\Rightarrow \phi_m = 60^\circ \quad (\omega_{gc} \approx 1.8 \text{ rad/s}) \end{aligned}$$

These results verify the actual phase margin values that we previously computed using the Nyquist plot. Note that G_m is infinity for both cases.

Now let's illustrate the approximate bode plots (dashed lines), which are illustrated in the Figure below on top of the actual ones

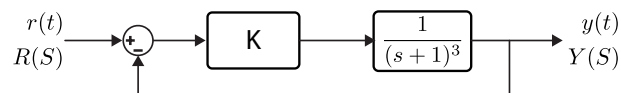


If we approximately compute the phase margins based on the approximate bode plots we obtain that

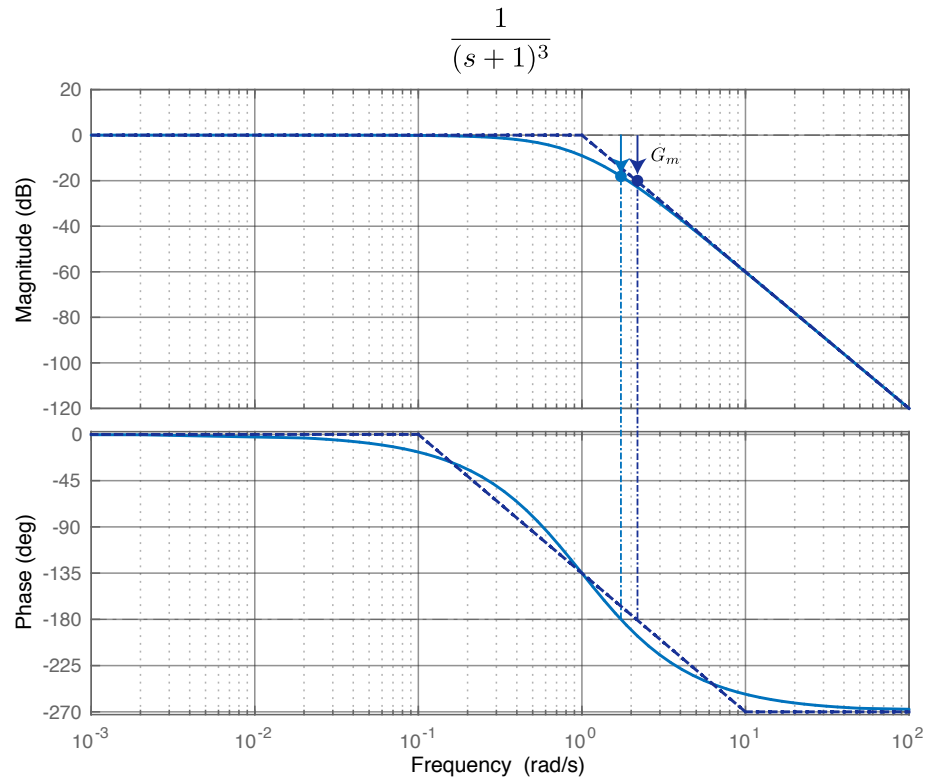
$$K = 2 \Rightarrow \phi_m \approx 80^\circ$$

$$K = 4 \Rightarrow \phi_m \approx 63^\circ$$

Ex: Compute the gain margin and phase margin for the following closed-loop system for $K = 1$ and $K = 8$ both from the actual and approximate bode plots.



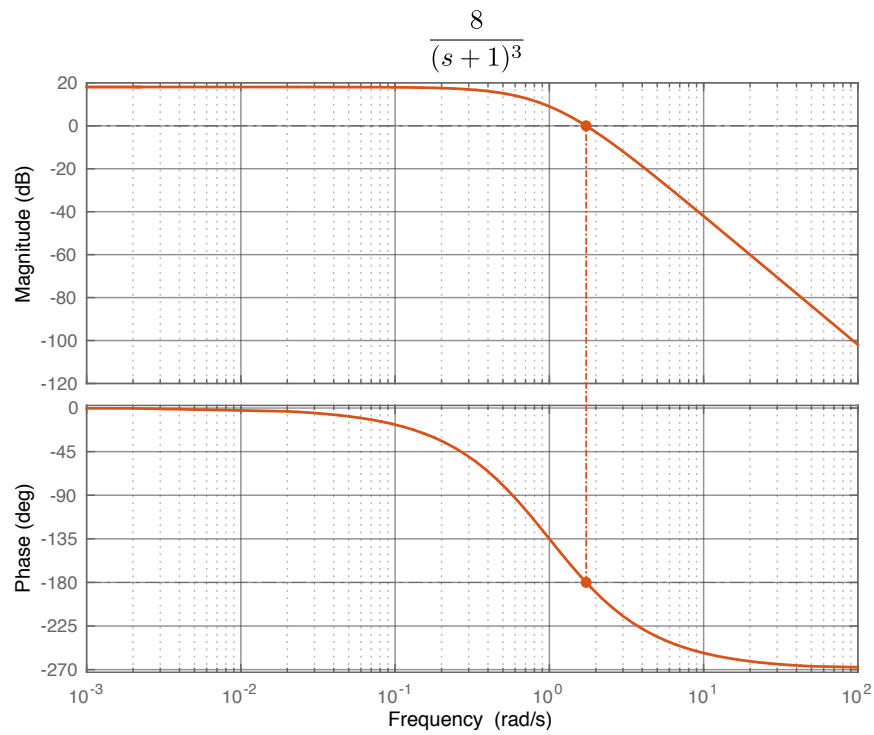
First let's analyze $K = 1$, the Figure below illustrates the actual and approximate bode plots of $G(s) = \frac{1}{(s+1)^3}$.



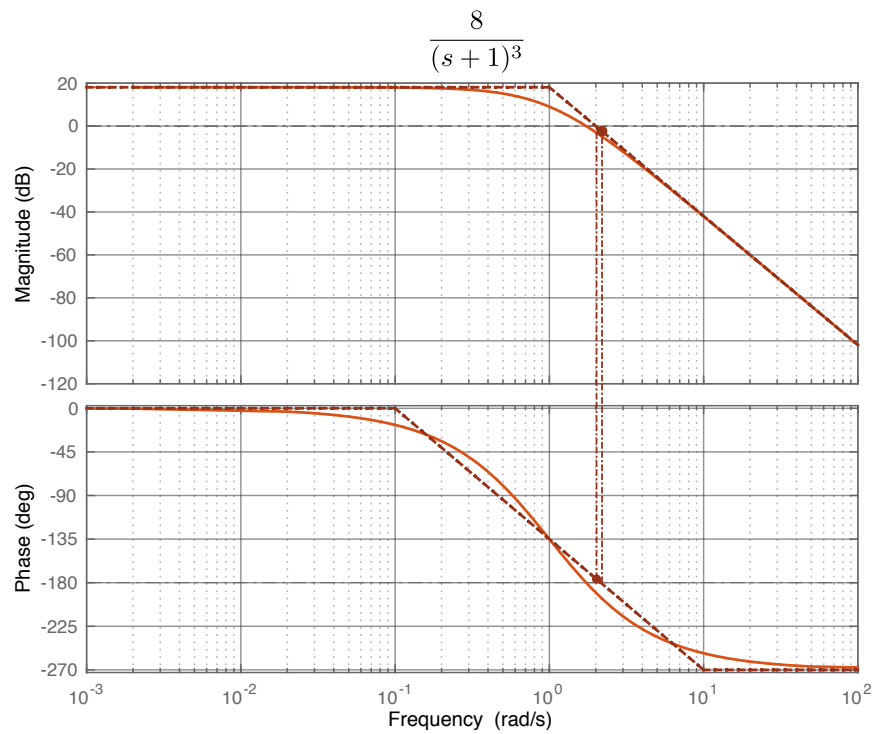
We can see that phase margin is 180° for the given system, since $\omega_{gc} = 0 \rightarrow \phi(\omega_{gc}) = 0^\circ$. On the other hand we can derive the following gain margin estimates from the actual and approximate bode plots

$$\begin{aligned} \text{Actual : } G_m &\approx 18 \text{ dB} \rightarrow g_m \approx 8 \\ \text{Approximate : } G_m &\approx 20 \text{ dB} \rightarrow g_m \approx 10 \end{aligned}$$

Now let's analyze $K = 8$, the Figure below illustrates the actual Bode plots of $G(s) = \frac{8}{(s+1)^3}$.



We can see from the actual bode-plot that $G_m = 0$ dB and $\phi_m = 0^\circ$. However, if we draw the approximate bode plots

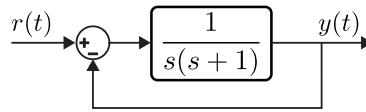


and estimate these margins we obtain

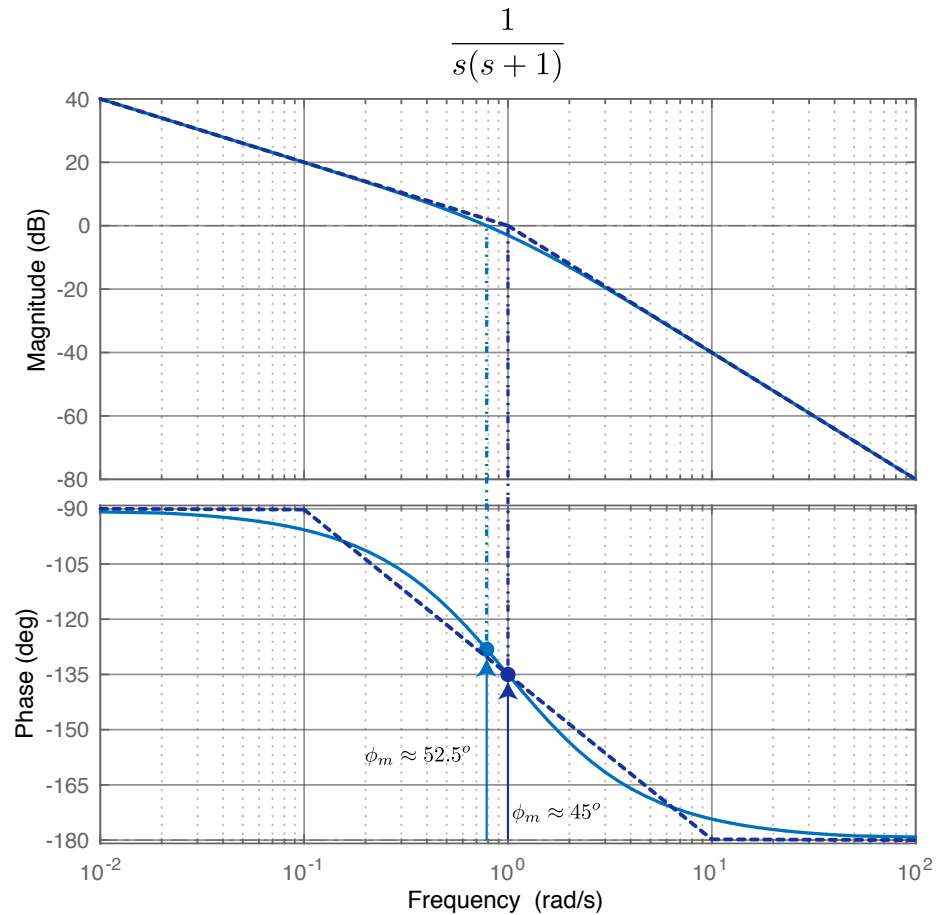
$$\begin{aligned} G_m &\approx 2 \text{ dB} \quad \rightarrow \quad g_m \approx 1.2 \\ \phi_m &= 2^\circ \end{aligned}$$

If we consider the bode plots that we draw based on asymptotic approximations and compute the phase and gain margins, we can assume that system is indeed stable. However, these margins are rather low. In this respect, we can conclude that if one analyzes the absolute stability of a system using approximately bode plots, in order to comment on stability, he/she should observe significant phase and gain margins.

Ex: Compute phase margin for the following closed-loop system using the actual and approximate bode plots



The Figure below illustrates the actual and approximate bode plots.



We can derive the following phase margin computations from the actual and approximate bode plots

$$\text{Actual : } \phi_m \approx 52.5^\circ \quad (\omega_{gc} \approx 0.8 \text{ rad/s})$$

$$\text{Approximate : } \phi_m \approx 45^\circ \quad (\omega_{gc} = 1 \text{ rad/s})$$

Lecture 19

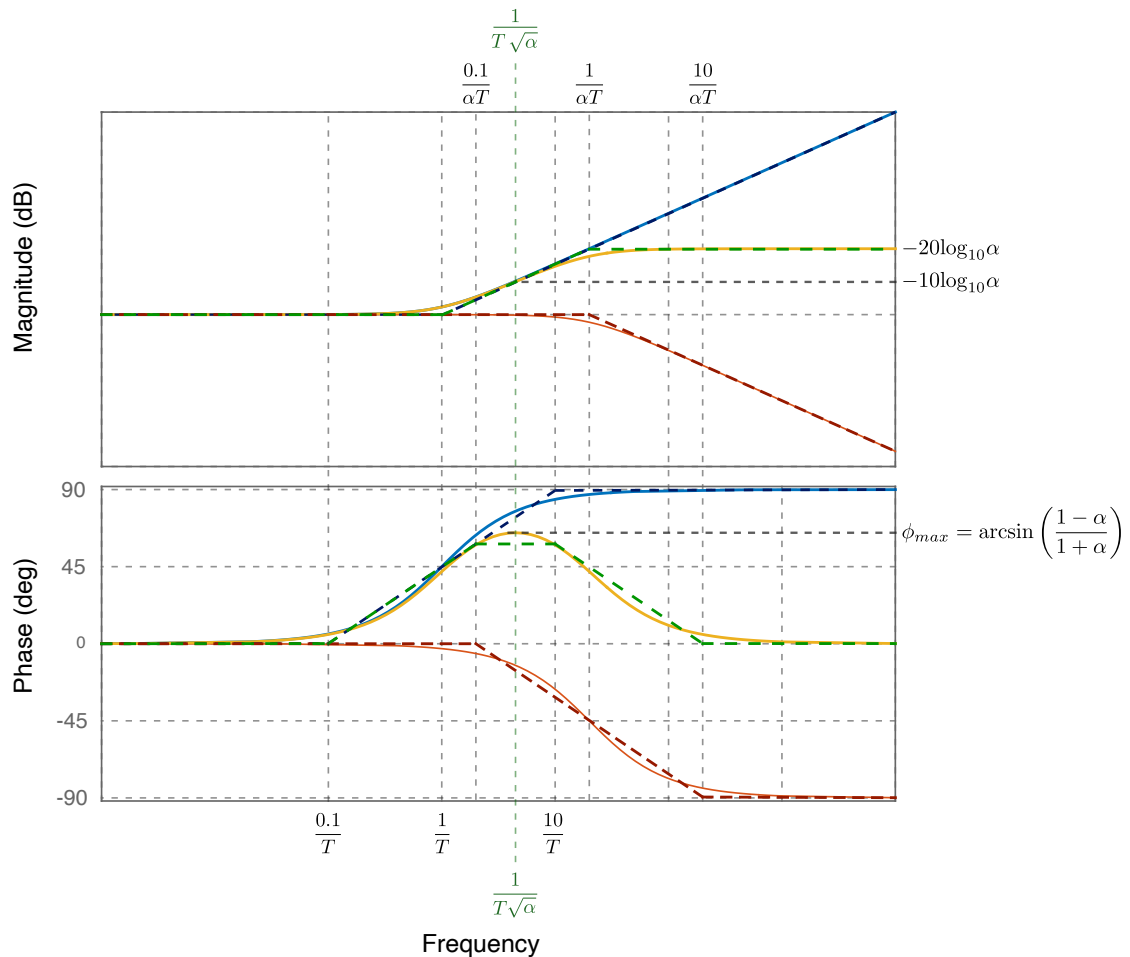
Lecturer: Asst. Prof. M. Mert Ankarali

19.1 Lead Compensator Design

The lead-compensator is a controller which has the form of a first-order high-pass filter

$$G_c(s) = K_c \frac{Ts + 1}{T\alpha s + 1} \quad \alpha \in (0, 1)$$

In general, we first design K_c based on the steady-state requirements of the system, and then design T and α based on the phase-margin requirement. First let's illustrate the bode-plots of a unity gain lead-compensator to understand how we can utilize its properties for the design process.



We can see that (both from the actual and approximate bode-plots), phase-lead compensator is a type of high-pass filter for which the cut-off (mid) frequency is $\omega_c = \frac{1}{\sqrt{\alpha}T}$. The low and high frequency gains are respectively 0 dB and $-20\log_{10}\alpha$ dB. The lead-compensator phase, ϕ_{max} , peaks at its cut-off frequency, and basically we will try to use this positive maximum phase shift to improve the phase margin of the feedback system.

Let's derive a formula for ϕ_{max} , which we will need during the design phase

$$\begin{aligned}\phi_{max} &= \angle[G_c(j\omega_c)] = \angle[1 + T\omega_c j] - \angle[1 + \alpha T\omega_c j] = \arctan(T\omega_c) - \arctan(\alpha T\omega_c) \\ &= \arctan\left(\frac{1}{\sqrt{\alpha}}\right) - \arctan(\sqrt{\alpha}) \\ &= \pi/2 - 2\arctan(\sqrt{\alpha}) \\ \sin \phi_{max} &= \cos(2\arctan(\sqrt{\alpha})) = 1 - 2[\sin(\arctan(\sqrt{\alpha}))]^2 = 1 - 2\frac{\alpha}{1+\alpha} \\ &= \frac{1-\alpha}{1+\alpha} \\ \phi_{max} &= \arcsin\left(\frac{1-\alpha}{1+\alpha}\right) \Rightarrow \alpha = \frac{1 - \sin \phi_{max}}{1 + \sin \phi_{max}}\end{aligned}$$

As expected when $\alpha = 1$, $\phi_{max} = 0$ since numerator and denominator time constants becomes equal in this case. Accordingly, we can see that

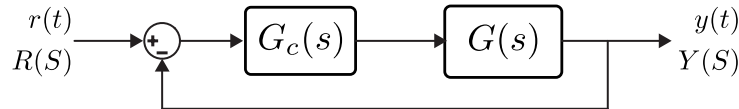
$$\alpha \searrow \Rightarrow \phi_{max} \nearrow$$

Theoretical maximum value for ϕ_{max} is 90° , however practically $\phi_{max} < 75^\circ$ for analog lead compensator circuits. Another important factor that we will need to pay attention is the gain-shift of the lead-compensator at the cut-off frequency

$$M_{dB}(j\omega_c) = -10\log_{10}\alpha \quad \& \quad |G(j\omega_c)| = \frac{1}{\sqrt{\alpha}}$$

We will illustrate the design process on an example

Ex: Consider the following feedback system illustrated below. It is given that $G(s) = \frac{1}{s(s+1)}$ and we want to design a lead-compensator, $G_c(s) = K_c \frac{Ts+1}{T\alpha s+1}$, such that unit-ramp steady-state error satisfies, $e_{ss} = 0.1$ and phase-margin of the compensated system satisfies $\phi_m^* = [45^\circ, 55^\circ]$.



Solution:

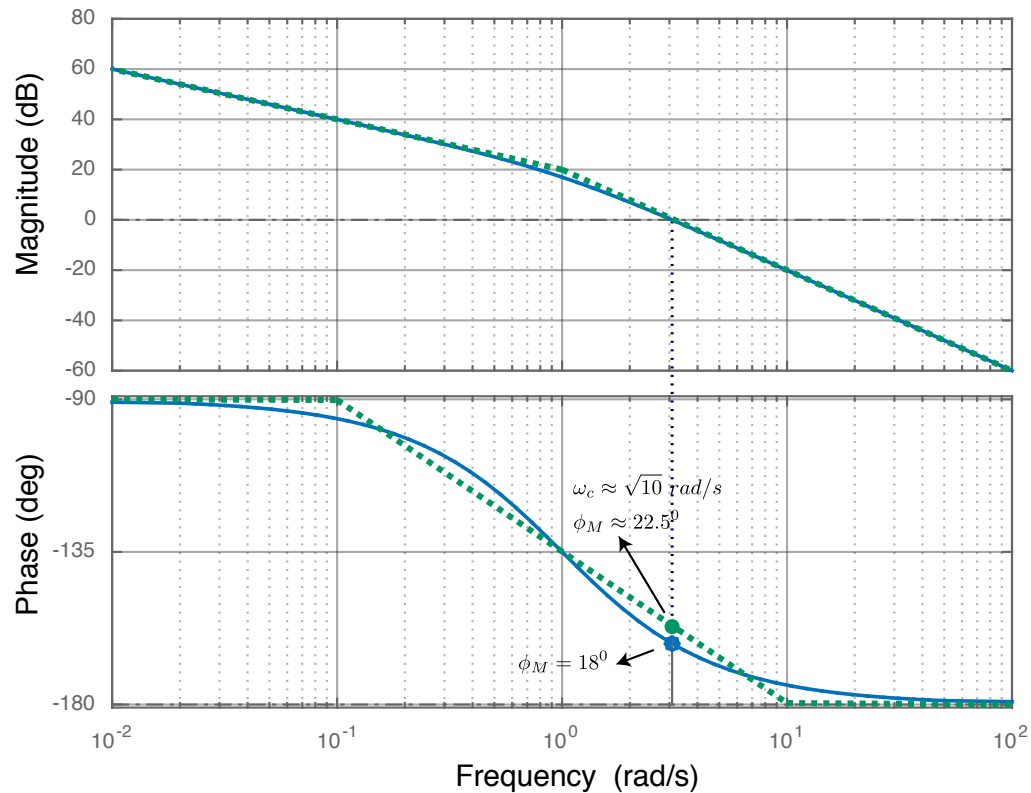
Step 1: We design/compute K_c based on the steady-state requirement on the unit-ramp error.

$$e_{ss} = \frac{1}{K_v} = \frac{1}{K_c} = 1 \rightarrow K_c = 10 \quad (19.1)$$

Step 2: Embed K_c into $G(s)$, i.e.

$$\bar{G}(s) = K_c G(s) = \frac{10}{s(s+1)},$$

draw the bode-plot for $\bar{G}(s)$, and compute the gain-crossover frequency, ω_{gc} , and the phase margin, ϕ_M , for the uncompensated $\bar{G}(s)$. The bode plot of $\bar{G}(s)$ and its approximation is illustrated in the Figure below.



If we concentrate on the approximate bode-plots, we can estimate the gain-crossover frequency and the phase-margin as

$$\begin{aligned}\omega_{gc} &\approx \sqrt{10} \text{ rad/s} = 3.16 \text{ rad/s} \\ \phi_m &\approx 22.5^\circ\end{aligned}$$

If we compute the actual values from the actual Bode plot, we obtain

$$\begin{aligned}\omega_{gc} &= 3.08 \text{ rad/s} \\ \phi_m &\approx 18^\circ\end{aligned}$$

We can see that approximation is pretty good for the ω_c , however a little crude for ϕ_m . So let's compute $\angle[G(j\omega_{gc})]$ for $\omega_{gc} \approx \sqrt{10}$ and estimate the phase-margin based on this frequency

$$\begin{aligned}\angle[G(j\omega_{gc})] &= \angle[G(j\sqrt{10})] = -90^\circ - \arctan(\sqrt{10}) = -162.5^\circ \\ \phi_M &\approx 17.5^\circ\end{aligned}$$

Step 3: Compute the required phase increment, $\Delta\phi$ to be added by the compensator and compute α

$$\begin{aligned}\phi_{max} &\approx \Delta\phi = \phi_M^* - \phi_M + \epsilon \\ \epsilon &= 5^0 - 10^0\end{aligned}$$

So for the given problem, we can compute ϕ_{max} and α as

$$\begin{aligned}\phi_{max} &\approx 47.5^0 - 17.5^0 + 7^0 = 37^0 \\ \alpha &= \frac{1 - \sin \phi_{max}}{1 + \sin \phi_{max}} \approx \frac{1}{4}\end{aligned}$$

Step 4: Estimate the “new” gain-crossover frequency, $\hat{\omega}_{gc}$, and place the peak point of the lead-compensator, at this estimated $\hat{\omega}_{gc}$.

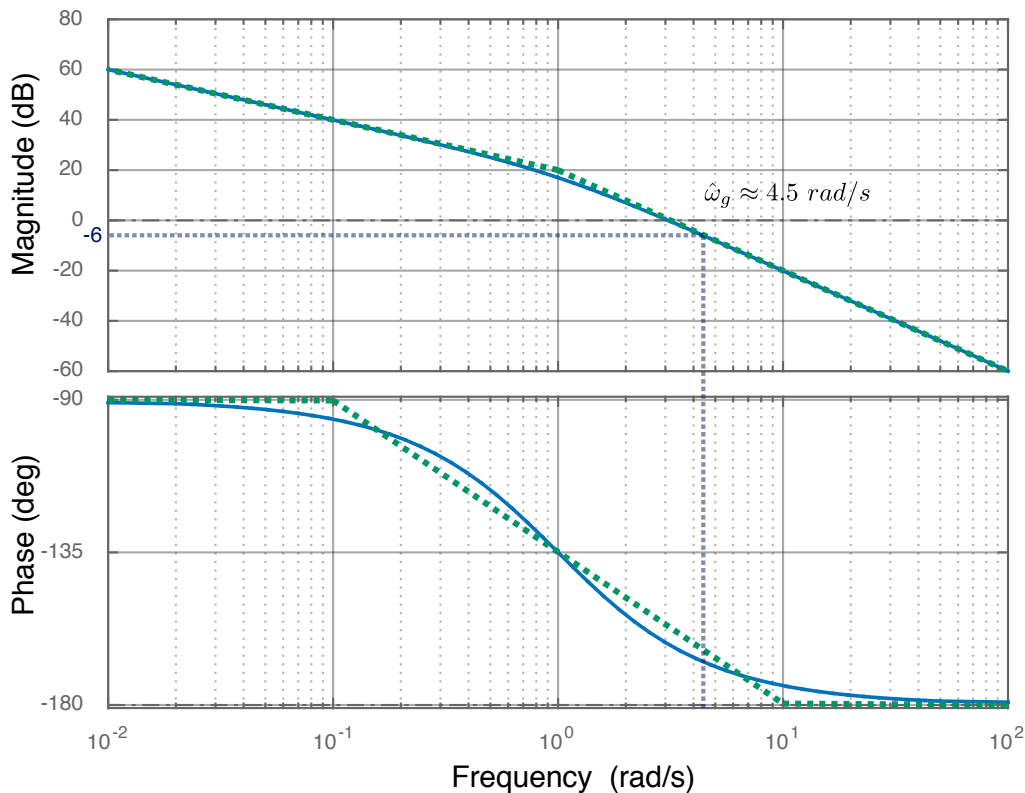
We already know that, lead-compensator at its center/cutoff frequency shifts the bode magnitude by $-10\log_{10}\alpha$ (or increases the gain by $\frac{1}{\sqrt{6}}$) and this causes a shift in gain-crossover frequency. For this reason, we can estimate the new gain-crossover frequency as the point where the bode magnitude of $\bar{G}(s)$ crosses $10 \log_{10}\alpha$, i.e.

$$M_{dB}(j\hat{\omega}_{gc}) = 10 \log_{10}\alpha \quad \text{or} \quad |G(j\hat{\omega}_{gc})| = \sqrt{\alpha}$$

In our problem,

$$M_{dB}(j\hat{\omega}_{gc}) \approx -6 \text{ dB} \quad \text{or} \quad |G(j\hat{\omega}_{gc})| = \frac{1}{2}$$

We can indeed estimate the new gain-crossover frequency graphically from the bode-plot. The figure below, illustrates how we can find the new-gain crossover frequency graphically



where $\hat{\omega}_{gc} \approx 4.5 \text{ rad/s}$. We can also estimate the new-gain crossover frequency numerically

$$\begin{aligned} |G(j\hat{\omega}_{gc})| &= \frac{1}{2} \\ \frac{100}{\hat{\omega}_{gc}^2 (\hat{\omega}_{gc}^2 + 1)} &= \frac{1}{4} \\ \hat{\omega}_{gc}^4 + \hat{\omega}_{gc}^2 - 400 &= 0 \\ \hat{\omega}_{gc} &\approx 400^{1/4} = 4.47 \text{ rad/s} \end{aligned}$$

which is very close to the graphical estimation.

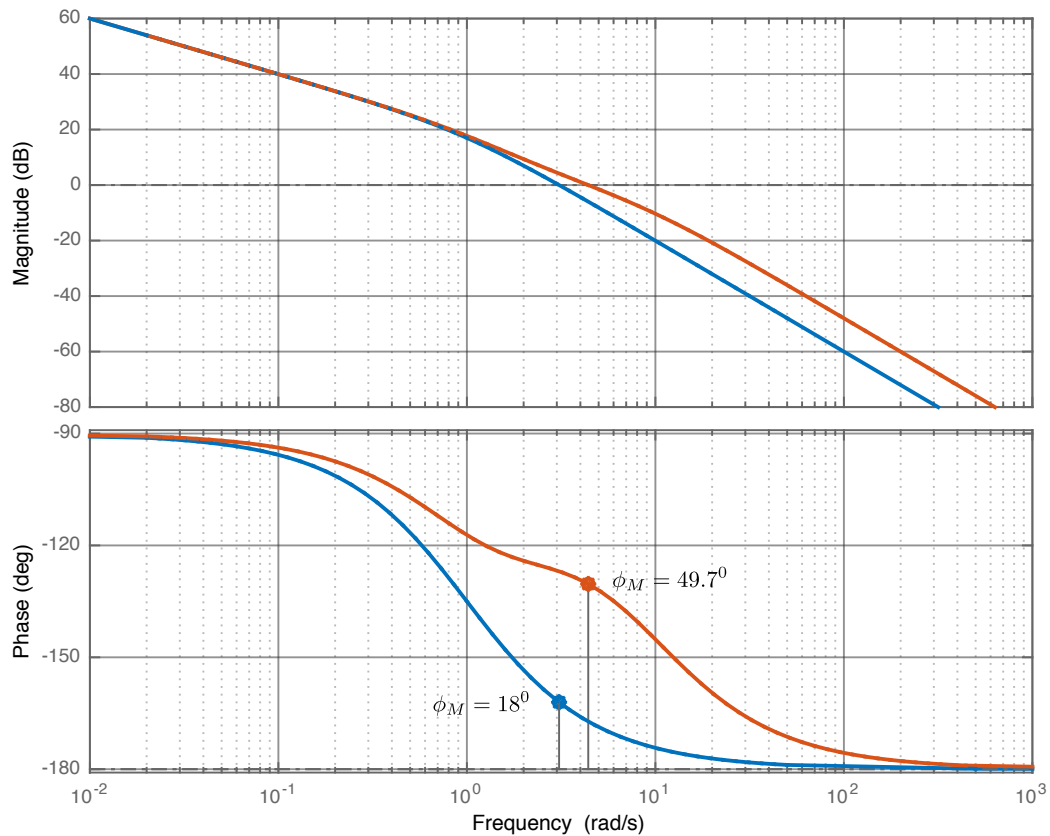
Step 5: Compute T using $\hat{\omega}_{gc}$ and check if the lead-compensator meets the phase-margin requirement. Otherwise, repeat the process with a higher $\Delta\phi$ angle.

$$\hat{\omega}_{gc} = \omega_c = \frac{1}{\sqrt{\alpha}T} \quad \Rightarrow \quad T = \frac{1}{\sqrt{\alpha}\hat{\omega}_{gc}}$$

In our example

$$T = \frac{1}{\frac{4.5}{2}} \approx 0.45 \quad \Rightarrow \quad G_c(s) = 10 \frac{0.45s + 1}{0.1125s + 1} = 40 \frac{s + 2.25}{s + 9}$$

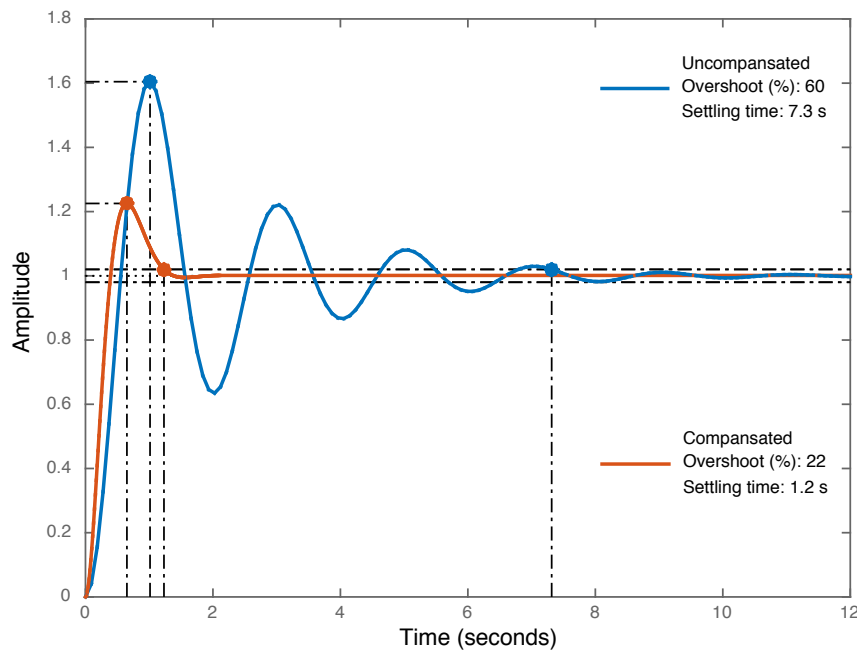
The Figure below illustrates the bode plots of both compensated and uncompensated ($\bar{G}(s) = K_c G(s)$) systems. Compensated systems has a phase margin of $\phi_m = 49.7^\circ$ which meets the requirements.



Now let's compute new phase margin directly using the new gain-crossover frequency

$$\begin{aligned}\phi_M &= -180^\circ + \angle[G_c(j4.45 \text{ rad/s})\bar{G}(j4.45 \text{ rad/s})] = -180^\circ + 37^\circ + \angle[\bar{G}(j4.45 \text{ rad/s})] \\ &= -180^\circ + 37^\circ - 90^\circ - \angle[j4.45 + 1] = -180^\circ + 37^\circ - 90^\circ - 77^\circ \\ &= 50^\circ\end{aligned}$$

which is consistent with the phase-margin on the bode-plot. In Figure below, we compare the closed-loop step responses of both uncompensated and compensated closed-loop transfer functions. We can clearly see that, the lead-compensator substantially improves both the settling time and over-shoot performance.



Phase-Lead Design Guideline:

1. Design/compute K_c based on the steady-state requirements
2. Embed K_c into $G(s)$, i.e. $\bar{G}(s) = K_c G(s)$, then draw the bode-plots for $\bar{G}(s)$, and compute the gain-crossover frequency, ω_{gc} , and the phase margin, ϕ_M , for the uncompensated $\bar{G}(s)$.
3. Compute the required phase increment, $\Delta\phi$ to be added by the compensator and design/compute α
4. Estimate the “new” gain-crossover frequency, $\hat{\omega}_{gc}$, and place the peak point of the lead-compensator, at this estimated $\hat{\omega}_{gc}$.
5. Check the phase-margin if it does not meet the requirements increase $\Delta\phi$ and repeat the process.

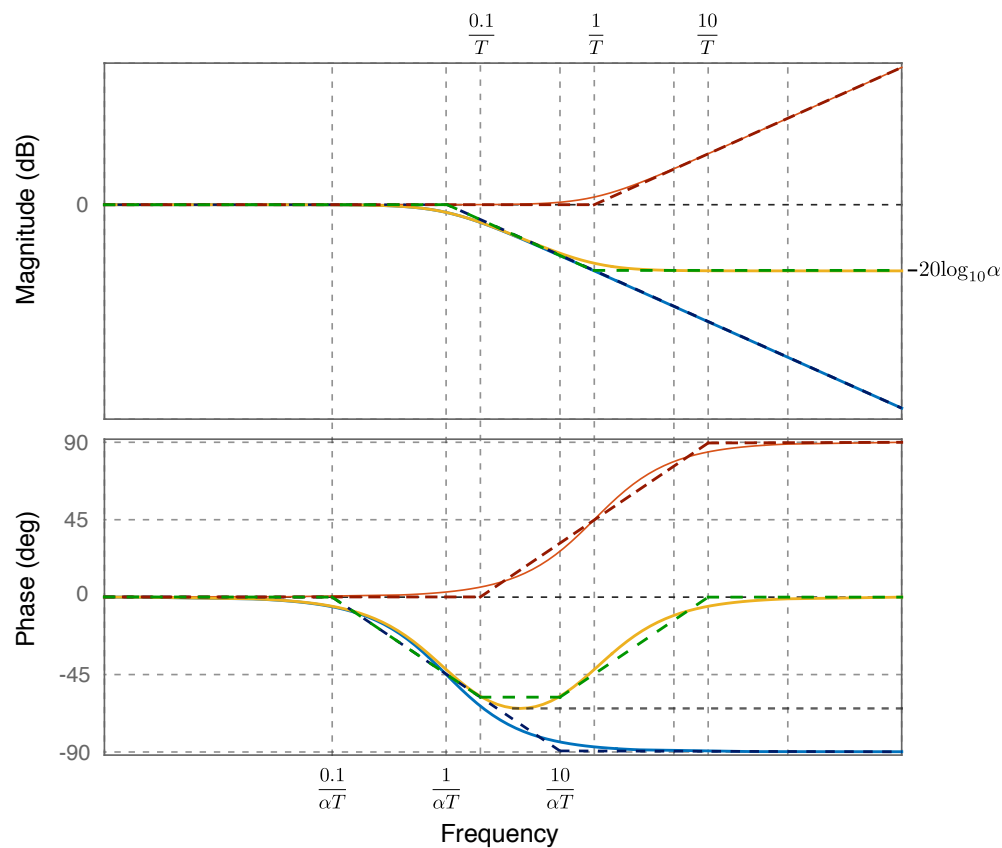
19.2 Lag Compensator Design

The lag-compensator is a controller which has the form of a first-order low-pass filter

$$G_c(s) = K_c \frac{Ts + 1}{T\alpha s + 1} \quad \alpha \in (1, \infty)$$

In general, we first design K_c based on the steady-state requirements of the system, then design α based on the phase-margin requirement, and finally choose a T such that phase-lag of the compensator does not interfere with the gain-crossover frequency.

First let's illustrate the bode-plots of a unity gain lag-compensator to understand how we can utilize its properties for the design process.



In lag-compensator design, we basically use the negative-gain shift of the compensator in the high-frequency region and we try to push the low and mid frequency region to the left (in frequency axis) such that they don't interfere with the gain-crossover frequency. For this reason, design process easier compared to the lead-compensator.

We will illustrate the lag-compensator design process on an the same example

Ex: Consider the feedback system that we analyzed previously in lead compensatory case. Plant is same, $G(s) = \frac{1}{s(s+1)}$. However now we want to design a lag-compensator, $G_c(s) = K_c \frac{Ts+1}{T\alpha s+1}$, $\alpha \in (1, \infty)$, such that unit-ramp steady-state error satisfies, $e_{ss} < 0.1$ and phase-margin of the compensated system satisfies $\phi_m^* > 40^\circ$.

Solution:

Step 1: Same as the lead-design, we design/compute K_c based on the steady-state requirement on the unit-ramp error.

$$e_{ss} = \frac{1}{K_v} = \frac{1}{K_c} = 1 \rightarrow K_c = 10 \quad (19.2)$$

Step 2: Embed K_c into $G(s)$, i.e.

$$\bar{G}(s) = K_c G(s) = \frac{10}{s(s+1)},$$

draw the bode-plot for $\bar{G}(s)$.

Step 3: Compute/find the required gain-crossover frequency, ω_{gc}^* , based on the required phase-margin, ϕ_M^* , and compute/find the magnitude of $G(s)$ at ω_{gc}^* , i.e. $|\bar{G}(j\omega_{gc}^*)|$ or $20\log_{10}|\bar{G}(j\omega_{gc}^*)|$.

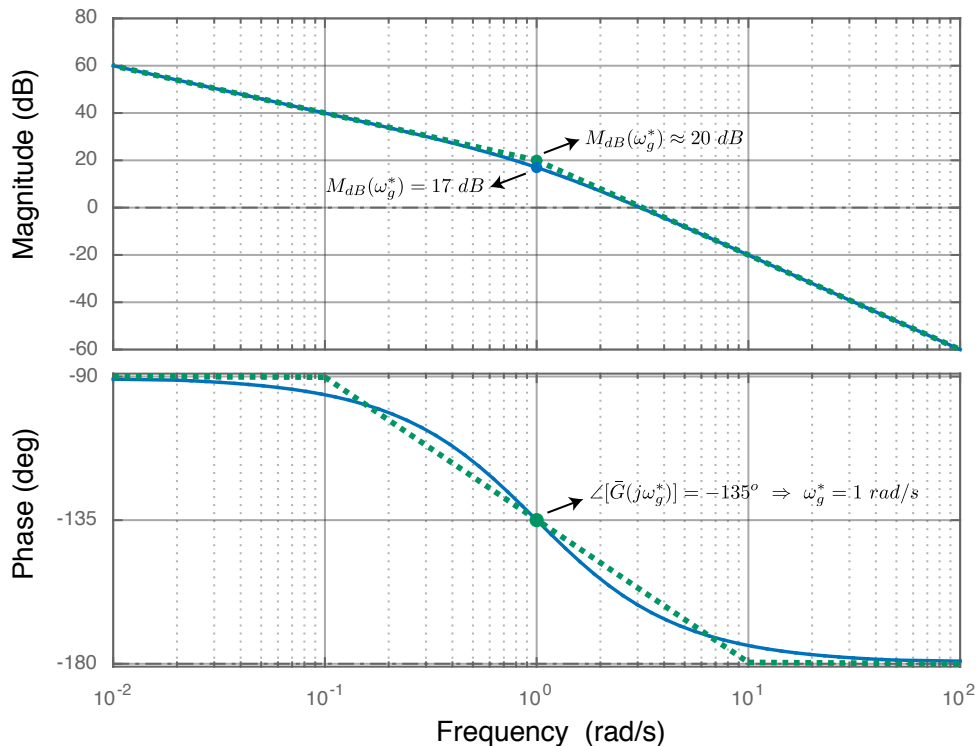
$$\begin{aligned} \angle[\bar{G}(j\omega_{gc}^*)] &= -180^\circ + \phi_M^* + \epsilon \\ \epsilon &\approx 5^\circ \end{aligned}$$

Note that at high-frequency region, lag-compensator acts like negative gain shift on magnitude plot while not affecting the phase response. Such a gain shift technically changes the gain-crossover frequency, which can be potentially used for changing the phase margin.

For our example, compensated gain-crossover frequency can be computed as

$$\angle[\bar{G}(j\omega_{gc}^*)] = -135^\circ \Rightarrow \omega_{gc}^* \approx 1 \text{ rad/s}$$

The bode plot of $\bar{G}(s)$, required gain-crossover frequency, and the magnitude of $\bar{G}(s)$ at the desired gain-crossover frequency are illustrated in the Figure below.



From the bode-plots we can observe that at the desired $\omega_{gc}^* \approx 1 \text{ rad/s}$, bode-plot approximation has a magnitude of 20 dB , whereas the magnitude in the actual bode plot is approximately 17 dB . In this example, let's use the magnitude of the approximation in the next Step.

Step 4: Compute α to compensate the the magnitude at the new-gain crossover frequency

$$20\log_{10}\alpha = 20\log_{10}|\bar{G}(j\omega_{gc}^*)| \quad \text{or} \quad \alpha = |\bar{G}(j\omega_{gc}^*)|$$

In our example, α , can be computed as

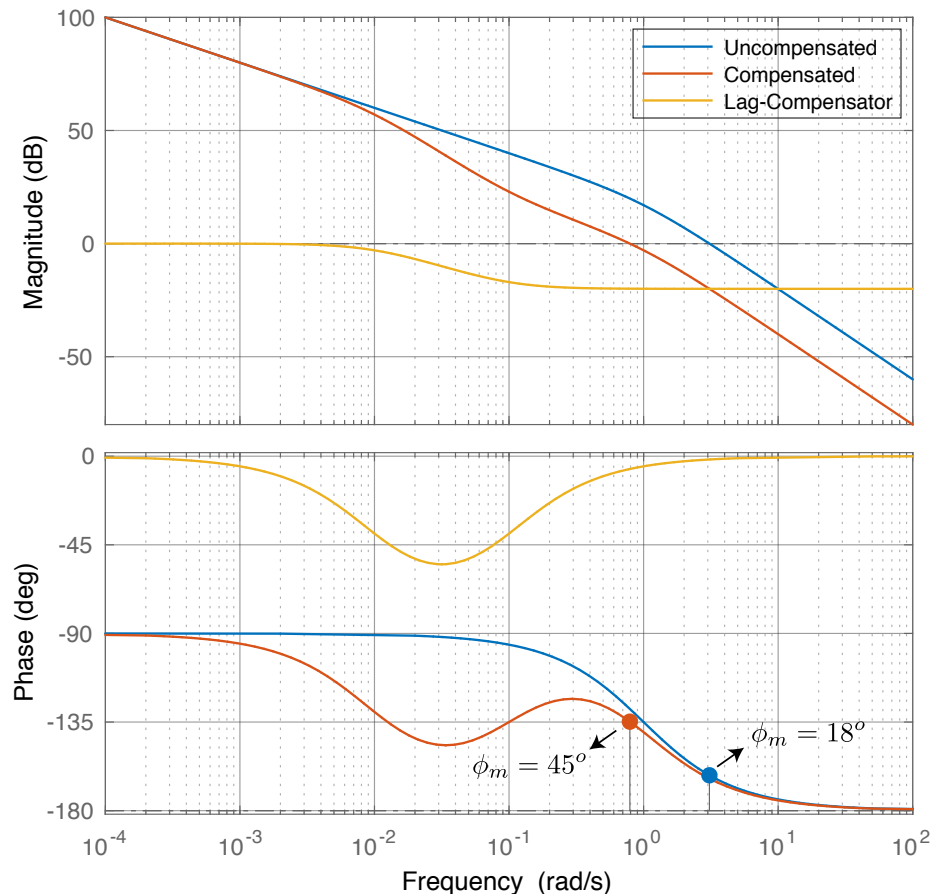
$$20\log_{10}\alpha \approx 20 \text{ dB} \Rightarrow \alpha \approx 10$$

Step 4: Choose T such that " $10/T \leq \omega_{gc}^*$ ". Note that $10/T$ is the frequency where the phase of the compensator approximately re-approaches to zero. This is required in order the negative phase bump of the compensator does not affect the phase-margin.

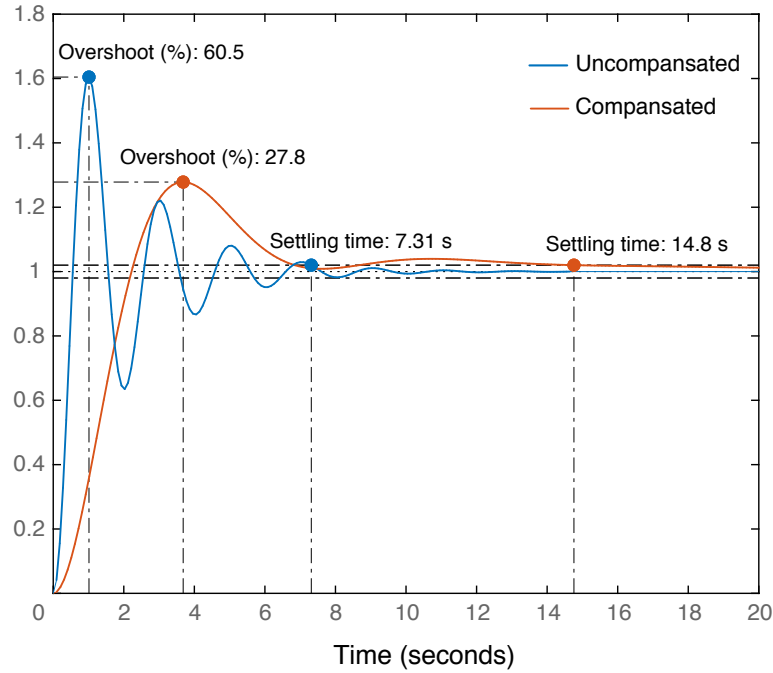
$$\frac{1}{T} \approx 0.1\omega_{gc}^* \Rightarrow T \approx 10$$

$$G_c(s) = 10 \frac{10s + 1}{100s + 1} = \frac{s + 0.1}{s + 0.01}$$

The Figure below illustrates the bode plots of the uncompensated ($\bar{G}(s) = K_c G(s)$) system, designed lag-compensator, and the compensated system. Compensated systems has a phase margin of $\phi_m = 45^\circ$ which meets the requirements.



In Figure below, we compare the closed-loop step responses of both uncompensated and compensated closed-loop transfer functions. We can see that, similar to the the lead-compensator, lag-compensator improves the over-shoot performance. However, the settling-time of the compensated system is approximately two times of the original system. This is major the drawback of the lag-compensator.



The reason behind the reduced convergence speed performance is that new gain-crossover frequency is less than the original gain-crossover frequency. PM is closely related with over-shoot performance. On the other hand gain-crossover frequency determines the band-width of the system which is closely related with the rise and settling times.

Phase-Lag Design Guideline:

1. Design/compute K_c based on the steady-state requirements
2. Compute/find the required gain-crossover frequency, ω_{gc}^* , based on the required phase-margin, ϕ_M^* , and compute/find the magnitude of $G(s)$ at ω_{gc}^*
3. Compute α to compensate the the magnitude at the new-gain crossover frequency
4. Chose T at such that " $10/T \leq \omega_{gc}^*$ ". This is required in order the negative phase bump of the compensator does not affect the phase-margin. Note that $10/T$ is the frequency where the phase of the compensator approximately re-approaches to zero.
5. Check the phase-margin if it does not meet the requirements change $\Delta\phi$ and repeat the process.

Lecture 20

Lecturer: Asst. Prof. M. Mert Ankarali

20.1 State-Space Analysis of LTI Systems

20.1.1 State-Space to TF

Let's first re-visit the conversion from a state-space representation to the transfer function representations for LTI systems.

Note that a SS representation of an n^{th} order LTI system has the form below.

$$\begin{aligned} \text{Let } x(t) &\in \mathbb{R}^n, \quad y(t) \in \mathbb{R}, \quad u(t) \in \mathbb{R}, \\ \dot{x}(t) &= Ax(t) + Bu(t), \\ y(t) &= Cx(t) + Du(t), \\ \text{where } A &\in \mathbb{R}^{n \times n}, \quad B \in \mathbb{R}^{n \times 1}, \quad C \in \mathbb{R}^{1 \times n}, \quad D \in \mathbb{R} \end{aligned}$$

In order to convert state-space to transfer function, we start with taking the Laplace transform of the both sides of the state-equation

$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t) \\ sX(s) &= AX(s) + BU(s) \\ sX(s) - AX(s) &= BU(s) \\ (sI - A)X(s) &= BU(s) \\ X(s) &= (sI - A)^{-1} BU(s) \end{aligned}$$

Now let's concentrate on the output equation

$$\begin{aligned} y(t) &= Cx(t) + Du(t) \\ Y(s) &= \left[C(sI - A)^{-1} B + D \right] U(s) \\ G(s) &= C(sI - A)^{-1} B + D \end{aligned}$$

Example: Let p be a pole of $G(s)$, then show that p is also an eigenvalue of A .

Solution: Let

$$G(s) = \frac{n(s)}{d(s)}$$

If p is a pole of $G(s)$, then $d(s)|_p = 0$. Now let's analyze the dependence of $G(s)$ to the state-space form.

$$\begin{aligned} G(s) &= [C(sI - A)^{-1}B + D] \\ (sI - G)^{-1} &= \frac{\text{Adj}(sI - A)}{\det(sI - A)} \\ G(s) &= \frac{C\text{Adj}(sI - G)B + D\det(sI - A)}{\det(sI - A)} \end{aligned}$$

If p is a pole of $G(s)$, then

$$\det(sI - A)|_{s=p} = 0$$

Obviously p is an eigenvalue of A .

20.1.2 Similarity Transformations

Now let's define a new "state-vector" $\hat{x}$ such that

$$\begin{aligned} Px(t) &= \hat{x}(t) \quad \text{where} \\ P &\in \mathbb{R}^{n \times n}, \det(P) \neq 0 \end{aligned}$$

Then we can transform the state-space equations using P as

$$\begin{aligned} P^{-1}\dot{\hat{x}}(t) &= AP^{-1}\hat{x}(t) + Bu(t) \quad , \quad y(t) = CP^{-1}\hat{x}(t) + Du(t) \\ \dot{\hat{x}} &= PAP^{-1}\hat{x}(t) + PBu(t) \quad , \quad y(t) = CP^{-1}\hat{x}(t) + Du(t) \end{aligned}$$

The "new" state-space representation is obtained as

$$\begin{aligned} \dot{\hat{x}}(t) &= \hat{A}\hat{x}(t) + \hat{B}u(t) \\ y(t) &= \hat{C}\hat{x}(t) + \hat{D}u(t) \\ \hat{A} &= PAP^{-1} \quad , \quad \hat{B} = PB \quad , \quad \hat{C} = CP^{-1} \quad , \quad \hat{D} = D \end{aligned}$$

Since there exist infinitely many non-singular $n \times n$ matrices, for a given LTI system, there exist infinitely many different but equivalent state-space representations.

Example: Show that $A \in \mathbb{R}^{n \times n}$ and $P^{-1}AP$, where $P \in \mathbb{R}^{n \times n}$ and $\det(P) \neq 0$, have the same characteristic equation

Solution:

$$\begin{aligned} \det(\lambda I - P^{-1}AP) &= \det(\lambda P^{-1}IP - P^{-1}AP) \\ &= \det(P^{-1}(\lambda I - A)P) \\ &= \det(P^{-1})\det(\lambda I - A)\det(P) \\ &= \det(P^{-1})\det(P)\det(\lambda I - A) \\ \det(\lambda I - P^{-1}AP) &= \det(\lambda I - A) \end{aligned}$$

20.1.2.1 Invariance of Transfer Functions Under Similarity Transformation

Consider the two different state-space representations

$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t) & \dot{\hat{x}}(t) &= \hat{A}\hat{x}(t) + \hat{B}u(t) \\ y(t) &= Cx(t) + Du(t) & y(t) &= \hat{C}\hat{x}(t) + \hat{D}u(t) \end{aligned}$$

where they are related with the following similarity transformation

$$Px(t) = \hat{x}(t), \quad \hat{G} = PAP^{-1}, \quad \hat{B} = PB, \quad \hat{C} = CP^{-1}, \quad \hat{D} = D$$

Let's compute the transfer function for the second representation

$$\begin{aligned} \hat{G}(s) &= \left[\hat{C} (sI - \hat{A})^{-1} \hat{B} + \hat{D} \right] \\ &= \left[CP^{-1} (sI - PAP^{-1})^{-1} PB + D \right] \\ &= \left[CP^{-1} (P(sI - A)P^{-1})^{-1} PB + D \right] \\ &= \left[CP^{-1}P(sI - A)^{-1}P^{-1}PB + D \right] \\ &= \left[C(sI - A)^{-1}B + D \right] \\ \hat{G}(s) &= G(s) \end{aligned}$$

20.1.3 Canonical State-Space Realizations

We know that for a given LTI system, there exist infinitely many different SS representations. We previously learnt some methods to convert a TF/ODE into State-Space form. We will now re-visit them and talk about the canonical state-space forms.

For the sake of clarity, derivations are given for a general 3^{rd} order LTI system.

20.1.4 Controllable Canonical Form

In this method of realization, we use the fact the system is LTI. Let's consider the transfer function of the system and let's perform some LTI operations.

$$\begin{aligned} Y(s) &= \frac{b_3s^3 + b_2s^2 + b_1s + b_0}{s^3 + a_2s^2 + a_1s + a_0}U(s) \\ &= (b_3s^3 + b_2s^2 + b_1s + b_0) \frac{1}{s^3 + a_2s^2 + a_1s + a_0}U(s) \\ &= G_2(s)G_1(s)U(s) \text{ where} \\ G_1(s) &= \frac{H(s)}{U(s)} = \frac{1}{s^3 + a_2s^2 + a_1s + a_0} \\ G_2(s) &= \frac{Y(s)}{H(s)} = b_3s^3 + b_2s^2 + b_1s + b_0 \end{aligned}$$

As you can see we introduced an intermediate variable $h(t)$ or with a Laplace transform of $H(s)$. First transfer function has static input dynamics, operates on $u(t)$, and produces an output, i.e. $h(t)$. Second

transfer function is a “non-causal” system and operates on $h(t)$ and produces output $y(t)$. If we write the ODEs of both systems we obtain

$$\begin{aligned}\ddot{h} &= -a_2\ddot{h} - a_1\dot{h} - a_0h + u \\ y &= b_3\ddot{h} + b_2\dot{h} + b_1\dot{h} + b_0h\end{aligned}$$

Now let the state-variables be $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} h \\ \dot{h} \\ \ddot{h} \end{bmatrix}$. Then, individual state equations take the form

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ \dot{x}_3 &= -a_2x_3 - a_1x_2 - a_0x_1 + u\end{aligned}$$

and the output equation take the form

$$\begin{aligned}y &= b_3(-a_2x_3 - a_1x_2 - a_0x_1 + u) + b_2x_3 + b_1x_2 + b_0x_1 \\ &= (b_0 - b_3a_0)x_1 + (b_1 - b_3a_1)x_2 + (b_2 - b_3a_2)x_3 + b_3u\end{aligned}$$

If we re-write the equations in matrix form we obtain the state-space representation as

$$\begin{aligned}\dot{x} &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \\ y &= \begin{bmatrix} (b_0 - b_3a_0) & (b_1 - b_3a_1) & (b_2 - b_3a_2) \end{bmatrix} x + [b_3] u\end{aligned}$$

If we obtain a state-space model from this approach, the form will be in *controllable canonical form*.

For a general n^{th} order transfer function controllable canonical form has the following A , B , C , & D matrices

$$\begin{aligned}A &= \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \\ C &= \begin{bmatrix} (b_0 - b_na_0) & (b_1 - b_na_1) & \cdots & (b_{n-1} - b_na_{n-1}) \end{bmatrix}, \quad D = b_n\end{aligned}$$

20.1.5 Observable Canonical Form

In this method will obtain a different minimal state-space realization, the form is called observable canonical form. The process is different and state-space structure will have a different topology. Let's start with a 3^{rd} transfer function and perform some grouping based on the s elements.

$$\begin{aligned}Y(s) &= \frac{b_3s^3 + b_2s^2 + b_1s + b_0}{s^3 + a_2s^2 + a_1s + a_0}U(s) \\ Y(s)(s^3 + a_2s^2 + a_1s + a_0) &= (b_3s^3 + b_2s^2 + b_1s + b_0)U(s) \\ s^3Y(s) &= b_3s^3U(s) + s^2(-a_2Y(s) + b_2U(s)) + s(-a_1Y(s) + b_1U(s)) + (-a_0Y(s) + b_0U(s))\end{aligned}$$

Let's multiply both sides with $\frac{1}{s^3}$ and perform further grouping

$$Y(s) = b_3 U(s) + \frac{1}{s} (-a_2 Y(s) + b_2 U(s)) + \frac{1}{s^2} (-a_1 Y(s) + b_1 U(s)) + \frac{1}{s^3} (-a_0 Y(s) + b_0 U(s))$$

$$Y(s) = b_3 U(s) + \frac{1}{s} \left[(-a_2 Y(s) + b_2 U(s)) + \frac{1}{s} \left\{ (-a_1 Y(s) + b_1 U(s)) + \frac{1}{s} (-a_0 Y(s) + b_0 U(s)) \right\} \right]$$

Let the Laplace domain representations of state variables $X(s) = \begin{bmatrix} X_1(s) \\ X_2(s) \\ X_3(s) \end{bmatrix}$ defined as

$$X_1(s) = \frac{1}{s} (-a_0 Y(s) + b_0 U(s))$$

$$X_2(s) = \frac{1}{s} \left\{ (-a_1 Y(s) + b_1 U(s)) + \frac{1}{s} (-a_0 Y(s) + b_0 U(s)) \right\}$$

$$X_3(s) = \frac{1}{s} \left[(-a_2 Y(s) + b_2 U(s)) + \frac{1}{s} \left\{ (-a_1 Y(s) + b_1 U(s)) + \frac{1}{s} (-a_0 Y(s) + b_0 U(s)) \right\} \right]$$

In this context output equation in s and time domains simply takes the form

$$Y(s) = X_3(s) + b_3 U(s) \quad \rightarrow \quad y(t) = x_3(t) + b_3 u(t)$$

Dependently the state equations (in s and time domains) take the form

$$sX_1(s) = -a_0 X_3(s) + (b_0 - a_0 b_3) U(s) \quad \rightarrow \quad \dot{x}_1 = -a_0 x_3 + (b_0 - a_0 b_3) u$$

$$sX_2(s) = X_1(s) - a_1 X_3(s) + (b_1 - a_1 b_3) U(s) \quad \rightarrow \quad \dot{x}_2 = x_1 - a_1 x_3 + (b_1 - a_1 b_3) u$$

$$sX_3(s) = X_2(s) - a_2 X_3(s) + (b_2 - a_2 b_3) U(s) \quad \rightarrow \quad \dot{x}_3 = x_2 - a_2 x_3 + (b_2 - a_2 b_3) u$$

If we re-write all equations in matrix form, we obtain the state-space representation as

$$\dot{x} = \begin{bmatrix} 0 & 0 & -a_0 \\ 1 & 0 & -a_1 \\ 0 & 1 & -a_2 \end{bmatrix} x + \begin{bmatrix} (b_0 - b_3 a_0) \\ (b_1 - b_3 a_1) \\ (b_2 - b_3 a_2) \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} x + [b_3] u$$

If we obtain a state-space model from this method, the form will be in *observable canonical form*. Thus we can call this representation also as *observable canonical realization*. This form and representation is the dual of the previous representation.

For a general n^{th} order system controllable canonical form has the following A , B , C , & D matrices

$$A = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & -a_{n-2} \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{bmatrix}, \quad B = \begin{bmatrix} (b_0 - b_n a_0) \\ (b_1 - b_n a_1) \\ \vdots \\ (b_{n-2} - b_n a_{n-2}) \\ (b_{n-1} - b_n a_{n-1}) \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 0 & \cdots & 0 & 1 \end{bmatrix}, \quad D = b_n$$

Diagonal Canonical Form

If the transfer function of the LTI system has distinct poles, we can expand it using partial fraction expansion

$$Y(s) = \left[b_3 + \frac{c_1}{s - p_1} + \frac{c_2}{s - p_2} + \frac{c_3}{s - p_3} \right] U(s)$$

Now let's concentrate on the candidate "state variables" and try to write state evaluation equations

$$X_1(s) = \frac{1}{s - p_1} U(s) \quad \rightarrow \quad \dot{x}_1 = p_1 x_1 + u$$

$$X_2(s) = \frac{1}{s - p_2} U(s) \quad \rightarrow \quad \dot{x}_2 = p_2 x_2 + u$$

$$X_3(s) = \frac{1}{s - p_3} U(s) \quad \rightarrow \quad \dot{x}_3 = p_3 x_3 + u$$

where as output equation can be derived as

$$y(t) = b_3 u(t) + c_1 x_1(t) + c_2 x_2(t) + c_3 x_3(t)$$

If we combine the state and output equations, we can obtain the state space form as

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \begin{bmatrix} p_1 & 0 & 0 \\ 0 & p_2 & 0 \\ 0 & 0 & p_3 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u(t) \\ y(t) &= \begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix} \mathbf{x}(t) + b_3 u(t) \end{aligned}$$

where

$$\mathbf{x} = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}, \quad A = \begin{bmatrix} p_1 & 0 & 0 \\ 0 & p_2 & 0 \\ 0 & 0 & p_3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix}, \quad D = b_3$$

The form obtained with this approach is called diagonal canonical form. Obviously, this form is not applicable for "some" systems that has repeated roots.

For a general n^{th} order system with distinct roots diagonal canonical form has the following A , B , C , & D matrices

$$\begin{aligned} A &= \begin{bmatrix} p_1 & 0 & \cdots & 0 & 0 \\ 0 & p_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & p_{n-1} & 0 \\ 0 & 0 & \cdots & 0 & p_n \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \\ 1 \end{bmatrix} \\ C &= \begin{bmatrix} c_1 & c_2 & \cdots & c_{n-1} & c_n \end{bmatrix}, \quad D = b_n \end{aligned}$$

Example: Given that

$$G(s) = \frac{s^2 + 8s + 10}{s^2 + 3s + 2}$$

find a controllable, observable, and diagonal canonical state-space representation of the given TF.

Solution:

If we follow the derivation of controllable canonical form for a second order system we obtain the following structure

$$\begin{aligned}\dot{x} &= \begin{bmatrix} 0 & 1 \\ -a_0 & -a_1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \\ y &= \begin{bmatrix} (b_0 - b_2 a_0) & (b_1 - b_2 a_1) \end{bmatrix} x + [b_2] u\end{aligned}$$

where

$$a_0 = 2, \quad a_1 = 3, \quad b_0 = 10, \quad b_1 = 8, \quad \& \quad b_2 = 1$$

Thus, the state-space representation takes the form

$$\begin{aligned}\dot{x} &= \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \\ y &= \begin{bmatrix} 8 & 5 \end{bmatrix} x + [1]u\end{aligned}$$

Observable canonical form is the dual of the controllable canonical form thus for the given system, we know that

$$\begin{aligned}A_{OCF} &= A_{CCF}^T = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix} \\ B_{OCF} &= C_{CCF}^T = \begin{bmatrix} 8 \\ 5 \end{bmatrix} \\ C_{OCF} &= B_{CCF}^T = \begin{bmatrix} 0 & 1 \end{bmatrix} \\ D_{OCF} &= D_{CCF} = [1]\end{aligned}$$

In order to find the diagonal canonical form, we need to perform partial fraction expansion

$$G(s) = \frac{s^2 + 8s + 10}{s^2 + 3s + 2} = 1 + \frac{3}{s+1} + \frac{2}{s+2}$$

then SS matrices for the diagonal canonical form can be simply derived as

$$\begin{aligned}A_{DCF} &= \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \\ B_{DCF} &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ C_{DCF} &= \begin{bmatrix} 3 & 2 \end{bmatrix} \\ D_{DCF} &= [1]\end{aligned}$$

Example: Consider the following general state-space representation

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t), \\ y(t) &= Cx(t) + Du(t)\end{aligned}$$

Now let's consider the following state-space representation

$$\begin{aligned}\dot{\bar{x}}(t) &= A^T \bar{x}(t) + C^T u(t), \\ y(t) &= B^T \bar{x}(t) + Du(t)\end{aligned}$$

Show that these two state-space representations results in same transfer function form

Solution: For the second representation we have

$$\begin{aligned}\bar{G}(s) &= \bar{C} (sI - \bar{A})^{-1} \bar{B} + D \\ &= B^T (sI - A^T)^{-1} C^T + D\end{aligned}$$

Since $\bar{G}(s)$ is a scalar quantity we can take its transpose

$$\begin{aligned}\bar{G}(s) &= [\bar{G}(s)]^T = [B^T (sI - A^T)^{-1} C^T + D]^T \\ &= (C^T)^T \left((sI - A^T)^{-1} \right)^T (B^T)^T + D \\ &= C \left((sI - A^T)^T \right)^{-1} B + D \\ &= C (sI - A)^{-1} B + D \\ \bar{G}(s) &= G(s)\end{aligned}$$

This result also shows that controllable and observable canonical representations are similar.

20.2 Stability & State-Space Representations

Let's consider the state-representation of an LTI system

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t)\end{aligned}$$

Given LTI system is called *asymptotically stable* if, with $u(t) = 0$ and $\forall x(0) \in \mathbb{R}^n$, we have

$$\lim_{t \rightarrow \infty} \|x(t)\| = 0$$

Theorem: A state-space representation is asymptotically stable if and only if all of the eigenvalues of the system matrix, A , have negative real parts, i.e.

$$\forall x_0 \in \mathbb{R}^n, \lim_{t \rightarrow \infty} \|x(t)\| = 0 \iff \forall i \in \{1, \dots, n\}, \operatorname{Re}\{\lambda_i\} < 0$$

Ex: Show that if a state-space representation is asymptotically stable then its transfer function representation is BIBO stable.

Solution: Previously we showed that if p is a pole of $G(s)$, then it is also an eigenvalue of A , since we can write $G(s)$ as

$$G(s) = \frac{CA\operatorname{Adj}(sI - A)B + D\det(sI - A)}{\det(sI - A)}$$

If the state-space representation is asymptotically stable then we know that for each pole, p_i of $G(s)$ we have $\operatorname{Re}\{p_i\} < 0$ which makes the input-output dynamics BIBO stable. In conclusion,

Asymptotically stable $\Rightarrow$ BIBO stable

Example: Consider the following state-space form of a CT system

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t) \\ y(t) &= \begin{bmatrix} 1 & -1 \end{bmatrix} x(t)\end{aligned}$$

- Is this system asymptotically stable?
- Is this system BIBO stable?

Solution: Let's compute the eigenvalues of A

$$\det\left(\begin{bmatrix} \lambda & -1 \\ -1 & \lambda \end{bmatrix}\right) = \lambda^2 - 1$$

$$\lambda_{1,2} = \pm 1$$

Thus the system is NOT Asymptotically Stable. Now let's check BIBO stability condition. First, compute the $G(s)$

$$\begin{aligned}
G(s) &= \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} s & -1 \\ -1 & s \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} s & 1 \\ 1 & s \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \frac{1}{s^2 - 1} \\
&= \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} s & 1 \\ 1 & s \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \frac{1}{s^2 - 1} \\
&= \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ s \end{bmatrix} \frac{1}{s^2 - 1} \\
&= \frac{-(s-1)}{s^2 - 1} \\
&= \frac{-1}{s+1}
\end{aligned}$$

Indeed, the system is BIBO Stable.

In conclusion

- Asymptotically Stable $\Rightarrow$ BIBO Stable
- BIBO Stable $\nRightarrow$ Asymptotically Stable

Lecture 21

Lecturer: Asst. Prof. M. Mert Ankarali

21.1 Reachability/Controllability

For a LTI continuous time state-space representation

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t)\end{aligned}$$

- A state x_d is said to be **reachable** if there exist a finite time interval $t \in [0, t_f]$ and an input signal defined on this interval, $u(t)$, that transfers the state vector $x(t)$ from the origin (i.e. $x(0) = 0$) to the state x_d within this time interval, i.e. $x(t_f) = x_d$.
- A state x_d is said to be **controllable** if there exist a finite time interval $t \in [0, t_f]$ and an input signal defined on this interval, $u(t)$, that transfers the state vector $x(t)$ from the initial state x_d (i.e. $x(0) = x_d$) to the origin this time interval, i.e. $x(t_f) = 0$.
- The set $\mathcal{R}$ of all reachable states is a linear (sub)space: $\mathcal{R} \subset \mathbb{R}^n$
- The set $\mathcal{C}$ of all controllable states is a linear (sub)space: $\mathcal{C} \subset \mathbb{R}^n$

For CT systems $x_d \in \mathcal{R}$ if and only if $x_d \in \mathcal{C}$, the Reachability and Controllability conditions are equivalent.

- If the reachable (or controllable) set is the entire state space, i.e., if $\mathcal{R} = \mathbb{R}^n$, then the system is called fully reachable/controllable.

One way of testing reachability/controllability is checking the rank (or the range space) the of reachability/controllability matrix

$$Q = [\ B \ AB \ \cdots \ A^{n-1}B \]$$

A CT system is reachable/controllable if and only of

$$\text{rank}(Q) = n$$

or equivalently

$$\text{Ra}(\mathbf{M}) = \mathbb{R}^n$$

Example: Consider the following system

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x + Bu$$

Analyze the controllability for two different input, B , matrices

$$B_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$B_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Solution:

Let's start with B_1 and derive the controllability matrix

$$Q_1 = [B_1 \quad AB_1] = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\det(Q_1) = 0$$

Thus the system is not fully controllable. Now let's get some intuition by writing the ODEs for individual states x_1 and x_2 , where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.

$$\dot{x}_1 = x_2 + u$$

$$\dot{x}_2 = 0$$

Let's go with reachability definition, which is based on starting from zero initial conditions and going to a desired state, we can derive the following relations

$$x_1(T) = \int_0^T u(t) dt$$

$$x_2(T) = 0$$

Neither the input nor the first state has an affect on the second state, so $x_2 = \alpha$ for $\alpha \neq 0$ is not controllable (or reachable). However it is also easy to see that we can always find $u(t)$ that will drive the x_1 to a desired state, $x_1^*(T)$.

Now let's analyze the case with B_2

$$Q_2 = [B_2 \quad AB_2] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\det(Q_2) = -1 \neq 0$$

Now the system is fully controllable.

Remark: A state-space representation that is in controllable canonical form is always fully controllable.

21.2 State-Feedback & Pole Placement

Given a continuous-time state-evolution equation

$$\dot{x}(t) = Ax(t) + Bu(t)$$

If direct measurements of all of the states of the system (e.g. $y(t) = x(t)$) are available, one of the most popular and powerful control method is the linear state feedback control,

$$u(t) = -Kx(t)$$

or when there is a reference input

$$u(t) = r(t) - Kx(t)$$

which can be thought as a generalization of P controller to the vector form. Under this control law, state-equations for the closed-loop system takes the form

$$\begin{aligned}\dot{x}(t) &= Ax(t) + B(r(t) - Kx(t)) \\ \dot{x}(t) &= (A - BK)x(t) + Br(t)\end{aligned}$$

The system matrix of this autonomous system is $\hat{A} = A - BK$. Important questions is how to choose K . Note that

$$K \in \mathbb{R}^n \quad \text{Single - Input}$$

As in all of the control design techniques, the most critical criterion is stability, thus we want all of the eigenvalues to be in the open-left-half s-plane. However, we know that there could be different requirements on the poles/eigenvalues of the system.

The fundamental principle of “pole-placement” design is that we first define a desired closed-loop eigenvalue set $\mathcal{E}^* = \{\lambda_1^*, \dots, \lambda_n^*\}$, and then if possible we choose K^* such that the closed-loop eigenvalues match the desired ones.

The necessary and sufficient condition on arbitrary pole-placement is that the system should be fully Controllable/Reachable.

In Pole-Placement, first step is computing the desired characteristic polynomial.

$$\begin{aligned}\mathcal{E}^* &= \{\lambda_1^*, \dots, \lambda_n^*\} \\ p^*(s) &= (s - \lambda_1^*) \cdots (s - \lambda_n^*) \\ &= s^n + a_1^* s^{n-1} + \cdots + a_{n-1}^* s + a_n^*\end{aligned}$$

Then we tune K such that

$$\det(sI - (A - BK)) = p^*(s)$$

21.2.1 Direct Design of State-Feedback Gain

If n is small, the most efficient method could be the direct design.

Example: Consider the following system

$$\dot{x} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

Design a state-feedback rule such that poles are located at $\lambda_{1,2} = -1$

Solution: Desired characteristic equation can be computed as

$$p^*(s) = s^2 + 2s + 1$$

Let $K = \begin{bmatrix} k_1 & k_2 \end{bmatrix}$, then the characteristic equation of $\hat{A}$ can be computed as

$$\begin{aligned} \det(sI - (A - BK)) &= \det\left(\begin{bmatrix} s - 1 + k_1 & k_1 \\ k_2 & s - 2 + k_2 \end{bmatrix}\right) \\ &= s^2 + s(k_1 + k_2 - 3) + (2 - 2k_1 - k_2) \end{aligned}$$

If we match the equations

$$\begin{aligned} s^2 + s(k_1 + k_2 - 3) + (2 - 2k_1 - k_2) &= s^2 + 2s + 1 \\ k_1 + k_2 &= 5 \\ 2k_1 + k_2 &= 1 \\ k_1 &= -4 \\ k_2 &= 9 \end{aligned}$$

Thus $K = \begin{bmatrix} -4 & 9 \end{bmatrix}$.

21.2.2 Design of State-Feedback Gain Using Controllable Canonical Form

Let's assume that the state-space representation is in controllable canonical form and we have access to the all states of this form

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u$$

Let $K = \begin{bmatrix} k_n & \cdots & k_1 \end{bmatrix}$, then closed-loop system takes the form

$$\begin{aligned} \dot{x} &= \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} (r - \begin{bmatrix} k_n & \cdots & k_1 \end{bmatrix} x) \\ \dot{x} &= \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -(a_n + k_n) & -(a_{n-1} + k_{n-1}) & -(a_{n-2} + k_{n-2}) & \cdots & -(a_1 + k_1) \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} r \end{aligned}$$

Let $p^*(s) = s^2 + a_1^*s + \cdots + a_{n-1}^*s + a_n^*$, then computation of K is super straight-forward

$$K = \begin{bmatrix} (a_n^* - a_n) & \cdots & (a_1^* - a_1) \end{bmatrix}$$

However, what if the system is not in controllable canonical form. We can find a transformation which finds the controllable canonical representation.

The controllability matrix of a state-space representation is given as

$$Q = [B \mid AB \mid \cdots \mid A^{n-1}B]$$

Let's define a transformation matrix T as follows:

$$\begin{aligned} T &= QW \quad , \quad x(t) = T\hat{x}(t) \\ \dot{\hat{x}} &= [T^{-1}AT] \hat{x} + T^{-1}Bu \end{aligned}$$

where

$$W = \begin{bmatrix} a_{n-1} & a_{n-2} & \cdots & a_1 & 1 \\ a_{n-2} & a_{n-3} & \cdots & 1 & 0 \\ \vdots & \vdots & \ddots & & \vdots \\ a_1 & 1 & & & 0 \\ 1 & 0 & \cdots & & 0 \end{bmatrix}$$

Then it is given that

$$\begin{aligned} T^{-1}AT = \hat{A} &= \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix} \\ T^{-1}B = \hat{B} &= \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \end{aligned}$$

We know how to design a state-feedback gain $\hat{K}$ for the controllable canonical form. Given $\hat{K} u(t)$ is given as

$$\begin{aligned} u(t) &= r(t) - \hat{K}\hat{x}(t) \\ &= r(t) - \hat{K}T^{-1}\hat{x}(t) \\ K &= \hat{K}T^{-1} \end{aligned}$$

Example 2: Consider the following system

$$\dot{x} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

Design a state-feedback rule using the controllable canonical form approach, such that poles are located at $\lambda_{1,2} = -1$

Solution: Characteristic equation of A can be derived as

$$\det \left(\begin{bmatrix} s-1 & 0 \\ 0 & s-2 \end{bmatrix} \right) = s^2 - 3s + 2$$

The controllability matrix can be computed as

$$Q = [\ B \ | \ AB \] = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

The matrix W can be computed as

$$W = \begin{bmatrix} -3 & 1 \\ 1 & 0 \end{bmatrix}$$

Transformation matrix, T and its inverse T^{-1} can be computed as

$$T = QW = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -3 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ -1 & 1 \end{bmatrix}$$

$$T^{-1} = \begin{bmatrix} -1 & 1 \\ -1 & 2 \end{bmatrix}$$

Given that desired characteristic polynomial is $p^*(s) = s^2 + 2s + 1$, $\hat{K}$ of controllable canonical form can be computed as

$$\begin{aligned} \hat{K} &= [\ -a_2 \quad -a_1 \] \\ &= [\ 1 - 2 \quad 2 - (-3) \] = [\ -1 \quad 5 \] \end{aligned}$$

Finally K can be computed as

$$\begin{aligned} K &= \hat{K}T^{-1} = [\ -1 \quad 5 \] \begin{bmatrix} -1 & 1 \\ -1 & 2 \end{bmatrix} \\ &= [\ -4 \quad 9 \] \end{aligned}$$

As expected this is the same result with the one found with Direct-Method.

21.3 Observability

It turns out that it is more natural to think in terms of “un-observability” as reflected in the following definitions.

- A state x_o of a finite dimensional CT linear dynamical system is said to be unobservable, if with $x(0) = x_o$ and for every $u(t)$ we get the same $y(t)$ as we would with $x(0) = 0$.

The set $\bar{\mathcal{O}}$ of all unobservable states is a linear (sub)space: $\bar{\mathcal{O}} \subset \mathbb{R}^n$

- If the unobservable set only contains the origin, i.e., if $\bar{\mathcal{O}} = \{0\}$,
- If the dimension of unobservable subspace is equal to 0, $\dim(\bar{\mathcal{O}}) = 0$,
- If any initial condition, $x(0)$ or $x[0]$, can be uniquely determined from input-output measurement,

then the system is called Observable.

21.3.0.1 Test of Observability on CT Systems

One way of testing Observability of CT systems is checking the rank (or the range space, or null space) the of the Observability matrix

$$\mathbf{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

A CT system is Observable if and only of

$$\text{rank}(\mathbf{O}) = n$$

or equivalently

$$\text{Ra}(\mathbf{O}) = \mathbb{R}^n$$

or equivalently

$$\dim(\mathcal{N}(\mathbf{O})) = 0$$

Remark: A state-space representation that is in observable canonical form is always fully Observable

Remark: A state-space representation is called minimal if it is both fully Controllable and Observable.

21.4 Luenberger Observer

In general the state, $x(t)$, of a system is not accessible and *observers, estimators, filters*) have to be used to extract this information. The output, $y(t)$, represents the measurements which is a function of $x(t)$ and $u(t)$.

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du\end{aligned}$$

A Luenberger observers is built using a “simulated” model of the system and the errors caused by the mismatched initial conditions $x_0 \neq \hat{x}_0$ (or other types of perturbations) are reduced by introducing output error feedback.

Let’s assume that the state vector of the simulated system is $\hat{x}$, then the state space equation of this synthetic system takes the form

$$\begin{aligned}\dot{\hat{x}} &= A\hat{x} + Hu \\ \hat{y} &= C\hat{x} + Du\end{aligned}$$

Note that since u is the input that is supplied by the controller, we assume that it is known apriori. If $x(0) = \hat{x}(0)$ and when there is no model mismatch or uncertainty in the system then we expect that $x(t) = \hat{x}(t)$ and $y(t) = \hat{y}(t)$ for all $t \in \mathbb{R}^+$. When $x(0) \neq \hat{x}(0)$, then we should observe a difference between the measured and predicted output $y(t) \neq \hat{y}(t)$ (if the initial condition is not in the unobservable sub-space). The core idea in Luenberger observer is feeding the error in the output prediction $y(t) - \hat{y}(t)$ to the simulated system via a linear feedback gain.

$$\begin{aligned}\dot{\hat{x}} &= A\hat{x} + Hu + L(y - \hat{y}) \\ \hat{y} &= C\hat{x} + Du\end{aligned}$$

In order to understand how a Luenberger observer works and to choose a proper observer gain L , we define an error signal $e = x - \hat{x}$. The dynamics w.r.t e can be derived as

$$\begin{aligned}\dot{e} &= \dot{x} - \dot{\hat{x}} \\ &= (Ax + Hu) - (A\hat{x} + Hu + L(y - \hat{y})) \\ \dot{e} &= (A - LC)e\end{aligned}$$

where $e(0) = x(0) - \hat{x}(0)$ denotes the error in the initial condition.

If the matrix $(A - LC)$ is stable then the errors in initial condition will diminish eventually. Moreover, in order to have a good observer/estimator performance the observer convergence should be sufficiently fast.

Observer Gain & Pole Placement

Similar to the state-feedback gain design, the fundamental principle of “pole-placement” Observer design is that we first define a desired closed-loop eigenvalue set and compute the associated desired characteristic polynomial.

$$\begin{aligned}\mathcal{E}^* &= \{\lambda_1^*, \dots, \lambda_n^*\} \\ p^*(s) &= (s - \lambda_1^*) \cdots (s - \lambda_n^*) \\ &= s^n + a_0^* z^{n-1} + \cdots + a_{n-2}^* z + a_{n-1}^*\end{aligned}$$

The necessary and sufficient condition on arbitrary observer pole-placement is that the system should be fully Observable. Then, we can tune L such that

$$\det(sI - (A - LC)) = p^*(s)$$

Direct Design of Observer Gain

If n is small, the most efficient method could be the direct design.

Example: Consider the following DT system

$$\begin{aligned}\dot{x} &= \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u \\ y &= \begin{bmatrix} 1 & -1 \end{bmatrix} x\end{aligned}$$

Design an observer such that estimator poles are located at $\lambda_{1,2} = -5$ (Dead-beat Observer)

Solution: Desired characteristic equation can be computed as

$$p^*(s) = s^2 + 10s + 25$$

Let $L = \begin{bmatrix} l_2 \\ l_1 \end{bmatrix}$, then the characteristic equation of $(G - LC)$ can be computed as

$$\begin{aligned}\det(sI - (A - LC)) &= \det\left(\begin{bmatrix} s - 1 + l_2 & -l_2 \\ l_1 & s - 2 - l_1 \end{bmatrix}\right) \\ &= s^2 + s(l_2 - l_1 - 3) + (l_1 - 2l_2 + 2)\end{aligned}$$

If we match the equations

$$\begin{aligned}l_2 - l_1 &= 13 \\ l_1 - 2l_2 &= 23 \\ l_2 &= -49 \\ l_1 &= -36\end{aligned}$$

Thus $L = \begin{bmatrix} -49 \\ -36 \end{bmatrix}$